$\mathcal{N} = 4$ super Yang–Mills theory in Costello's formalism

Step 1: one-loop counterterms, local operators and their dilatation at the UV boundary

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Chapter 0

Overview

This manuscript constructs the one-loop quantum Batalin–Vilkovisky theory of four-dimensional $\mathcal{N} = 4$ super Yang–Mills theory on Euclidean $\mathbb{R}^4$, together with the transformations of its local operators under the whole superconformal algebra, in the renormalisation formalism of Costello [C07] and Costello–Gwilliam [CG2]. The single regulator is a lower cutoff $\varepsilon > 0$ on Schwinger time, the ultraviolet (UV) boundary; the upper end of every Schwinger integral is ∞, so that ε is the single scale of the construction. This overview states what is constructed, where the construction needs care, how the chapters are organised, and the results of this first step.

0.1 What is constructed

The fields are those of ten-dimensional super Yang–Mills theory reduced to $\mathbb{R}^4$ [ESW, Prop 2.20, §10.3], complexified. Write V for the complexification of $\mathbb{R}^4$ and $W_{\mathbb{R}}^{\perp}$ for the six transverse directions, and $\mathfrak{g}$ for the complexified gauge Lie algebra with its invariant pairing $(-,-)$, which carries the coupling. The gauge field A is a ten-component one-form with values in $\mathfrak{g}$: its components along V form the connection $A|_V$, and its components $\iota_w A$, $w \in W_{\mathbb{R}}^{\perp}$, are the six scalars. The gaugino λ takes values in the ten-dimensional semi-spin representation Σ, which in four dimensions is $S_+ \otimes W \oplus S_- \otimes W^*$, with $S_\pm$ the two Weyl spinor representations and W the four-dimensional R-symmetry space [ESW, §10]; spinors stay index free throughout. The ghost c and the antifields A^*, λ^*, c^* complete the fields [ESW, Def 3.2, Rem 1.36]. The action S is [ESW, (31)] reduced to four dimensions; its quadratic part defines the linear differential Q, the rest is the interaction I, and $\{-,-\}$ is the odd Poisson bracket [C07, Lemma 4.3.2].

Costello's construction needs three more objects. A gauge-fixing operator Q^{GF} [C07, §1.1, Def 4.3.3], here of Feynman type, makes $H = [Q, Q^{GF}]$ [C07, Def 4.3.3(4)] the Laplacian on every field. The heat kernel K_t of H at Schwinger time t [C07, §6.5] gives the propagator

$$P(\varepsilon,\infty) = \int_\varepsilon^\infty (Q^{GF} \otimes 1) K_t \, dt,$$

and Feynman graphs with these edges assemble into the renormalisation group flow of an even functional J,

$$W\big(P(\varepsilon,\infty), J\big) = \hbar \log\Big(e^{\hbar \partial_{P(\varepsilon,\infty)}} e^{J/\hbar}\Big) \qquad \text{[CG2, §7.2.7]},$$

with $\partial_{P(\varepsilon,\infty)}$ the second-order operator contracting two fields with the kernel $P(\varepsilon,\infty)$ [C07, §6.4]; this is the functional called Γ in [C07, Def 6.6.1]. As in Theorem B of [C07], whose analogue on $\mathbb{R}^4$ with upper scale ∞ Chapter 4 proves at one loop, the counterterm $I^{CT}[\varepsilon]$ is the singular part of the graph expansion, extracted graph by graph, with coefficients in the span of ε^{-1} and $\log \varepsilon$ at one loop; it is local by power counting in the Schwinger times, it is unique among the translation-invariant and

$\mathfrak{g}$-invariant local functionals with such coefficients, and the order-$\hbar$ part of $W\big(P(\varepsilon,\infty), I - I^{CT}[\varepsilon]\big)$ has a limit as $\varepsilon \to 0$, as plane-wave values at generic leg momenta. A local operator O, for instance $\int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$ with ω a compactly supported four-form [CG2, Lemma 10.1.1.3], enters through a square-zero parameter δ [CG2, Def 8.3.1.1]: the counterterm $\partial_\delta I^{CT}[\varepsilon]$ of the family $I + \delta O$ defines the local operator at the UV end, $O(\varepsilon) = O - \partial_\delta I^{CT}[\varepsilon]$, and its quantum observable is

$$O[\infty] = \lim_{\varepsilon\to 0} \partial_\delta W\big(P(\varepsilon,\infty),\, I - I^{CT}[\varepsilon] + \delta\, O(\varepsilon)\big) \qquad \text{[CG2, Def 8.3.1.3].}$$

The superconformal algebra $\mathfrak{psl}(4|4)$ [SW, (7), (17)] acts through an equivariant action $S+S^{(1)}+S^{(2)}$, a Maurer–Cartan element of the Chevalley–Eilenberg complex of [ESW, Def 1.26–1.27] with differential d_{CE}. Building $O[\infty]$ from the equivariant interaction, the transformation of O along a generator $x_1 \in \mathfrak{psl}(4|4)$ is the component $[(d_{CE}+Q)O[\infty]](x_1)$. Along the dilatation Δ, the generator $2E$ of [SW, (17)] with E the Euler vector field [SW, (3)], it is the anomalous dimension; along the supercharges Q^i_α, $\bar{Q}_{\dot\alpha i}$ and the special supercharges S^α_i, $\bar{S}^{\dot\alpha i}$ (the names of [SW, (8)]) it is the quantum-corrected supersymmetry; along the special conformal generators it is the primary condition. The program therefore has two outputs: the quantised interaction, with its counterterms and its quantum master equation, and, for every local operator, its observable and its transformations along all of $\mathfrak{psl}(4|4)$.

0.2 Three points that need care

Operators. Inserting a composite operator produces divergences, and the limit defining $O[\infty]$ exists for many choices of the functional $O(\varepsilon)$. At scale ∞, however, $K_\infty = \lim_{t\to\infty} K_t = 0$ on $\mathbb{R}^4$, and $P(0,\infty) = \lim_{\varepsilon\to 0} P(\varepsilon,\infty)$ [C07, §1.2] is a contracting homotopy for Q: for compactly supported α,

$$Q\big(P(0,\infty)\star\alpha\big) + P(0,\infty)\star(Q\alpha) = \alpha,$$

where $P(0,\infty)\star$ is the operator with kernel $P(0,\infty)$ [C07, §6.3]. On moment-free Schwartz sections, whose Fourier transforms vanish to infinite order at $p = 0$, every Q-closed continuous polynomial functional without constant term is Q-exact (Chapter 3); on compactly supported sections Q has in addition nonzero classes, among them the pairings with polynomial zero modes of Q, finitely many in each polynomial degree and infinitely many in all. An arbitrary $O(\varepsilon)$ therefore gives the zero class on moment-free sections. The content of an operator is the locality of $O(\varepsilon)$: $O(\varepsilon) = O - \partial_\delta I^{CT}[\varepsilon]$ with local coefficients in the span of ε^{-1} and $\log\varepsilon$. Locality and the existence of the limit fix the singular part uniquely, and the finite part is the scheme. An equation expressing $[(d_{CE}+Q)O[\infty]](x_1)$ as the scale-∞ observable of a second functional has content because that functional is required to be local, and for an operator whose observable is a cocycle, $Q\,O[\infty] = 0$ (Chapter 5), anomalous dimensions are read off in the cohomology of local functionals.

Counterterms. The singular part $I^{CT}[\varepsilon]$ is unique among the translation-invariant and $\mathfrak{g}$-invariant local functionals (Chapter 4). The cutoff on every edge breaks gauge invariance and superconformal symmetry, and $I^{CT}[\varepsilon]$ contains ε^{-1} mass terms with nonzero image under Q. The quantum master equation, transported to the UV boundary, then forces finite local counterterms. They are constrained by the master equation and fixed up to a choice; this choice is the renormalisation condition, part of the definition of the quantum theory and of every operator.

The UV boundary. The lower end $t = \varepsilon$ of the Schwinger integral carries two kernels,

$$\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty) = -\varepsilon L_\varepsilon, \qquad (Q\otimes 1 + 1\otimes Q)P(\varepsilon,\infty) = K_\varepsilon, \qquad L_\varepsilon = (Q^{GF}\otimes 1)K_\varepsilon \quad \text{[C07, §6.5],}$$

the second being [C07, Lemma 10.2.1] with $K_\infty = 0$. Edges pinned at $t = \varepsilon$ and carrying L_ε produce the singular coefficients and the scaling behaviour. Edges carrying K_ε produce the defects of the quantum

master equation and the transformations of observables, through

$$(d_{CE}+Q)\,e^{\hbar\partial_{P(\varepsilon,\infty)}}=e^{\hbar\partial_{P(\varepsilon,\infty)}}\big(d_{CE}+Q+\hbar\Delta_\varepsilon\big),\qquad \Delta_\varepsilon=-\partial_{K_\varepsilon}\quad\text{[C07, §10.1]}.$$

Every singular or anomalous coefficient of this manuscript is carried by edges pinned at $t=\varepsilon$ and carrying L_ε, by special edges carrying $K_\varepsilon-K_0$, whose Schwinger time lies in $[0,\varepsilon]$, and by the counterterm vertices $-QI^{CT}[\varepsilon]-\{I,I^{CT}[\varepsilon]\}_\varepsilon$.

0.3 Plan of the chapters

Chapter 1 fixes the fields, the odd pairing, the reduced action with its linear differential and vertices, and the classical master equation. Chapter 2 realises $\mathfrak{psl}(4|4)$ on Euclidean $\mathbb{R}^4$ and constructs the equivariant action $S+S^{(1)}+S^{(2)}$ in the Chevalley–Eilenberg complex. Chapter 3 constructs the Feynman-gauge Q^{GF}, records which axioms of [C07, Def 4.3.3] it satisfies, and proves the UV-boundary identity, the acyclicity at scale ∞ and the one-loop formula. Chapter 4 computes the one-loop counterterms of the action in the two-, three- and four-point sectors and the quantum master equation at the UV boundary. Chapter 5 computes $O(\varepsilon)$ for the single letters and the scalar bilinears, Chapter 6 their dilatation through the scaling covariance of the regulator, and Chapter 7 the selection rules for the other superconformal components and the program of Steps 2–5.

0.4 Results of Step 1

All one-loop coefficients are rational multiples of

$$\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2\,(X,Y)},$$

which is independent of $X,Y\in\mathfrak g$ for simple $\mathfrak g$; below it is called the unit and written out in every formula.

- Chapter 2. The equivariant action $S+S^{(1)}+S^{(2)}$, with $S^{(k)}=0$ for $k\geq 3$, satisfies its master equation for every Lie algebra $\mathfrak g$ with a nondegenerate invariant symmetric pairing: in arity 0, in arities ≥ 4 and in arity 3 with at most one or with three odd arguments by a structural argument; in arities 1, 2 and in arity 3 with two odd arguments by exact evaluation on every tuple of basis elements of $\mathfrak{psl}(4|4)$ for $\mathfrak g=\mathfrak{su}(2)$ and $\mathfrak{su}(3)$, carried to every $\mathfrak g$ by a colour-reduction lemma.
- Chapter 3. With the Feynman-gauge operator, H is the Hodge Laplacian d^*d+dd^* of flat $\mathbb{R}^4$, acting on every component as the positive Laplacian of functions, and

  $$H=[Q,Q^{GF}]=d^*d+dd^*,\qquad (Q\otimes 1+1\otimes Q)P(\varepsilon,\infty)=K_\varepsilon,\qquad (Q^{GF})^2\neq 0;$$

  conditions (1), (2), (4) of [C07, Def 4.3.3] hold and (3), (5) fail, and every identity used later rests on the conditions that hold.
- Chapter 4. The part of $I^{CT}[\varepsilon]$ quadratic in the fields is, with dvol the Euclidean volume form and $\not{d}$ the Dirac operator [ESW, §2.1],

  $$\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int\mathrm{dvol}\Big\{\frac{9}{4\varepsilon}\Big[(A|_V,A|_V)+(A|_{W_{\mathbb R}^\perp},A|_{W_{\mathbb R}^\perp})\Big]$$
  $$+\log\varepsilon\Big[-(dA|_V,dA|_V)-(dA|_{W_{\mathbb R}^\perp},dA|_{W_{\mathbb R}^\perp})+4(\lambda,\not{d}\lambda)-(dc,A^*)\Big]\Big\}.$$

- Chapter 4. Through the cubic and quartic sectors the $\log\varepsilon$ part of $I^{CT}[\varepsilon]$ is

$$\log\varepsilon\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,\big(Q+\{I,-\}\big)\int\mathrm{dvol}\Big(2(A,A^*)-4(\lambda^*,\lambda)-(c^*,c)\Big),$$

which multiplies the tree vertices $(A|_V)^3$, $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$, $\lambda\lambda A|_V$, $\lambda\lambda A|_{W_{\mathbb{R}}^\perp}$, $cA|_VA^*|_V$, $cA|_{W_{\mathbb{R}}^\perp}A^*|_{W_{\mathbb{R}}^\perp}$, $c\lambda\lambda^*$, ccc^*, $(A|_V)^4$, $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$, $(A|_{W_{\mathbb{R}}^\perp})^4$ by $6,6,10,10,1,1,1,1,8,8,8$ units; its component along the coupling vanishes, so the coupling is unrenormalised at one loop, and the ε^{-1} terms lie in the quadratic sector.

- Chapter 4. In the sector $(c,A|_V)$ of the quantum master equation the ε^{-1} parts cancel against $-Q$ of the mass counterterm, no $\log\varepsilon$ appears, and the finite obstruction is

$$-\frac{27}{8}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int\mathrm{dvol}\,(d^*dc,d^*A|_V)=\frac{27}{16}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,Q\int\mathrm{dvol}\,(d^*A|_V,d^*A|_V),$$

removed by the finite change $I\mapsto I-\frac{27}{16}\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int\mathrm{dvol}\,(d^*A|_V,d^*A|_V)$.

- Chapter 5. The single letters $\int\omega\,(X,\iota_wA)$, $\int\omega\,(X,\iota_v(A|_V))$, $\int\omega\,(\zeta,\lambda)$, $\int\omega\,(X,c)$, with fixed $X\in\mathfrak{g}$, $v\in V$ and $\zeta\in\Sigma^*\otimes\mathfrak{g}$, Σ^* the dual semi-spin representation [ESW, §2.1], have $O(\varepsilon)=O$. For $O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$, with w running over an orthonormal basis of $W_{\mathbb{R}}^\perp$,

$$\partial_\delta I^{CT}[\varepsilon]=\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\Big[4\,O-2(w_1,w_2)\sum_w\int\omega\,(\iota_wA,\iota_wA)\Big]+\frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2\varepsilon}\int\omega;$$

for $(w_1,w_2)=0$, where O lies in the traceless part, and for the trace $\sum_w\int\omega\,(\iota_wA,\iota_wA)$ this gives

$$\begin{aligned}\text{traceless:}\quad O(\varepsilon)&=\Big(1-4\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\Big)O,\\ \text{trace:}\quad O(\varepsilon)&=\Big(1+8\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\Big)O-\frac{12\hbar\dim\mathfrak{g}}{16\pi^2\varepsilon}\int\omega.\end{aligned}$$

- Chapter 6. Let O have Weyl weight w_O (the sum of the weights 1 of A, $\frac{3}{2}$ of λ, 0 of c, plus the number of derivatives), so that $\Delta=2E$ acts on it classically by $2(E+w_O)$, E acting on its density. The scaling covariance of the regulator, together with the identification of the Weyl scaling with the dilatation cochain that Chapter 6 states as the task of Step 2, gives

$$[(d_{CE}+Q)O[\infty]](\Delta)=2\Big(E+w_O+\text{coefficient}\cdot\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big)O[\infty]+\cdots,$$

the dots being terms built from $(Q+\{I,-\})O$, and the coefficient given in the table. In the normalisation $\Delta=-E$ of [SW, (3)] the same statement reads $-\Big(E+w_O+\text{coefficient}\cdot\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big)O[\infty]+\cdots$.

operator	w_O	coefficient
$\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$, $(w_1,w_2)=0$	2	0
$\sum_w\int\omega\,(\iota_wA,\iota_wA)$	2	-24
letter ι_wA	1	-4
letter $A\|_V$, modulo terms built from $(Q+\{I,-\})O$	1	-4
letter λ	$\frac{3}{2}$	-8
letter c (degree 1)	0	-2

The coefficient multiplying the unit in the dilatation component (Chapter 6). The entries for the letters are those of the Feynman gauge used throughout.

For $\mathfrak{g} = \mathfrak{su}(N)$ with $(X, Y) = -\frac{4\hbar}{g^2}\operatorname{tr}(XY)$, tr the trace in the defining representation and g the Yang–Mills coupling (Chapter 6), the unit equals $-g^2N/(32\pi^2)$, and -24 units equal $3g^2N/(4\pi^2)$, the one-loop anomalous dimension of the Konishi operator; this serves as a consistency check.

- Chapter 7. Translations, rotations and the R-symmetry commute with Q^{GF} and H; for gauge-invariant operators their components can be chosen with no one-loop correction, a choice to be realised in Step 2, and the counterterms of the traceless and the trace scalar bilinears stay in their own representations. Along the supercharges, the special supercharges and the special conformal generators the regulator moves, and the one-loop output lies in an explicit list of local structures; under the hypotheses of Proposition 7.13, a local representative and equivariant counterterms, the supercharge component of the traceless scalar bilinear is a multiple of its classical structure.

Chapter 1

The classical theory

Every number computed in the following chapters is a coefficient in front of a local functional, and its sign and normalisation are inherited from three classical data: the odd pairing on the fields, the BV action and the linear differential Q [C07, §4.3]. The sources supply the ten-dimensional action [ESW, (31)] and its dimensional reduction [ESW, Prop 2.20, §10.3], while the pairing on the antifields stays implicit in the phrase "natural (-1)-shifted symplectic structure" [ESW, Def 1.34]. This chapter fixes all three data for the reduction to Euclidean $\mathbb{R}^4$, writes the interaction vertex by vertex, and shows that the classical master equation determines the relative signs of the pairing.

1.1 Spaces and the dimensional reduction

Definition 1.1 (Spaces)**.** The ten-dimensional space is $V_{10} = V_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$, where $V_{\mathbb{R}} = \mathbb{R}^{10}$ carries its Euclidean form and the orthogonal decomposition $V_{\mathbb{R}} = W_{\mathbb{R}} \oplus W_{\mathbb{R}}^{\perp}$ of [ESW, §2.2, §2.6]: $W_{\mathbb{R}} = \mathbb{R}^4$ is spacetime and $W_{\mathbb{R}}^{\perp}$ consists of the six transverse directions. With $V = W_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ the complexification of $\mathbb{R}^4$,

$$V_{10} = V \oplus \big(W_{\mathbb{R}}^{\perp} \otimes_{\mathbb{R}} \mathbb{C}\big),$$

and from here on $W_{\mathbb{R}}^{\perp}$ and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ denote these complexifications. The nondegenerate symmetric form $(-,-)$ of V_{10} is the complex-bilinear extension of the Euclidean form. The letter W alone denotes the multiplicity space of [ESW, §10].

The gauge Lie algebra $\mathfrak{g}$ is complex, with bracket $[-,-]$ and a nondegenerate invariant symmetric pairing $(-,-)$ [ESW, §3.1], chosen positive definite on the compact real form, so that $\langle e, Qe\rangle$ is negative definite on the real slice of [C07, §6.6] (Remark 1.10). The coupling constant sits in $(-,-)$. The inverse pairing is the element $(-,-)^{-1} \in \mathfrak{g} \otimes \mathfrak{g}$.

The two sixteen-dimensional semi-spin representations Σ and Σ^* of $\mathfrak{so}(V_{10})$ form the $\mathbb{Z}/2$-graded Clifford module $\Sigma \oplus \Sigma^*$ with Clifford action ρ and pairing $(-,-)\colon \Sigma \otimes \Sigma^* \to \mathbb{C}$ [ESW, §2.1]:

$$\rho(v_1)\rho(v_2) + \rho(v_2)\rho(v_1) = 2(v_1, v_2), \qquad (\rho(v)Q_1, Q_2) = (Q_1, \rho(v)Q_2), \qquad v, v_1, v_2 \in V_{10}.$$

The Γ-pairing $\Gamma\colon \mathrm{Sym}^2(\Sigma) \to V_{10}$, $\Gamma\colon \mathrm{Sym}^2(\Sigma^*) \to V_{10}$ is defined by [ESW, (28)]

$$(\rho(v)Q_1, Q_2) = (v, \Gamma(Q_1, Q_2)).$$

The map $q\colon \wedge^{\bullet} V_{10} \to \mathrm{Cl}(V_{10})$ is antisymmetrisation, $q(v_1 \wedge v_2) = v_1 v_2 - (v_1, v_2)$ [ESW, (29)]. Applied to a two-form, such as F_A, the symbol ρ means $\rho \circ q$, as in $\rho(F_A)$ [ESW, (33)].

The exterior derivative d acts along $\mathbb{R}^4$, and every field is constant along $W_{\mathbb{R}}^{\perp}$. The operator d^* is the formal adjoint of d for the Euclidean metric, dvol is the Euclidean volume density of $\mathbb{R}^4$, and $\not{d} = \rho \circ d$ is the Dirac operator from Σ-valued to Σ^*-valued functions [ESW, §2.1]; the same formula on Σ^*-valued functions is written $\not{d}|_{\Sigma^*}$.

The four-dimensional content of Σ is [ESW, §10, §10.3]

$$\Sigma \cong S_+ \otimes W \oplus S_- \otimes W^*, \qquad \dim W = 4, \qquad W \cong S_+^6 \text{ as a representation of } \mathrm{Spin}(6;\mathbb{C}) \cong SL(4,\mathbb{C}),$$

with $S_\pm$ the two-dimensional semi-spin representations of $\mathrm{Spin}(4,\mathbb{C})$.

Proposition 1.2 (Four-dimensional chirality)**.** *Fix an orientation of V. For an oriented orthonormal basis $v_1,\dots,v_4$ of V the element $q(v_1\wedge v_2\wedge v_3\wedge v_4) = v_1v_2v_3v_4$ of $\mathrm{Cl}(V_{10})$ depends on the orientation alone. The operator $\rho(v_1v_2v_3v_4)$ squares to 1 on Σ and on Σ^*, with eigenvalues $+1$ and -1 each of multiplicity 8. Writing*

$$S_+\otimes W = \{+1 \text{ eigenspace on } \Sigma\}, \qquad S_-\otimes W^* = \{-1 \text{ eigenspace on } \Sigma\},$$
$$S_-\otimes W = \{-1 \text{ eigenspace on } \Sigma^*\}, \qquad S_+\otimes W^* = \{+1 \text{ eigenspace on } \Sigma^*\},$$

one has, for $v\in V$ and $w\in W_{\mathbb{R}}^{\perp}$,

$$\rho(v)\colon S_+\otimes W \to S_-\otimes W, \qquad \rho(w)\colon S_+\otimes W \to S_+\otimes W^*,$$

and correspondingly on the other summands: $\rho(v)$ exchanges the four-dimensional chiralities and $\rho(w)$ preserves them.

Proof. The element $v_1v_2v_3v_4$ is the antisymmetrisation q of the oriented unit element of $\wedge^4 V$, hence depends on the orientation alone. By the Clifford relation the v_i anticommute pairwise and square to 1, hence

$$(v_1v_2v_3v_4)^2 = (-1)^{3+2+1}\, v_1^2v_2^2v_3^2v_4^2 = 1.$$

Moving v_1 through $v_1v_2v_3v_4$ meets v_2,v_3,v_4 once each, so $v_1\,(v_1v_2v_3v_4) = -(v_1v_2v_3v_4)\,v_1$; the same holds for each v_i, hence for every $v\in V$ by linearity. For $w\in W_{\mathbb{R}}^{\perp}$, w anticommutes with all four v_i, so $w\,(v_1v_2v_3v_4) = (+1)(v_1v_2v_3v_4)\,w$. Since $\rho(v)$ maps $\Sigma\to\Sigma^*$, the first relation reads $\rho(v)\rho(v_1v_2v_3v_4)|_\Sigma = -\rho(v_1v_2v_3v_4)|_{\Sigma^*}\rho(v)$, so $\rho(v)$ maps the $+1$ eigenspace on Σ to the -1 eigenspace on Σ^*; the second relation gives $\rho(w)$ mapping the $+1$ eigenspace on Σ to the $+1$ eigenspace on Σ^*. For the multiplicities, take a unit vector $w\in W_{\mathbb{R}}^{\perp}$. The even element v_1w satisfies $(v_1w)^2 = -v_1^2w^2 = -1$ and $v_1w\,(v_1v_2v_3v_4) = -(v_1v_2v_3v_4)\,v_1w$, so $\rho(v_1w)$ is an invertible endomorphism of Σ (and of Σ^*) exchanging the two eigenspaces of $\rho(v_1v_2v_3v_4)$; hence each eigenspace has dimension $16/2 = 8$. □

Remark 1.3. The names of the eigenspaces depend on the orientation of V; the opposite orientation exchanges $S_+\leftrightarrow S_-$ in all four of them. This orientation is fixed once; Chapter 2 uses it to name the supercharges.

1.2 Fields, degrees and parities

Definition 1.4 (Fields)**.** The BRST fields of [ESW, Def 3.2], reduced along $W_{\mathbb{R}}^{\perp}$ [ESW, Def 1.55, Prop 1.56, Prop 2.20], form

$$F_{\text{gauge}} = \Omega^0(\mathbb{R}^4;\mathfrak{g}[1]) \oplus \Omega^0(\mathbb{R}^4;V_{10}^\vee\otimes\mathfrak{g}) \oplus \Omega^0(\mathbb{R}^4;\Pi\Sigma\otimes\mathfrak{g}),$$

with sections (c,A,λ). The space of BV fields is $\mathscr{E}$ [C07, §4.1],

$$\mathscr{E} = F_{\text{gauge}} \oplus F_{\text{gauge}}^!{}[-1] \qquad \text{[ESW, Rem 1.36]},$$

with antifields (c^*,A^*,λ^*); duals are identified through $(-,-)$ on $\mathfrak{g}$, the metric of V_{10}, the pairing $\Sigma\otimes\Sigma^*\to\mathbb{C}$ and dvol. The component of A along V is the connection $A|_V$, the component along $W_{\mathbb{R}}^{\perp}$ is $A|_{W_{\mathbb{R}}^{\perp}}$, and $\iota_w A = A(w)$ for $w\in W_{\mathbb{R}}^{\perp}$ are the six scalars. The compactly supported elements of $\mathscr{E}$ form $\mathscr{E}_c$ ($\mathcal{E}_c$ in [ESW, §1.1]).

The degrees are those of [ESW, Rem 1.33]: the ghost c is a section in degree -1, which forces A^* in degree $+1$ and c^* in degree $+2$. Signs follow the Koszul rule for the total parity, the sum of the $\mathbb{Z}$-degree and the $\mathbb{Z}/2$-degree modulo 2 [ESW, §1.1].

field	fibre	$\mathbb{Z}$-degree	$\mathbb{Z}/2$-degree	total parity	$\mathrm{Spin}(4)\times\mathrm{Spin}(6)$ content
c	$\mathfrak{g}$	-1	even	odd	scalar
A	$V_{10}^{\vee}\otimes\mathfrak{g}$	0	even	even	$A\vert_V$ vector of Spin(4), $A\vert_{W_{\mathbb{R}}^{\perp}}$ vector of Spin(6)
λ	$\Pi\Sigma\otimes\mathfrak{g}$	0	odd	odd	$S_+\otimes W\oplus S_-\otimes W^*$
A^*	$V_{10}\otimes\mathfrak{g}$	$+1$	even	odd	as A
λ^*	$\Pi\Sigma^*\otimes\mathfrak{g}$	$+1$	odd	even	$S_-\otimes W\oplus S_+\otimes W^*$
c^*	$\mathfrak{g}$	$+2$	even	even	scalar

The BV fields of the reduced theory. Coordinate functionals on $\mathscr{E}$ carry the opposite $\mathbb{Z}$-degree.

1.3 Pairing and bracket

Definition 1.5 (Odd pairing)**.** The odd pairing $\langle\,,\rangle$ on $\mathscr{E}$ [C07, Def 4.3.1; ESW, Def 1.13, Def 1.16] is nonzero exactly between a field and its antifield. For each of the three pairs, call the member of even total parity the even member: A, λ^*, c^*; their partners A^*, λ, c are odd. For a section e_0 in the slot of an even member and a section e_1 in the slot of its partner,

$$\langle e_0, e_1\rangle = \int \mathrm{dvol}\,(e_0, e_1), \qquad \langle e_1, e_0\rangle = -\langle e_0, e_1\rangle,$$

where (e_0, e_1) is the fibre pairing of Definition 1.4; for the pair (λ^*, λ) it is (e_1, e_0) with $(-,-)\colon \Sigma\otimes\Sigma^*\to\mathbb{C}$ tensored with $(-,-)$ on $\mathfrak{g}$.

Definition 1.6 (Bracket and the kernel of the identity)**.** The kernel K_0 of the identity [CG2, §7.2.3] has nonzero components exactly in the slot pairs

$$A\otimes A^*, \quad A^*\otimes A, \quad \lambda^*\otimes\lambda, \quad \lambda\otimes\lambda^*, \quad c^*\otimes c, \quad c\otimes c^*,$$

each equal to $-\delta(x-y)$ times the inverse of the fibre pairing: $(-,-)^{-1}$ tensored with the inverse metric of V_{10} for A, A^*, with the canonical element of $\Sigma\otimes\Sigma^*$ for λ, λ^*, and with 1 for c, c^*; here $x, y\in\mathbb{R}^4$ are the two positions and $\delta(x-y)$ is the delta distribution of the diagonal, always written with its argument. The bracket is the odd Poisson bracket of [C07, Lemma 4.3.2], defined for f local and g arbitrary; it is given by the formula of [C07, §10.1] for the manifold a point,

$$\{f, g\} = \Delta_0(fg) - \Delta_0(f)g - (-1)^{|f|}f\Delta_0(g), \qquad \Delta_0 = -\partial_{K_0},$$

with ∂_ϕ the order-two operator of [C07, §6.4] and f local. It satisfies

$$\{g, f\} = (-1)^{|f||g|}\{f, g\},$$

the graded symmetry of [ESW, Def 1.15].

Remark 1.7. The overall sign of $\langle\,,\rangle$ is a convention: reversing it reverses K_0 and the bracket, and leaves the master equation $\{S, S\} = 0$ invariant. The relative signs of the three pairs are fixed by the master equation (Proposition 1.14).

1.4 The action

Definition 1.8 (Action)**.** The action is the dimensional reduction of [ESW, (31)]:

$$S = \int_{\mathbb{R}^4} \mathrm{dvol}\Big(-\tfrac{1}{4}(F_A, F_A) + \tfrac{1}{2}(\lambda,\, \not{d}_A\lambda) + (d_A c, A^*) + ([c,\lambda],\lambda^*) + \tfrac{1}{2}([c,c],c^*)\Big), \tag{1.1}$$

with

$$F_A = dA + \tfrac{1}{2}[A\wedge A], \qquad d_A = d + [A,-], \qquad \not{d}_A = \rho\circ d_A.$$

Here A is regarded as a one-form on V_{10} constant along $W_{\mathbb{R}}^{\perp}$; d acts along $\mathbb{R}^4$, and d_A, $\not{d}_A$ use all ten components of A. The pairing (F_A, F_A) is the Hodge pairing on $\wedge^2 V_{10}^{\vee}$ tensored with $(-,-)$ on $\mathfrak{g}$, normalised by

$$\big((v_1,-)\wedge(v_2,-),\,(v_1,-)\wedge(v_2,-)\big) = (v_1,v_1)(v_2,v_2) - (v_1,v_2)^2, \qquad v_1, v_2 \in V_{10}.$$

Every pairing in (1.1) is read with its left entry first, as printed in [ESW, (31)]. In [ESW] the full action is S and the interaction is I; in [C07] the letter S denotes the interaction. Here S is always the full action and I the interaction.

Splitting $A = A|_V + A|_{W_{\mathbb{R}}^{\perp}}$,

$$(F_A,F_A) = (F_{A|_V},F_{A|_V}) + \big(d_{A|_V}A|_{W_{\mathbb{R}}^{\perp}},\, d_{A|_V}A|_{W_{\mathbb{R}}^{\perp}}\big) + \big(\tfrac{1}{2}[A|_{W_{\mathbb{R}}^{\perp}}\wedge A|_{W_{\mathbb{R}}^{\perp}}],\, \tfrac{1}{2}[A|_{W_{\mathbb{R}}^{\perp}}\wedge A|_{W_{\mathbb{R}}^{\perp}}]\big),$$

$$\not{d}_A\lambda = \not{d}\lambda + \rho([A|_V,\lambda]) + \rho([A|_{W_{\mathbb{R}}^{\perp}},\lambda]), \qquad (d_A c, A^*) = (dc, A^*|_V) + ([A,c],A^*),$$

where $\rho([A,\lambda])$ is Clifford multiplication by the $V_{10}^{\vee}$ factor of A (through the metric) together with $[-,-]$ on the $\mathfrak{g}$ factors, i.e. $(\not{d}_A - \not{d})\lambda$.

Proposition 1.9 (Normalisation of the gauge kinetic term)**.** *For $Q_1 \in \Sigma$ put*

$$S^{(1)}(Q_1) = \int \mathrm{dvol}\Big(-(\Gamma(Q_1,\lambda),A^*) + \tfrac{1}{2}(\rho(F_A)Q_1,\lambda^*)\Big) \qquad [\textit{ESW, (33)}].$$

For abelian $\mathfrak{g}$ the bracket $\{S, S^{(1)}(Q_1)\}$ is an affine function of the coefficient of the Hodge pairing (F_A,F_A) in (1.1)*, and the identity $\{S,S^{(1)}(Q_1)\} = 0$ of [ESW, Lemma 3.4] fixes this coefficient to $-\frac{1}{4}$. The coefficients $+\frac{1}{4}$ and $-\frac{1}{2}$ (the reading of (F_A,F_A) as a sum over ordered pairs of directions, twice the Hodge pairing) give a nonzero bracket.*

Proof. For abelian $\mathfrak{g}$ the bracket is affine in the coefficient of (F_A,F_A): this coefficient times (dA,dA) meets $-(\Gamma(Q_1,\lambda),A^*)$ through the (A,A^*) pair, $\frac{1}{2}(\lambda,\not{d}\lambda)$ meets $\frac{1}{2}(\rho(dA)Q_1,\lambda^*)$ through the (λ^*,λ) pair, and (dc,A^*) meets $\frac{1}{2}(\rho(dA)Q_1,\lambda^*)$ in $(\rho(d\,dc)Q_1,\lambda^*) = 0$. With the coefficient $-\frac{1}{4}$ the bracket vanishes for every $Q_1\in\Sigma$ and every $\mathfrak{g}$: this is [ESW, Lemma 3.4] in ten dimensions, pulled back along the reduction [ESW, Prop 2.20], and for $\mathfrak{g}=\mathfrak{su}(2)$ and $\mathfrak{g}=\mathfrak{su}(3)$ it holds on every element of a basis of Σ (Remark 1.15); the part of this bracket quadratic in the fields is the abelian bracket. With the coefficient $+\frac{1}{4}$ the abelian bracket is a nonzero local functional at an element $Q_1\in\Sigma$ (Remark 1.15). An affine function that vanishes at $-\frac{1}{4}$ and is nonzero at $+\frac{1}{4}$ has $-\frac{1}{4}$ as its unique zero; in particular it is nonzero at $-\frac{1}{2}$. □

The free part of S is

$$\tfrac{1}{2}\langle e, Qe\rangle = \int \mathrm{dvol}\Big(-\tfrac{1}{4}(dA,dA) + \tfrac{1}{2}(\lambda,\not{d}\lambda) + (dc,A^*)\Big) \qquad [\text{C07, §1, §4.3}], \tag{1.2}$$

and $I = S - \frac{1}{2}\langle e, Qe\rangle$ is the interaction.

Fields and kernels on $\mathbb{R}^4 \times \mathbb{R}^4$ are Fourier transformed as

$$\int \frac{d^4p}{(2\pi)^4} e^{i(p,x)}(\cdots)(p), \qquad \int \frac{d^4p}{(2\pi)^4} e^{i(p,x-y)}(\cdots)(p),$$

with $p \in V$ and $|p|^2 = (p,p)$; thus d becomes $i(p,-)$.

Remark 1.10 (Real slice)**.** The weight is $e^{+S/\hbar}$, with $\langle e, Qe\rangle$ negative definite on the real slice [C07, §6.6]. For A real with respect to the compact real form of $\mathfrak{g}$,

$$\langle A, QA\rangle = -\tfrac{1}{2}\int \mathrm{dvol}\,(dA, dA) \le 0,$$

and for a fixed nonzero momentum p the quadratic form on the fibre $V_{10}^\vee$ has eigenvalues, in units of $(-,-)$, equal to $-\frac{1}{2}|p|^2$ with multiplicity 9 and 0 with multiplicity 1; the zero eigenvector is the longitudinal part of $A|_V$, which lies in $\operatorname{Im} Q$.

1.5 The linear differential

Proposition 1.11 (The differential Q)**.** *The linear part of $\{S,-\}$ is the derivation $\{\frac{1}{2}\langle e, Qe\rangle, -\}$ of functionals, induced as in [C07, §6.2] by the operator $Q\colon \mathscr{E} \to \mathscr{E}$ given in the following table, and* (1.2) *holds.*

on the slot of	Q	*lands in the slot of*	*momentum space*
c	$c \mapsto -dc$	$A\|_V$	$-i(p,-)\,c$
A	$A \mapsto -\frac{1}{2}\,d^*dA$	A^*	$-\frac{1}{2}\big(\|p\|^2 A\|_V - (p,-)\,\iota_p A\big)$ *and* $-\frac{1}{2}\|p\|^2 A\|_{W_{\mathbb{R}}^\perp}$
A^*	$A^* \mapsto d^*(A^*\|_V)$	c^*	$-i\,(p, A^*\|_V)$
λ	$\lambda \mapsto \not{d}\lambda$	λ^*	$i\rho(p)\lambda$
λ^*, c^*	0		

The operator Q. On $A|_{W_{\mathbb{R}}^\perp}$, d^*d is the positive Laplacian of each scalar $\iota_w A$; on $A|_V$ it is d^*d on one-forms.

Moreover:

- $Q^2 = 0$*;*
- Q *preserves the pairing:* $\langle Qe_0, e_1\rangle + (-1)^{|e_0|}\langle e_0, Qe_1\rangle = 0$ *for all* $e_0, e_1 \in \mathscr{E}_c$*; equivalently* $(Q\otimes 1 + 1\otimes Q)K_0 = 0$.

Proof. $Q^2 = 0$: the composable pairs of arrows of Q are $c \to A|_V \to A^*$ and $A \to A^* \to c^*$. On the first,

$$Q^2 c = -\tfrac{1}{2}\,d^*d(-dc) = \tfrac{1}{2}\,d^*(d\,dc) = 0;$$

on the second, $d^*(A^*|_V)$ applied to $-\frac{1}{2}d^*dA$ gives $-\frac{1}{2}\,d^*d^*(dA) = 0$, since $d^*d^* = 0$.
Preservation of the pairing is checked on the three pairs; all other pairings vanish. For e_0, e_1 in the A slot (even),

$$\langle Qe_0, e_1\rangle = -\langle e_1, -\tfrac{1}{2}d^*de_0\rangle = \tfrac{1}{2}\int \mathrm{dvol}\,(de_1, de_0), \qquad \langle e_0, Qe_1\rangle = -\tfrac{1}{2}\int \mathrm{dvol}\,(de_0, de_1),$$

and the sum vanishes. For e_0 in the A^* slot (odd) and e_1 in the c slot,

$$\langle Qe_0, e_1\rangle = \int \mathrm{dvol}\,(d^*(e_0|_V), e_1) = \int \mathrm{dvol}\,(e_0, de_1), \qquad \langle e_0, Qe_1\rangle = -\langle -de_1, e_0\rangle = \int \mathrm{dvol}\,(de_1, e_0),$$

so $\langle Qe_0, e_1\rangle - \langle e_0, Qe_1\rangle = 0$. For e_0, e_1 in the λ slot (odd), with $\not{d} = \rho \circ d$ and the symmetry $(\rho(v)Q_1, Q_2) = (Q_1, \rho(v)Q_2)$,

$$\langle Qe_0, e_1\rangle - \langle e_0, Qe_1\rangle = \int \mathrm{dvol}\big((e_1, \not{d}e_0) + (e_0, \not{d}e_1)\big) = 0,$$

by integration by parts. The kernel form is the same statement for the kernel of the identity. □

Remark 1.12 (A sign conflict in the sources)**.** [ESW, Def 1.16] prints the adjointness of the BV differential as $\int \omega(e_0, Q_{BV}e_1) = (-1)^{|e_0|}\int \omega(Q_{BV}e_0, e_1)$, where ω and Q_{BV} are the [ESW] names of $\langle\,,\rangle$ and Q. Combined with the graded antisymmetry $\langle e_1, e_0\rangle = -\langle e_0, e_1\rangle$ of Definition 1.5 for e_0 even and e_1 odd, the printed relation gives, for $e_0 = e_1 = A$ in the A slot,

$$\langle A, QA\rangle = \langle QA, A\rangle = -\langle A, QA\rangle,$$

hence $\langle A, QA\rangle = 0$, which contradicts the kinetic term $\frac{1}{2}\langle A, QA\rangle = -\frac{1}{4}\int \mathrm{dvol}\,(dA, dA)$ of (1.2). The relation of Proposition 1.11, $\langle Qe_0, e_1\rangle + (-1)^{|e_0|}\langle e_0, Qe_1\rangle = 0$, is the one satisfied by the action, and it is used throughout.

1.6 The interaction, vertex by vertex

Expanding (1.1) around (1.2),

$$\begin{aligned} I = &\underbrace{\int \mathrm{dvol}\Big(-\tfrac{1}{2}\big(dA, \tfrac{1}{2}[A\wedge A]\big) + \tfrac{1}{2}\big(\lambda, (\not{d}_A - \not{d})\lambda\big) + ([A,c], A^*) + ([c,\lambda],\lambda^*) + \tfrac{1}{2}([c,c],c^*)\Big)}_{\text{cubic part of } I} \\ &+ \underbrace{\int \mathrm{dvol}\Big(-\tfrac{1}{4}\big(\tfrac{1}{2}[A\wedge A], \tfrac{1}{2}[A\wedge A]\big)\Big)}_{\text{quartic part of } I}. \end{aligned} \tag{1.3}$$

term	component types
$-\frac{1}{2}(dA, \frac{1}{2}[A\wedge A])$	$(A\|_V)^3$; $A\|_V\,(A\|_{W_{\mathbb{R}}^\perp})^2$
$-\frac{1}{4}(\frac{1}{2}[A\wedge A], \frac{1}{2}[A\wedge A])$	$(A\|_V)^4$; $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$; $(A\|_{W_{\mathbb{R}}^\perp})^4$
$\frac{1}{2}(\lambda, (\not{d}_A - \not{d})\lambda)$	$\lambda\lambda\, A\|_V$ (gauge); $\lambda\lambda\, \iota_w A$ (Yukawa)
$([A,c], A^*)$	$A\|_V\, c\, A^*\|_V$; $A\|_{W_{\mathbb{R}}^\perp}\, c\, A^*\|_{W_{\mathbb{R}}^\perp}$
$([c,\lambda],\lambda^*)$	$c\,\lambda\,\lambda^*$
$\frac{1}{2}([c,c],c^*)$	$c\,c\,c^*$

Component types of the vertices. The cubic gauge vertex has the component types $(A|_V)^3$ and $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$, because d acts along $\mathbb{R}^4$.

Definition 1.13 (Vertex functions)**.** Number the legs of a vertex $1,\dots,n$, give leg j an incoming momentum $p_j \in V$ and a fibre element written $A(p_j)$, $\lambda(p_j)$, $c(p_j)$, $A^*(p_j)$, $\lambda^*(p_j)$ or $c^*(p_j)$ according to its field. Evaluate the corresponding term of I on $\sum_j r_j$ (fibre element of leg j) $e^{i(p_j,x)}$, where $r_j = 1$ for even legs and r_j is an odd element of an auxiliary nilpotent supercommutative ring for odd legs [C07, Lemma 6.4.1, proof]. The vertex function is the coefficient of the product of the r_j of the odd legs, in leg order, in the part multilinear in the legs, with the factor $(2\pi)^4\delta(\sum_j p_j)$ removed.

In the following formulas, $\iota_p A(p_j) \in \mathfrak{g}$ is the contraction of the $V_{10}^\vee$ factor with $p \in V$; $[A(p_1), A(p_2)] \in \mathfrak{g}$ contracts the $V_{10}^\vee$ factors with the metric and brackets the $\mathfrak{g}$ factors; $[A(p_1)\wedge A(p_2)]$ is the $\mathfrak{g}$-valued two-form $(v_1, v_2) \mapsto [A(p_1)(v_1), A(p_2)(v_2)] - [A(p_1)(v_2), A(p_2)(v_1)]$, paired by the Hodge pairing.

The gauge vertices, with legs $1,2,3$ (resp. $1,\dots,4$) all of type A:

$$A\,A\,A: \qquad -\frac{i}{2}\Big[\big(\iota_{p_2-p_1}A(p_3),[A(p_1),A(p_2)]\big)+\big(\iota_{p_3-p_2}A(p_1),[A(p_2),A(p_3)]\big) + \big(\iota_{p_1-p_3}A(p_2),[A(p_3),A(p_1)]\big)\Big], \tag{1.4}$$

$$A\,A\,A\,A: \qquad -\frac{1}{2}\Big[\big([A(p_1)\wedge A(p_2)],[A(p_3)\wedge A(p_4)]\big)+\big([A(p_1)\wedge A(p_3)],[A(p_2)\wedge A(p_4)]\big) + \big([A(p_1)\wedge A(p_4)],[A(p_2)\wedge A(p_3)]\big)\Big]. \tag{1.5}$$

The vertices with fermions, ghosts and antifields:

vertex (legs in order)	vertex function
$\lambda\,\lambda\,A$	$r_1r_2\,(\lambda(p_1),\rho([A(p_3),\lambda(p_2)]))$
$A\,c\,A^*$	$r_2r_3\,([A(p_1),c(p_2)],A^*(p_3))$
$c\,\lambda\,\lambda^*$	$r_1r_2\,([c(p_1),\lambda(p_2)],\lambda^*(p_3))$
$c\,c\,c^*$	$r_1r_2\,([c(p_1),c(p_2)],c^*(p_3))$

Momentum-space vertex functions of I with odd legs (all momenta incoming); these four are independent of the momenta.

Each entry is the multilinear part of the corresponding term of (1.3) on plane waves, including all permutations of identical legs; these are the vertex labels of every graph in the later chapters.

1.7 The classical master equation

Proposition 1.14 (Classical master equation)**.** *For every $\mathfrak{g}$ as in Definition 1.1:*

- $\{S,S\}=0$ *[ESW, Def 1.16];*
- *equivalently,* $\{\frac12\langle e,Qe\rangle,\frac12\langle e,Qe\rangle\}=0$ *(that is, $Q^2=0$) and* $QI+\frac12\{I,I\}=0$.

If moreover $[\mathfrak{g},\mathfrak{g}]\neq 0$, the master equation fixes the relative signs of the pairing and the sign of the ghost term:

- *if the sign of the (λ^*,λ) pairing, or of the (c^*,c) pairing, or of both, is reversed relative to the (A,A^*) pairing in Definition 1.5, then $\{S,S\}\neq 0$;*
- *with the pairing of Definition 1.5 and the sign of the term $\frac12([c,c],c^*)$ reversed, $\{S,S\}\neq 0$.*

Proof. Equivalence. Write $S=\frac12\langle e,Qe\rangle+I$. Both summands are even, so by graded symmetry $\{I,\frac12\langle e,Qe\rangle\}=\{\frac12\langle e,Qe\rangle,I\}=QI$ (Proposition 1.11), and

$$\{S,S\}=\{\tfrac12\langle e,Qe\rangle,\tfrac12\langle e,Qe\rangle\}+2\,QI+\{I,I\}.$$

The first term is Q applied to $\frac12\langle e,Qe\rangle$; by the preservation of the pairing in Proposition 1.11, $\langle Qe,Qe\rangle=-(-1)^{|e|}\langle e,Q^2e\rangle$, so this term is a multiple of $\langle e,Q^2e\rangle$ and vanishes by $Q^2=0$. This gives the equivalence of the first two items.
$\{S,S\}=0$ for every $\mathfrak{g}$. The action (1.1) is the BV action of the BRST theory of [ESW, Def 3.2], built as in [ESW, Def 1.34] with the natural pairing of the shifted cotangent bundle; it satisfies $\{S,S\}=0$ for every $\mathfrak{g}$, and the dimensional reduction preserves it [ESW, Def 1.55, Prop 1.56]. The natural pairing differs from Definition 1.5 at most by a sign on each of the three pairs, and these signs are independent of $\mathfrak{g}$. An overall sign leaves $\{S,S\}=0$ invariant. For $\mathfrak{g}=\mathfrak{su}(2)$ the density of $\{S,S\}$ computed with the pairing of Definition 1.5 vanishes as a local functional, together with

$2QI + \{I, I\}$, while $QI \neq 0$; for each of the three other relative sign patterns it is nonzero (Remark 1.15). Hence the natural pairing coincides with Definition 1.5 up to an overall sign, and $\{S, S\} = 0$ holds for every $\mathfrak{g}$. The same exact computation gives $\{S, S\} = 0$ for $\mathfrak{g} = \mathfrak{su}(3)$.
The relative signs. The action is linear in the antifields, so $\{S, S\}$ has antifield number 0 and 1. Each term of $\{S, S\}$ pairs a derivative along a field with a derivative along its partner and carries the sign of that pair. The derivatives of S along the antifields are the gauge transformations $A \mapsto d_A c$, $\lambda \mapsto [c, \lambda]$, $c \mapsto \frac{1}{2}[c, c]$.

- Antifield number 0: the gauge variation of $\int \mathrm{dvol}\big(-\frac{1}{4}(F_A, F_A) + \frac{1}{2}(\lambda, \not{d}_A \lambda)\big)$. The A-variation enters through the (A, A^*) pair and the λ-variation through the (λ^*, λ) pair.
- Linear in A^*: $[d_A c, c]$ through the (A, A^*) pair against $d_A(\frac{1}{2}[c, c])$ through the (c^*, c) pair, related by the Leibniz rule $d_A[c, c] = 2[d_A c, c]$.
- Linear in λ^*: $[c, [c, \lambda]]$ through the (λ^*, λ) pair against $\frac{1}{2}[[c, c], \lambda]$ through the (c^*, c) pair, related by the Jacobi identity $2[c, [c, \lambda]] = [[c, c], \lambda]$.
- Linear in c^*: $[c, [c, c]] = 0$.

Consider the part of $\{S, S\}$ with field content (c, λ, λ), no A and no antifield. The (c^*, c) pair contributes nothing to it, since $\frac{1}{2}[c, c]$ contains two ghosts. The (λ^*, λ) pair contributes the variation of $\frac{1}{2}\int \mathrm{dvol}\,(\lambda, \not{d}\lambda)$ along $\lambda \mapsto [c, \lambda]$, and the (A, A^*) pair contributes, up to the sign of that pair, a nonzero multiple of the variation of $\frac{1}{2}\int \mathrm{dvol}\,(\lambda, \rho([A, \lambda]))$ along $A \mapsto dc$, which is a nonzero multiple of

$$\int \mathrm{dvol}\,\big(\lambda, \rho([dc, \lambda])\big).$$

With the pairing of Definition 1.5 these two contributions cancel, because $\{S, S\} = 0$; reversing the sign of the (λ^*, λ) pair relative to the (A, A^*) pair therefore turns this part of $\{S, S\}$ into twice the (A, A^*) contribution. This functional is nonzero when $[\mathfrak{g}, \mathfrak{g}] \neq 0$: choose $X, Y, Z \in \mathfrak{g}$ with $(X, [Z, Y]) \neq 0$, compactly supported Σ-valued functions s_1, s_2 and a compactly supported function f with $\int \mathrm{dvol}\,(df, \Gamma(s_1, s_2)) \neq 0$, and put $\lambda = r_1 s_1 X + r_2 s_2 Y$, $c = r_3 f Z$ with r_1, r_2, r_3 odd as in Definition 1.13. Using $r_2 r_3 r_1 = -r_1 r_3 r_2$, $(Y, [Z, X]) = -(X, [Z, Y])$ and the symmetry of Γ,

$$\int \mathrm{dvol}\,\big(\lambda, \rho([dc, \lambda])\big) = 2\, r_1 r_3 r_2\, (X, [Z, Y]) \int \mathrm{dvol}\,(df, \Gamma(s_1, s_2)) \neq 0.$$

This proves the first of the last two items when the (λ^*, λ) sign is reversed, alone or together with the (c^*, c) sign. Next consider the part with field content (c, c, A^*) and no A. The (λ^*, λ) pair contributes nothing to it. The (A, A^*) pair contributes the variation of $\int \mathrm{dvol}\,([A, c], A^*)$ along $A \mapsto dc$, which is $\int \mathrm{dvol}\,([dc, c], A^*)$ up to the sign of the pair, and the (c^*, c) pair contributes the variation of $\int \mathrm{dvol}\,(dc, A^*)$ along $c \mapsto \frac{1}{2}[c, c]$, which is $\int \mathrm{dvol}\,([dc, c], A^*)$ up to the product of the sign of the pair and the sign of the term $\frac{1}{2}([c, c], c^*)$. They cancel with the data of Definitions 1.5 and 1.8, so reversing the (c^*, c) sign alone, or the sign of $\frac{1}{2}([c, c], c^*)$ alone, turns this part into twice $\pm \int \mathrm{dvol}\,([dc, c], A^*)$. On $c = r_1 f_1 X + r_2 f_2 Y$ with $[X, Y] \neq 0$,

$$\int \mathrm{dvol}\,([dc, c], A^*) = r_1 r_2 \int \mathrm{dvol}\,\big([X, Y],\, (d(f_1 f_2), A^*|_V)\big),$$

which is nonzero for a suitable compactly supported $A^*|_V$. □

Remark 1.15 (Exact verification of identities of local functionals)**.** Propositions 1.9 and 1.14 use the following exact computation. For $\mathfrak{g} = \mathfrak{su}(2)$ and $\mathfrak{g} = \mathfrak{su}(3)$ fix bases of $\mathfrak{g}$, of V_{10} and of $\Sigma \oplus \Sigma^*$ in

which the bracket, the pairings, ρ and Γ have Gaussian-rational entries. The density of each local functional of this chapter, and of each bracket of two of them, is then a polynomial in the jets of the components of the six fields with Gaussian-rational coefficients, computed in exact arithmetic. A local functional $\int \mathrm{dvol}\, D$ without constant term vanishes on $\mathscr{E}_c$ when all Euler–Lagrange derivatives of D vanish identically: for $e \in \mathscr{E}_c$, after integration by parts with compact support, $\frac{d}{ds} \int \mathrm{dvol}\, D(se)$ is the integral of the Euler–Lagrange derivatives of D at se paired with e (odd components multiplied by odd elements of an auxiliary nilpotent ring, as in Definition 1.13), so $\int \mathrm{dvol}\, D(se)$ is independent of s and equals its value 0 at $s = 0$. Conversely, a nonzero Euler–Lagrange derivative shows that the functional is nonzero. Statements that a bracket vanishes for every element of Σ are established on every element of a basis of Σ; this suffices, because the bracket is linear in that element.

Chapter 2

Superconformal symmetry as an equivariant action

The dilatation of a composite operator, and every other superconformal transformation, acts on the fields of $\mathcal{N} = 4$ super Yang–Mills theory [ESW, Def 2.4, §10.3] up to gauge transformations and equations of motion. In the BV formalism this structure is a Maurer–Cartan element of a Chevalley–Eilenberg complex, an L_∞ action in the sense of [ESW, Def 1.26–1.27], and it is the classical input of the equivariant observables used later. [ESW, (32)–(34), Thm 3.3] construct it for the Poincaré supersymmetry algebra; the special conformal transformations and the special supercharges require Weyl weights, a compensating term with the six scalars, and an R-symmetry rotation in the odd bracket. This chapter realises $\mathfrak{psl}(4|4)$, the complexification of $\mathfrak{psu}(2,2|4)$ [SW, (7)], on Euclidean $\mathbb{R}^4$, fixes the sign conventions of the complex, writes the action $S + S^{(1)} + S^{(2)}$, and proves its master equation for every gauge algebra.

2.1 The algebra $\mathfrak{psl}(4|4)$ in complexified Euclidean form

Throughout, $x \in V$ denotes the position on $\mathbb{R}^4$ and E the Euler vector field $E(x) = x$ of [SW, (3)]; in this chapter E always means this vector field. For a vector field v on V_{10}, $(v, -)$ is its metric dual one-form, L_v the Lie derivative [ESW, (32)], $v(-)$ the derivative along v, and $\operatorname{div} v$ the divergence, $L_v \operatorname{dvol} = (\operatorname{div} v) \operatorname{dvol}$.

Definition 2.1 (Even part)**.** The even part of $\mathfrak{psl}(4|4)$ is the Lie algebra of vector fields on $V_{10} = V \oplus W_{\mathbb{R}}^{\perp}$ spanned by

- translations $x \mapsto v_1$, $v_1 \in V$;
- rotations, the linear vector fields given by elements of $\mathfrak{so}(V)$;
- the dilatation $\Delta = 2E$, the generator $\Delta = \operatorname{diag}(1, -1 \,|\, 0)$ of [SW, (17)], with $\operatorname{ad}_\Delta = -2$ on translations (Remark 2.6);
- special conformal fields $x \mapsto (x,x)v_1 - 2(v_1, x)x$, $v_1 \in V$, the special conformal field of [SW, (3)] along v_1;
- linear vector fields on $W_{\mathbb{R}}^{\perp}$ given by elements of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$,

with the bracket $[v, v'] = v(v') - v'(v)$. The first four kinds are the conformal Killing vector fields of V; together with $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ the even part has dimension $15 + 15 = 30$.

Definition 2.2 (Odd part)**.** The odd part of $\mathfrak{psl}(4|4)$ is $\Pi(\Sigma \oplus \Sigma^*)$. An element $Q_1 \in \Sigma$ is realised by the constant spinor Q_1, and an element $Q_2 \in \Sigma^*$ by the Σ-valued function $x \mapsto \rho(x)Q_2$. Together these span the conformal Killing spinors

$$x \mapsto Q_1 + \rho(x)Q_2, \qquad Q_1 \in \Sigma,\ Q_2 \in \Sigma^*.$$

Supercharge spinors always carry a subscript, $Q_1, Q_2, \ldots$, and each formula states whether they lie in Σ or in Σ^*; the unsubscripted Q is the differential of Chapter 1.

Definition 2.3 (Bracket)**.** On two even elements the bracket of $\mathfrak{psl}(4|4)$ is the bracket of vector fields, $[v, v'] = v(v') - v'(v)$. For an even element v and an odd element,

$$[v,\, Q_1 + \rho(x)Q_2] = v(Q_1 + \rho(x)Q_2) + \tfrac{1}{4}\rho(d(v,-))(Q_1 + \rho(x)Q_2) - \tfrac{1}{8}(\operatorname{div} v)(Q_1 + \rho(x)Q_2), \qquad (2.1)$$

where ρ of the two-form $d(v,-)$ means $\rho \circ q$ [ESW, (29)]; the right-hand side is again a conformal Killing spinor. For two odd elements, with $Q_1, Q_3 \in \Sigma$ and $Q_2, Q_4 \in \Sigma^*$,

$$[Q_1 + \rho(x)Q_2,\ Q_3 + \rho(x)Q_4] = \Big(x \mapsto \Gamma\big(Q_1 + \rho(x)Q_2,\, Q_3 + \rho(x)Q_4\big)\big|_V\Big) + \big(\text{rotation of } W_{\mathbb{R}}^{\perp}\big), \qquad (2.2)$$

where $|_V$ is the orthogonal projection $V_{10} \to V$ and the rotation of $W_{\mathbb{R}}^{\perp}$ is the element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ with matrix elements

$$\big(w, (\text{rotation of } W_{\mathbb{R}}^{\perp})\, w'\big) = (\rho(q(w \wedge w'))Q_1, Q_4) + (\rho(q(w \wedge w'))Q_3, Q_2), \qquad w, w' \in W_{\mathbb{R}}^{\perp}.$$

The $W_{\mathbb{R}}^{\perp}$-components of Γ drop out of the bracket, as in the dimensional reduction of the super-translation algebra [ESW, §2.6, Prop 2.20]; they reappear in $S^{(2)}$ (Definition 2.15). The coefficient 1 in front of the rotation of $W_{\mathbb{R}}^{\perp}$ is fixed by the super-Jacobi identity on triples in $\Sigma \times \Sigma \times \Sigma^*$: each term of the Jacobi sum of three odd elements contains exactly one odd–odd bracket, so the sum is affine in this coefficient; it vanishes at 1 and is nonzero at 0 (Remark 2.18), so 1 is its unique zero.

Proposition 2.4 (Brackets of the named generators)**.**

- *For $Q_1, Q_2 \in \Sigma$: $[Q_1, Q_2] = \Gamma(Q_1, Q_2)|_V$, a translation; this is [ESW, Def 2.5] after reduction.*
- *For $Q_1, Q_2 \in \Sigma^*$: $[Q_1, Q_2] = -\big(x \mapsto (x,x)v_1 - 2(v_1,x)x\big)$ with $v_1 = \Gamma(Q_1, Q_2)|_V$, a special conformal field.*
- *For $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$: $[Q_1, Q_2] = \big(x \mapsto \Gamma(Q_1, \rho(x)Q_2)|_V\big) + (\text{rotation of } W_{\mathbb{R}}^{\perp})$ as in Definition 2.3; the dilatation part of the first summand is $\frac{1}{2}(Q_1, Q_2)\,\Delta$, and the rest is a rotation of V.*
- *For a translation $x \mapsto v_1$: $[v_1, Q_1] = 0$ for $Q_1 \in \Sigma$ and $[v_1, Q_2] = \rho(v_1)Q_2 \in \Sigma$ for $Q_2 \in \Sigma^*$.*
- *For a special conformal field $x \mapsto (x,x)v_1 - 2(v_1,x)x$: its bracket with $Q_1 \in \Sigma$ is $\rho(v_1)Q_1 \in \Sigma^*$, and with $Q_2 \in \Sigma^*$ it is 0.*
- *$[\Delta, Q_1] = -Q_1$ for $Q_1 \in \Sigma$ and $[\Delta, Q_2] = +Q_2$ for $Q_2 \in \Sigma^*$.*

Proof. The first two items follow from (2.2): for $Q_2 = Q_4 = 0$ the rotation of $W_{\mathbb{R}}^{\perp}$ vanishes; for $Q_1 = Q_3 = 0$ it vanishes as well, and for $v \in V$

$$(v, \Gamma(\rho(x)Q_1, \rho(x)Q_2)) = (\rho(x)\rho(v)\rho(x)Q_1, Q_2) = 2(v,x)(x, \Gamma(Q_1, Q_2)) - (x,x)(v, \Gamma(Q_1, Q_2)),$$

using $\rho(x)\rho(v)\rho(x) = 2(v,x)\rho(x) - (x,x)\rho(v)$ from the Clifford relation and $(\rho(x)Q_1, Q_2) = (Q_1, \rho(x)Q_2)$. Projecting to V gives $-\big((x,x)v_1 - 2(v_1,x)x\big)$.

For $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$ and $v \in V$, $(v, \Gamma(Q_1, \rho(x)Q_2)) = (\rho(x)\rho(v)Q_1, Q_2)$; the part symmetric under $x \leftrightarrow v$ is $\frac{1}{2}(\rho(x)\rho(v) + \rho(v)\rho(x)) = (x, v)$, so the linear vector field $x \mapsto \Gamma(Q_1, \rho(x)Q_2)|_V$ equals $(Q_1, Q_2)\, E = \frac{1}{2}(Q_1, Q_2)\, \Delta$ plus an antisymmetric linear field, i.e. a rotation.
For the even–odd brackets use (2.1). For $v = 2E$: $(v, -) = d(x, x)$ is closed, $\operatorname{div}(2E) = 8$, $2E(Q_1) = 0$ and $2E(\rho(x)Q_2) = 2\rho(x)Q_2$, so $[\Delta, Q_1] = -Q_1$ and $[\Delta, \rho(x)Q_2] = (2-1)\rho(x)Q_2$. For a translation, $(v, -)$ is constant and $\operatorname{div} v = 0$, so (2.1) reduces to $v_1(\rho(x)Q_2) = \rho(v_1)Q_2$. For $v = (x \mapsto (x, x)v_1 - 2(v_1, x)x)$,

$$\begin{aligned} d(v, -) &= d(x, x) \wedge (v_1, -) - 2\, d(v_1, x) \wedge (x, -) = 4\, (x, -) \wedge (v_1, -), \\ \operatorname{div} v &= 2(x, v_1) - 2(v_1, x) - 8(v_1, x) = -8(v_1, x), \end{aligned}$$

so $\frac{1}{4}\rho(d(v, -)) - \frac{1}{8} \operatorname{div} v = \rho(x)\rho(v_1) - (x, v_1) + (v_1, x) = \rho(x)\rho(v_1)$. On a constant $Q_1 \in \Sigma$ this gives $x \mapsto \rho(x)\rho(v_1)Q_1$, the realisation of $\rho(v_1)Q_1 \in \Sigma^*$. On $x \mapsto \rho(x)Q_2$ it gives

$$\begin{aligned} &\rho\big((x, x)v_1 - 2(v_1, x)x\big)Q_2 + \rho(x)\rho(v_1)\rho(x)Q_2 \\ &\quad = (x, x)\rho(v_1)Q_2 - 2(v_1, x)\rho(x)Q_2 + 2(v_1, x)\rho(x)Q_2 - (x, x)\rho(v_1)Q_2 = 0. \end{aligned}$$

□

subspace	realisation	block of [SW, (17)]	dimension	ad_Δ eigenvalue
translations	$x \mapsto v_1$, $v_1 \in V$	$(2, 1)$	4	-2
$\mathfrak{so}(V)$, part acting on S_+	linear vector field of $\mathfrak{so}(V)$	$(1, 1)$	3	0
$\mathfrak{so}(V)$, part acting on S_-	linear vector field of $\mathfrak{so}(V)$	$(2, 2)$	3	0
dilatation	$\Delta = 2E$	$\operatorname{diag}(1, -1 \,\vert\, 0)$	1	0
special conformal	$x \mapsto (x, x)v_1 - 2(v_1, x)x$	$(1, 2)$	4	$+2$
$\mathfrak{so}(W_{\mathbb{R}}^{\perp})$	linear vector fields on $W_{\mathbb{R}}^{\perp}$	$(3, 3)$	15	0
$S_+ \otimes W \subset \Sigma$	constant spinor	$(3, 1)$	8	-1
$S_- \otimes W^* \subset \Sigma$	constant spinor	$(2, 3)$	8	-1
$S_- \otimes W \subset \Sigma^*$	$x \mapsto \rho(x)Q_2$	$(3, 2)$	8	$+1$
$S_+ \otimes W^* \subset \Sigma^*$	$x \mapsto \rho(x)Q_2$	$(1, 3)$	8	$+1$

The ten named subspaces of $\mathfrak{psl}(4|4)$; rows and columns of the blocks are ordered as in [SW, (17)]. The eigenspaces of Proposition 1.2 fix the names; in particular the block $(3, 1)$ of [SW, (17)] is $S_+ \otimes W$.

Proposition 2.5 (Identification with $\mathfrak{psl}(4|4)$)**.** *The bracket of Definition 2.3 has the following properties.*

- *The dimension is* $30|32$*, and the bracket of two elements of the ten named subspaces lies in their span.*
- *The bracket is graded antisymmetric and satisfies the super-Jacobi identity.*
- *The* V*-part of every odd–odd bracket is a conformal Killing vector field of* V*; the rotation of* $W_{\mathbb{R}}^{\perp}$ *is nonzero for* $\Sigma \times \Sigma^*$ *and vanishes for* $\Sigma \times \Sigma$ *and* $\Sigma^* \times \Sigma^*$*.*
- *The block rule of the supermatrix presentation [SW, (17), (19)] holds for every pair of the ten named subspaces: their bracket lies in the block with the row of the first subspace and the column of the second when the column of the first equals the row of the second, plus the block with the row of the second and the column of the first when the column of the second equals the row of the first, and vanishes otherwise.*
- ad_Δ *is diagonal on the named subspaces with the eigenvalues of the table above, which are those of* $\Delta = \operatorname{diag}(1, -1 \,\vert\, 0)$ *in [SW, (17)].*

- *The Killing form* $\mathrm{str}(\mathrm{ad}_{x_1}\mathrm{ad}_{x_2})$ *for* $x_1, x_2 \in \mathfrak{psl}(4|4)$*, with* str *the supertrace [SW, after (17)], vanishes identically.*
- *The algebra equals its derived algebra and has trivial centre; its even part is the sum of the two commuting ideals formed by the conformal Killing vector fields of* V *and by* $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$*, each of dimension* 15 *with nondegenerate Killing form.*

These are the properties by which the algebra is identified with the complexification $\mathfrak{psl}(4|4)$ *of* $\mathfrak{psu}(2,2|4)$ *[SW, (7), (17)].*

Proof. The dimension count is $4+6+1+4+15 = 30$ and $16+16 = 32$. Every other item is a bilinear or trilinear identity in the arguments, or a statement about the structure constants; each is established on all ordered pairs and all ordered triples of elements of a basis adapted to the ten named subspaces (Remark 2.18), hence for all arguments. □

Remark 2.6 (Dilatation normalisation)**.** The dilatation is the Δ of [SW, (17)], the vector field $2E$, with ad_{Δ} eigenvalue -2 on translations, $+2$ on special conformal fields, -1 on Σ, $+1$ on Σ^* and 0 on rotations. The generator called Δ in [SW, (3)] is the vector field $-E$, and the bosonic part of the Δ of [SW, (12)] is E. Every cochain component is linear in its arguments, so a component evaluated on $2E$ is -2 times the same component evaluated on $-E$:

$$S^{(1)}(2E) = -2\,S^{(1)}(-E),$$

and a coefficient computed in the normalisation of [SW, (3)] is converted to the normalisation used here by the factor -2.

2.2 The Chevalley–Eilenberg complex and its sign conventions

Definition 2.7 (Cochains)**.** Following [ESW, Def 1.26] with the Lie algebra of loc. cit. taken to be $\mathfrak{psl}(4|4)$,

$$C^{\bullet}(\mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E})) = \bigoplus_n \mathrm{Hom}(\wedge^n \mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E}))[-n],$$

with $\mathscr{O}_l(\mathscr{E})$ the local functionals [C07, Def 4.2.2] ($\mathcal{O}_{\mathrm{loc}}$ in [ESW, Def 1.26]). The elements of this complex are multilinear maps f of elements $x_1, x_2, \dots$ of $\mathfrak{psl}(4|4)$ (the notation of [ESW, Def 1.26]; the position x carries no subscript) with the antisymmetry

$$f(\dots, x_i, x_{i+1}, \dots) = (-1)^{|x_i||x_{i+1}|+1} f(\dots, x_{i+1}, x_i, \dots);$$

[ESW, Def 1.26] prints the exponent $|x_1||x_2|+1$ [sic]. Here $|f|$ denotes the degree of f in $C^{\bullet}$, and an element of degree 0 has values $f(x_1, \dots, x_k)$ of parity $k + \sum_i |x_i|$.

Definition 2.8 (Cochain bracket and differential)**.** The cochain bracket is the formula of [ESW, Def 1.26]: $\{f, g\}(x_1, \dots, x_{k+l})$ is the sum over (k,l)-shuffles σ of

$$\mathrm{sgn}(\sigma) \cdot (\text{Koszul sign of } \sigma) \cdot (-1)^{|g|k + \sum_{i=1}^{k} |x_{\sigma(i)}|(l+|g|)} \{f(x_{\sigma(1)}, \dots, x_{\sigma(k)}),\, g(x_{\sigma(k+1)}, \dots, x_{\sigma(k+l)})\},$$

with $\{-,-\}$ the bracket of Definition 1.6 and $|g|$ the degree of g in $C^{\bullet}$. The Chevalley–Eilenberg differential is

$$(d_{CE}f)(x_1, \dots, x_n) = \sum_{i<j} (-1)^{|x_i| \sum_{p<i} |x_p| + |x_j| \sum_{p<j,\, p\neq i} |x_p| + i + j + |f| + 1} f([x_i, x_j], x_1, \dots, \widehat{x_i}, \dots, \widehat{x_j}, \dots, x_n), \tag{2.3}$$

that is, the formula printed in [ESW, Def 1.26] multiplied by -1.

Lemma 2.9 (The sign of the Chevalley–Eilenberg differential)**.** *For $Q_1, Q_2 \in \Sigma$ and $|f| = 0$, the formula printed in [ESW, Def 1.26] gives*

$$(d_{CE}S^{(1)})(Q_1, Q_2) = -S^{(1)}([Q_1, Q_2]),$$

and (2.3) *gives*

$$(d_{CE}S^{(1)})(Q_1, Q_2) = +S^{(1)}([Q_1, Q_2]);$$

the second is [ESW, Lemma 3.6, item 2]. With the bracket of Definition 1.6 and the functionals [ESW, (33), (34)], the equation of [ESW, Lemma 3.6]

$$\{S, S^{(2)}(Q_1, Q_2)\} + S^{(1)}(\Gamma(Q_1, Q_2)) - \{S^{(1)}(Q_1), S^{(1)}(Q_2)\} = 0$$

holds; this fixes the sign of the term $S^{(1)}(\Gamma(Q_1, Q_2))$, and with this bracket [ESW, proof of Lemma 3.6, item 3] holds term by term. The printed formula is the Chevalley–Eilenberg differential of the opposite algebra; the substitution $x_i \mapsto -x_i$ in every argument, $S^{(k)} \mapsto (-1)^k S^{(k)}$, exchanges the two conventions. Hence (2.3) *keeps [ESW, (32)–(34)] verbatim. Under the printed convention the arity-one functionals are -1 times those of [ESW, (32)–(33)], so the sign of every one-generator component is tied to the choice of d_{CE}.*

Proof. For $n = 2$ the pair $(i, j) = (1, 2)$ is the single term, both sums over p are empty, and the exponent in (2.3) is $1 + 2 + |f| + 1 \equiv |f|$; for $|f| = 0$ this gives $+f([x_1, x_2])$, and the printed formula gives $-f([x_1, x_2])$. The displayed equation of [ESW, Lemma 3.6] is the arity-two component of the master equation on $\Sigma \times \Sigma$ (Lemma 2.11 with $|x_1| = 1$), which holds for every $\mathfrak{g}$ by Theorem 2.20. With $-S^{(1)}(\Gamma(Q_1, Q_2))$ in place of $+S^{(1)}(\Gamma(Q_1, Q_2))$ the left side becomes $-2S^{(1)}(\Gamma(Q_1, Q_2))$, which contains $-2(L_v A, A^*)$ for the translation $v = \Gamma(Q_1, Q_2)|_V$ and is nonzero when $\Gamma(Q_1, Q_2)|_V \neq 0$; with the bracket replaced by its negative the left side becomes $2S^{(1)}(\Gamma(Q_1, Q_2))$, nonzero as well. The expression for $\{S, S^{(2)}(Q_1, Q_2)\}$ in [ESW, proof of Lemma 3.6, item 3] is reproduced term by term with the bracket of Definition 1.6 by the exact evaluation of Remark 2.18. For the last statement: d_{CE} is linear in the bracket of $\mathfrak{psl}(4|4)$, so the printed formula is (2.3) for the opposite bracket $-[-,-]$; the linear map $x_1 \mapsto -x_1$ is an isomorphism from $\mathfrak{psl}(4|4)$ to its opposite, since $-[x_1, x_2] = -[-x_1, -x_2]$, and it transports a cochain of arity k to $(-1)^k$ times itself. □

Definition 2.10 (Action of $\mathfrak{psl}(4|4)$)**.** An action of $\mathfrak{psl}(4|4)$ on $(\mathscr{E}, \langle\,,\rangle, S)$ [ESW, Def 1.27] is an element

$$S_{\mathfrak{psl}(4|4)} = \sum_{k \geq 0} S^{(k)}_{\mathfrak{psl}(4|4)} \in C^\bullet(\mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E}))$$

of degree 0, with $S^{(0)}_{\mathfrak{psl}(4|4)} = S$, each $S^{(k\geq 1)}_{\mathfrak{psl}(4|4)}(x_1, \ldots, x_k)$ at least quadratic in the fields, and

$$d_{CE}S_{\mathfrak{psl}(4|4)} + \tfrac{1}{2}\{S_{\mathfrak{psl}(4|4)}, S_{\mathfrak{psl}(4|4)}\} = 0. \tag{2.4}$$

Equivalently [ESW, Rem 1.29], $I_{\mathfrak{psl}(4|4)} = I + \sum_{k\geq 1} S^{(k)}_{\mathfrak{psl}(4|4)}$ satisfies

$$(d_{CE} + Q)I_{\mathfrak{psl}(4|4)} + \tfrac{1}{2}\{I_{\mathfrak{psl}(4|4)}, I_{\mathfrak{psl}(4|4)}\} = 0.$$

We abbreviate $S^{(k)} = S^{(k)}_{\mathfrak{psl}(4|4)}$.

Lemma 2.11 (Component equations)**.** *With the conventions of Definitions 2.7–2.8, the component of* (2.4) *on $(x_1, \ldots, x_n)$ reads as follows. In arity* 0, *$\{S, S\} = 0$. In arity* 1, *$\{S, S^{(1)}(x_1)\} = 0$. In arity* 2,

$$\{S, S^{(2)}(x_1, x_2)\} + S^{(1)}([x_1, x_2]) + (-1)^{|x_1|}\{S^{(1)}(x_1), S^{(1)}(x_2)\} = 0.$$

In arity 3, with x_1 even and x_2, x_3 odd,

$$\{S, S^{(3)}(x_1,x_2,x_3)\} + S^{(2)}([x_1,x_2],x_3) + S^{(2)}([x_1,x_3],x_2) + S^{(2)}([x_2,x_3],x_1) \\ + \{S^{(1)}(x_1), S^{(2)}(x_2,x_3)\} - \{S^{(1)}(x_2), S^{(2)}(x_1,x_3)\} - \{S^{(1)}(x_3), S^{(2)}(x_1,x_2)\} = 0.$$

In arity 3 with x_1, x_2, x_3 odd the component is the same expression with all seven signs +; this is the cyclic form of [ESW, Lemma 3.7]. In arity 4, with all arguments odd,

$$\{S, S^{(4)}(x_1,\dots,x_4)\} + \sum_{i<j} S^{(3)}([x_i,x_j],\dots) - \sum_i \{S^{(1)}(x_i), S^{(3)}(\dots)\} \\ + \{S^{(2)}(x_1,x_2), S^{(2)}(x_3,x_4)\} + \{S^{(2)}(x_1,x_3), S^{(2)}(x_2,x_4)\} + \{S^{(2)}(x_1,x_4), S^{(2)}(x_2,x_3)\} = 0,$$

where the dots are the remaining arguments in increasing order. For $\mathfrak{psl}(4|4)$ replaced by the super-translation algebra these are the four equations of [ESW, Thm 3.3].

Proof. Every component of $S_{\mathfrak{psl}(4|4)}$ has degree 0 in $C^\bullet$, so $|f| = |g| = 0$ in Definition 2.8, the shuffle sign reduces to $\mathrm{sgn}(\sigma)$ times the Koszul sign times $(-1)^{l\sum_{i\le k}|x_{\sigma(i)}|}$, and the value $S^{(k)}(x_1,\dots,x_k)$ has parity $k + \sum|x_i|$.
Arity 1: d_{CE} of $S^{(0)}$ has no pairs; the shuffles $(0,1)$ and $(1,0)$ give $\{S, S^{(1)}(x_1)\}$ and $\{S^{(1)}(x_1), S\} = \{S, S^{(1)}(x_1)\}$, and $\frac{1}{2}$ of their sum is $\{S, S^{(1)}(x_1)\}$.
Arity 2: by Lemma 2.9, $(d_{CE}S^{(1)})(x_1,x_2) = S^{(1)}([x_1,x_2])$. The shuffles $(0,2)$ and $(2,0)$ give $2\{S, S^{(2)}(x_1,x_2)\}$. The $(1,1)$-shuffle $\sigma = \mathrm{id}$ gives $(-1)^{|x_1|}\{S^{(1)}(x_1), S^{(1)}(x_2)\}$; the transposition gives $-(-1)^{|x_1||x_2|+|x_2|}\{S^{(1)}(x_2), S^{(1)}(x_1)\}$, and graded symmetry with $|S^{(1)}(x_i)| = 1 + |x_i|$ turns it into

$$-(-1)^{|x_1||x_2|+|x_2|+(1+|x_1|)(1+|x_2|)}\{S^{(1)}(x_1), S^{(1)}(x_2)\} = (-1)^{|x_1|}\{S^{(1)}(x_1), S^{(1)}(x_2)\}.$$

Half the sum is the stated equation.
Arity 3 with x_1 even, x_2, x_3 odd. In (2.3) the pairs $(1,2)$, $(1,3)$, $(2,3)$ have exponents 4, $|x_3||x_2|+5 = 6$ and 6, so all three $S^{(2)}([x_i,x_j],x_k)$ enter with +. The $(1,2)$-shuffles give

$$\{S^{(1)}(x_1), S^{(2)}(x_2,x_3)\} - \{S^{(1)}(x_2), S^{(2)}(x_1,x_3)\} - \{S^{(1)}(x_3), S^{(2)}(x_1,x_2)\},$$

the second sign from $\mathrm{sgn} = -1$, and the third from the Koszul sign $|x_3||x_2| = 1$. The $(2,1)$-shuffles $(12|3)$, $(13|2)$, $(23|1)$ carry the signs $-$, $(-)(-)(-) = -$ and $+$; moving $S^{(1)}$ to the left costs $(-1)^{|S^{(2)}||S^{(1)}|} = +1$ in each case, because $S^{(1)}(x_3)$, $S^{(1)}(x_2)$ are even and $S^{(2)}(x_2,x_3)$ is even. The $(2,1)$-shuffles therefore reproduce the same three terms, and half the total is the stated equation; the $(0,3)$ and $(3,0)$ shuffles give $\{S, S^{(3)}\}$. The remaining patterns follow from the same expansion. □

2.3 The Noether current [ESW, Rem 1.28] in arity one

Definition 2.12 (Arity one)**.** For a conformal Killing vector field v of V, or v the linear vector field of an element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, regarded as a vector field on V_{10}:

$$S^{(1)}(v) = \int \mathrm{dvol}\Big(\big(L_v A + \tfrac{1}{4}(\mathrm{div}\, v)\, A|_{W_{\mathbb{R}}^{\perp}},\, A^*\big) - \big(v.\lambda, \lambda^*\big) - \big(v.c,\, c^*\big)\Big), \\ v.\lambda = v(\lambda) + \tfrac{1}{4}\rho(d(v,-))\lambda + \tfrac{3}{8}(\mathrm{div}\, v)\lambda, \tag{2.5}$$

where $v.c = v(c)$ is the derivative of c along v [ESW, (32)]. Here A is a one-form on V_{10} constant along $W_{\mathbb{R}}^{\perp}$ [ESW, Def 1.55, Prop 1.56] and L_v is its Lie derivative on V_{10}: for v on V, $L_v(A|_V)$ is the Lie derivative of the one-form and $L_v(\iota_w A) = v(\iota_w A)$; for v the linear vector field of an element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, $\iota_w(L_v A) = \iota_{v(w)}A$.

For $Q_1 \in \Sigma$ [ESW, (33)], reduced:

$$S^{(1)}(Q_1) = \int \mathrm{dvol}\Big(-(\Gamma(Q_1,\lambda), A^*) + \tfrac{1}{2}(\rho(F_A)Q_1, \lambda^*)\Big). \tag{2.6}$$

For $Q_2 \in \Sigma^*$, with conformal Killing spinor $x \mapsto \rho(x)Q_2$:

$$S^{(1)}(Q_2) = \int \mathrm{dvol}\Big(-(\Gamma(\rho(x)Q_2,\lambda), A^*) + \tfrac{1}{2}(\rho(F_A)\rho(x)Q_2, \lambda^*) - (\rho(A|_{W_{\mathbb{R}}^\perp})Q_2, \lambda^*)\Big), \tag{2.7}$$

where $\rho(A|_{W_{\mathbb{R}}^\perp})$ is Clifford multiplication by the $W_{\mathbb{R}}^\perp$-component of A, with its $\mathfrak{g}$ factor kept.

The divergences are $\operatorname{div} v = 0$ for translations, rotations and $\mathfrak{so}(W_{\mathbb{R}}^\perp)$, $\operatorname{div}(2E) = 8$, and $\operatorname{div} v = -8(v_1, x)$ for $v = (x \mapsto (x,x)v_1 - 2(v_1,x)x)$. For $v \in \mathfrak{iso}(V) \oplus \mathfrak{so}(W_{\mathbb{R}}^\perp)$, (2.5) is [ESW, (32)], whose "$\mathfrak{so}(V)$ action on Σ" is $\frac{1}{4}\rho(d(v,-))$. Since $\not{d}(\rho(x)Q_2) = (\dim V)\, Q_2 = 4Q_2$ by the Clifford relation, the last term of (2.7) equals $-\frac{1}{4}(\rho(A|_{W_{\mathbb{R}}^\perp})\not{d}(\rho(x)Q_2), \lambda^*)$; in this form one formula covers the whole conformal Killing spinor $x \mapsto Q_1 + \rho(x)Q_2$, the term vanishing for constant spinors.

Example 2.13 (The dilatation)**.** For $\Delta = 2E$: $(2E, -) = d(x,x)$ is closed, so $\rho(d(v,-)) = 0$, and $\operatorname{div}(2E) = 8$. On the one-form $A|_V$, $L_{2E}A|_V = 2E(A|_V) + 2A|_V$ with $E(-)$ acting componentwise, and on the scalars $L_{2E}\iota_w A + \frac{1}{4}\cdot 8\,\iota_w A = 2E(\iota_w A) + 2\iota_w A$. Hence

$$S^{(1)}(\Delta) = \int \mathrm{dvol}\Big(\big(2E(A) + 2A, A^*\big) - \big(2E(\lambda) + 3\lambda, \lambda^*\big) - \big(2E(c), c^*\big)\Big) :$$

in the normalisation of Remark 2.6, A, λ, c carry the weights 2, 3, 0, i.e. the Weyl weights 1, $\frac{3}{2}$, 0 with respect to E.

Proposition 2.14 (The coefficients are forced)**.** *Replace the coefficients $\frac{1}{4}$ (spin), $\frac{3}{8}$ (Weyl weight of λ) and $\frac{1}{4}$ (Weyl weight of $A|_{W_{\mathbb{R}}^\perp}$) in* (2.5)*, and the coefficient -1 of the last term of* (2.7)*, by unknowns. For $\mathfrak{g} = \mathfrak{su}(2)$ the arity-one equation $\{S, S^{(1)}(x_1)\} = 0$ at one element x_1 of each type determines the unknowns uniquely:*

- *a rotation in $\mathfrak{so}(V)$, and an element of $\mathfrak{so}(W_{\mathbb{R}}^\perp)$: spin coefficient $\frac{1}{4}$;*
- *Δ: $(\frac{3}{8}, \frac{1}{4})$ for the two Weyl weights;*
- *a special conformal field: $(\frac{1}{4}, \frac{3}{8}, \frac{1}{4})$;*
- *$Q_2 \in \Sigma^*$: -1.*

With these values the arity-one equation holds for every element of $\mathfrak{psl}(4|4)$ and every $\mathfrak{g}$ (Theorem 2.20). The Weyl weight 1 of $A|_V$ is carried by L_v and the weight 0 of c by v.c.

Proof. For each type the bracket $\{S, S^{(1)}(x_1)\}$ is affine in the unknowns. For $\mathfrak{g} = \mathfrak{su}(2)$ and one element x_1 of each type, its density is a polynomial in the jets of the fields with coefficients affine in the unknowns, and the conditions that all Euler–Lagrange derivatives vanish form a linear system whose unique solution is the one stated (Remark 2.18). For translations, rotations, $\mathfrak{so}(W_{\mathbb{R}}^\perp)$ and $Q_1 \in \Sigma$ the equation is [ESW, (32), Lemma 3.4] after reduction. □

The functionals $S^{(1)}(Q_1)$ and $S^{(1)}(Q_2)$ contain no c^*; the c^* terms of $S^{(1)}$ are $-(v.c, c^*)$ for even v.

2.4 Quadratic-antifield terms and the end of the tower

Definition 2.15 (Arity two)**.** $S^{(2)}(x_1, x_2) = 0$ when x_1 or x_2 is even. For Q_1, Q_2 each in Σ or in Σ^*, $S^{(2)}(Q_1, Q_2)$ is [ESW, (34)] evaluated pointwise on the conformal Killing spinors, i.e. every $Q_i \in \Sigma^*$ is replaced by the Σ-valued function $x \mapsto \rho(x)Q_i$ in

$$S^{(2)}(Q_1, Q_2) = \int \mathrm{dvol}\Big(\tfrac{1}{4}\big(\Gamma(Q_1, Q_2), \Gamma(\lambda^*, \lambda^*)\big) - \tfrac{1}{2}(Q_1, \lambda^*)(Q_2, \lambda^*) - \big(\iota_{\Gamma(Q_1,Q_2)}A,\, c^*\big)\Big), \tag{2.8}$$

with $\Gamma(Q_1, Q_2) \in V_{10}$ the full ten-component Γ-pairing, so that the gauge parameter is $\iota_{\Gamma(Q_1,Q_2)}A = \iota_{\Gamma(Q_1,Q_2)|_V}A|_V + \iota_{\Gamma(Q_1,Q_2)|_{W_{\mathbb{R}}^\perp}}A$. For $k \geq 3$, $S^{(k)} = 0$.

The field-dependent gauge parameter $\iota_{\Gamma(Q_1,Q_2)}A$ of the c^* term is:

- $Q_1, Q_2 \in \Sigma$: $\Gamma(Q_1, Q_2)$ constant, as in [ESW, (34)];
- $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$: $\Gamma(Q_1, \rho(x)Q_2)$, linear in x;
- $Q_1, Q_2 \in \Sigma^*$: $\Gamma(\rho(x)Q_1, \rho(x)Q_2)$, with V-part $-\big((x,x)v_1 - 2(v_1, x)x\big)$, $v_1 = \Gamma(Q_1, Q_2)|_V$, and $W_{\mathbb{R}}^\perp$-part $-(x,x)\,\Gamma(Q_1, Q_2)|_{W_{\mathbb{R}}^\perp}$ (proof of Proposition 2.4).

Both parts of $S^{(2)}$ are required by the arity-two equation: with $S^{(2)} = 0$ its component on every pair of odd elements of a basis is a nonzero local functional, and with the $W_{\mathbb{R}}^\perp$-components of Γ removed from (2.8) its components on $\Sigma \times \Sigma$ and on $\Sigma \times \Sigma^*$ are nonzero (Remark 2.18).

Proposition 2.16 (The tower stops at arity two)**.** *For every $\mathfrak{g}$ and all arguments, with $S^{(k\geq 3)} = 0$:*

- $\{S^{(2)}(x_1, x_2), S^{(2)}(x_3, x_4)\} = 0$*;*
- *the components of* (2.4) *of arity ≥ 4 hold;*
- *the arity-three components with at most one odd argument hold;*
- *the arity-three components with three odd arguments hold.*

Proof. $S^{(2)}$ contains the fields A, λ^*, c^* and none of their partners A^*, λ, c, and the bracket pairs each field with its partner (Definition 1.6); hence the first item. In arity 4, with $S^{(3)} = S^{(4)} = 0$, Lemma 2.11 leaves the three brackets $\{S^{(2)}, S^{(2)}\}$, which vanish; when an argument is even, every surviving term contains $S^{(2)}$ with an even argument and vanishes. In arity $n \geq 5$ every term contains $S^{(k)}$ with $k \geq 3$ or $\{S^{(k)}, S^{(l)}\}$ with $k + l = n$, hence $k \geq 3$ or $l \geq 3$, and vanishes. In arity 3 with at most one odd argument, every term contains $S^{(2)}$ with an even entry, because among $[x_i, x_j]$ and x_k, and among any two of the arguments, at most one entry is odd; so every term vanishes. In arity 3 with three odd arguments, every $[x_i, x_j]$ is even, so the equation of Lemma 2.11 reduces to the cyclic sum $\sum_{\mathrm{cyc}}\{S^{(1)}(x_1), S^{(2)}(x_2, x_3)\}$. In this bracket, $S^{(2)}$ supplies A and λ^*, whose partners A^* and λ occur in $S^{(1)}(x_1)$ exclusively through the term $-(\Gamma(Q_1, \lambda), A^*)$, with Q_1 replaced by the conformal Killing spinor of x_1; the last term of (2.7) contains A and λ^* and has vanishing bracket with $S^{(2)}$. No derivative acts on the spinors, so the cyclic sum is computed pointwise in x exactly as in the proof of [ESW, Lemma 3.7], with Q_1, Q_2, Q_3 replaced by the values of the three conformal Killing spinors at x; it vanishes by [ESW, Assumption 3.1], which holds for the ten-dimensional $\Sigma = S_+$ by [ESW, Thm 2.10] (list after [ESW, Assumption 3.1]), and the Clifford relation. □

2.5 The master equation

External input 2.17 (Poincaré supersymmetry, ten dimensions)**.** [ESW, Thm 3.3, Lemmas 3.4–3.7, (32)–(34)] and [ESW, Prop 2.20]: for every $\mathfrak{g}$, the functionals (2.5) on $\mathfrak{iso}(V) \oplus \mathfrak{so}(W_{\mathbb{R}}^{\perp})$, (2.6) and (2.8) on $\Sigma \times \Sigma$ satisfy every component of (2.4) with arguments in the subalgebra generated by translations, $\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ and Σ, with d_{CE} the differential (2.3) used in the proof of [ESW, Lemma 3.6].

Remark 2.18 (Exact verification on a basis)**.** The computer-assisted statements of this chapter are exact identities established on a basis. Fix a basis of $\mathfrak{psl}(4|4)$ adapted to the ten named subspaces, with $4+3+3+1+4+15$ even and $8+8+8+8$ odd elements, in which ρ, Γ and the pairing $\Sigma \otimes \Sigma^* \to \mathbb{C}$ have Gaussian-rational entries. The bracket of Definition 2.3 is computed in exact arithmetic on every ordered pair of basis elements; its values are expanded in the basis, and graded antisymmetry, the super-Jacobi identity on every ordered triple, the Killing form and the block rule on every ordered pair, the eigenvalues of ad_{Δ}, the derived algebra, the centre and the Killing forms of the two even ideals are computed from these structure constants. With the coefficient 0 in place of 1 in front of the rotation of $W_{\mathbb{R}}^{\perp}$ in (2.2), the super-Jacobi identity fails. For $\mathfrak{g} = \mathfrak{su}(2)$ and $\mathfrak{g} = \mathfrak{su}(3)$, with the data of Remark 1.15, every component of (2.4) of arity 1, 2 and 3 is evaluated on every unordered tuple of basis elements, repetitions allowed, with at least two odd elements in arity 3; each is a local functional whose density is a polynomial in the jets of the fields with Gaussian-rational coefficients, and it vanishes when all its Euler–Lagrange derivatives vanish identically (Remark 1.15). All these identities are multilinear in their arguments, and the components of (2.4) are graded antisymmetric under exchange of arguments (Definition 2.7); the identities on basis tuples therefore hold for all arguments.

Lemma 2.19 (Colour reduction)**.** *Consider local functionals that are finite sums of terms, each the product of a polynomial in the jets of the V_{10}-, spinor- and form-components of the fields, independent of $\mathfrak{g}$, and a tree tensor in the $\mathfrak{g}$-components of at most five fields: a multilinear form on $\mathfrak{g}^k$, $k \leq 5$, obtained from copies of $([-,-],-)$ by contraction with the inverse pairing $(-,-)^{-1}$ along a tree, or the pairing $(-,-)$ itself for $k = 2$. If the Euler–Lagrange derivatives of the density of such a functional vanish identically for $\mathfrak{g} = \mathfrak{su}(2)$, they vanish identically for every Lie algebra $\mathfrak{g}$ with a nondegenerate invariant symmetric pairing.*

Proof. By invariance of the pairing, every tree tensor with k legs is a sign times $(b(X_1, \dots, X_{k-1}), X_k)$ for a bracket monomial b of $X_1, \dots, X_{k-1} \in \mathfrak{g}$, each used once. Hence the tree tensors with k legs are the values of a linear map, natural in $\mathfrak{g}$ and equivariant under permutations of the legs, from the space of formal trees modulo antisymmetry and the Jacobi identity, which is the multilinear part of the free Lie algebra on $k-1$ letters, of dimension $(k-2)!$. For $\mathfrak{g} = \mathfrak{su}(2)$ this map has rank 1, 1, 2, 6 for $k = 2, 3, 4, 5$, computed exactly, so it is injective for $k \leq 5$, and averaging a left inverse over the permutations of the legs gives an equivariant left inverse. A functional of the stated form is therefore the evaluation in $\mathfrak{g}$ of a formal expression, a sum of jet polynomials times formal trees. Its Euler–Lagrange derivative along a field, paired with a fixed element of $\mathfrak{g}$ treated as one more leg, is again of this form, because differentiation acts on the jet polynomials and leaves the tree. The equivariant left inverse acts on the trees alone; applied to the vanishing $\mathfrak{su}(2)$ evaluation it shows that the formal Euler–Lagrange derivatives vanish, hence their evaluations vanish for every $\mathfrak{g}$. □

Theorem 2.20 (Master equation of the equivariant action)**.** *For every Lie algebra $\mathfrak{g}$ with a nondegenerate invariant symmetric pairing, $S_{\mathfrak{psl}(4|4)} = S + S^{(1)} + S^{(2)}$, with $S^{(1)}$, $S^{(2)}$ of Definitions 2.12–2.15 and $S^{(k \geq 3)} = 0$, is an action of $\mathfrak{psl}(4|4)$ on $(\mathscr{E}, \langle\,,\rangle, S)$: it satisfies (2.4), and $I_{\mathfrak{psl}(4|4)} = I + S^{(1)} + S^{(2)}$ satisfies $(d_{CE} + Q)I_{\mathfrak{psl}(4|4)} + \frac{1}{2}\{I_{\mathfrak{psl}(4|4)}, I_{\mathfrak{psl}(4|4)}\} = 0$.*

Proof. By Lemma 2.11 it suffices to prove the components. Arity 0 is Proposition 1.14. The components of arity ≥ 4, and of arity 3 with at most one or with three odd arguments, hold for every $\mathfrak{g}$ by Proposition 2.16. The remaining components, of arities 1 and 2 and of arity 3 with exactly two odd arguments, vanish for $\mathfrak{g} = \mathfrak{su}(2)$ and $\mathfrak{g} = \mathfrak{su}(3)$ for all arguments (Remark 2.18). Each of them is of the form of Lemma 2.19: every term of S, $S^{(1)}$, $S^{(2)}$ is a tree tensor in at most four, three and two $\mathfrak{g}$-components respectively, built from $(-,-)$ and $[-,-]$; the bracket of Definition 1.6 contracts one leg of each factor with the inverse pairing and joins two trees into a tree; so every term has at most $4 + 3 - 2 = 5$ legs. By Lemma 2.19 these components vanish for every $\mathfrak{g}$. The equation for $I_{\mathfrak{psl}(4|4)}$ is [ESW, Rem 1.29]. On the subalgebra of Axiom 2.17 the statement is [ESW, Thm 3.3]. □

Remark 2.21 (Physical reading)**.** For an even v and Q_1 odd, $S^{(2)}(v, Q_1) = 0$ and the arity-two equation is

$$S^{(1)}([v, Q_1]) = -\{S^{(1)}(v), S^{(1)}(Q_1)\} :$$

the conformal and R-symmetry transformations, with the Weyl weights of Example 2.13, commute with the supersymmetry transformations off shell. For two odd elements,

$$\{S^{(1)}(Q_1), S^{(1)}(Q_2)\} = S^{(1)}([Q_1, Q_2]) + \{S, S^{(2)}(Q_1, Q_2)\},$$

where $[Q_1, Q_2]$ is a conformal Killing vector field plus an element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$; $\{S, S^{(2)}\}$ contains the fermion equation of motion (the $\lambda^*\lambda^*$ terms) and the gauge transformation with field-dependent parameter $\iota_{\Gamma(Q_1,Q_2)}A$ (the c^* term).

Chapter 3

Costello's data with a UV boundary

Costello's renormalisation needs a gauge-fixing operator Q^{GF}, the heat kernel of $H = [Q, Q^{GF}]$ and the propagator $P(\varepsilon, \infty) = \int_\varepsilon^\infty (Q^{GF} \otimes 1) K_t \, dt$ [C07, §6.5], and its axioms [C07, Def 4.3.3] are formulated on a compact manifold; [C07, §5.2] records that Yang–Mills theory in second-order form lies outside them. On Euclidean $\mathbb{R}^4$ we keep a single regulator, the lower Schwinger limit ε, and set the upper limit to ∞, so that ε is the single scale of the construction. This chapter constructs a Feynman-gauge operator Q^{GF} that satisfies three of the five axioms, identifies the two that fail, and proves the identities that hold with these data: $QP(\varepsilon, \infty) = K_\varepsilon$, the UV-boundary identity, the acyclicity of Q at scale ∞ on sections with vanishing moments together with the zero-mode classes that Q has on compactly supported sections, and the one-loop formula for the graph expansion.

3.1 The Feynman-gauge operator

Definition 3.1 (Gauge-fixing operator)**.** $Q^{GF} \colon \mathscr{E} \to \mathscr{E}$ is the odd operator of degree -1 given by the table; it is zero on the slots of c and λ.

on the slot of	Q^{GF}	lands in the slot of	momentum space
A^*	$A^* \mapsto -2A^*$ (all ten components, through the metric of V_{10})	A	-2
A	$A \mapsto -d^*(A\|_V)$	c	$i\,\iota_p A$
c^*	$c^* \mapsto dc^*$	$A^*\|_V$	$i\,p\,c^*$
λ^*	$\lambda^* \mapsto -\not{d}\|_{\Sigma^*}\lambda^*$	λ	$-i\rho(p)\lambda^*$

The Feynman-gauge operator Q^{GF}.

Proposition 3.2 (The Laplacian)**.**

- $H := [Q, Q^{GF}] = QQ^{GF} + Q^{GF}Q = d^*d + dd^*$*, the Hodge Laplacian of flat* $\mathbb{R}^4$*, which acts on every component function of every slot as the positive Laplacian of functions; in momentum space* $H = |p|^2$*.*
- $[Q^{GF}, H] = 0$ *and* $[Q, H] = 0$*.*
- *Among operators with the arrows of Definition 3.1 and arbitrary coefficients in place of the four coefficients* -2*,* -1*,* 1*,* -1 *of the table, these are the unique ones for which* H *is the componentwise Laplacian.*

Proof. Using Proposition 1.11, slot by slot:

$$\begin{aligned}
c:&\quad Q^{GF}Qc = Q^{GF}(-dc) = -\big(-d^*dc\big) = d^*dc, \qquad QQ^{GF}c = 0;\\
A:&\quad QQ^{GF}A = Q\big(-d^*(A|_V)\big) = dd^*(A|_V), \qquad Q^{GF}QA = Q^{GF}\big(-\tfrac{1}{2}d^*dA\big) = d^*dA;\\
A^*:&\quad QQ^{GF}A^* = Q(-2A^*) = d^*dA^*, \qquad Q^{GF}QA^* = Q^{GF}\big(d^*(A^*|_V)\big) = dd^*(A^*|_V);\\
c^*:&\quad QQ^{GF}c^* = Q(dc^*) = d^*dc^*, \qquad Q^{GF}Qc^* = 0;\\
\lambda:&\quad Q^{GF}Q\lambda = -\not{d}|_{\Sigma^*}\not{d}\lambda = d^*d\lambda, \qquad QQ^{GF}\lambda = 0;\\
\lambda^*:&\quad QQ^{GF}\lambda^* = -\not{d}\,\not{d}|_{\Sigma^*}\lambda^* = d^*d\lambda^*, \qquad Q^{GF}Q\lambda^* = 0,
\end{aligned}$$

where on a plane wave $e^{i(p,x)}$ the Dirac operators become $i\rho(p)$, so that $-\not{d}|_{\Sigma^*}\not{d}$ and $-\not{d}\,\not{d}|_{\Sigma^*}$ become $\rho(p)\rho(p) = (p,p) = |p|^2$ by the Clifford relation, the symbol of d^*d. On $A|_V$ and on $A^*|_V$ the sum $d^*d + dd^*$ is the Hodge Laplacian on one-forms of flat $\mathbb{R}^4$, which is the Laplacian d^*d of functions on each component; on $A|_{W_{\mathbb{R}}^\perp}$ and $A^*|_{W_{\mathbb{R}}^\perp}$ the term dd^* is absent and d^*d is the Laplacian of each scalar. Since H is a constant-coefficient scalar operator, it commutes with Q and Q^{GF}. For the uniqueness, each coefficient is fixed by one slot: the c slot fixes the coefficient of $A \mapsto -d^*(A|_V)$, the transverse part of A fixes the coefficient of $A^* \mapsto -2A^*$ (because Q contributes $-\frac{1}{2}$), the c^* slot fixes the coefficient of $c^* \mapsto dc^*$, and the λ slot fixes the coefficient of $\lambda^* \mapsto -\not{d}|_{\Sigma^*}\lambda^*$. □

Proposition 3.3 (Adjointness)**.** *For all $e_0, e_1 \in \mathscr{E}_c$,*

$$\langle Q^{GF}e_0, e_1\rangle = (-1)^{|e_0|}\langle e_0, Q^{GF}e_1\rangle, \qquad \langle Qe_0, e_1\rangle = -(-1)^{|e_0|}\langle e_0, Qe_1\rangle;$$

equivalently $(Q^{GF}\otimes 1 - 1\otimes Q^{GF})K_0 = 0$ and $(Q\otimes 1 + 1\otimes Q)K_0 = 0$. Thus Q^{GF} has the adjointness opposite to that of Q.

Proof. The statement for Q is Proposition 1.11. For Q^{GF} the pairing vanishes outside the following cases. For e_0, e_1 in the A^* slot (odd),

$$\langle Q^{GF}e_0, e_1\rangle = -2\int \mathrm{dvol}\,(e_0, e_1), \qquad \langle e_0, Q^{GF}e_1\rangle = -\langle -2e_1, e_0\rangle = 2\int \mathrm{dvol}\,(e_0, e_1).$$

For e_0 in the A slot and e_1 in the c^* slot (both even),

$$\langle Q^{GF}e_0, e_1\rangle = -\langle e_1, -d^*(e_0|_V)\rangle = \int \mathrm{dvol}\,(de_1, e_0), \qquad \langle e_0, Q^{GF}e_1\rangle = \int \mathrm{dvol}\,(e_0, de_1).$$

For e_0, e_1 in the λ^* slot (even), with the symmetry of $\rho(v)$ and integration by parts,

$$\langle Q^{GF}e_0, e_1\rangle = \int \mathrm{dvol}\,(\not{d}e_0, e_1), \qquad \langle e_0, Q^{GF}e_1\rangle = -\int \mathrm{dvol}\,(\not{d}e_1, e_0) = \int \mathrm{dvol}\,(\not{d}e_0, e_1).$$

The case e_0 in the c^* slot, e_1 in the A slot follows by graded antisymmetry. In each case $\langle Q^{GF}e_0, e_1\rangle = (-1)^{|e_0|}\langle e_0, Q^{GF}e_1\rangle$. □

Remark 3.4 (Names of the adjointness relations in the sources)**.** [C07, §4.3] calls Q "self adjoint" and [C07, Def 4.3.3(1)] calls Q^{GF} "skew self adjoint"; [C07, §1, §1.1, §2.1] and [CG2, §7.2.1] use the opposite names, Q skew self adjoint and Q^{GF} self adjoint. The relations of Proposition 3.3 fix the content, and condition (1) of [C07, Def 4.3.3] is read here as: Q^{GF} has the adjointness opposite to that of Q. The printed adjointness relation of [ESW, Def 1.16] is discussed in Remark 1.12.

Proposition 3.5 (Axioms of a gauge-fixing operator)**.** *Of the conditions of [C07, Def 4.3.3], three hold:*

- *(1), read as in Remark 3.4: Proposition 3.3;*
- *(2): every component of Q^{GF} has order 0 or 1;*
- *(4): $H = d^*d + dd^*$, the componentwise Laplacian, is a generalised Laplacian (Proposition 3.2).*

Condition (3) fails: $(Q^{GF})^2$ vanishes on the slots of c, A, λ, λ^*, and

$$(Q^{GF})^2 A^* = 2\, d^*(A^*|_V) \ \ (\textit{in the slot of } c), \qquad (Q^{GF})^2 c^* = -2\, dc^* \ \ (\textit{in the slot of } A|_V).$$

Condition (5) fails: $\operatorname{Im} Q \cap \operatorname{Im} Q^{GF} \neq 0$, since for every nonzero $c \in \mathscr{E}_c$ in the ghost slot

$$Qc = -dc = Q^{GF}\big(\tfrac{1}{2} dc \textit{ in the slot of } A^*|_V\big).$$

Moreover $\operatorname{Im} Q^{GF}$ pairs nontrivially with itself: for e_0 in the A^* slot and e_1 in the c^* slot,

$$\langle Q^{GF} e_0, Q^{GF} e_1 \rangle = -2 \int \operatorname{dvol}(e_0, de_1).$$

Proof. $(Q^{GF})^2 A^* = Q^{GF}(-2A^*) = -2\big(-d^*(A^*|_V)\big)$ and $(Q^{GF})^2 c^* = Q^{GF}(dc^*) = -2\,dc^*$; on the other slots the image of Q^{GF} lies in the slots of c and λ, where Q^{GF} vanishes. For (5), $Q^{GF}(\frac{1}{2}dc) = -2 \cdot \frac{1}{2} dc$. The last formula is $\langle -2e_0, de_1 \rangle$ with $-2e_0$ in the A slot. □

Remark 3.6 (Consequences of the two failures)**.** Three statements of [C07] use (3) or (5): [C07, Lemma 6.5.1] ($P(0,\infty)$ inverts Q: $\operatorname{Im} Q^{GF} \to \operatorname{Im} Q$; its proof uses $(Q^{GF})^2 = 0$), the real-slice Gaussian over $(\operatorname{Im} Q^{GF})_{\mathbb{R}}$ of [C07, §6.6], and [C07, Lemma 6.6.2]. Indeed $\operatorname{Im} Q^{GF}$ contains the whole A slot, on which $\langle e, Qe \rangle$ has the longitudinal zero eigenvector (Remark 1.10). The arguments below use $Q^2 = 0$, $H = [Q, Q^{GF}]$, $[Q, H] = 0$ and the adjointness relations; Theorem 3.19 takes the place of [C07, Lemma 6.5.1].

Remark 3.7 (Feynman gauge)**.** In momentum space the $A \otimes A$ component of the propagator $P(0,\infty)$ (Proposition 3.11 at $\varepsilon = 0$) is $2(-,-)^{-1} \otimes$ (inverse metric of V_{10}) $|p|^{-2}$. It is the inverse of the second variation of $-\big(\frac{1}{2}\langle A, QA\rangle - \frac{1}{4}\int \operatorname{dvol}(d^*A|_V, d^*A|_V)\big) = \frac{1}{4}\int \operatorname{dvol}(A, HA)$, i.e. the covariance of the weight $e^{+S/\hbar}$ for the Yang–Mills kinetic term completed by the Feynman-gauge term $-\frac{1}{4}(d^*A|_V, d^*A|_V)$; it has the eigenvalue $2(-,-)^{-1}|p|^{-2}$, positive, in all ten directions of V_{10}. Restricted to λ, the propagator $P(0,\infty)$ is minus the inverse of the second variation of $\frac{1}{2}\int \operatorname{dvol}(\lambda, \not{d}\lambda)$.

3.2 Heat kernel and propagator

Definition 3.8 (Heat kernel)**.** For $t > 0$, $K_t \in \mathscr{E} \otimes \mathscr{E}$ is the kernel with the components of K_0 (Definition 1.6) with $\delta(x-y)$ replaced by $(4\pi t)^{-2} e^{-|x-y|^2/4t}$; in momentum space this factor is $e^{-t|p|^2}$. Then $K_t \star e = e^{-tH} e$ [C07, §6.5; C07, §15.2]. The kernel

$$L_t = (Q^{GF} \otimes 1) K_t$$

represents $Q^{GF} e^{-tH}$ [C07, §6.5]. On $\mathbb{R}^4$,

$$K_\infty := \lim_{t\to\infty} K_t = 0,$$

pointwise in $x - y$, and in momentum space for every $p \neq 0$.

Proposition 3.9 (Heat-kernel identities)**.**

$$(Q \otimes 1 + 1 \otimes Q) K_t = 0, \qquad \frac{d}{dt} K_t + (Q \otimes 1 + 1 \otimes Q) L_t = 0 \qquad [\textit{C07, §6.5}].$$

Proof. $K_t = (e^{-tH}\otimes 1)K_0$, $[Q,H]=0$ and $(Q\otimes 1+1\otimes Q)K_0=0$ give the first identity. For the second, odd operators on different tensor factors anticommute, $(1\otimes Q)(Q^{GF}\otimes 1) = -(Q^{GF}\otimes 1)(1\otimes Q)$, and $(1\otimes Q)K_t = -(Q\otimes 1)K_t$, so

$$(Q\otimes 1+1\otimes Q)L_t = (QQ^{GF}\otimes 1)K_t + (Q^{GF}Q\otimes 1)K_t = (H\otimes 1)K_t = -\frac{d}{dt}K_t.$$

□

Definition 3.10 (Propagator)**.**

$$P(\varepsilon,\infty) = \int_\varepsilon^\infty L_t\,dt = \int_\varepsilon^\infty (Q^{GF}\otimes 1)K_t\,dt \qquad \text{[C07, §6.5]}, \qquad \varepsilon>0.$$

The component in a row labelled by two slots in the table below is the value of $\partial_{P(\varepsilon,\infty)}$ [C07, §6.4] on the product of a coordinate functional of the first slot at x and one of the second slot at y, in this order.

Proposition 3.11 (Components of the propagator)**.** *The scalar factor of K_t integrates to*

$$\int_\varepsilon^\infty (4\pi t)^{-2}e^{-|x-y|^2/4t}\,dt = \frac{1-e^{-|x-y|^2/4\varepsilon}}{4\pi^2|x-y|^2}, \qquad \int_\varepsilon^\infty e^{-t|p|^2}\,dt = \frac{e^{-\varepsilon|p|^2}}{|p|^2},$$

and the nonzero components of $P(\varepsilon,\infty)$ are:

component	*fibre factor*	*position space*	*momentum space*
$A\otimes A$	$2(-,-)^{-1}\otimes$ (*inverse metric of* V_{10})	$\dfrac{1-e^{-\lvert x-y\rvert^2/4\varepsilon}}{4\pi^2\lvert x-y\rvert^2}$	$\dfrac{e^{-\varepsilon\lvert p\rvert^2}}{\lvert p\rvert^2}$
$c\otimes A^*\lvert_V$	$-(-,-)^{-1}$	$d_x\dfrac{1-e^{-\lvert x-y\rvert^2/4\varepsilon}}{4\pi^2\lvert x-y\rvert^2}$	$i\,p\,\dfrac{e^{-\varepsilon\lvert p\rvert^2}}{\lvert p\rvert^2}$
$A^*\lvert_V\otimes c$	$-(-,-)^{-1}$	$d_x\dfrac{1-e^{-\lvert x-y\rvert^2/4\varepsilon}}{4\pi^2\lvert x-y\rvert^2}$	$i\,p\,\dfrac{e^{-\varepsilon\lvert p\rvert^2}}{\lvert p\rvert^2}$
$\lambda\otimes\lambda$	$(-,-)^{-1}$	$\not{d}_x\lvert_{\Sigma^*}\dfrac{1-e^{-\lvert x-y\rvert^2/4\varepsilon}}{4\pi^2\lvert x-y\rvert^2}$	$i\rho(p)\lvert_{\Sigma^*\to\Sigma}\dfrac{e^{-\varepsilon\lvert p\rvert^2}}{\lvert p\rvert^2}$

The propagator $P(\varepsilon,\infty)$; the kernel $\rho(p)|_{\Sigma^*\to\Sigma}$ is read as an element of $\Sigma\otimes\Sigma$ through the pairing $\Sigma\otimes\Sigma^*\to\mathbb{C}$.

All other components vanish, among them $A\otimes A^*$, $\lambda\otimes\lambda^*$, $c\otimes c^*$ and $c\otimes A^*|_{W_{\mathbb{R}}^\perp}$. Explicitly,

$$d_x\frac{1-e^{-|x-y|^2/4\varepsilon}}{4\pi^2|x-y|^2} = (x-y)\,\frac{\frac{|x-y|^2}{2\varepsilon}e^{-|x-y|^2/4\varepsilon} - 2\big(1-e^{-|x-y|^2/4\varepsilon}\big)}{4\pi^2|x-y|^4}, \qquad \frac{1-e^{-|x-y|^2/4\varepsilon}}{4\pi^2|x-y|^2}\bigg|_{x=y} = \frac{1}{16\pi^2\varepsilon}.$$

Moreover $P(\varepsilon,\infty)$ is even and graded symmetric: exchanging the two factors together with $x\leftrightarrow y$ ($p\mapsto -p$) multiplies each component by the Koszul sign of the two slots.

Proof. Since $\frac{d}{dt}e^{-|x-y|^2/4t} = \frac{|x-y|^2}{4t^2}e^{-|x-y|^2/4t}$,

$$\int_\varepsilon^\infty \frac{e^{-|x-y|^2/4t}}{16\pi^2t^2}\,dt = \frac{1}{4\pi^2|x-y|^2}\Big[e^{-|x-y|^2/4t}\Big]_{t=\varepsilon}^{t=\infty} = \frac{1-e^{-|x-y|^2/4\varepsilon}}{4\pi^2|x-y|^2}.$$

The components follow from $L_t = (Q^{GF}\otimes 1)K_t$ and Definition 3.1, with Q^{GF} acting on the first factor at x (Koszul sign $+1$). The $A^*\otimes A$ component of K_t, $-(4\pi t)^{-2}e^{-|x-y|^2/4t}$ times the identity, is sent by $A^*\mapsto -2A^*$ to $+2(4\pi t)^{-2}e^{-|x-y|^2/4t}$ in $A\otimes A$. The $A|_V\otimes A^*|_V$ component is sent by $A\mapsto -d^*(A|_V)$, the divergence of $A|_V$ in the first factor, to $-d_x$ of the scalar factor in $c\otimes A^*|_V$.

The $c^* \otimes c$ component is sent by $c^* \mapsto dc^*$ to $-d_x$ of the scalar factor in $A^*|_V \otimes c$. The $\lambda^* \otimes \lambda$ component is sent by $\lambda^* \mapsto -\not{d}|_{\Sigma^*}\lambda^*$ to $+\not{d}_x|_{\Sigma^*}$ of the scalar factor in $\lambda \otimes \lambda$. The momentum-space column follows from $d_x \mapsto ip$, and the value at $x = y$ from

$$\lim_{x \to y} \frac{1 - e^{-|x-y|^2/4\varepsilon}}{|x-y|^2} = \frac{1}{4\varepsilon}.$$

Graded symmetry: for $c \otimes A^*|_V$ and $A^*|_V \otimes c$ the Koszul sign is -1 and the gradient of the even scalar factor is odd under $x \leftrightarrow y$; for $\lambda \otimes \lambda$ the Koszul sign is -1 and $\rho(-p) = -\rho(p)$. □

Proposition 3.12 (The two UV-boundary kernels)**.**

$$(Q \otimes 1 + 1 \otimes Q)P(\varepsilon, \infty) = K_\varepsilon, \tag{3.1}$$

$$\varepsilon \frac{d}{d\varepsilon} P(\varepsilon, \infty) = -\varepsilon L_\varepsilon. \tag{3.2}$$

Proof. By Proposition 3.9,

$$(Q \otimes 1 + 1 \otimes Q)P(\varepsilon, \infty) = \int_\varepsilon^\infty (Q \otimes 1 + 1 \otimes Q)L_t \, dt = -\int_\varepsilon^\infty \frac{d}{dt} K_t \, dt = K_\varepsilon - K_\infty = K_\varepsilon.$$

This is [C07, Lemma 10.2.1] with the upper scale at ∞ and $K_\infty = 0$. In position space the scalar core of (3.1) is $d^* d \, \frac{1-e^{-|x-y|^2/4\varepsilon}}{4\pi^2|x-y|^2} = (4\pi\varepsilon)^{-2} e^{-|x-y|^2/4\varepsilon}$. The second identity is the fundamental theorem of calculus applied to $\int_\varepsilon^\infty L_t \, dt$. □

Remark 3.13 (Two objects at $t = \varepsilon$)**.** The lower end of the Schwinger integral carries two different kernels. By (3.2), $L_\varepsilon = (Q^{GF} \otimes 1)K_\varepsilon$ governs the ε-dependence of every graph: for a graph γ whose value is the integral of f_γ over the Schwinger times $t_e \in [\varepsilon, \infty)$ of its edge set $E(\gamma)$ [C07, §15.5], $\varepsilon \frac{d}{d\varepsilon}$ of the value is minus the sum over $e \in E(\gamma)$ of εf_γ with $t_e = \varepsilon$, that is, with the edge e carrying L_ε. The singular coefficients (ε^{-1}, $\log \varepsilon$) and the scaling behaviour therefore sit on L_ε edges. By (3.1), K_ε governs the action of Q (Theorem 3.17): the obstruction to the quantum master equation and the Q-variation of observables sit on edges carrying K_ε, or $K_\varepsilon - K_0$, and on the counterterm vertices $-QI^{CT}[\varepsilon] - \{I, I^{CT}[\varepsilon]\}_\varepsilon$ (Definition 3.15; the counterterm $I^{CT}[\varepsilon]$ is constructed in Chapter 4), together with their δ-derivatives for observables (Chapter 5).

3.3 The UV-boundary identity

Remark 3.14 (Functionals on $\mathbb{R}^4$)**.** On $\mathbb{R}^4$ the operators ∂_{K_t} and $\partial_{P(\varepsilon,\infty)}$ [C07, §6.4] and the brackets below act on functionals of compactly supported sections, $\mathscr{O}(\mathscr{E}_c) \supset \mathscr{O}_l(\mathscr{E})$ [C07, Def 4.2.2]. The interaction, its counterterms and their flows are taken modulo constants [CG2, §7.2.6]; the observables of Chapter 5 keep their constant terms, which depend on the four-form ω and carry the subtraction of the expectation value. A local functional $\int \mathrm{dvol}\,(\text{density})$, such as S, I or a counterterm, is a density modulo total derivatives, and on it these operators are evaluated graph by graph (Proposition 3.22), with compactly supported legs. For the interaction and its counterterms constants are discarded because a contraction that uses every field of a translation-invariant functional is proportional to the volume of $\mathbb{R}^4$. If the Taylor component of a functional, evaluated on compactly supported sections in its remaining arguments, is an integrable kernel in the two contracted positions, its contraction with K_t is bounded by $(4\pi t)^{-2}$ times the integral of the absolute value of that kernel and tends to 0 as $t \to \infty$. This applies to every local functional $\int \omega\,(\text{density})$ with ω a compactly supported four-form

[CG2, Lemma 10.1.1.3], to every local functional $\int \mathrm{dvol}\,(\text{density})$ at least quadratic in the fields, and to the bracket of two such functionals: a contraction of two legs at one point is a density times $(4\pi t)^{-2}$, and a contraction between two such functionals pairs two compactly supported densities through K_t. For a linear functional $\int \mathrm{dvol}\,(c^*, X)$, $X \in \mathfrak{g}$, the contraction integrates K_t over one position, and the integral of the scalar factor $(4\pi t)^{-2} e^{-|x-y|^2/4t}$ over x equals 1 for every t; these linear functionals are the zero-mode classes of Theorem 3.19.

Definition 3.15 (Scale-ε Laplacian and bracket)**.**

$$\Delta_\varepsilon = -\partial_{K_\varepsilon} \quad \text{[C07, §10.1]}, \qquad \{f,g\}_\varepsilon = \Delta_\varepsilon(fg) - \Delta_\varepsilon(f)g - (-1)^{|f|} f\Delta_\varepsilon(g) \quad \text{[CG2, §7.2.8]},$$

the second with the Laplacian Δ_ε of [C07] in the product rule of [CG2]. At scale ∞ the kernel is $K_\infty := 0$ (Definition 3.8); accordingly $\Delta_\infty := -\partial_{K_\infty} = 0$ and $\{-,-\}_\infty = 0$ identically, on every functional, and this is the scale-∞ structure used throughout. On functionals whose contracted Taylor components are integrable kernels (Remark 3.14), in particular on local functionals with a compactly supported four-form, on local functionals $\int \mathrm{dvol}\,(\text{density})$ at least quadratic in the fields, and on brackets of two such, it is also the limit $t \to \infty$ of Δ_t and $\{-,-\}_t$.

Definition 3.16 (Renormalisation group flow)**.** For an even $\phi \in \mathrm{Sym}^2\mathscr{E}$ and an even functional J, at least cubic modulo $\hbar$,

$$W(\phi, J) = \hbar \log\left(e^{\hbar\partial_\phi} e^{J/\hbar}\right) \qquad \text{[CG2, §7.2.7]},$$

which is $\Gamma(\phi, J)$ of [C07, Def 6.6.1]. The letter W with two arguments always denotes this flow; W alone is the multiplicity space of [ESW, §10], and Γ is reserved for the Γ-pairing of [ESW, (28)].

Theorem 3.17 (UV-boundary identity)**.** *On functionals with values in $C^\bullet(\mathfrak{psl}(4|4))$ (Chapter 2), as formal power series in $\hbar$,*

$$(d_{CE} + Q)\, e^{\hbar\partial_{P(\varepsilon,\infty)}} = e^{\hbar\partial_{P(\varepsilon,\infty)}}\, (d_{CE} + Q + \hbar\Delta_\varepsilon). \tag{3.3}$$

Here Q^{GF} is the operator of Definition 3.1.

Proof. By [C07, §6.4], $[\partial_\phi, Q] = \partial_{Q\phi}$ for $\phi \in \mathrm{Sym}^2\mathscr{E}$, with Q acting on $\mathscr{E} \otimes \mathscr{E}$ as $Q \otimes 1 + 1 \otimes Q$. With (3.1),

$$\partial_{P(\varepsilon,\infty)} Q - Q\, \partial_{P(\varepsilon,\infty)} = \partial_{K_\varepsilon} = -\Delta_\varepsilon, \qquad \text{i.e.} \qquad Q\, \partial_{P(\varepsilon,\infty)} = \partial_{P(\varepsilon,\infty)} Q + \Delta_\varepsilon.$$

The operators ∂_ϕ mutually commute [C07, §6.4], so Δ_ε commutes with $\partial_{P(\varepsilon,\infty)}$, and by induction on n

$$Q\, \partial^n_{P(\varepsilon,\infty)} = \partial^n_{P(\varepsilon,\infty)} Q + n\, \partial^{n-1}_{P(\varepsilon,\infty)} \Delta_\varepsilon.$$

Multiplying by $\hbar^n/n!$ and summing over n,

$$Q\, e^{\hbar\partial_{P(\varepsilon,\infty)}} = e^{\hbar\partial_{P(\varepsilon,\infty)}} Q + \hbar\, e^{\hbar\partial_{P(\varepsilon,\infty)}} \Delta_\varepsilon.$$

Finally d_{CE} acts on the arguments in $\mathfrak{psl}(4|4)$ and $\partial_{P(\varepsilon,\infty)}$ on the fields; $P(\varepsilon,\infty)$ is a kernel on the fields, independent of the arguments of the cochains, so $[\partial_{P(\varepsilon,\infty)}, d_{CE}] = 0$. Each step uses $Q^2 = 0$, (3.1) and the algebra of the operators ∂_ϕ. □

Corollary 3.18 (Master equations at the two ends)**.** *Let J be an even functional with values in $C^\bullet(\mathfrak{psl}(4|4))[[\hbar]]$, at least cubic modulo $\hbar$ when the arguments of the cochains are counted as fields, as in [CG2, §13.2.2], and let $W(P(\varepsilon,\infty), J) = \hbar \log\left(e^{\hbar\partial_{P(\varepsilon,\infty)}} e^{J/\hbar}\right)$ be its flow (Definition 3.16). Then*

$$(d_{CE} + Q + \hbar\Delta_\varepsilon) e^{J/\hbar} = 0 \iff (d_{CE} + Q)\, e^{W(P(\varepsilon,\infty),J)/\hbar} = 0,$$

and the two sides are equivalent to

$$(d_{CE}+Q)J+\tfrac{1}{2}\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J=0,\qquad (d_{CE}+Q)\,W(P(\varepsilon,\infty),J)=0.$$

Proof. Apply (3.3) to $e^{J/\hbar}$; the operator $e^{\hbar\partial_{P(\varepsilon,\infty)}}$ is invertible. This is [C07, Lemma 10.2.2] with the second scale at ∞, where $\Delta_\infty=0$. For an even J, the product rule of Definition 3.15, applied to the powers J^n, gives $\Delta_\varepsilon e^{J/\hbar}=e^{J/\hbar}\big(\hbar^{-1}\Delta_\varepsilon J+\frac{1}{2}\hbar^{-2}\{J,J\}_\varepsilon\big)$, while $d_{CE}+Q$ is a derivation, $(d_{CE}+Q)e^{J/\hbar}=\hbar^{-1}e^{J/\hbar}(d_{CE}+Q)J$; this is the equivalence of [C07, §10.1] with Δ_ε in place of Δ_0. □

3.4 Acyclicity at scale infinity

For a kernel $\phi\in\mathscr{E}\otimes\mathscr{E}$ write $\phi\star$ for the operator it represents [C07, §6.3]. Thus $K_t\star=e^{-tH}$, and $P(\varepsilon,\infty)\star=\int_\varepsilon^\infty Q^{GF}e^{-tH}dt$; $P(0,\infty)\star$ is the limit $\varepsilon\to 0$. A Schwartz section of the BV bundle is a section whose components are Schwartz functions on $\mathbb{R}^4$; its moments are the integrals $\int\mathrm{dvol}\,f\,e$ with f a polynomial function on $\mathbb{R}^4$. A Schwartz section is moment-free when all its moments vanish, equivalently when its Fourier transform (Chapter 1) vanishes to infinite order at $p=0$. The moment-free Schwartz sections form a closed subspace of the Schwartz sections, with the induced topology.

Theorem 3.19 (Acyclicity at scale ∞)**.** *For every $\alpha\in\mathscr{E}_c$ the integral $P(0,\infty)\star\alpha=\int_0^\infty Q^{GF}e^{-tH}\alpha\,dt$ converges to a smooth section, bounded together with its first derivatives by a multiple of $(1+|x|)^{-2}$, and*

$$Q\big(P(0,\infty)\star\alpha\big)+P(0,\infty)\star(Q\alpha)=\alpha;\tag{3.4}$$

in kernel form, $(Q\otimes 1+1\otimes Q)P(0,\infty)=K_0$. Moreover:

- *The moment-free Schwartz sections form a space stable under Q, Q^{GF}, e^{-tH} and $P(0,\infty)\star$, on which $P(0,\infty)\star$ acts continuously and (3.4) holds. Thus $P(0,\infty)\star$ is a contracting homotopy for Q there: every moment-free Schwartz section e with $Qe=0$ equals $Q\big(P(0,\infty)\star e\big)$, with $P(0,\infty)\star e$ moment-free.*
- *Zero-mode classes. Let e_0 be a section in the slot of c, of $A|_{W_{\mathbb{R}}^\perp}$ or of λ whose components are polynomial functions on $\mathbb{R}^4$, with $Qe_0=0$: a constant ghost $e_0=X\in\mathfrak{g}$, a section of $A|_{W_{\mathbb{R}}^\perp}$ with harmonic polynomial components, or a polynomial solution of $\not{\partial}e_0=0$. The linear functional $e\mapsto\langle e_0,e\rangle$ on $\mathscr{E}_c$ is Q-closed and is Q-exact for no $e_0\neq 0$, and it vanishes on moment-free Schwartz sections. For a constant ghost X it is $-\int\mathrm{dvol}\,(c^*,X)$. For each bound on the polynomial degree of e_0 these functionals span a finite-dimensional space.*

Proof. Compactly supported sections. Each component of $e^{-tH}\alpha$ is the convolution of a component of α with $(4\pi t)^{-2}e^{-|x|^2/4t}$, so $e^{-tH}\alpha$ and its first derivatives are bounded by $(4\pi t)^{-2}$ times $\int\mathrm{dvol}\,|\alpha|$ and $\int\mathrm{dvol}\,|d\alpha|$, and by the suprema of $|\alpha|$ and $|d\alpha|$ for $t\le 1$. Since Q^{GF} has constant coefficients and order at most 1, $\sup_{x\in\mathbb{R}^4}|Q^{GF}e^{-tH}\alpha|$ is integrable on $(0,\infty)$; the same bounds for the derivatives allow Q to pass under the integral. At a point x with $\mathrm{dist}(x,\mathrm{supp}\,\alpha)\ge 1$, $|e^{-tH}\alpha(x)|\le(4\pi t)^{-2}e^{-\mathrm{dist}(x,\mathrm{supp}\,\alpha)^2/4t}\int\mathrm{dvol}\,|\alpha|$, and

$$\int_0^\infty(4\pi t)^{-2}e^{-\mathrm{dist}(x,\mathrm{supp}\,\alpha)^2/4t}\,dt=\frac{1}{4\pi^2\,\mathrm{dist}(x,\mathrm{supp}\,\alpha)^2},$$

which gives the bound by a multiple of $(1+|x|)^{-2}$; the first derivatives obey the same bound. Since

$[Q,H]=0$, $Q^{GF}e^{-tH}Q\alpha = Q^{GF}Qe^{-tH}\alpha$, and

$$Q\,P(0,\infty)\star\alpha + P(0,\infty)\star Q\alpha = \int_0^\infty (QQ^{GF}+Q^{GF}Q)e^{-tH}\alpha\,dt = \int_0^\infty He^{-tH}\alpha\,dt$$
$$= -\int_0^\infty \frac{d}{dt}e^{-tH}\alpha\,dt = \alpha - \lim_{t\to\infty} e^{-tH}\alpha.$$

The limit vanishes because $|e^{-tH}\alpha| \le (4\pi t)^{-2}\int \mathrm{dvol}\,|\alpha|$; this is $K_\infty = 0$. The kernel form is (3.1) at $\varepsilon\to 0$ with $K_\varepsilon \to K_0$.

Moment-free sections. In momentum space Q and Q^{GF} act by multiplication with the matrices of polynomials in p of Proposition 1.11 and Definition 3.1, e^{-tH} by $e^{-t|p|^2}$, and $P(0,\infty)\star$ by the matrix of Q^{GF} times $\int_0^\infty e^{-t|p|^2}dt = |p|^{-2}$. The first three preserve Schwartz functions of p that vanish to infinite order at $p=0$. For such a function h and every n, Taylor's formula at $p=0$ bounds every derivative $\partial^\beta h$ on $|p|\le 1$ by $|p|^n$ times a multiple of the supremum of the derivatives of h of order $n+|\beta|$ on the unit ball, while every derivative of order k of $|p|^{-2}$ is bounded by a multiple of $|p|^{-2-k}$; by the Leibniz rule every derivative of order k of $|p|^{-2}h$ is bounded on $0<|p|\le 1$ by $|p|^{n-2-k}$ times continuous seminorms of h. Hence $|p|^{-2}h$ extends smoothly by 0 at $p=0$ and vanishes there to infinite order; on $|p|\ge 1$ the derivatives of $|p|^{-2}$ are bounded, so rapid decay and the seminorm bounds persist. This proves stability and continuity. For $p\neq 0$ the matrices satisfy

$$(QQ^{GF}+Q^{GF}Q)\,|p|^{-2} = H\,|p|^{-2} = 1$$

by Proposition 3.2, and both sides of (3.4) vanish to infinite order at $p=0$; this gives (3.4) on moment-free Schwartz sections.

Zero-mode classes. The table of Proposition 1.11 gives $Qe_0 = -de_0$ on the ghost slot, $Qe_0 = -\frac12 d^*de_0$ on $A|_{W_{\mathbb{R}}^\perp}$ and $Qe_0 = \not{d}e_0$ on λ, which vanish exactly for the three kinds of e_0 listed. For $e\in\mathscr{E}_c^\circ$ the pairing $\langle e_0,e\rangle$ is defined, and integration by parts with compact support gives $\langle e_0,Qe\rangle = -(-1)^{|e_0|}\langle Qe_0,e\rangle = 0$, as in Proposition 1.11; so the functional is Q-closed. A Q-exact linear functional on $\mathscr{E}_c^\circ$ is, up to sign, $G\circ Q$ for a linear functional G. The operator Q vanishes on the slots of c^*, λ^* and $A^*|_{W_{\mathbb{R}}^\perp}$, so $G\circ Q$ is independent of the components of e in these slots, while $\langle e_0,e\rangle$ is the pairing of e_0 with the component of e in the partner slot of e_0, which is one of these three, and is a nonzero functional of it for $e_0\neq 0$. Hence $\langle e_0,-\rangle$ is Q-exact for no $e_0\neq 0$. Since e_0 has polynomial components, $\langle e_0,e\rangle$ is a finite combination of moments of e; it extends continuously to Schwartz sections and vanishes on moment-free ones. For $e_0 = X$ in the ghost slot, Definition 1.5 gives $\langle X,e\rangle = -\langle e,X\rangle = -\int \mathrm{dvol}\,(e_{c^*},X)$, with e_{c^*} the component of e in the slot of c^*. Polynomial sections of bounded degree form a finite-dimensional space. □

The derivation Q of [C07, §6.2] acts on an n-linear graded-symmetric functional F of parity $|F|$ by

$$QF = (-1)^{|F|}\,F\circ\Big(\sum_{j=1}^n 1^{\otimes(j-1)}\otimes Q\otimes 1^{\otimes(n-j)}\Big),$$

the operator in brackets being the tensor-product differential of [C07, §6.2], with the Koszul rule for odd operators on tensor factors: in $F(e+rQe,\dots)$ with e even and r an odd element of an auxiliary nilpotent ring [C07, Lemma 6.4.1, proof], the element r passes through F.

Corollary 3.20 (Functionals at scale ∞)**.** *Let $n\ge 1$ and let F be a continuous graded-symmetric n-linear functional on moment-free Schwartz sections with $F\circ\big(\sum_{j=1}^n 1^{\otimes(j-1)}\otimes Q\otimes 1^{\otimes(n-j)}\big)=0$. Then*

$$F = \Big(F\circ\frac1n\sum_{i=1}^n 1^{\otimes(i-1)}\otimes P(0,\infty)\star\otimes 1^{\otimes(n-i)}\Big)\circ\Big(\sum_{j=1}^n 1^{\otimes(j-1)}\otimes Q\otimes 1^{\otimes(n-j)}\Big),$$

and the functional $F \circ \frac{1}{n}\sum_{i=1}^{n} 1^{\otimes(i-1)} \otimes P(0,\infty)\star \otimes 1^{\otimes(n-i)}$ *is continuous, graded symmetric and of parity* $|F|+1$*; F equals* $(-1)^{|F|+1}$ *times Q of it. Hence the complex of continuous polynomial functionals on moment-free Schwartz sections, without constant term, is acyclic for Q: every Q-closed functional without constant term is Q-exact.*

Proof. By Theorem 3.19, $P(0,\infty)\star$ acts continuously on moment-free Schwartz sections, so the composite functional is continuous; averaging over the n factors keeps it graded symmetric, and $P(0,\infty)\star$ is odd, which gives the parity. Odd operators acting on different tensor factors anti-commute, so the graded commutator of Q on the j-th factor with $P(0,\infty)\star$ on the i-th factor vanishes for $i \neq j$, and for $i = j$ it is the identity of the i-th factor by (3.4). Summing over i, j and dividing by n, the graded commutator of the tensor-product differential with the averaged operator $\frac{1}{n}\sum_i 1^{\otimes(i-1)} \otimes P(0,\infty) \star \otimes 1^{\otimes(n-i)}$ is $\frac{1}{n}\sum_{i=1}^{n} \mathrm{id} = \mathrm{id}$. Composing with F and using $F \circ \big(\sum_j 1^{\otimes(j-1)} \otimes Q \otimes 1^{\otimes(n-j)}\big) = 0$ gives the displayed identity, and the action of Q on functionals of parity $|F|+1$ gives the last statement. □

Remark 3.21 (Where the content of an operator lies)**.** At scale ∞ the differential on functionals is $d_{CE} + Q$ ($\Delta_\infty = 0$, $\{-,-\}_\infty = 0$, Definition 3.15). On moment-free Schwartz sections the cohomology of Q on functionals without constant term is zero (Corollary 3.20): every Q-closed functional is Q of a nonlocal functional built with $P(0,\infty)\star$. On compactly supported sections Q has nonzero classes, among them the zero-mode classes of Theorem 3.19, the pairings with polynomial zero modes of Q, such as $-\int \mathrm{dvol}\,(c^*, X)$ for a constant ghost X. The list of Theorem 3.19 is a list of examples: the slot of $A|_V$ contributes further classes. For $a, b \in V$ with $(a,b) = 0$ and a, b linearly independent, the one-form $e_0 = (a,x)\,(b,-)$ satisfies $d^*e_0 = -(a,b) = 0$ and $d^*de_0 = 0$, hence $Qe_0 = -\frac{1}{2}d^*de_0 = 0$, while $de_0 = (a,-) \wedge (b,-) \neq 0$. The functional $e \mapsto \langle e_0, e\rangle$ is Q-closed; a Q-exact linear functional depends on the component of e in the slot of $A^*|_V$ through its image under d^* alone, which vanishes on $d^*\beta$ for every compactly supported two-form β, while $\langle e_0, d^*\beta\rangle$ is $\pm\int \mathrm{dvol}\,(de_0, \beta)$, nonzero for suitable β; so this class is nonzero. The zero-mode classes vanish on moment-free sections, sit at $p = 0$, are independent of ε, and are finitely many in each polynomial degree and infinitely many in all; products of them give nonlinear classes. On moment-free Schwartz sections every Q-closed functional without constant term therefore defines the zero class, whatever functional $O(\varepsilon)$ it is obtained from. The content of an operator is the local functional $O(\varepsilon)$ at the UV end whose flow $O[\infty]$ to scale ∞ [CG2, Def 8.3.1.3] has an $\varepsilon \to 0$ limit; both are defined in Chapter 5. This requirement fixes the singular part of $O(\varepsilon)$ uniquely; the finite part is the scheme freedom. Correspondingly, an equation expressing $Q\,O[\infty]$ as the scale-∞ observable of a second local functional has content because that functional is local, and for an operator whose observable is a cocycle, $Q\,O[\infty] = 0$ (Remark 5.6), the anomalous dimension is read off in the cohomology of local functionals.

3.5 The graph expansion and the one-loop formula

Proposition 3.22 (Graph expansion)**.** *For* $J = \sum_{i\geq 0} \hbar^i J_i$ *as in Definition 3.16,*

$$W(P(\varepsilon,\infty), J) = \sum_{\gamma} \hbar^{\,b_1(\gamma)+\sum_v \deg v}\, \frac{1}{|\operatorname{Aut}\gamma|}\, w_\gamma \qquad [C07,\ §15.4],$$

the sum over connected graphs γ *with legs, with vertex set* $V(\gamma)$ *and edge set* $E(\gamma)$ *[C07, §15.5] (these always carry the argument* γ*) and first Betti number* $b_1(\gamma) = |E(\gamma)| - |V(\gamma)| + 1$*, each vertex* v *labelled by a Taylor component of* $J_{\deg v}$ *(vertices of degree* 0 *at least trivalent), each edge by* $P(\varepsilon,\infty)$*, each leg by an argument;* $|\operatorname{Aut}\gamma|$ *counts the automorphisms fixing the legs, and* w_γ *is obtained from the full multilinear vertex functions, contracted along the edges and summed over all component labels. Edge*

rule: two contracted coordinate functionals, brought adjacent by Koszul reordering, are replaced by the component of $P(\varepsilon,\infty)$ in the same order (Definition 3.10). At order $\hbar^1$ either $b_1(\gamma)=1$ and all vertices have degree 0, or $b_1(\gamma)=0$ and exactly one vertex has degree 1, for instance a counterterm vertex.

Proof. Write $\phi = P(\varepsilon,\infty) = \sum \phi' \otimes \phi''$. Expanding the exponentials, a term of $e^{\hbar\partial_\phi} e^{J/\hbar}$ is a set of vertices, each a factor $\hbar^{\deg v - 1} J_{\deg v}$, with $|E(\gamma)|$ pairs of half-edges contracted by $\hbar\partial_\phi$; the factor $\frac{1}{2}$ in $\partial_\phi = \frac{1}{2}\sum \partial_{\phi''}\partial_{\phi'}$ [C07, §6.4] and the factorials of the exponentials combine to $1/|\operatorname{Aut}\gamma|$, and on two coordinate functionals ∂_ϕ returns the component of $P(\varepsilon,\infty)$ in their order. The logarithm keeps the connected graphs, and the overall factor $\hbar$ gives the power

$$\hbar^{1+|E(\gamma)|-|V(\gamma)|+\sum_v \deg v} = \hbar^{b_1(\gamma)+\sum_v \deg v}.$$

The last statement is $b_1(\gamma) + \sum_v \deg v = 1$. □

For the one-loop formula let $\mathscr{E}$ be a finite-dimensional super vector space, as when the manifold of [C07, §6.6, §10.1] is a point, and let $\phi = \sum \phi' \otimes \phi'' \in \operatorname{Sym}^2 \mathscr{E}$ be even with complex coefficients [C07, §6.4]. Let J be a quadratic functional on $\mathscr{E}$, even as a whole, whose coefficients lie in a supercommutative algebra: the complex numbers, or an auxiliary nilpotent ring with odd elements as in [C07, Lemma 6.4.1, proof]. The one-loop operator of (ϕ, J) is the linear map

$$e \mapsto \sum \phi' \, (-1)^{|\phi''|} \, \partial_{\phi''}\partial_e J,$$

i.e. ϕ composed with the parity operator $(-1)^{|\cdot|}$ composed with the second derivative of J, the right-hand slot differentiated first, with the coefficients written to the right and extended linearly over them. STr is the supertrace on such linear maps [CG2, §13.7.2]: on the map $e \mapsto e_0\,\ell(e)$, with $e_0 \in \mathscr{E}$ homogeneous and ℓ a linear functional with values in the coefficients, it is $(-1)^{|e_0|}\ell(e_0)$.

Proposition 3.23 (One-loop formula)**.** *As formal power series in J,*

$$\hbar \log\left(e^{\hbar\partial_\phi} e^{J/\hbar}\right)\Big|_{e=0} = \hbar \sum_{n\geq 1} \frac{1}{2n} \operatorname{STr}\big((\textit{one-loop operator})^n\big) = -\frac{\hbar}{2} \operatorname{STr}\log\big(1 - \textit{one-loop operator}\big). \quad (3.5)$$

The term $n=1$ is $\hbar\,\partial_\phi J$, and the term of order n is the sum of the cycles with n vertices, whose automorphism group has order $2n$.

Proof. Two identities for an even quadratic functional F with coefficients as above. The second derivative $\partial_e\partial_{e'}F$ is a coefficient of parity $|e|+|e'|$ with $\partial_e\partial_{e'}F = (-1)^{|e||e'|}\partial_{e'}\partial_e F$; since ϕ is even, $|\phi'| = |\phi''|$ in every term.
First, $\frac{1}{2}$STr of the one-loop operator of F is $\partial_\phi F$: by the rank-one formula,

$$\tfrac{1}{2}\operatorname{STr} = \tfrac{1}{2}\sum (-1)^{|\phi'|+|\phi''|}\,\partial_{\phi''}\partial_{\phi'}F = \tfrac{1}{2}\sum \partial_{\phi''}\partial_{\phi'}F = \partial_\phi F.$$

Second, since ∂_ϕ is of order two, $\partial_\phi(F_1F_2) - (\partial_\phi F_1)F_2 - F_1(\partial_\phi F_2)$ is the sum of the two terms with one derivative on each factor; for F_1 even, the graded symmetry $\sum \phi'\otimes\phi'' = \sum (-1)^{|\phi'||\phi''|}\phi''\otimes\phi'$ makes the two terms equal, so this sum is $\sum (\partial_{\phi''}F_1)(\partial_{\phi'}F_2)$. It is a derivation in each argument; hence, by induction on m,

$$\partial_\phi(F^m) = m\,(\partial_\phi F)\,F^{m-1} + \tfrac{1}{2}m(m-1)\,F^{m-2}\sum(\partial_{\phi''}F)(\partial_{\phi'}F),$$

$$\partial_\phi e^{F/\hbar} = e^{F/\hbar}\Big(\hbar^{-1}\partial_\phi F + \tfrac{1}{2}\hbar^{-2}\sum(\partial_{\phi''}F)(\partial_{\phi'}F)\Big).$$

The one-loop operator of the quadratic functional $\frac{1}{2}\sum(\partial_{\phi''}F)(\partial_{\phi'}F)$ is the square of the one-loop operator of F. Indeed, the Leibniz rule for the linear functionals $\partial_{\phi''}F$ and $\partial_{\phi'}F$ gives two terms for the second derivative, which coincide after the graded symmetry of ϕ and the exchange of the two coefficients, so that

$$\partial_e\partial_{e'}\Big(\tfrac{1}{2}\sum(\partial_{\phi''}F)(\partial_{\phi'}F)\Big) = \sum(-1)^{|e'||\phi''|}\,(\partial_e\partial_{\phi''}F)\,(\partial_{e'}\partial_{\phi'}F).$$

Writing $\phi = \sum_k \phi'_k\otimes\phi''_k = \sum_l \phi'_l\otimes\phi''_l$ for two copies of the same sum, the one-loop operator of this functional sends e to

$$\sum_{l,k}\phi'_l(-1)^{|\phi''_l|}(-1)^{|e||\phi''_k|}(\partial_{\phi''_l}\partial_{\phi''_k}F)(\partial_e\partial_{\phi'_k}F) = \sum_{l,k}\phi'_l(-1)^{|\phi''_l|}(\partial_{\phi''_l}\partial_{\phi'_k}F)(-1)^{|\phi''_k|}(\partial_{\phi''_k}\partial_e F),$$

where the graded symmetry of ϕ in the index k contributes $(-1)^{|\phi''_k|}$ and $\partial_e\partial_{\phi''_k}F = (-1)^{|e||\phi''_k|}\partial_{\phi''_k}\partial_e F$; the right side is the one-loop operator of F applied twice to e.

The flow in an auxiliary formal parameter s. Let J_s be the quadratic functional determined, as a formal power series in s and in J, by

$$\frac{d}{ds}J_s = \tfrac{1}{2}\sum(\partial_{\phi''}J_s)(\partial_{\phi'}J_s), \qquad J_0 = J.$$

By the formula for $\partial_\phi e^{F/\hbar}$ with $F = J_s$, the functional $\exp\big(\int_0^s \partial_\phi J_{s'}\,ds'\big)e^{J_s/\hbar}$ satisfies $\frac{d}{ds}(\cdots) = \hbar\partial_\phi(\cdots)$, as does $e^{s\hbar\partial_\phi}e^{J/\hbar}$; both equal $e^{J/\hbar}$ at $s=0$, and this equation determines the coefficient of each power of s from the previous one, so they coincide. At order m in J the powers s^k that occur have $k\le m$, because ∂_ϕ lowers the polynomial degree by two and J^m has degree $2m$; so $s=1$ may be substituted. At the origin $e^{J_1/\hbar}$ equals 1, hence

$$\hbar\log\big(e^{\hbar\partial_\phi}e^{J/\hbar}\big)\Big|_{e=0} = \hbar\int_0^1 \partial_\phi J_s\,ds.$$

The one-loop operator of J_s depends linearly on J_s, so by the second identity its derivative in s is its square; its value at $s=0$ is the one-loop operator of J, and the unique formal solution is $\sum_{n\ge1}s^{n-1}(\text{one-loop operator of } J)^n$. By the first identity,

$$\partial_\phi J_s = \tfrac{1}{2}\sum_{n\ge1}s^{n-1}\,\mathrm{STr}\big((\text{one-loop operator})^n\big),$$

$$\hbar\int_0^1\partial_\phi J_s\,ds = \hbar\sum_{n\ge1}\frac{1}{2n}\,\mathrm{STr}\big((\text{one-loop operator})^n\big),$$

which is (3.5), with $\mathrm{STr}\log(1-\text{one-loop operator}) = -\sum_{n\ge1}\frac{1}{n}\mathrm{STr}\big((\text{one-loop operator})^n\big)$. The term $n=1$ is $\hbar\,\partial_\phi J$ by the first identity. In the graph expansion with vertices J the graphs without legs and with bivalent vertices are the cycles, and a cycle with n vertices has $2n$ automorphisms. □

Remark 3.24 (Use in the field theory)**.** For the one-loop part of $W(P(\varepsilon,\infty),I)$ at a background field configuration, J is the part of $I(\text{background}+\text{fluctuation})$ quadratic in the fluctuation, $\phi = P(\varepsilon,\infty)$, STr includes the trace over $\mathfrak{g}$ and the integration over positions, and expanding in the background produces the one-loop one-particle-irreducible graphs with their external legs; the parity operator and the supertrace produce the signs of fermion and ghost loops. When the background contains odd fields, multiplied by odd elements of an auxiliary nilpotent ring as in Definition 1.13, the coefficients of J contain odd elements, and Proposition 3.23 applies with these coefficients.

Chapter 4

Counterterms at one loop and the quantum master equation at the UV boundary

The one-loop counterterm, the analogue on $\mathbb{R}^4$ at order $\hbar$ of the counterterm of [C07, Thm B] (Remark 4.9), is fixed by the requirement that the order-$\hbar$ part of the flow $W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ have a limit as $\varepsilon \to 0$ at generic leg momenta, and it is unique among the translation-invariant and $\mathfrak{g}$-invariant local functionals with coefficients in the span of ε^{-1} and $\log\varepsilon$ (Propositions 4.8 and 4.10). The regulator, however, cuts every edge at the Schwinger time ε and breaks the gauge symmetry, so the one-loop counterterm contains the mass terms $\varepsilon^{-1}\int \mathrm{dvol}\,(A,A)$, whose Q-variation is nonzero. Transported to the UV boundary by Theorem 3.17, the quantum master equation then fails by a finite local functional, and a finite local counterterm has to be chosen to remove it: this is the first renormalisation condition of the theory. This chapter computes $I^{CT}[\varepsilon]$ in every one-loop sector of superficial degree ≥ 0, identifies its $\log\varepsilon$ part as a rescaling of the fields with vanishing component along the coupling, and evaluates the master equation at the UV boundary in the sector with legs $(c, A|_V)$, where the first finite counterterm appears.

4.1 The one-loop counterterm and its boundary form

Throughout this chapter $\mathfrak{g}$ is simple. Then the Killing form $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)$ is a multiple of the invariant pairing, and the ratio $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(X,Y)$, for $X, Y \in \mathfrak{g}$ with $(X,Y) \neq 0$, is a number independent of X and Y. Every one-loop coefficient of this chapter is a rational multiple of

$$\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2\,(X,Y)},$$

except the coefficients of the fourth-order adjoint traces of the quartic graphs, which are rational multiples of $\hbar/16\pi^2$. The upper Schwinger limit is ∞ (Chapter 3).

The value of a translation-invariant functional with k legs on plane waves $e^{i(p_j,x)}$, $j = 1,\dots,k$, with fibre elements as in Definition 1.13, is $(2\pi)^4\delta(p_1+\cdots+p_k)$ times a function of $(p_1,\dots,p_k)$ on the subspace where $p_1+\cdots+p_k = 0$; this function is its plane-wave value. Leg momenta are generic when every leg momentum is nonzero and no sum of the momenta of a nonempty proper subset of the legs vanishes. For $k = 1$ the set of generic configurations is empty; for $k \geq 2$ it is the complement of finitely many proper linear subspaces of the subspace $p_1+\cdots+p_k = 0$, an open dense subset. A functional depending on ε has a limit as $\varepsilon \to 0$ at generic leg momenta when the plane-wave values of its Taylor

components converge as $\varepsilon \to 0$ at every generic configuration of the leg momenta. Every limit of a flow in this chapter is a limit in this sense.

Definition 4.1 (One-loop counterterm)**.** For a connected one-particle-irreducible graph γ with legs, $b_1(\gamma) = 1$ and all vertices labelled by I (Proposition 3.22), let $w_\gamma(\varepsilon)$ be its value with every edge labelled by $P(\varepsilon, \infty)$ [C07, §15.4]. Every vertex of γ lies on its cycle and carries a leg, so at generic leg momenta the momenta of distinct edges of γ differ, and by Lemma 4.6, as $\varepsilon \to 0$, the plane-wave value of $w_\gamma(\varepsilon)$ is the sum of multiples of negative integer powers of ε, a multiple of $\log \varepsilon$, and a function of ε with a limit; the first two are polynomials in the leg momenta, and $\mathrm{Sing}_\varepsilon w_\gamma(\varepsilon)$ denotes the local functional whose plane-wave value is their sum [CG2, §13.7.2]. The one-loop counterterm is

$$I^{CT}[\varepsilon] = \hbar \sum_\gamma \frac{1}{|\operatorname{Aut} \gamma|} \operatorname{Sing}_\varepsilon w_\gamma(\varepsilon), \tag{4.1}$$

the sum over the connected one-particle-irreducible graphs γ with $b_1(\gamma) = 1$ and at least two legs, modulo constants. Its coefficients lie in the span of ε^{-1} and $\log \varepsilon$ (Proposition 4.20), the one-loop part of the renormalisation scheme $\mathscr{A}_{<0}$ [C07, Def 1.4.1; CG2, §13.7.2].

The ε-dependence of a graph is carried by the lower ends of its Schwinger integrals. By (3.2), $\frac{d}{d\varepsilon} P(\varepsilon, \infty) = -L_\varepsilon$ with $L_\varepsilon = (Q^{GF} \otimes 1) K_\varepsilon$.

Proposition 4.2 (Boundary form of the scale derivative)**.** *Let γ be a graph with edge set $E(\gamma)$ [C07, §15.5], and let $w_\gamma(\varepsilon)$ be its value with every edge labelled by $P(\varepsilon, \infty)$. Then*

$$\varepsilon \frac{d}{d\varepsilon} w_\gamma(\varepsilon) = \sum_{e \in E(\gamma)} w_\gamma\big(\text{edge } e \text{ labelled by } -\varepsilon L_\varepsilon, \text{ the other edges by } P(\varepsilon, \infty)\big). \tag{4.2}$$

In the Schwinger form $w_\gamma(\varepsilon) = \int_{[\varepsilon,\infty)^{E(\gamma)}} f_\gamma \prod_e dt_e$ of [C07, §15.5], with the edge e carrying L_{t_e}, the e-th term of (4.2) *is*

$$-\varepsilon \int_{[\varepsilon,\infty)^{E(\gamma)\setminus\{e\}}} f_\gamma\big|_{t_e=\varepsilon} \prod_{e' \neq e} dt_{e'}.$$

Proof. The value w_γ is multilinear in the kernels labelling the edges, so the Leibniz rule gives

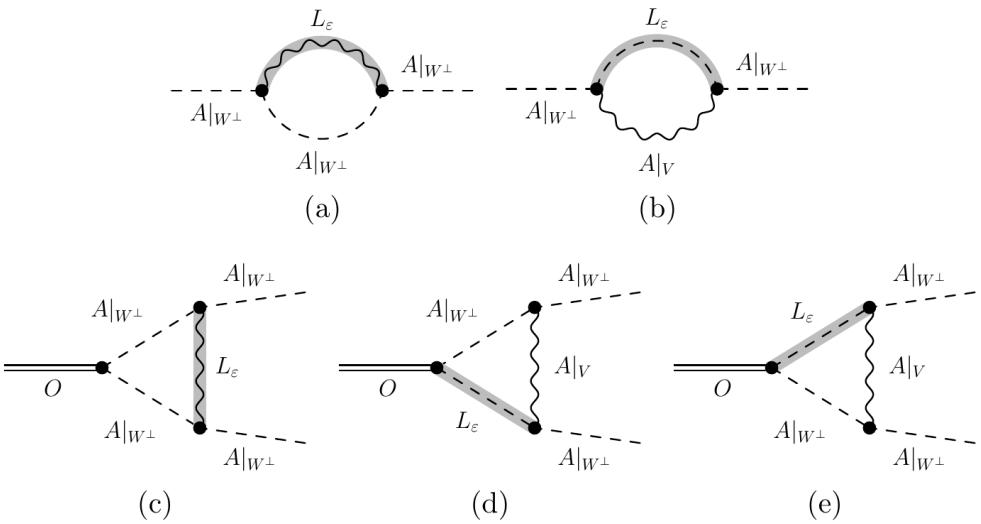
$$\frac{d}{d\varepsilon} w_\gamma = \sum_e w_\gamma\Big(\text{edge } e \text{ labelled by } \frac{d}{d\varepsilon} P(\varepsilon, \infty)\Big),$$

and $\frac{d}{d\varepsilon} P(\varepsilon, \infty) = -L_\varepsilon$ by (3.2). In the Schwinger form, $\frac{d}{d\varepsilon} \int_\varepsilon^\infty (\cdots)\, dt_e = -(\cdots)|_{t_e=\varepsilon}$ for each edge. □

Corollary 4.3 (Singular part from the boundary)**.** *At fixed leg momenta, if the right-hand side of* (4.2) *equals $-a\,\varepsilon^{-1} + b$ plus a term bounded by a multiple of ε, with a, b independent of ε, then*

$$w_\gamma(\varepsilon) = a\,\varepsilon^{-1} + b \log \varepsilon + (\textit{a function of } \varepsilon \textit{ with a limit}), \qquad \mathrm{Sing}_\varepsilon w_\gamma(\varepsilon) = a\,\varepsilon^{-1} + b \log \varepsilon.$$

Proof. $\frac{d}{d\varepsilon}\big(w_\gamma - a\varepsilon^{-1} - b \log \varepsilon\big) = \varepsilon^{-1}\big(\varepsilon \frac{d}{d\varepsilon} w_\gamma + a\varepsilon^{-1} - b\big)$ is ε^{-1} times a term bounded by a multiple of ε, hence bounded near $\varepsilon = 0$, so $w_\gamma - a\varepsilon^{-1} - b \log \varepsilon$ has a limit and differs from it by a term bounded by a multiple of ε. □



Boundary graphs. (a), (b): the two terms of (4.2) for the bubble with two vertices $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$ and external $A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp}$; the shaded edge is pinned at $t=\varepsilon$ and carries $L_\varepsilon = (Q^{GF}\otimes 1)K_\varepsilon$. (c)–(e): the same mechanism for a triangle with one insertion of a composite operator.

The singular coefficients of every graph are therefore computed from graphs with one edge pinned at $t=\varepsilon$. For one-loop graphs with at most two edges, the Schwinger integrals reduce to four elementary ones after a Gaussian integral over the loop momentum.

Lemma 4.4 (Gaussian integral over the loop momentum)**.** *Let $t_1, t_2 > 0$, $p \in V$, and let the numerator be a polynomial function of $p_1 \in V$ of degree at most two. Then*

$$\int \frac{d^4p_1}{(2\pi)^4}\, e^{-t_1|p_1|^2 - t_2|p_1-p|^2}\,(\textit{numerator}) \\ = \frac{e^{-t_1t_2|p|^2/(t_1+t_2)}}{16\pi^2(t_1+t_2)^2}\Big[\textit{numerator} - \frac{1}{4(t_1+t_2)}\big(\textit{positive Laplacian of the numerator}\big)\Big]_{p_1=\frac{t_2}{t_1+t_2}p},$$

the positive Laplacian, minus the sum of the second derivatives along an orthonormal basis, being taken in the variable p_1. For one edge, $\int \frac{d^4p_1}{(2\pi)^4} e^{-t|p_1|^2} = (16\pi^2t^2)^{-1}$.

Proof. Completing the square and integrating the Gaussian,

$$t_1|p_1|^2 + t_2|p_1-p|^2 = (t_1+t_2)\Big|p_1 - \tfrac{t_2}{t_1+t_2}p\Big|^2 + \tfrac{t_1t_2}{t_1+t_2}|p|^2,$$
$$\int \frac{d^4p_1}{(2\pi)^4} e^{-(t_1+t_2)|p_1|^2} = \frac{\pi^2}{(2\pi)^4(t_1+t_2)^2} = \frac{1}{16\pi^2(t_1+t_2)^2}.$$

The normalised Gaussian average in the shifted variable has vanishing odd moments, and its second moment sends $(v,p_1)(v',p_1)$ to $(v,v')/(2(t_1+t_2))$ for $v,v' \in V$. A polynomial of degree at most two is its value at the centre plus a linear term plus its quadratic part; the average of the quadratic part is $\frac{1}{4(t_1+t_2)}$ times the sum of the second derivatives along an orthonormal basis, i.e. $-\frac{1}{4(t_1+t_2)}$ times the positive Laplacian of the numerator in p_1. □

Lemma 4.5 (Elementary Schwinger integrals)**.** *For $p \neq 0$, modulo functions of ε with a limit as $\varepsilon \to 0$ ($\mathscr{A}_{\geq 0}$ of [C07, Def 1.4.1]),*

$$\int_\varepsilon^\infty \frac{dt}{t^2} = \varepsilon^{-1},$$
$$\iint_{[\varepsilon,\infty)^2} \frac{e^{-t_1 t_2 |p|^2/(t_1+t_2)}}{(t_1+t_2)^3}\, dt_1 dt_2 = \tfrac{1}{4}\varepsilon^{-1} + \tfrac{1}{6}|p|^2 \log\varepsilon,$$
$$\iint_{[\varepsilon,\infty)^2} \frac{e^{-t_1 t_2 |p|^2/(t_1+t_2)}}{(t_1+t_2)^2}\, dt_1 dt_2 = -\log\varepsilon,$$
$$\iint_{[\varepsilon,\infty)^2} \frac{t_1 t_2\, e^{-t_1 t_2 |p|^2/(t_1+t_2)}}{(t_1+t_2)^4}\, dt_1 dt_2 = -\tfrac{1}{6}\log\varepsilon.$$

Proof. The integrands are symmetric in t_1, t_2, so by Proposition 4.2 each $\varepsilon\frac{d}{d\varepsilon}$ equals -2ε times the integral over $t \in [\varepsilon,\infty)$ of the integrand at $t_1 = \varepsilon$, $t_2 = t$. On the pinned edge the exponent satisfies $0 \leq \frac{\varepsilon t|p|^2}{\varepsilon+t} \leq \varepsilon|p|^2$, so $e^{-\varepsilon t|p|^2/(\varepsilon+t)}$ equals $1 - \frac{\varepsilon t|p|^2}{\varepsilon+t}$ plus a remainder between 0 and $\frac{1}{2}\varepsilon^2|p|^4$, uniformly in t. With

$$\int_\varepsilon^\infty \frac{dt}{(\varepsilon+t)^2} = \frac{1}{2\varepsilon}, \qquad \int_\varepsilon^\infty \frac{dt}{(\varepsilon+t)^3} = \frac{1}{8\varepsilon^2},$$
$$\int_\varepsilon^\infty \frac{t\,dt}{(\varepsilon+t)^4} = \int_\varepsilon^\infty \Big(\frac{1}{(\varepsilon+t)^3} - \frac{\varepsilon}{(\varepsilon+t)^4}\Big)dt = \frac{1}{8\varepsilon^2} - \frac{1}{24\varepsilon^2} = \frac{1}{12\varepsilon^2},$$

the three pinned integrals are, up to terms bounded by a multiple of ε,

$$2\varepsilon\int_\varepsilon^\infty \frac{e^{-\varepsilon t|p|^2/(\varepsilon+t)}}{(\varepsilon+t)^3}dt = \frac{1}{4\varepsilon} - \frac{|p|^2}{6},$$
$$2\varepsilon\int_\varepsilon^\infty \frac{e^{-\varepsilon t|p|^2/(\varepsilon+t)}}{(\varepsilon+t)^2}dt = 1, \qquad 2\varepsilon\int_\varepsilon^\infty \frac{\varepsilon t\, e^{-\varepsilon t|p|^2/(\varepsilon+t)}}{(\varepsilon+t)^4}dt = \frac{1}{6}.$$

Corollary 4.3 with (a,b) equal to $(\frac{1}{4}, \frac{1}{6}|p|^2)$, $(0,-1)$, $(0,-\frac{1}{6})$ gives the three expansions. The one-edge integral is exact. □

Lemma 4.6 (Singular part of a one-loop graph)**.** *Let γ be a connected one-particle-irreducible graph with $b_1(\gamma) = 1$, so that every vertex lies on the cycle and $|E(\gamma)| = |V(\gamma)|$. Let its vertex functions be polynomials in the momenta, up to a factor depending on the leg momenta alone (such as the Fourier transform of a four-form at minus the total leg momentum, Chapter 5), and let its numerator, the product of the vertex functions and of the polynomial factors of the edge components of $P(\varepsilon,\infty)$ (Proposition 3.11), be homogeneous in the loop momentum and the leg momenta; for vertices labelled by I and legs in fixed slots this holds by Lemma 4.18. Its superficial degree is the degree of the numerator plus $4 - 2|E(\gamma)|$. Let the leg momenta be such that the momenta of any two distinct edges differ by a nonzero vector. Then, at these leg momenta and as $\varepsilon \to 0$,*

$$w_\gamma(\varepsilon) = \sum_{1 \leq k \leq \frac{1}{2}(\textit{superficial degree})} \varepsilon^{-k}\, a_k + b\,\log\varepsilon + (\textit{a function of } \varepsilon \textit{ with a limit}),$$

where a_k and b are polynomials in the leg momenta, a_k homogeneous of degree (superficial degree)$-2k$ and b homogeneous of degree equal to the superficial degree, and $\varepsilon\frac{d}{d\varepsilon}$ of the function with a limit tends to 0. In particular the sum of the terms with ε^{-k} and $\log\varepsilon$ is the plane-wave value of a local functional, $\mathrm{Sing}_\varepsilon w_\gamma(\varepsilon)$*, and it vanishes when the superficial degree is negative.*

Proof. Let the edge $e \in E(\gamma)$ carry the momentum $p_1 - q_e$, with p_1 the loop momentum and q_e a sum of leg momenta, so that $q_e \neq q_f$ for $e \neq f$. By Proposition 3.11, $w_\gamma(\varepsilon)$ is the integral over p_1, with the measure $d^4p_1/(2\pi)^4$, of the numerator times $\prod_e e^{-\varepsilon|p_1-q_e|^2}/|p_1-q_e|^2$, and $e^{-\varepsilon|k|^2}/|k|^2 = \int_\varepsilon^\infty e^{-t|k|^2}dt$ introduces a Schwinger time t_e for each edge. By Tonelli's theorem,

$$\int \frac{d^4p_1}{(2\pi)^4} \int_{[\varepsilon,\infty)^{E(\gamma)}} \big|\text{numerator}\big|\, e^{-\sum_e t_e|p_1-q_e|^2} \prod_e dt_e = \int \frac{d^4p_1}{(2\pi)^4} \big|\text{numerator}\big| \prod_e \frac{e^{-\varepsilon|p_1-q_e|^2}}{|p_1-q_e|^2},$$

and the right side is finite: the points $p_1 = q_e$ are distinct, near each of them exactly one factor is singular, $|p_1 - q_e|^{-2}$ being integrable near $p_1 = q_e$ in four dimensions, and the Gaussian factors bound the polynomial numerator at large $|p_1|$. By Fubini's theorem, $w_\gamma(\varepsilon)$ is the integral over $t \in [\varepsilon,\infty)^{E(\gamma)}$ of the integral over p_1, with the measure $\prod_e dt_e\, d^4p_1/(2\pi)^4$, of $\exp\big(-\sum_e t_e|p_1-q_e|^2\big)$ times the numerator. Write $\tau = \sum_e t_e$. Completing the square,

$$\sum_e t_e|p_1-q_e|^2 = \tau\Big|p_1 - \sum_e \frac{t_e}{\tau}\, q_e\Big|^2 + \tau \cdot \frac{1}{2} \sum_{e,f} \frac{t_e}{\tau}\frac{t_f}{\tau}|q_e - q_f|^2,$$

and the Gaussian integral over p_1, with the moments of Lemma 4.4 (each contracted pair of loop momenta gives $(2\tau)^{-1}$ times the metric), gives

$$w_\gamma(\varepsilon) = \sum_{j\geq 0} \int_{[\varepsilon,\infty)^{E(\gamma)}} \frac{(\text{numerator})_j}{16\pi^2\,\tau^{2+j}}\, \exp\Big(-\tau\cdot\frac{1}{2}\sum_{e,f}\frac{t_e}{\tau}\frac{t_f}{\tau}|q_e-q_f|^2\Big) \prod_e dt_e,$$

where $(\text{numerator})_j$, the part with j contracted pairs, is a polynomial in the ratios t_e/τ and the leg momenta, homogeneous of degree (degree of the numerator)$-2j$, and zero when $2j$ exceeds the degree of the numerator. Put $m_j = j+2-|E(\gamma)|$; then $(\text{numerator})_j \neq 0$ implies $2m_j \leq$ (superficial degree). The ratios t_e/τ range over the simplex $\sum_e t_e/\tau = 1$, and $\prod_e dt_e$ is $\tau^{|E(\gamma)|-1}d\tau$ times the Lebesgue measure of the simplex. Since $\sum_{e\neq f} t_e t_f/\tau^2 = 1 - \sum_e t_e^2/\tau^2 \geq 1 - \max_e t_e/\tau$, the factor multiplying τ in the exponent is at least one half of the smallest of the numbers $|q_e - q_f|^2$, $e \neq f$, times $1 - \max_e t_e/\tau$, and $\tau(1-\max_e t_e/\tau)$ is the sum of the Schwinger times other than the largest. The region $\tau \geq 1$. The integrand of the j-th term is bounded by a multiple of τ^{-2-j} times the exponential of minus a positive multiple of the sum of the Schwinger times other than the largest. Integrating these times against the exponential, and the largest time, which is at least $\tau/|E(\gamma)| \geq 1/|E(\gamma)|$, against τ^{-2-j}, gives a bound independent of ε; by dominated convergence this part has a limit as $\varepsilon \to 0$.

The region $\tau \leq 1$. If $m_j < 0$, the integrand times $\tau^{|E(\gamma)|-1}$ is bounded by a multiple of τ^{-1-m_j}, which is integrable, and the part has a limit. If $m_j \geq 0$, expand the exponential to order m_j. The remainder is bounded by $1/(m_j+1)!$ times the (m_j+1)-th power of the exponent, so its contribution is bounded by a multiple of $\tau^{-2-j+m_j+1}\tau^{|E(\gamma)|-1} = \tau^0$ and has a limit. The term of order $i \leq m_j$ is the integral, over the part of the simplex where $\min_e t_e/\tau \geq \varepsilon$, of $(16\pi^2\, i!)^{-1}(\text{numerator})_j$ times the i-th power of minus the factor multiplying τ in the exponent, times

$$\int_{\varepsilon/\min_e(t_e/\tau)}^1 \tau^{-1-(m_j-i)}\, d\tau = \begin{cases} \dfrac{\varepsilon^{-(m_j-i)}\big(\min_e t_e/\tau\big)^{m_j-i} - 1}{m_j - i}, & i < m_j, \\ \log\big(\min_e t_e/\tau\big) - \log\varepsilon, & i = m_j. \end{cases}$$

The part of the simplex where $\min_e t_e/\tau < \varepsilon$ has measure at most a multiple of ε, the function $(\min_e t_e/\tau)^{m_j-i}$ is at most ε^{m_j-i} there, and $\log(\min_e t_e/\tau)$ is integrable on the simplex. Hence the term of order $i < m_j$ is $\varepsilon^{-(m_j-i)}$ times an integral over the whole simplex plus a function with a

limit, and the term of order m_j is $-\log\varepsilon$ times an integral over the whole simplex plus a function with a limit. These integrals over the simplex, of polynomials in the ratios and the leg momenta times $(\min_e t_e/\tau)^{m_j-i}$, are polynomials in the leg momenta, homogeneous of degree (degree of the numerator)$-2j+2i$, which equals (superficial degree)$-2(m_j-i)$. Collecting the powers $k=m_j-i$ gives the expansion, with $1\le k\le m_j\le\frac{1}{2}$(superficial degree); for negative superficial degree every m_j with $(\text{numerator})_j\neq 0$ is negative, and the singular part vanishes.
Finally, $\varepsilon\frac{d}{d\varepsilon}$ of the function with a limit tends to 0. That function is the sum, over j and i, of the following pieces; in the second and third item the simplex integrand of the term of order i is $(16\pi^2\, i!)^{-1}(\text{numerator})_j$ times the i-th power of minus the factor multiplying τ in the exponent, a polynomial in the ratios and the leg momenta.

- The integrals over $t\in[\varepsilon,\infty)^{E(\gamma)}$ restricted to $\tau\ge 1$, those restricted to $\tau\le 1$ with $m_j<0$, and the remainders of the expansion of the exponential on $\tau\le 1$. By Proposition 4.2, $\varepsilon\frac{d}{d\varepsilon}$ of each is minus ε times its integrals over the faces $t_e=\varepsilon$; by the bounds above these face integrals are bounded on $\tau\ge 1$ and bounded by a multiple of $1+|\log\varepsilon|$ on $\tau\le 1$, the integrand being at most a multiple of $\tau^{1-|E(\gamma)|}$ there.

- For the term of order $i<m_j$, with $k=m_j-i$: $-k^{-1}\varepsilon^{-k}h(\varepsilon)$, where $h(\varepsilon)$ is the integral of the simplex integrand times $(\min_e t_e/\tau)^k$ over the part of the simplex where $\min_e t_e/\tau<\varepsilon$; and $-k^{-1}$ times the integral of the simplex integrand over the part where $\min_e t_e/\tau\ge\varepsilon$, which comes from the -1 in the integral over τ.

- For the term of order m_j: $h(\varepsilon)\log\varepsilon$, where $h(\varepsilon)$ is the integral of the simplex integrand over the part where $\min_e t_e/\tau<\varepsilon$; and the integral of the simplex integrand times $\log(\min_e t_e/\tau)$ over the part where $\min_e t_e/\tau\ge\varepsilon$.

The part of the simplex where $\min_e t_e/\tau<s$ is the union over e of the parts where $t_e/\tau<s$. For a function continuous on the part where $\min_e t_e/\tau\ge s_0$, its integral over the part where $\min_e t_e/\tau\ge s$, $s\ge s_0$, is therefore differentiable in s, with derivative minus a sum of integrals of the function over the faces where $\min_e t_e/\tau=s$, each face having $(|E(\gamma)|-2)$-dimensional measure at most that of a face of the simplex; the same holds, with the opposite sign and $s_0=0$, for the integral of a function continuous on the simplex over the part where $\min_e t_e/\tau<s$, and that integral is at most the maximum of the absolute value of the function there times a multiple of s. In the second item the integrand of h is at most a multiple of ε^k where $\min_e t_e/\tau<\varepsilon$, so $h(\varepsilon)$ and $\varepsilon\frac{dh}{d\varepsilon}$ are bounded by multiples of ε^{k+1}, and $\varepsilon\frac{d}{d\varepsilon}\big(\varepsilon^{-k}h(\varepsilon)\big)=-k\varepsilon^{-k}h(\varepsilon)+\varepsilon^{1-k}\frac{dh}{d\varepsilon}$ is bounded by a multiple of ε; $\varepsilon\frac{d}{d\varepsilon}$ of the integral over the part where $\min_e t_e/\tau\ge\varepsilon$ is bounded by a multiple of ε. In the third item $h(\varepsilon)$ and $\varepsilon\frac{dh}{d\varepsilon}$ are bounded by multiples of ε, so $\varepsilon\frac{d}{d\varepsilon}\big(h(\varepsilon)\log\varepsilon\big)=h(\varepsilon)+\varepsilon\frac{dh}{d\varepsilon}\log\varepsilon$ is bounded by a multiple of $\varepsilon(1+|\log\varepsilon|)$; and $\log(\min_e t_e/\tau)=\log\varepsilon$ on the faces where $\min_e t_e/\tau=\varepsilon$, so $\varepsilon\frac{d}{d\varepsilon}$ of the integral over the part where $\min_e t_e/\tau\ge\varepsilon$ is bounded by a multiple of $\varepsilon|\log\varepsilon|$. Every piece has $\varepsilon\frac{d}{d\varepsilon}$ tending to 0. □

Lemma 4.7 (One-loop graphs as trees with a one-loop vertex)**.** *Consider the graphs of Proposition 3.22, with labelled legs, whose vertices are labelled by Taylor components of I, except at most one vertex of valence at least 1, the marked vertex, labelled by a Taylor component of a further functional (in Chapter 5, the operator O). For a connected graph Γ with $b_1(\Gamma)=1$, let γ be its cycle together with the vertices on the cycle, the remaining half-edges at these vertices being the legs of γ, and let T be the tree obtained from Γ by contracting γ to one vertex s, whose half-edges are the legs of γ.*

- *γ is connected and one-particle-irreducible with $b_1(\gamma)=1$. The automorphisms of T fixing the legs are the identity, and so are those of every tree with the given legs whose vertices are labelled as above except one distinguished vertex s. Restriction to γ is a bijection from the automorphisms*

of Γ *fixing the legs to the automorphisms of* γ *fixing the legs of* γ*. The value* w_Γ *is* w_T *with* s *labelled by the multilinear form* w_γ *in the half-edges of* s*.*

- *The sum of* $w_\Gamma/|\operatorname{Aut}\Gamma|$ *over the connected graphs* Γ *with* $b_1(\Gamma)=1$ *and the given legs equals the sum of* $w_T/|\operatorname{Aut}T| = w_T$ *over the trees* T *with the given legs and one vertex* s*, labelled by the Taylor component, with legs the half-edges of* s*, of the sum of* $w_\gamma/|\operatorname{Aut}\gamma|$ *over the connected one-particle-irreducible graphs* γ *with* $b_1(\gamma)=1$*.*
- *A connected one-particle-irreducible graph with* $b_1=1$*, one leg and all vertices labelled by* I *is a cubic vertex of* I *with one edge from the vertex to itself, and its value is* 0 *before any integration over positions or momenta.*

Proof. First item. A connected graph with $b_1=1$ contains exactly one cycle. Every vertex of γ lies on it, so removing an edge of γ leaves γ connected, and $b_1(\gamma)=|E(\gamma)|-|V(\gamma)|+1=1$. Contracting the cycle lowers b_1 by one, so T is a tree.
Let T be such a tree, with distinguished vertex s. An automorphism α of T fixing the legs preserves the labels of the vertices, so it fixes s and the marked vertex. For a vertex $v\neq s$, the vertices beyond v are v and the vertices separated from s by v. A vertex beyond v at maximal distance from s has one half-edge towards s and its other half-edges are legs; if it is labelled by I it has at least two legs. Hence the vertices beyond v carry a leg or include the marked vertex. Let $v'\neq v$ be a second vertex different from s, the two being exchanged if v lies beyond v'. If v' lies beyond v and v is labelled by I, then v has, besides its half-edges towards s and towards v', a further half-edge, which is a leg or leads to vertices beyond v and not beyond v' that carry a leg or include the marked vertex; if v' lies beyond v and v is the marked vertex, the vertices beyond v include the marked vertex and those beyond v' exclude it; if neither lies beyond the other, the two sets of vertices beyond them are disjoint, and each carries a leg or includes the marked vertex. In every case v and v' differ in the set of legs carried by the vertices beyond them or in whether these include the marked vertex. Both data are preserved by α, so α fixes every vertex. A tree has no two edges with the same ends and no edge from a vertex to itself, so α fixes every edge, and, with the legs, every half-edge.
An automorphism of Γ fixing the legs maps the cycle to itself and therefore induces an automorphism of T fixing the legs, which is the identity. It fixes every vertex outside the cycle and every half-edge outside the edges of the cycle, the legs of γ included, so it restricts to an automorphism of γ fixing the legs of γ, and this restriction determines it. Conversely, an automorphism of γ fixing its legs extends by the identity to an automorphism of Γ fixing the legs.
The value w_Γ is the contraction of the tensor product of the vertex functions along the edges of Γ, with the edge rule of Proposition 3.22. The contractions along different edges commute [C07, §6.4], so contracting the edges of the cycle first gives w_γ, a multilinear form in the half-edges of s, and the remaining contractions give w_T with s labelled by w_γ.
Second item. By the first item the half-edges of s are determined by T and its legs, and $\Gamma\mapsto(T,\gamma)$, with the legs of γ identified with the half-edges of s, is a bijection from the isomorphism classes of connected graphs with $b_1=1$ and the given legs to the pairs consisting of an isomorphism class of trees T with the given legs and a vertex s, and an isomorphism class of connected one-particle-irreducible graphs γ with $b_1(\gamma)=1$ whose legs are the half-edges of s; the inverse glues γ into T in place of s. With $|\operatorname{Aut}\Gamma|=|\operatorname{Aut}\gamma|$, $|\operatorname{Aut}T|=1$ and the linearity of w_T in the label of s,

$$\sum_\Gamma \frac{w_\Gamma}{|\operatorname{Aut}\Gamma|} = \sum_T\sum_\gamma \frac{1}{|\operatorname{Aut}\gamma|}\, w_T\big(s \text{ labelled by } w_\gamma\big) = \sum_T \frac{1}{|\operatorname{Aut}T|}\, w_T\Big(s \text{ labelled by } \sum_\gamma \frac{w_\gamma}{|\operatorname{Aut}\gamma|}\Big).$$

Third item. A cycle with n vertices of valence at least three has at least n legs, so one leg forces $n=1$ and a cubic vertex, two of whose slots are joined by the edge. By Definition 1.13, every cubic vertex function of I, namely $A\,A\,A$ (which contains $(A|_V)^3$ and $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$), $\lambda\,\lambda\,A$, $A\,c\,A^*$, $c\,\lambda\,\lambda^*$

and $c\,c\,c^*$, is the form $(X_1,[X_2,X_3])$ in the $\mathfrak{g}$-factors X_1,X_2,X_3 of its three legs, up to sign, times a factor independent of them; by the invariance of $(-,-)$ and the antisymmetry of the bracket, $(X_1,[X_2,X_3])$ is totally antisymmetric. The $\mathfrak{g}$-factor of every component of $P(\varepsilon,\infty)$ is a multiple of $(-,-)^{-1}$ (Proposition 3.11), which is symmetric. The contraction of two arguments of a totally antisymmetric form with a symmetric tensor is 0, so the vertex with two of its slots joined is the zero functional, before any integration. □

Proposition 4.8 (Existence of the limit)**.** *The order-$\hbar$ part of $W(P(\varepsilon,\infty),I-I^{CT}[\varepsilon])$ has a limit as $\varepsilon\to 0$ at generic leg momenta.*

Proof. By Proposition 3.22 with $J=I-I^{CT}[\varepsilon]$, the order-$\hbar$ part is $\hbar$ times the sum of $w_\Gamma/|\operatorname{Aut}\Gamma|$ over the connected graphs Γ with $b_1(\Gamma)=1$ and vertices labelled by I, plus $\hbar$ times the sum of $w_T/|\operatorname{Aut}T|$ over the trees T with one vertex s labelled by a Taylor component of $-\hbar^{-1}I^{CT}[\varepsilon]$ and the other vertices labelled by I. By Lemma 4.7 (without marked vertex), $|\operatorname{Aut}T|=1$ for these trees, and the first sum is the sum over the same trees of w_T with s labelled by the sum of $w_\gamma/|\operatorname{Aut}\gamma|$ over the connected one-particle-irreducible graphs γ with $b_1(\gamma)=1$. A cycle of vertices of valence at least three carries legs, so every such γ has legs; those with one leg have value 0 (Lemma 4.7), and $I^{CT}[\varepsilon]$ has no Taylor component with one leg (Definition 4.1). By (4.1), the order-$\hbar$ part is therefore $\hbar$ times the sum over the trees T whose vertex s has at least two half-edges of w_T with s labelled by

$$\sum_\gamma \frac{1}{|\operatorname{Aut}\gamma|}\Big(w_\gamma(\varepsilon)-\operatorname{Sing}_\varepsilon w_\gamma(\varepsilon)\Big),$$

the sum over the γ whose legs are the half-edges of s. For a fixed number of legs this sum is finite: a tree with r vertices labelled by I and a vertex s with m half-edges has r edges and at least $3r+m-2r=r+m$ legs, and a cycle with m legs has at most m vertices.

Fix generic leg momenta. Each edge of T separates T into two parts, and each part carries a leg: in the part without s, a vertex at maximal distance from s carries at least two legs; in the part with s, the vertex s has, besides the half-edge on the path from s to the edge, a further half-edge, which is a leg or leads to vertices separated from s and from the edge, among which a vertex at maximal distance from s carries at least two legs. Hence the edge carries the sum of the momenta of a nonempty proper subset of the legs, a nonzero momentum k, and its edge factor, a polynomial times $e^{-\varepsilon|k|^2}/|k|^2$ (Proposition 3.11), converges to the polynomial times $|k|^{-2}$. The half-edges of s carry the sums of the momenta of disjoint nonempty sets of legs, one set for each half-edge. Two distinct edges of the cycle of γ cut it into two arcs, each containing a vertex, and every vertex of γ carries a leg of γ; the momenta of the two edges differ by the sum of the momenta entering γ along one arc, the sum of the momenta of a nonempty proper subset of the legs of T, which is nonzero. By Lemma 4.6, with the homogeneity of Lemma 4.18, the plane-wave value of $w_\gamma(\varepsilon)-\operatorname{Sing}_\varepsilon w_\gamma(\varepsilon)$ at these momenta has a limit as $\varepsilon\to 0$. The plane-wave value of w_T is the sum, over the components of the internal slots, of the products of the vertex functions of T, of its edge factors and of this value, a finite sum of products of functions of ε with limits. □

Remark 4.9 (Relation with Theorem B)**.** [C07, Thm B] constructs, on a compact manifold with a finite upper scale, a counterterm at all orders in $\hbar$, unique among the local functionals with coefficients in $\mathscr{A}_{<0}$, for which the flow of the interaction minus the counterterm has a limit as $\varepsilon\to 0$ as a functional; at one loop, its part with k legs is the singular part of the order-$\hbar$, k-leg part of the flow of the interaction minus the counterterms with fewer legs, and it is independent of the upper scale [C07, §1.5]. Definition 4.1 and Propositions 4.8 and 4.10 give its analogue on $\mathbb{R}^4$ with upper scale ∞, for the even, local, at least cubic interaction I of (1.3), in the following scope: the order is $\hbar$; the singular parts are computed graph by graph on the one-particle-irreducible graphs, and the one-particle-reducible graphs

carry them through their cycles (Lemma 4.7); the limit is the limit of plane-wave values at generic leg momenta; and uniqueness holds among the translation-invariant and $\mathfrak{g}$-invariant local functionals whose coefficients lie in a space of functions of ε containing ε^{-1} and $\log\varepsilon$ in which every function with a limit as $\varepsilon \to 0$ is zero.

Proposition 4.10 (Uniqueness of the one-loop counterterm)**.** *Let I' be a local functional of order $\hbar$, without constant term, invariant under translations and under the adjoint action of $\mathfrak{g}$, whose coefficients lie in a space of functions of ε containing ε^{-1} and $\log\varepsilon$ in which every function with a limit as $\varepsilon \to 0$ is zero, for instance the span of ε^{-1} and $\log\varepsilon$, or a complement $\mathscr{A}_{<0}$ of $\mathscr{A}_{\geq 0}$ containing them [C07, Def 1.4.1]. If the order-$\hbar$ part of $W(P(\varepsilon,\infty), I - I')$ has a limit as $\varepsilon \to 0$ at generic leg momenta, then $I' = I^{CT}[\varepsilon]$.*

Proof. The vertex functions of I and the components of $P(\varepsilon,\infty)$ are translation invariant, and their $\mathfrak{g}$-factors are built from the bracket, $(-,-)$ and $(-,-)^{-1}$; so $I^{CT}[\varepsilon]$, and with it the difference $I^{CT}[\varepsilon] - I'$, is invariant under translations and under the adjoint action of $\mathfrak{g}$, with coefficients in the given space of functions. A translation-invariant local functional with one leg is, modulo total derivatives, the integral of a constant linear form on the fibre of the bundle of fields. The fibre is $\mathfrak{g}$ tensored with a representation of $\mathrm{Spin}(4,\mathbb{C}) \times \mathrm{Spin}(6,\mathbb{C})$ (Chapter 1), so a $\mathfrak{g}$-invariant linear form on it lies in the tensor product of the invariant linear forms on $\mathfrak{g}$ with the dual of that representation; the invariant linear forms on $\mathfrak{g}$ are, through $(-,-)$, the centre of $\mathfrak{g}$, which is 0 for $\mathfrak{g}$ simple. Hence the difference has no Taylor component with one leg, and by hypothesis none without legs.
The order-$\hbar$ parts of $W(P(\varepsilon,\infty), I - I')$ and $W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ differ by $\hbar$ times the sum of the trees with one vertex labelled by a Taylor component of $\hbar^{-1}(I^{CT}[\varepsilon] - I')$ and all other vertices labelled by I (Proposition 3.22). That vertex has at least two half-edges, so, as in the proof of Proposition 4.8, every edge of such a tree carries a nonzero momentum at generic leg momenta. By the hypothesis and Proposition 4.8, the sum of the trees has a limit at generic leg momenta. Suppose the difference is nonzero, and let $k \geq 2$ be the smallest number of legs of its nonzero Taylor components. A tree with $r \geq 1$ vertices labelled by I, each with at least three half-edges, has r edges, and its vertex labelled by the difference has at least k half-edges, so it has at least $k+3r-2r = k+r$ legs; the k-leg part of the sum of the trees is therefore the k-leg Taylor component of $\hbar^{-1}(I^{CT}[\varepsilon]-I')$ itself, which then has a limit at generic leg momenta. Its plane-wave value is a finite sum of products of linearly independent coefficient functions of ε with polynomials in $(p_1,\dots,p_k)$ on the subspace where $p_1 + \cdots + p_k = 0$. At a generic configuration this sum is a function of ε in the given space with a limit, hence zero, and by linear independence every polynomial vanishes there. The generic configurations are dense in the subspace, so the polynomials vanish on it. A polynomial vanishing on that subspace is a combination of the components of $p_1 + \cdots + p_k$ with polynomial coefficients, the plane-wave value of a total derivative; so the k-leg component is zero, a contradiction. □

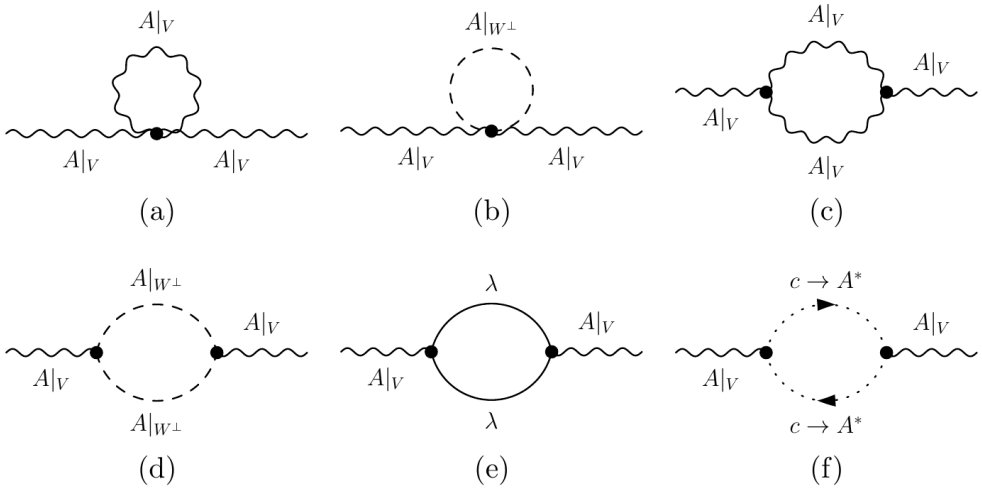
4.2 Two-point graphs

A quadratic functional is evaluated on plane waves. For two legs of $A|_V$ the background is $A(p)e^{i(p,x)} + A(-p)e^{-i(p,x)}$ with $A(\pm p) \in V^\vee \otimes \mathfrak{g}$, and the plane-wave value of a functional is the coefficient of $(2\pi)^4\delta(0)$ in its part bilinear in $A(p)$ and $A(-p)$; for two odd legs the background is $r_1(\cdot)e^{i(p,x)} + r_2(\cdot)e^{-i(p,x)}$ and the plane-wave value is the coefficient of $r_1 r_2$, as in Definition 1.13. Since d becomes

$i(p,-)$ and d^* becomes $-i\,\iota_p$,

$$\int \mathrm{dvol}\,(A|_V, A|_V) \mapsto 2\,(A(p), A(-p)), \qquad \int \mathrm{dvol}\,(d^*A|_V, d^*A|_V) \mapsto 2\,(\iota_p A(p), \iota_p A(-p)),$$
$$\int \mathrm{dvol}\,(dA|_V, dA|_V) \mapsto 2\,\big(|p|^2(A(p), A(-p)) - (\iota_p A(p), \iota_p A(-p))\big), \tag{4.3}$$
$$\int \mathrm{dvol}\,(dc, A^*) \mapsto i\,(p\,c(-p), A^*(p)) \quad \text{for the legs } r_1 A^*(p)e^{i(p,x)} + r_2 c(-p)e^{-i(p,x)},$$

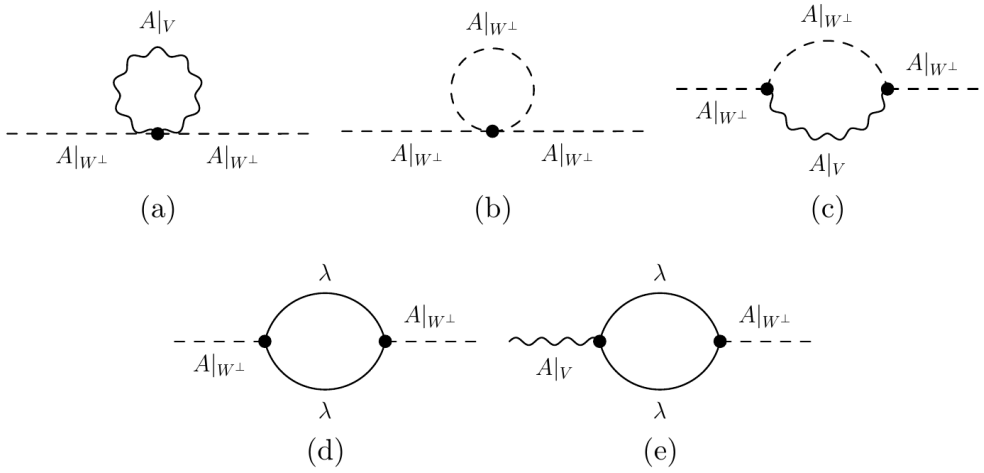
where $(A(p), A(-p))$ pairs the $V_{10}^\vee$ factors by the metric and the $\mathfrak{g}$ factors by $(-,-)$, $(p\,c(-p), A^*(p))$ is the fibre pairing of $p \otimes c(-p) \in V \otimes \mathfrak{g}$ with $A^*(p)$, and the Hodge normalisation of Definition 1.8 gives $((p,-)\wedge A(p), (p,-)\wedge A(-p)) = |p|^2(A(p), A(-p)) - (\iota_p A(p), \iota_p A(-p))$.



The one-loop one-particle-irreducible graphs with external $A|_V, A|_V$; every edge carries $P(\varepsilon,\infty)$. (a), (b): quartic vertex with an $A|_V$, resp. $A|_{W_{\mathbb{R}}^\perp}$, loop. (c): gauge bubble with $A|_V$ lines. (d): bubble with $A|_{W_{\mathbb{R}}^\perp}$ lines. (e): gaugino loop. (f): ghost loop of c–A^* lines.

graph	$1/\lvert\operatorname{Aut}\gamma\rvert$	ε^{-1} part	$\log\varepsilon$ part
(a)	$\frac{1}{2}$	$\frac{3}{2}\int \mathrm{dvol}\,(A\|_V, A\|_V)$	0
(b)	$\frac{1}{2}$	$3\int \mathrm{dvol}\,(A\|_V, A\|_V)$	0
(c)	$\frac{1}{2}$	$-\frac{9}{16}\int \mathrm{dvol}\,(A\|_V, A\|_V)$	$\frac{19}{24}\int \mathrm{dvol}\,(dA\|_V, dA\|_V) - \frac{1}{8}\int \mathrm{dvol}\,(d^*A\|_V, d^*A\|_V)$
(d)	$\frac{1}{2}$	$-\frac{3}{4}\int \mathrm{dvol}\,(A\|_V, A\|_V)$	$-\frac{1}{2}\int \mathrm{dvol}\,(dA\|_V, dA\|_V)$
(e)	$\frac{1}{2}$	$-\int \mathrm{dvol}\,(A\|_V, A\|_V)$	$-\frac{4}{3}\int \mathrm{dvol}\,(dA\|_V, dA\|_V)$
(f)	1	$\frac{1}{16}\int \mathrm{dvol}\,(A\|_V, A\|_V)$	$\frac{1}{24}\int \mathrm{dvol}\,(dA\|_V, dA\|_V) + \frac{1}{8}\int \mathrm{dvol}\,(d^*A\|_V, d^*A\|_V)$
total		$\frac{9}{4}\int \mathrm{dvol}\,(A\|_V, A\|_V)$	$-\int \mathrm{dvol}\,(dA\|_V, dA\|_V)$

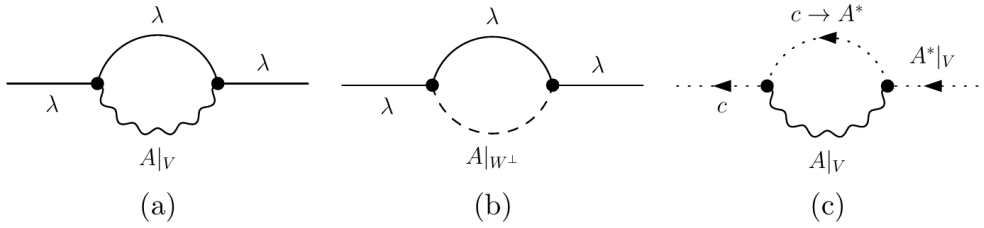
Contributions $\hbar\,|\operatorname{Aut}\gamma|^{-1}\mathrm{Sing}_\varepsilon w_\gamma(\varepsilon)$ of the graphs with external $A|_V, A|_V$ to $I^{CT}[\varepsilon]$, in units of $\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(16\pi^2(X,Y))$; the ε^{-1} part multiplies ε^{-1} and the $\log\varepsilon$ part multiplies $\log\varepsilon$. The coefficients of $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$ cancel between (c) and (f).



The one-loop one-particle-irreducible graphs with external $A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp}$: (a), (b) quartic vertex with an $A|_V$, resp. $A|_{W_{\mathbb{R}}^\perp}$, loop; (c) bubble with one $A|_V$ and one $A|_{W_{\mathbb{R}}^\perp}$ line; (d) gaugino loop with two vertices $\lambda\lambda A|_{W_{\mathbb{R}}^\perp}$. (e): the gaugino loop with external $A|_V, A|_{W_{\mathbb{R}}^\perp}$.

graph	$1/\lvert \operatorname{Aut}\gamma\rvert$	ε^{-1} part	$\log\varepsilon$ part
(a)	$\frac{1}{2}$	$2\int \operatorname{dvol}(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	0
(b)	$\frac{1}{2}$	$\frac{5}{2}\int \operatorname{dvol}(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	0
(c)	1	$-\frac{1}{4}\int \operatorname{dvol}(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	$\int \operatorname{dvol}(dA\|_{W_{\mathbb{R}}^\perp}, dA\|_{W_{\mathbb{R}}^\perp})$
(d)	$\frac{1}{2}$	$-2\int \operatorname{dvol}(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	$-2\int \operatorname{dvol}(dA\|_{W_{\mathbb{R}}^\perp}, dA\|_{W_{\mathbb{R}}^\perp})$
total		$\frac{9}{4}\int \operatorname{dvol}(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	$-\int \operatorname{dvol}(dA\|_{W_{\mathbb{R}}^\perp}, dA\|_{W_{\mathbb{R}}^\perp})$
(e)	$\frac{1}{2}$	0	0

Contributions of the graphs with external $A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp}$ (rows (a)–(d)) and $A|_V, A|_{W_{\mathbb{R}}^\perp}$ (row (e)) to $I^{CT}[\varepsilon]$, in the units of the previous table.



The one-loop one-particle-irreducible graphs with external λ, λ: (a) with an $A|_V$ line, (b) with an $A|_{W_{\mathbb{R}}^\perp}$ line; and with external $c, A^*|_V$: (c) with one $A|_V$ line and one c–A^* line.

graph	$1/\lvert \operatorname{Aut}\gamma\rvert$	ε^{-1} part	$\log\varepsilon$ part
(a)	1	0	$\int \operatorname{dvol}(\lambda, \not{d}\lambda)$
(b)	1	0	$3\int \operatorname{dvol}(\lambda, \not{d}\lambda)$
total λ, λ		0	$4\int \operatorname{dvol}(\lambda, \not{d}\lambda)$
(c)	1	0	$-\int \operatorname{dvol}(dc, A^*)$

Contributions of the graphs with external λ, λ and $c, A^*|_V$ to $I^{CT}[\varepsilon]$, in the units of the previous tables.

Five representative graphs are computed below in five steps: the amplitude, from the one-loop formula (Proposition 3.23), the vertex functions of Definition 1.13 and the propagator components of

Proposition 3.11; the numerator algebra; the Schwinger integral, by Lemma 4.4; the ε-expansion, by Lemma 4.5; and the local part, by (4.3). For two legs, the order-n term of (3.5) enters through its part bilinear in the legs: for $n=2$,

$$\frac{\hbar}{4}\mathrm{STr}\big((\text{one-loop operator})^2\big)\Big|_{\text{bilinear}} = \frac{\hbar}{2}\mathrm{STr}\big((\text{part linear in leg 1})(\text{part linear in leg 2})\big),$$

and for $n=1$ it is $\frac{\hbar}{2}\mathrm{STr}$ of the part of the one-loop operator bilinear in the legs. In the bubbles the two edges carry the momenta p_1 and p_1-p, with Schwinger times t_1 and t_2:

$$\frac{e^{-\varepsilon|p_1|^2}}{|p_1|^2} = \int_\varepsilon^\infty e^{-t_1|p_1|^2}dt_1, \qquad \frac{e^{-\varepsilon|p_1-p|^2}}{|p_1-p|^2} = \int_\varepsilon^\infty e^{-t_2|p_1-p|^2}dt_2.$$

The colour trace of the loop is $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}) = \frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{(X,Y)}(X_1,X_2)$ for the $\mathfrak{g}$-components X_1, X_2 of the two legs, and the factor (X_1,X_2) is part of the pairings $(A(p),A(-p))$ and $(\iota_pA(p),\iota_pA(-p))$.

Example 4.11 (Gauge bubble, graph (c) of the $A|_V$ two-point function)**.** Amplitude. The two vertices are (1.4) at the incoming momenta $(p,-p_1,p_1-p)$ and $(-p,p_1,p-p_1)$; each edge carries the $A|_V\otimes A|_V$ component $2(-,-)^{-1}\otimes$(inverse metric of V) of $P(\varepsilon,\infty)$, times $\frac{e^{-\varepsilon|p_1|^2}}{|p_1|^2}$, resp. $\frac{e^{-\varepsilon|p_1-p|^2}}{|p_1-p|^2}$. The plane-wave value of $\hbar|\operatorname{Aut}\gamma|^{-1}w_\gamma(\varepsilon)$ is

$$\frac{\hbar}{2}\frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{(X,Y)}\int\frac{d^4p_1}{(2\pi)^4}\,\frac{e^{-\varepsilon|p_1|^2}}{|p_1|^2}\,\frac{e^{-\varepsilon|p_1-p|^2}}{|p_1-p|^2}\,(\text{numerator}).$$

Numerator algebra. Contracting the two vertex functions along the two edges (the exchange of the edges is the automorphism counted by $1/|\operatorname{Aut}\gamma| = \frac12$),

$$\begin{aligned}\text{numerator} = &-\big(|p_1|^2+|p_1-p|^2+4|p|^2\big)(A(p),A(-p)) + 2(\iota_pA(p),\iota_pA(-p))\\ &-5\big[(\iota_{p_1}A(p),\iota_{p_1-p}A(-p)) + (\iota_{p_1-p}A(p),\iota_{p_1}A(-p))\big].\end{aligned}$$

At $p_1=\frac{t_2}{t_1+t_2}p$, $|p_1|^2+|p_1-p|^2 = \big(1-\frac{2t_1t_2}{(t_1+t_2)^2}\big)|p|^2$, and each of the two bracketed pairings equals $-\frac{t_1t_2}{(t_1+t_2)^2}(\iota_pA(p),\iota_pA(-p))$; the positive Laplacian in p_1 of the numerator is $36\,(A(p),A(-p))$.
Schwinger integral. By Lemma 4.4 the plane-wave value is

$$\begin{aligned}\frac{\hbar}{2}\frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\iint_{[\varepsilon,\infty)^2}\frac{e^{-\frac{t_1t_2|p|^2}{t_1+t_2}}}{(t_1+t_2)^2}\Big[&-\frac{9\,(A(p),A(-p))}{t_1+t_2}+\Big(\frac{2t_1t_2}{(t_1+t_2)^2}-5\Big)|p|^2(A(p),A(-p))\\ &+\Big(\frac{10t_1t_2}{(t_1+t_2)^2}+2\Big)(\iota_pA(p),\iota_pA(-p))\Big]dt_1dt_2.\end{aligned}$$

ε-expansion. By Lemma 4.5 the double integral of the bracket is, modulo $\mathscr{A}_{\geq 0}$,

$$\begin{aligned}&-9\Big(\tfrac14\varepsilon^{-1}+\tfrac16|p|^2\log\varepsilon\Big)(A(p),A(-p)) + \Big(-\tfrac13+5\Big)|p|^2\log\varepsilon\,(A(p),A(-p))\\ &+\Big(-\tfrac{10}{6}-2\Big)\log\varepsilon\,(\iota_pA(p),\iota_pA(-p))\\ &=-\tfrac94\varepsilon^{-1}(A(p),A(-p)) + \tfrac{19}{6}|p|^2\log\varepsilon\,(A(p),A(-p)) - \tfrac{11}{3}\log\varepsilon\,(\iota_pA(p),\iota_pA(-p)).\end{aligned}$$

Local part. The plane-wave coefficients are $\frac12\cdot(-\frac94)=-\frac98$, $\frac12\cdot\frac{19}{6}=\frac{19}{12}$, $\frac12\cdot(-\frac{11}{3})=-\frac{11}{6}$. By (4.3) these are the plane-wave coefficients of

$$\begin{aligned}\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big[&-\tfrac{9}{16}\varepsilon^{-1}\int\mathrm{dvol}\,(A|_V,A|_V)\\ &+\log\varepsilon\Big(\tfrac{19}{24}\int\mathrm{dvol}\,(dA|_V,dA|_V)-\tfrac18\int\mathrm{dvol}\,(d^*A|_V,d^*A|_V)\Big)\Big],\end{aligned}$$

since $2\cdot(-\frac{9}{16})=-\frac{9}{8}$, $2\cdot\frac{19}{24}=\frac{19}{12}$ and $2\big(-\frac{19}{24}-\frac{1}{8}\big)=-\frac{11}{6}$. This is row (c) of the first table.

Example 4.12 (Gaugino loop, graph (e) of the $A|_V$ two-point function)**.** Amplitude. Two vertices $\lambda\lambda A|_V$ (Definition 1.13); each edge carries the $\lambda\otimes\lambda$ component $(-,-)^{-1}\otimes i\rho(\cdot)|_{\Sigma^*\to\Sigma}$ of $P(\varepsilon,\infty)$ at the momentum p_1, resp. p_1-p; the parity operator in STr gives the sign of the loop of odd fields. The prefactor and the scalar propagator factors are those of Example 4.11.
Numerator algebra. The trace over Σ of the products of Clifford multiplications, evaluated with $\rho(v)\rho(v')+\rho(v')\rho(v)=2(v,v')$, gives

$$\text{numerator}=-16\big[(p_1,p_1-p)(A(p),A(-p))-(\iota_{p_1}A(p),\iota_{p_1-p}A(-p))-(\iota_{p_1-p}A(p),\iota_{p_1}A(-p))\big],$$

with value $\frac{16t_1t_2}{(t_1+t_2)^2}\big(|p|^2(A(p),A(-p))-2(\iota_pA(p),\iota_pA(-p))\big)$ at the centre and positive Laplacian $64\,(A(p),A(-p))$.
Schwinger integral.

$$\frac{\hbar}{2}\,\frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\iint_{[\varepsilon,\infty)^2}\frac{e^{-\frac{t_1t_2|p|^2}{t_1+t_2}}}{(t_1+t_2)^2}\Big[-\frac{16\,(A(p),A(-p))}{t_1+t_2}+\frac{16t_1t_2}{(t_1+t_2)^2}|p|^2(A(p),A(-p))-\frac{32t_1t_2}{(t_1+t_2)^2}(\iota_pA(p),\iota_pA(-p))\Big]dt_1dt_2.$$

ε-expansion. Modulo $\mathscr{A}_{\geq 0}$, the double integral of the bracket is $-4\varepsilon^{-1}(A(p),A(-p))-\frac{16}{3}\log\varepsilon\big(|p|^2(A(p),A(-p))-(\iota_pA(p),\iota_pA(-p))\big)$, the $|p|^2$ terms being $-16\cdot\frac{1}{6}-16\cdot\frac{1}{6}$.
Local part. $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big[-\varepsilon^{-1}\int\mathrm{dvol}\,(A|_V,A|_V)-\frac{4}{3}\log\varepsilon\int\mathrm{dvol}\,(dA|_V,dA|_V)\big]$: the $\log\varepsilon$ part of the gaugino loop is transverse.

Example 4.13 (Ghost loop, graph (f) of the $A|_V$ two-point function)**.** Amplitude. Two vertices $A|_V\,c\,A^*|_V$ (Definition 1.13); the edges carry the components $c\otimes A^*|_V$ and $A^*|_V\otimes c$ of $P(\varepsilon,\infty)$, equal to $-(-,-)^{-1}\,i$ (edge momentum) times the scalar factor (Proposition 3.11); the parity operator gives the sign of the loop of odd fields.
Numerator algebra. The two orientations of the ghost line contribute

$$\text{numerator}=(\iota_{p_1}A(p),\iota_{p_1-p}A(-p))+(\iota_{p_1-p}A(p),\iota_{p_1}A(-p)),$$

with value $-\frac{2t_1t_2}{(t_1+t_2)^2}(\iota_pA(p),\iota_pA(-p))$ at the centre and positive Laplacian $-4\,(A(p),A(-p))$.
Schwinger integral.

$$\frac{\hbar}{2}\,\frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\iint_{[\varepsilon,\infty)^2}\frac{e^{-\frac{t_1t_2|p|^2}{t_1+t_2}}}{(t_1+t_2)^2}\Big[\frac{(A(p),A(-p))}{t_1+t_2}-\frac{2t_1t_2}{(t_1+t_2)^2}(\iota_pA(p),\iota_pA(-p))\Big]dt_1dt_2.$$

ε-expansion. Modulo $\mathscr{A}_{\geq 0}$, the double integral of the bracket is $\frac{1}{4}\varepsilon^{-1}(A(p),A(-p))+\log\varepsilon\big(\frac{1}{6}|p|^2(A(p),A(-p))+\frac{1}{3}(\iota_pA(p),\iota_pA(-p))\big)$.
Local part.

$$\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big[\tfrac{1}{16}\varepsilon^{-1}\int\mathrm{dvol}\,(A|_V,A|_V)+\log\varepsilon\Big(\tfrac{1}{24}\int\mathrm{dvol}\,(dA|_V,dA|_V)+\tfrac{1}{8}\int\mathrm{dvol}\,(d^*A|_V,d^*A|_V)\Big)\Big],$$

since $2\cdot\frac{1}{16}=\frac{1}{2}\cdot\frac{1}{4}$, $2\cdot\frac{1}{24}=\frac{1}{2}\cdot\frac{1}{6}$ and $2\big(-\frac{1}{24}+\frac{1}{8}\big)=\frac{1}{2}\cdot\frac{1}{3}$. The longitudinal coefficient $\frac{1}{8}$ cancels the $-\frac{1}{8}$ of Example 4.11.

Example 4.14 (Seagull, graph (a) of the $A|_V$ two-point function)**.** Amplitude. One quartic vertex (1.5) with two legs and one $A|_V$ edge closed into a loop; the term $n = 1$ of (3.5).
Numerator algebra. Contracting two legs of (1.5) with $2(-,-)^{-1} \otimes$ (inverse metric of V) and summing over the pairs of legs gives the numerator $6\,(A(p), A(-p))$, independent of p_1.
Schwinger integral. By Lemma 4.4 with one edge,

$$\frac{\hbar}{2}\,\frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X,Y)} \int_\varepsilon^\infty \frac{6\,(A(p), A(-p))}{t^2}\, dt.$$

ε-expansion. The integral equals $6\,\varepsilon^{-1}(A(p), A(-p))$ exactly; its $\varepsilon \frac{d}{d\varepsilon}$ is the graph with the edge pinned at $t = \varepsilon$, $-\varepsilon \cdot 6\,\varepsilon^{-2}(A(p), A(-p))$, as in Proposition 4.2.
Local part. $\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X,Y)}\ \frac{3}{2}\,\varepsilon^{-1} \int \mathrm{dvol}\,(A|_V, A|_V)$, a mass term with vanishing $\log \varepsilon$ part.

Example 4.15 (The graph with external $c, A^*|_V$)**.** Amplitude. Legs $r_1 A^*(p) e^{i(p,x)}$ and $r_2 c(-p) e^{-i(p,x)}$; two vertices $A|_V\, c\, A^*|_V$; one $A|_V$ edge and one c–A^* edge (graph (c) of the third figure).
Numerator algebra. The coefficient of $r_1 r_2$ is $2i\,(p\,c(-p), A^*(p))$, independent of p_1.
Schwinger integral.

$$\frac{\hbar}{2}\,\frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X,Y)} \iint_{[\varepsilon,\infty)^2} \frac{e^{-\frac{t_1 t_2 |p|^2}{t_1+t_2}}}{(t_1+t_2)^2}\, 2i\,(p\,c(-p), A^*(p))\, dt_1 dt_2.$$

ε-expansion. Modulo $\mathscr{A}_{\geq 0}$, $-2i \log \varepsilon\,(p\,c(-p), A^*(p))$.
Local part. The plane-wave coefficient is $\frac{1}{2} \cdot (-2i) = -i$, and the plane-wave value of $\int \mathrm{dvol}\,(dc, A^*)$ is $i\,(p\,c(-p), A^*(p))$ by (4.3); the contribution is $\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X,Y)} \big(-\log \varepsilon\big) \int \mathrm{dvol}\,(dc, A^*)$.

Theorem 4.16 (The quadratic counterterm)**.** *The part of $I^{CT}[\varepsilon]$ quadratic in the fields is*

$$\begin{aligned}\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X,Y)} \Big\{ \tfrac{9}{4}\,\varepsilon^{-1} \Big[\int \mathrm{dvol}\,(A|_V, A|_V) + \int \mathrm{dvol}\,(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp}) \Big] \\ + \log \varepsilon \Big[- \int \mathrm{dvol}\,(dA|_V, dA|_V) - \int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^\perp}, dA|_{W_{\mathbb{R}}^\perp}) \\ + 4 \int \mathrm{dvol}\,(\lambda, \not{d}\lambda) - \int \mathrm{dvol}\,(dc, A^*) \Big] \Big\}.\end{aligned} \tag{4.4}$$

The coefficient of $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$ is 0, and every quadratic local functional mixing $A|_V$ with $A|_{W_{\mathbb{R}}^\perp}$, or c with $A^*|_{W_{\mathbb{R}}^\perp}$, has coefficient 0. In terms of the ten-component field A,

$$\begin{aligned}\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X,Y)} \Big\{ \tfrac{9}{4}\,\varepsilon^{-1} \int \mathrm{dvol}\,(A, A) \\ + \log \varepsilon \Big[- \int \mathrm{dvol}\,(dA, dA) + 4 \int \mathrm{dvol}\,(\lambda, \not{d}\lambda) - \int \mathrm{dvol}\,(dc, A^*) \Big] \Big\}.\end{aligned}$$

Proof. $I^{CT}[\varepsilon]$ is even and of degree 0, so by the degrees of Chapter 1 its quadratic part has legs (A, A), (λ, λ), (c, A^*) or (c, λ^*). The regulator is covariant under $\mathrm{Spin}(4, \mathbb{C}) \times \mathrm{Spin}(6, \mathbb{C})$, since $H = d^*d + dd^*$ and Q^{GF} commute with it, so $I^{CT}[\varepsilon]$ is invariant. The invariant pairings of a vector of V with a vector of $W_{\mathbb{R}}^\perp$, of a scalar with a spinor, and of a scalar with a vector of $W_{\mathbb{R}}^\perp$ vanish; this gives the vanishing mixings and the vanishing of the sectors (c, λ^*) and $(c, A^*|_{W_{\mathbb{R}}^\perp})$. The remaining

sectors are the sums of the three tables, whose rows are all one-particle-irreducible one-loop graphs with these legs; the representative rows are derived in Examples 4.11–4.15. The sums are

$$\begin{aligned} A|_V: &\quad \tfrac{3}{2}+3-\tfrac{9}{16}-\tfrac{3}{4}-1+\tfrac{1}{16}=\tfrac{9}{4}, \qquad \tfrac{19}{24}-\tfrac{1}{2}-\tfrac{4}{3}+\tfrac{1}{24}=-1, \qquad -\tfrac{1}{8}+\tfrac{1}{8}=0; \\ A|_{W_{\mathbb{R}}^{\perp}}: &\quad 2+\tfrac{5}{2}-\tfrac{1}{4}-2=\tfrac{9}{4}, \qquad 1-2=-1; \qquad \lambda: \quad 1+3=4; \qquad (c,A^*): \quad -1. \end{aligned}$$

The ten-component form uses $(dA,dA)=(dA|_V,dA|_V)+(dA|_{W_{\mathbb{R}}^{\perp}},dA|_{W_{\mathbb{R}}^{\perp}})$, the cross term vanishing by the orthogonality of V and $W_{\mathbb{R}}^{\perp}$. □

Remark 4.17 (Two properties of the quadratic counterterm)**.** The per-graph entries of the $A|_V$ and $A|_{W_{\mathbb{R}}^{\perp}}$ sectors differ, and their totals agree: (4.4) is a functional of the ten-component field A. The $\log\varepsilon$ part is transverse, the longitudinal coefficients $-\frac{1}{8}$ of the gauge bubble and $+\frac{1}{8}$ of the ghost loop cancelling, and it is a sum of multiples of the three terms of the free action (1.2):

$$\begin{aligned} &-\int \mathrm{dvol}\,(dA,dA)+4\int \mathrm{dvol}\,(\lambda,\not{d}\lambda)-\int \mathrm{dvol}\,(dc,A^*) \\ &\quad =4\Big(-\tfrac{1}{4}\int \mathrm{dvol}\,(dA,dA)\Big)+8\Big(\tfrac{1}{2}\int \mathrm{dvol}\,(\lambda,\not{d}\lambda)\Big)-1\cdot\int \mathrm{dvol}\,(dc,A^*). \end{aligned}$$

The ε^{-1} part is a mass term for all ten components of A.

4.3 Cubic and quartic graphs

The cubic and quartic parts of $I^{CT}[\varepsilon]$ are organised by the Weyl weights of Example 2.13: A has weight 1, λ weight $\frac{3}{2}$, c weight 0, and the antifields A^*,λ^*,c^* the complementary weights $3,\frac{5}{2},4$, so that each pairing of a field with its antifield has weight 4. The weight of a local functional $\int \mathrm{dvol}\,(\cdots)$ is the sum of the weights of its fields plus the number of its derivatives, and every term of S has weight 4. The superficial degree of a graph is 4 minus the sum of the weights of its legs: 2 for (A,A), 1 for (λ,λ), (c,A^*), $(A|_V,A|_V,A|_V)$, $(A|_V,A|_{W_{\mathbb{R}}^{\perp}},A|_{W_{\mathbb{R}}^{\perp}})$, and 0 for the other cubic and the quartic leg sets below.

Lemma 4.18 (Weights of the singular coefficients)**.** *Let γ be a one-loop one-particle-irreducible graph with vertices labelled by I. Its superficial degree in the sense of Lemma 4.6 is 4 minus the sum of the weights of its legs. The coefficient of ε^{-k} in its singular part is a local functional of weight $4-2k$, and the coefficient of $\log\varepsilon$ a local functional of weight 4; in particular the coefficient of ε^{-1} has weight 2.*

Proof. In momentum space, a vertex with fields of total weight $4-k$ is a polynomial of degree k in the momenta, since every term of S has weight 4; an edge component of $P(\varepsilon,\infty)$ between slots of total weight $2+k$ is a polynomial of degree k times $\int_\varepsilon^\infty e^{-t_e|\mathrm{momentum}|^2}dt_e$ (Proposition 3.11: degree 0 for $A\otimes A$, 1 for $c\otimes A^*$ and $\lambda\otimes\lambda$). Each internal half-edge contributes its weight once to a vertex and once to an edge, so, for legs in fixed slots and every assignment of slots to the internal half-edges, the numerator of γ is homogeneous of degree $4|V(\gamma)|-2|E(\gamma)|$ minus the sum of the weights of the legs, in the loop momentum p_1 and the leg momenta, which is the homogeneity required in Lemma 4.6; with $|E(\gamma)|=|V(\gamma)|$ at one loop, its degree plus $4-2|E(\gamma)|$ is 4 minus the sum of the weights of the legs. By Lemma 4.6 the coefficient of ε^{-k} is a polynomial in the leg momenta, homogeneous of degree (superficial degree)$-2k$, and that of $\log\varepsilon$ is homogeneous of degree equal to the superficial degree. The weight of such a local coefficient is the sum of the leg weights, 4−(superficial degree), plus its degree in the momenta, which gives $4-2k$ and 4. □

Lemma 4.19 (Local functionals of weight at most four)**.** *Let a local functional of degree 0, without constant term, have weight at most 4 and be invariant under $\mathrm{Spin}(4,\mathbb{C})\times\mathrm{Spin}(6,\mathbb{C})$ and the adjoint action*

of $\mathfrak{g}$*. If it is even, it is a linear combination of: in weight* 2, $\int \mathrm{dvol}\,(A|_V, A|_V)$ *and* $\int \mathrm{dvol}\,(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$*; in weight* 4*, functionals whose monomials have one of the field contents*

$$(A|_V)^2 \text{ and } (A|_{W_{\mathbb{R}}^\perp})^2 \text{ with two derivatives}; \qquad \lambda\lambda \text{ and } c\,A^*|_V \text{ with one derivative};$$
$$(A|_V)^3 \text{ and } A|_V (A|_{W_{\mathbb{R}}^\perp})^2 \text{ with one derivative}; \qquad \lambda\lambda A|_V,\ \lambda\lambda A|_{W_{\mathbb{R}}^\perp},$$
$$c\,A|_V\,A^*|_V,\ c\,A|_{W_{\mathbb{R}}^\perp} A^*|_{W_{\mathbb{R}}^\perp},\ c\,\lambda\,\lambda^*,\ c\,c\,c^*; \qquad (A|_V)^4,\ (A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2,\ (A|_{W_{\mathbb{R}}^\perp})^4,$$

In weights 1 *and* 3 *there are none.*

Proof. As functionals, c has degree $+1$, A^* and λ^* degree -1, c^* degree -2, and A, λ degree 0 (Chapter 1). Degree 0 requires the number of c to equal the number of A^* and λ^* plus twice the number of c^*, and weight at most 4 allows at most one antifield. With c^* (weight 4): $c\,c\,c^*$. With A^* (weight 3): one c and weight at most 1 more, i.e. $c\,A^*$ with one derivative or $c\,A\,A^*$; invariance pairs $A^*|_V$ with the derivative or with $A|_V$, and $A^*|_{W_{\mathbb{R}}^\perp}$ with $A|_{W_{\mathbb{R}}^\perp}$. With λ^* (weight $\frac{5}{2}$): one c and, by $\mathrm{Spin}(4,\mathbb{C})$, one spinor λ. Without antifields there is no c; the monomials are built from A, λ and derivatives, with an even number of spinors. $\mathrm{Spin}(6,\mathbb{C})$ admits invariants of an even number of vectors of $W_{\mathbb{R}}^\perp$ (the invariant volume form of $W_{\mathbb{R}}^\perp$ needs six) and $\mathrm{Spin}(4,\mathbb{C})$ of an even number of vector indices from $A|_V$ and the derivatives. In weight 2 this leaves $(A|_V)^2$, $(A|_{W_{\mathbb{R}}^\perp})^2$; in weight 4 the listed contents, a single A with three derivatives being a total derivative; in weight 1 the single A, without invariant; in weight 3 the contents A^3, $A\,A$ with one derivative, $c\,A^*$, each with an odd number of vector indices of V or of $W_{\mathbb{R}}^\perp$, and $\lambda\lambda$, whose components lie in $S_+ \otimes S_-$ (a vector of V) or in $W \otimes W$, $W^* \otimes W^*$ (without $SL(4,\mathbb{C})$-invariant); all of these have vanishing invariants. □

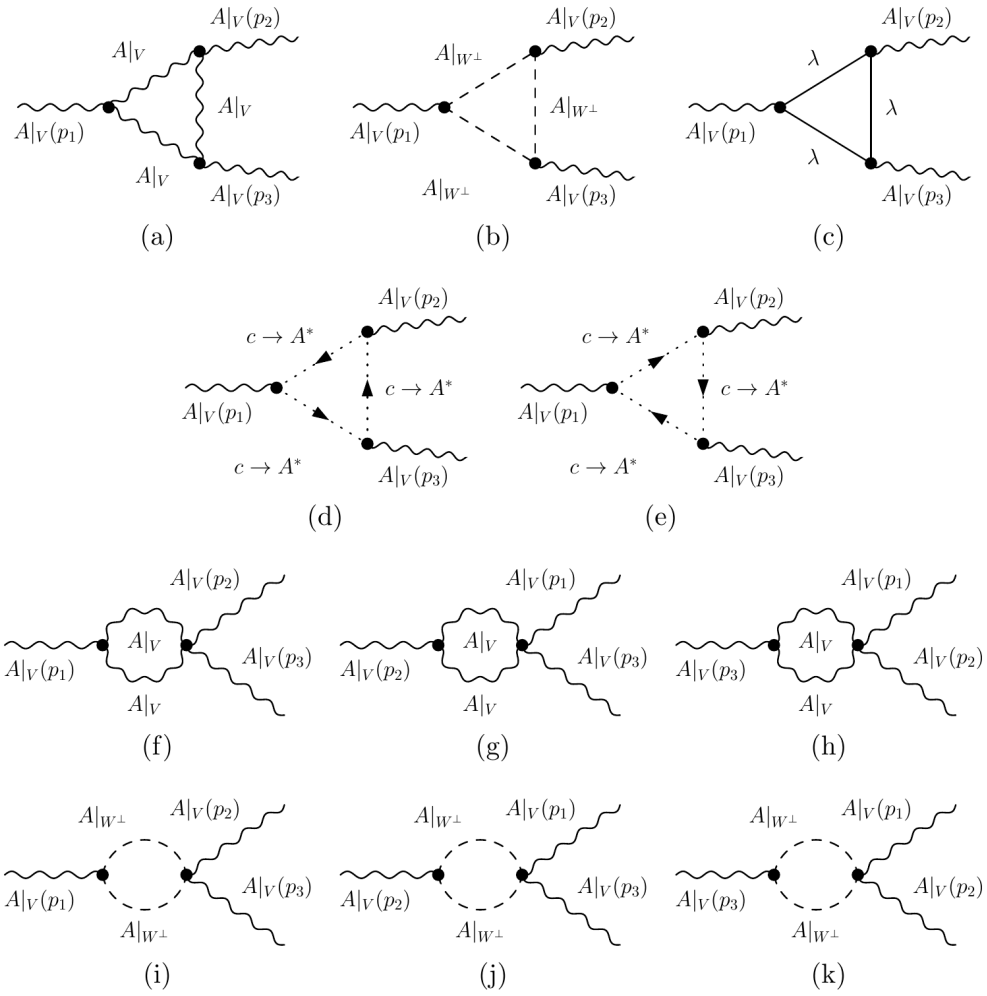
Proposition 4.20 (Structure of the one-loop counterterm)**.** $I^{CT}[\varepsilon]$ *is the sum of* ε^{-1} *times the mass term of Theorem 4.16 and* $\log\varepsilon$ *times a local functional of weight* 4 *with the field contents of Lemma 4.19. Its* $\log\varepsilon$ *part is therefore determined by the one-particle-irreducible one-loop graphs with the leg sets* $(A|_V, A|_V)$*,* $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$*,* (λ,λ)*,* $(c, A^*|_V)$*, the eight cubic sets* $(A|_V, A|_V, A|_V)$*,* $(A|_V, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$*,* $(\lambda,\lambda, A|_V)$*,* $(\lambda,\lambda, A|_{W_{\mathbb{R}}^\perp})$*,* $(c, A|_V, A^*|_V)$*,* $(c, A|_{W_{\mathbb{R}}^\perp}, A^*|_{W_{\mathbb{R}}^\perp})$*,* (c,λ,λ^*)*,* (c,c,c^*)*, and the three quartic sets* $(A|_V)^4$*,* $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$*,* $(A|_{W_{\mathbb{R}}^\perp})^4$*.*

Proof. By Lemmas 4.6 and 4.18, the singular part of a one-particle-irreducible one-loop graph with vertices labelled by I consists of ε^{-k} times a local functional of weight $4-2k$, $1 \le k \le \frac{1}{2}$(superficial degree), and $\log\varepsilon$ times a local functional of weight 4, and it vanishes when the superficial degree is negative. The slots of every vertex of I have degrees adding to 0, and so have the two slots joined by every nonzero component of $P(\varepsilon,\infty)$ (Proposition 3.11: $A\otimes A$, $c\otimes A^*|_V$, $A^*|_V \otimes c$, $\lambda\otimes\lambda$); hence the legs of every graph have degrees adding to 0. With the degrees of the proof of Lemma 4.19, a set of n legs with degrees adding to 0 has weight at least n: every leg other than c has weight at least 1, every c has weight 0 and degree $+1$, and the antifields exceed weight 1 by $\frac{3}{2}$ per unit of negative degree or more (A^*: $3 = 1+2$, λ^*: $\frac{5}{2} = 1+\frac{3}{2}$, c^*: $4 = 1+\frac{3}{2}\cdot 2$), so the weight is at least $n - (\text{number of } c) + \frac{3}{2}(\text{number of } c) \ge n$. The graphs with one leg have value 0 (Lemma 4.7). Hence the superficial degree, 4 minus the weight of the legs, is at most 2, and negative for five or more legs; by Lemma 4.6 no power ε^{-k} with $k \ge 2$ occurs, and the coefficient of ε^{-1} has weight 2. $I^{CT}[\varepsilon]$ is invariant under $\mathrm{Spin}(4,\mathbb{C}) \times \mathrm{Spin}(6,\mathbb{C})$ and the adjoint action of $\mathfrak{g}$ (proof of Theorem 4.16), so by Lemma 4.19 the coefficient of ε^{-1} is a combination of the two mass terms, computed in Theorem 4.16, and the coefficient of $\log\varepsilon$ has the listed contents. □

4.3.1 Cubic graphs

Every cubic graph below has vanishing coefficient of ε^{-1}, in accordance with Lemma 4.18. Its $\log\varepsilon$ coefficient, as a multilinear form in the three legs, is a rational multiple of the tree vertex function of Def-

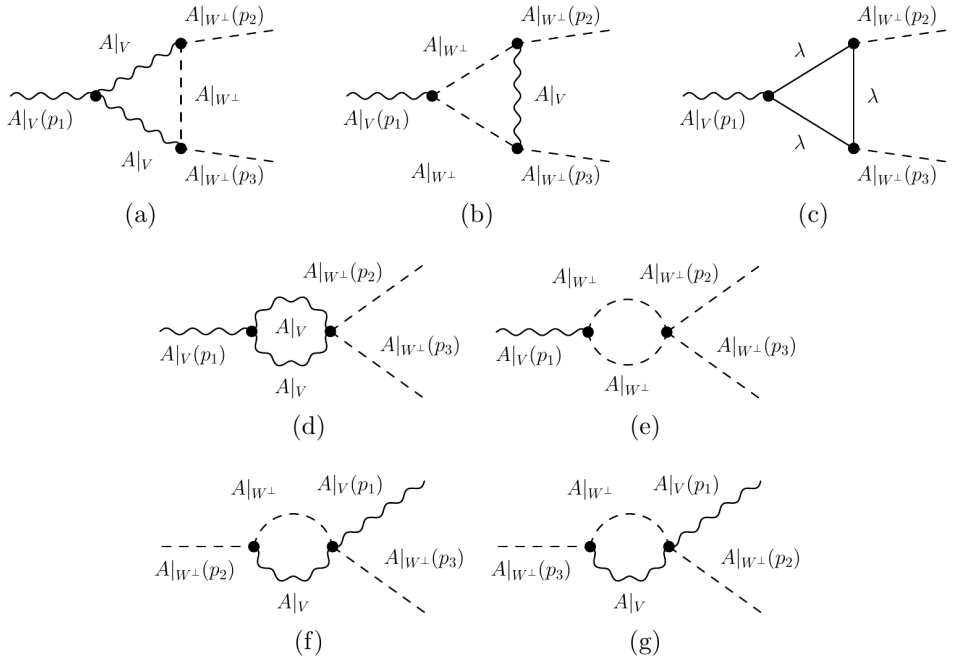
inition 1.13 with the same legs; the tables give these multiples, in units of $\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)/(16\pi^2(X,Y))$. The tree vertices are: the $(A|_V)^3$ and $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$ parts of $\int \mathrm{dvol}\,\big(-\frac{1}{2}(dA, \frac{1}{2}[A\wedge A])\big)$; the $A|_V$ and $A|_{W_{\mathbb{R}}^\perp}$ parts of $\int \mathrm{dvol}\,\frac{1}{2}(\lambda, (\not{d}_A - \not{d})\lambda)$; the $A|_V$ and $A|_{W_{\mathbb{R}}^\perp}$ parts of $\int \mathrm{dvol}\,([A,c],A^*)$; $\int \mathrm{dvol}\,([c,\lambda],\lambda^*)$; $\int \mathrm{dvol}\,\frac{1}{2}([c,c],c^*)$.



One-loop one-particle-irreducible graphs with external $(A|_V, A|_V, A|_V)$: triangles with (a) $A|_V$, (b) $A|_{W_{\mathbb{R}}^\perp}$, (c) λ lines, (d), (e) the two orientations of the c–A^* loop; bubbles with one quartic vertex and (f)–(h) $A|_V$, (i)–(k) $A|_{W_{\mathbb{R}}^\perp}$ lines, one for each leg at the cubic vertex.

graphs	vertices	lines	number	$1/\lvert\mathrm{Aut}\,\gamma\rvert$	$\log\varepsilon$ coefficient
(a)	$(A\|_V)^3$, $(A\|_V)^3$, $(A\|_V)^3$	$A\|_V$, $A\|_V$, $A\|_V$	1	1	$\frac{13}{4}$
(b)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	1	1	2
(c)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	λ, λ, λ	1	1	$\frac{16}{3}$
(d), (e)	$A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$	c–A^*, c–A^*, c–A^*	2	1	$-\frac{1}{12}$
(f)–(h)	$(A\|_V)^3$, $(A\|_V)^4$	$A\|_V$, $A\|_V$	3	$\frac{1}{2}$	$-\frac{9}{2}$
(i)–(k)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	3	$\frac{1}{2}$	0
total					6

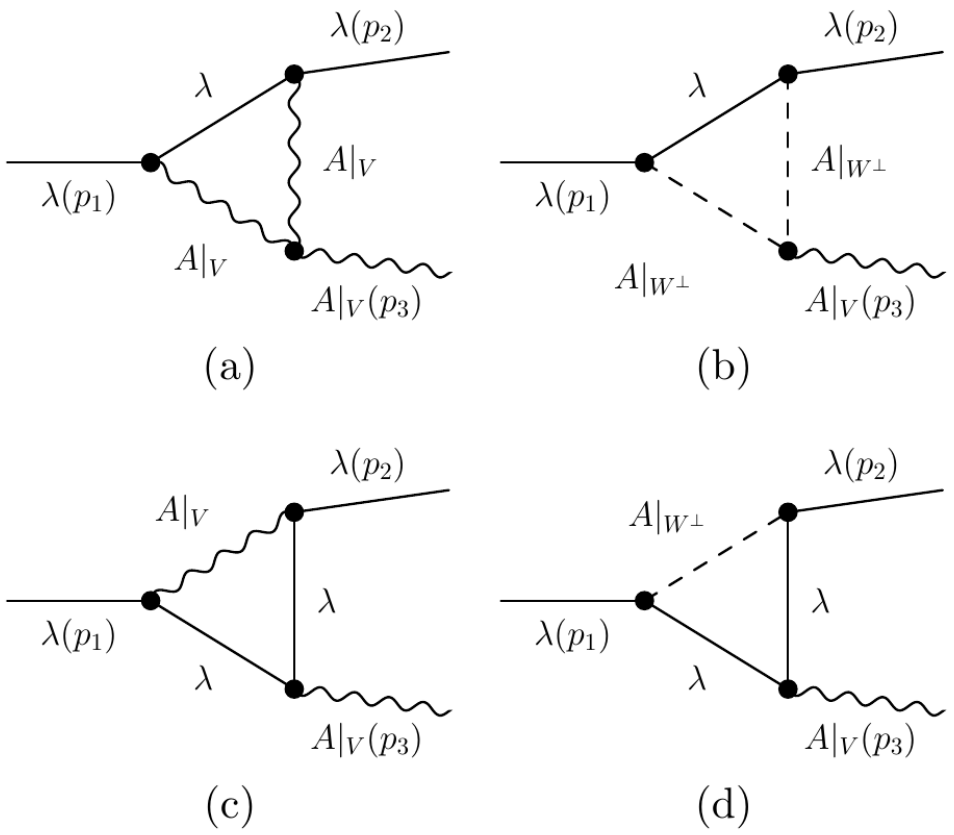
The $\log\varepsilon$ coefficients of the graphs with external $(A|_V, A|_V, A|_V)$, as multiples of the tree vertex.



One-loop one-particle-irreducible graphs with external $(A|_V, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$: triangles (a)–(c); bubbles with one quartic vertex (d)–(g).

graphs	vertices	lines	number	$1/\lvert\operatorname{Aut}\gamma\rvert$	$\log\varepsilon$ coefficient
(a)	$(A\|_V)^3$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	1	1	$\frac{3}{2}$
(b)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	1	1	1
(c)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	λ, λ, λ	1	1	8
(d)	$(A\|_V)^3$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$	1	$\frac{1}{2}$	0
(e)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_{W_{\mathbb{R}}^\perp})^4$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	1	$\frac{1}{2}$	0
(f), (g)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	2	1	$-\frac{9}{2}$
total					6

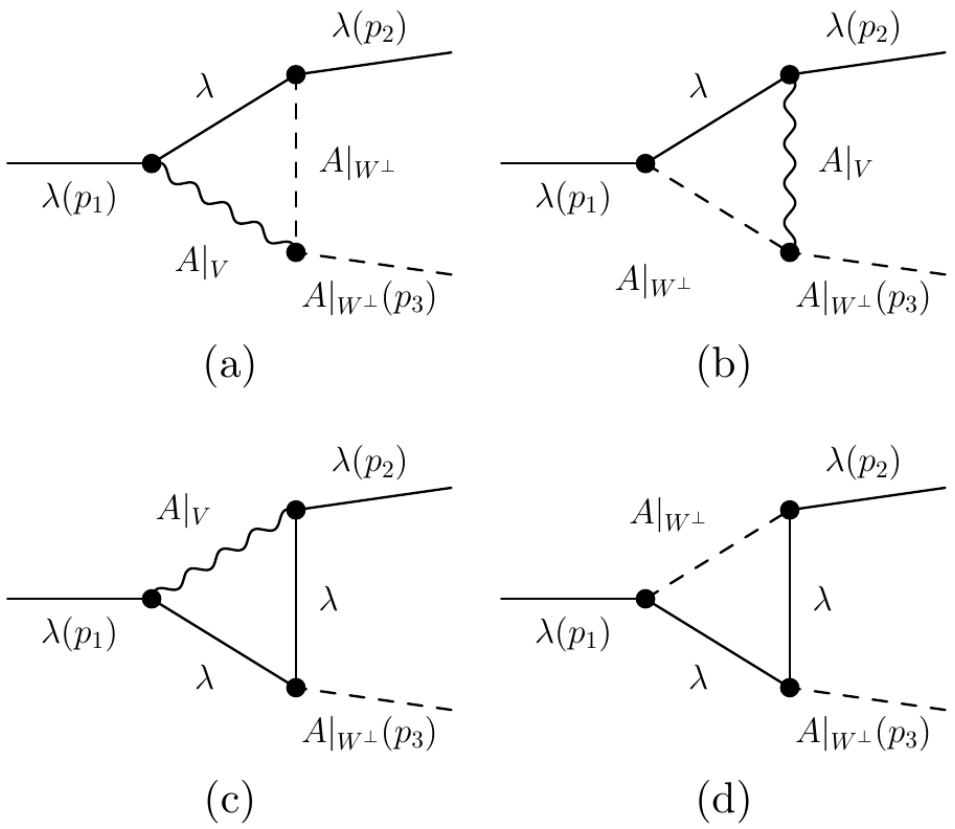
The $\log\varepsilon$ coefficients of the graphs with external $(A|_V, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, as multiples of the tree vertex.



One-loop one-particle-irreducible graphs with external $(\lambda, \lambda, A|_V)$.

graphs	vertices	lines	number	$1/\lvert \operatorname{Aut}\gamma \rvert$	$\log\varepsilon$ coefficient
(a)	$(A\|_V)^3$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	$A\|_V$, $A\|_V$, λ	1	1	3
(b)	$A\|_V(A\|_{W_\mathbb{R}^\perp})^2$, $\lambda\lambda A\|_{W_\mathbb{R}^\perp}$, $\lambda\lambda A\|_{W_\mathbb{R}^\perp}$	$A\|_{W_\mathbb{R}^\perp}$, $A\|_{W_\mathbb{R}^\perp}$, λ	1	1	3
(c)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	$A\|_V$, λ, λ	1	1	1
(d)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_{W_\mathbb{R}^\perp}$, $\lambda\lambda A\|_{W_\mathbb{R}^\perp}$	$A\|_{W_\mathbb{R}^\perp}$, λ, λ	1	1	3
total					10

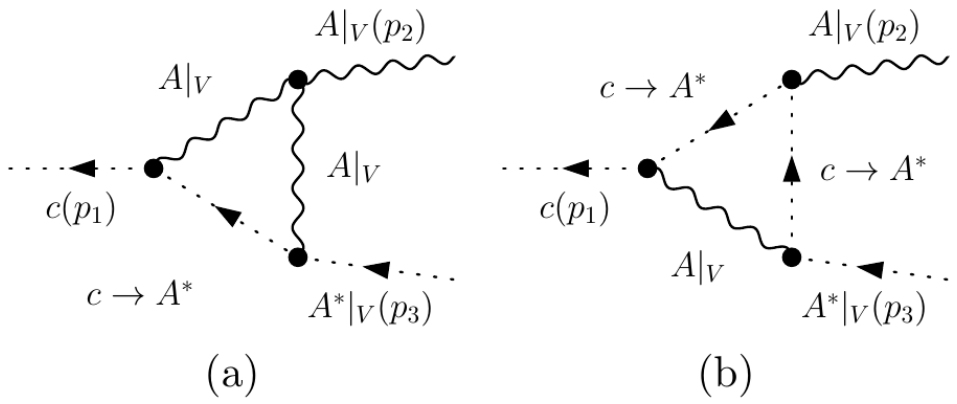
The $\log\varepsilon$ coefficients of the graphs with external $(\lambda, \lambda, A|_V)$, as multiples of the tree vertex; the tree vertex pairs the two four-dimensional chiralities $S_+ \otimes W$ and $S_- \otimes W^*$ of λ.



One-loop one-particle-irreducible graphs with external $(\lambda, \lambda, A|_{W_{\mathbb{R}}^\perp})$; (a), (b) differ by the leg attached to the vertex $A|_V (A|_{W_{\mathbb{R}}^\perp})^2$.

graphs	vertices	lines	number	$1/\lvert \operatorname{Aut}\gamma \rvert$	$\log\varepsilon$ coefficient
(a), (b)	$A\|_V (A\|_{W_{\mathbb{R}}^\perp})^2$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, λ	2	1	2
(c)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	$A\|_V$, λ, λ	1	1	4
(d)	$\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	$A\|_{W_{\mathbb{R}}^\perp}$, λ, λ	1	1	4
total					10

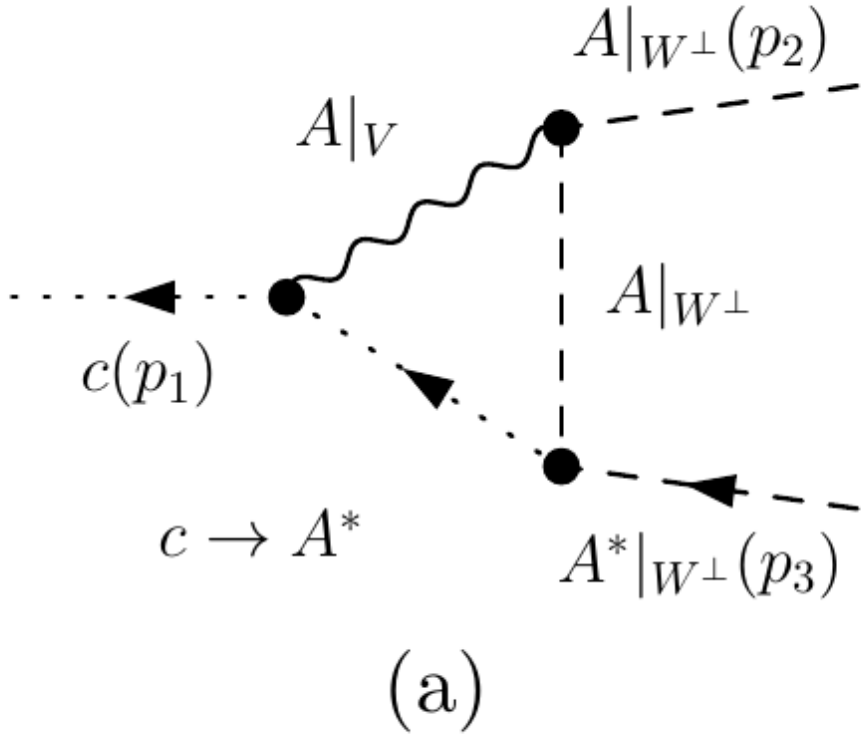
The $\log\varepsilon$ coefficients of the graphs with external $(\lambda, \lambda, A|_{W_{\mathbb{R}}^\perp})$, as multiples of the tree vertex. The tree vertex has two structures, both λ in $S_+ \otimes W$ or both in $S_- \otimes W^*$ (Proposition 1.2); the total is 10 times the tree in each of them.



One-loop one-particle-irreducible graphs with external $(c, A|_V, A^*|_V)$.

graphs	vertices	lines	number	$1/\lvert \operatorname{Aut} \gamma \rvert$	$\log \varepsilon$ coefficient
(a)	$(A\|_V)^3$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$	$A\|_V$, $A\|_V$, c–A^*	1	1	$\frac{3}{4}$
(b)	$A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$	$A\|_V$, c–A^*, c–A^*	1	1	$\frac{1}{4}$
total					1

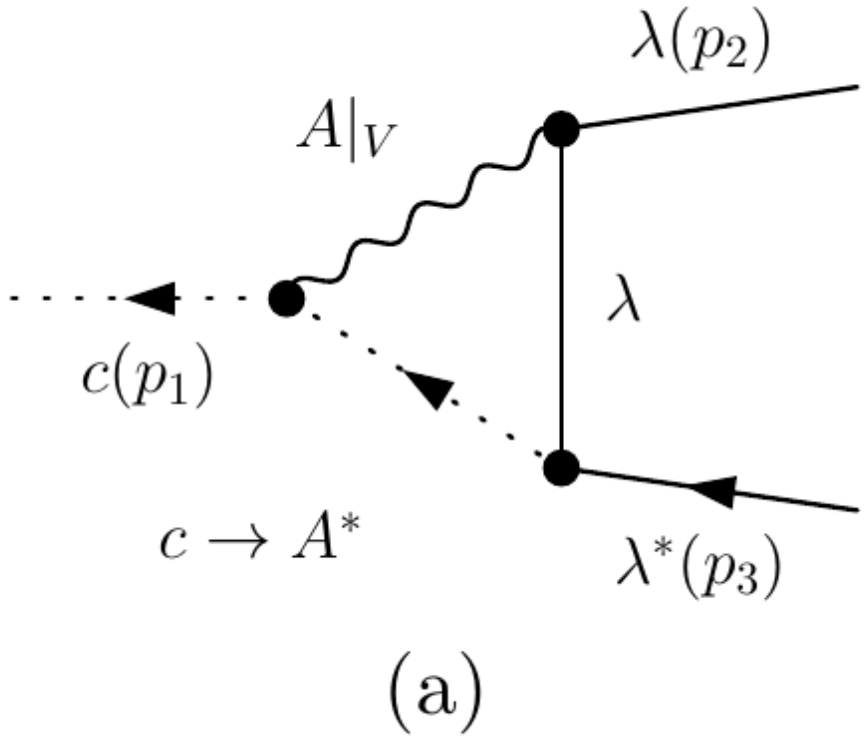
The $\log \varepsilon$ coefficients of the graphs with external $(c, A|_V, A^*|_V)$, as multiples of the tree vertex.



The one-loop one-particle-irreducible graph with external $(c, A|_{W_{\mathbb{R}}^\perp}, A^*|_{W_{\mathbb{R}}^\perp})$.

graphs	vertices	lines	number	$1/\lvert \operatorname{Aut}\gamma \rvert$	$\log\varepsilon$ coefficient
(a)	$A\lvert_V (A\lvert_{W_{\mathbb{R}}^\perp})^2$, $A\lvert_V \, c\, A^*\lvert_V$, $A\lvert_{W_{\mathbb{R}}^\perp} \, c\, A^*\lvert_{W_{\mathbb{R}}^\perp}$	$A\lvert_V$, $A\lvert_{W_{\mathbb{R}}^\perp}$, c–A^*	1	1	1
total					1

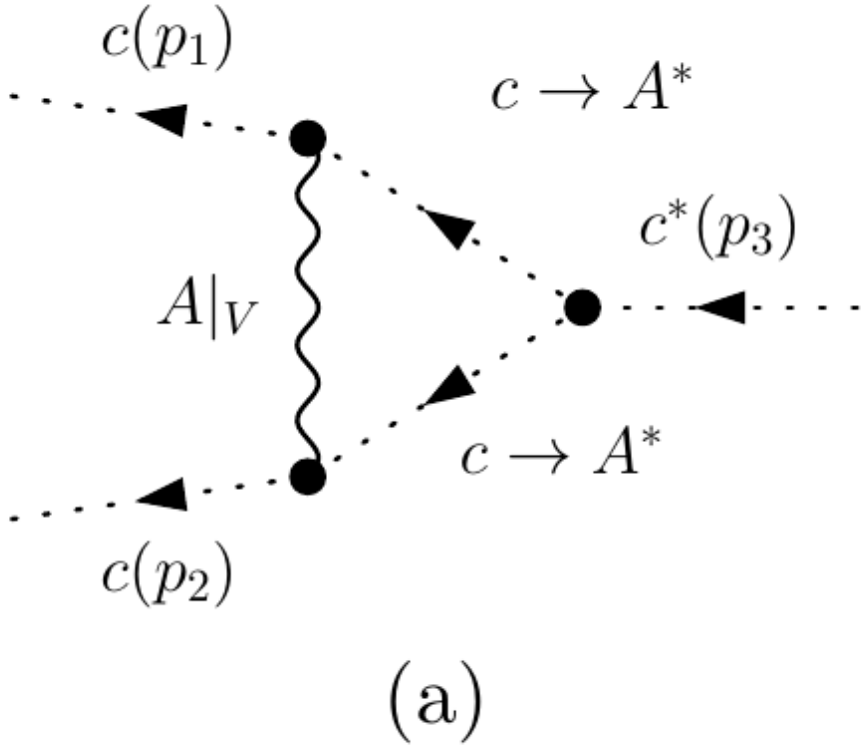
The $\log\varepsilon$ coefficient of the graph with external $(c, A\lvert_{W_{\mathbb{R}}^\perp}, A^*\lvert_{W_{\mathbb{R}}^\perp})$, as a multiple of the tree vertex.



The one-loop one-particle-irreducible graph with external (c, λ, λ^*).

graphs	vertices	lines	number	$1/\lvert \operatorname{Aut}\gamma \rvert$	$\log\varepsilon$ coefficient
(a)	$c\,\lambda\,\lambda^*$, $A\lvert_V \, c\, A^*\lvert_V$, $\lambda\lambda A\lvert_V$	$A\lvert_V$, c–A^*, λ	1	1	1
total					1

The $\log\varepsilon$ coefficient of the graph with external (c, λ, λ^*), as a multiple of the tree vertex, for λ in either chirality.



The one-loop one-particle-irreducible graph with external (c, c, c^*).

graphs	vertices	lines	number	$1/\lvert \operatorname{Aut}\gamma \rvert$	$\log\varepsilon$ coefficient
(a)	$c\,c\,c^*$, $A\lvert_V\,c\,A^*\lvert_V$, $A\lvert_V\,c\,A^*\lvert_V$	$A\lvert_V$, c–A^*, c–A^*	1	1	1
total					1

The $\log\varepsilon$ coefficient of the graph with external (c, c, c^*), as a multiple of the tree vertex.

4.3.2 Quartic graphs

For the quartic sets leg $j \in \{1, 2, 3, 4\}$ carries a decomposable fibre element $(v_j, -) \otimes X_j$ with $X_j \in \mathfrak{g}$, $v_j \in V$ for a leg $A|_V$ and $v_j \in W_{\mathbb{R}}^{\perp}$ for a leg $A|_{W_{\mathbb{R}}^{\perp}}$; in the set $(A|_V)^2(A|_{W_{\mathbb{R}}^{\perp}})^2$ legs $1, 2$ are $A|_V$ and legs $3, 4$ are $A|_{W_{\mathbb{R}}^{\perp}}$. A multilinear form is determined by its values on such legs, and by Lemma 4.18 the $\log\varepsilon$ coefficients are independent of the momenta. They are written in the basis of products of a colour factor and a Lorentz factor:

- colour factors: the pairing $([X_1, X_2], [X_3, X_4])$ of the bracket of the colour elements of legs $1, 2$ with the bracket of those of legs $3, 4$; the pairing $([X_1, X_4], [X_2, X_3])$; and the averaged trace $\frac{1}{24}\sum_{\text{orderings}} \operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_{X_1}\operatorname{ad}_{X_2}\operatorname{ad}_{X_3}\operatorname{ad}_{X_4})$, the fourth-order adjoint trace averaged over the 24 orderings of $X_1, \dots, X_4$. The third pairing $([X_1, X_3], [X_2, X_4])$ is the sum of the first two, by the Jacobi identity and the invariance of $(-, -)$.
- Lorentz factors: $(v_1, v_2)(v_3, v_4)$, $(v_1, v_3)(v_2, v_4)$, $(v_1, v_4)(v_2, v_3)$, products of metric pairings.

The two bracket pairings carry the unit $\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(16\pi^2(X,Y))$, the averaged trace the unit $\hbar/16\pi^2$. On such legs, by the Hodge normalisation of Definition 1.8,

$$\begin{aligned}&([A(p_1)\wedge A(p_2)],[A(p_3)\wedge A(p_4)])\\&\quad=([X_1,X_2],[X_3,X_4])\big((v_1,v_3)(v_2,v_4)-(v_1,v_4)(v_2,v_3)\big),\end{aligned}$$

and similarly for the other two terms of (1.5). With the Jacobi identity in the form

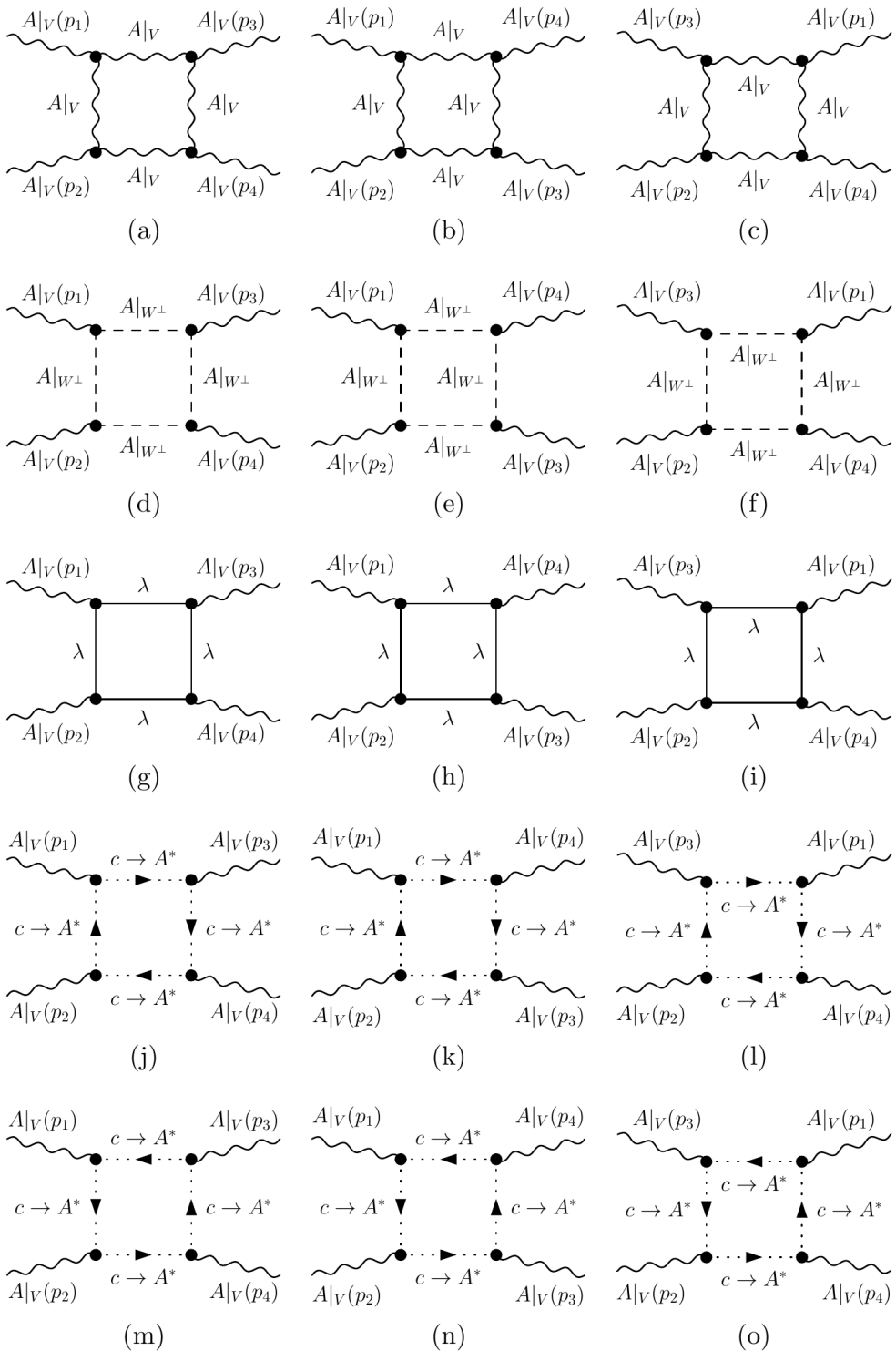
$$([X_1,X_3],[X_2,X_4])=([X_1,X_2],[X_3,X_4])+([X_1,X_4],[X_2,X_3]),$$

the tree vertex function (1.5) is

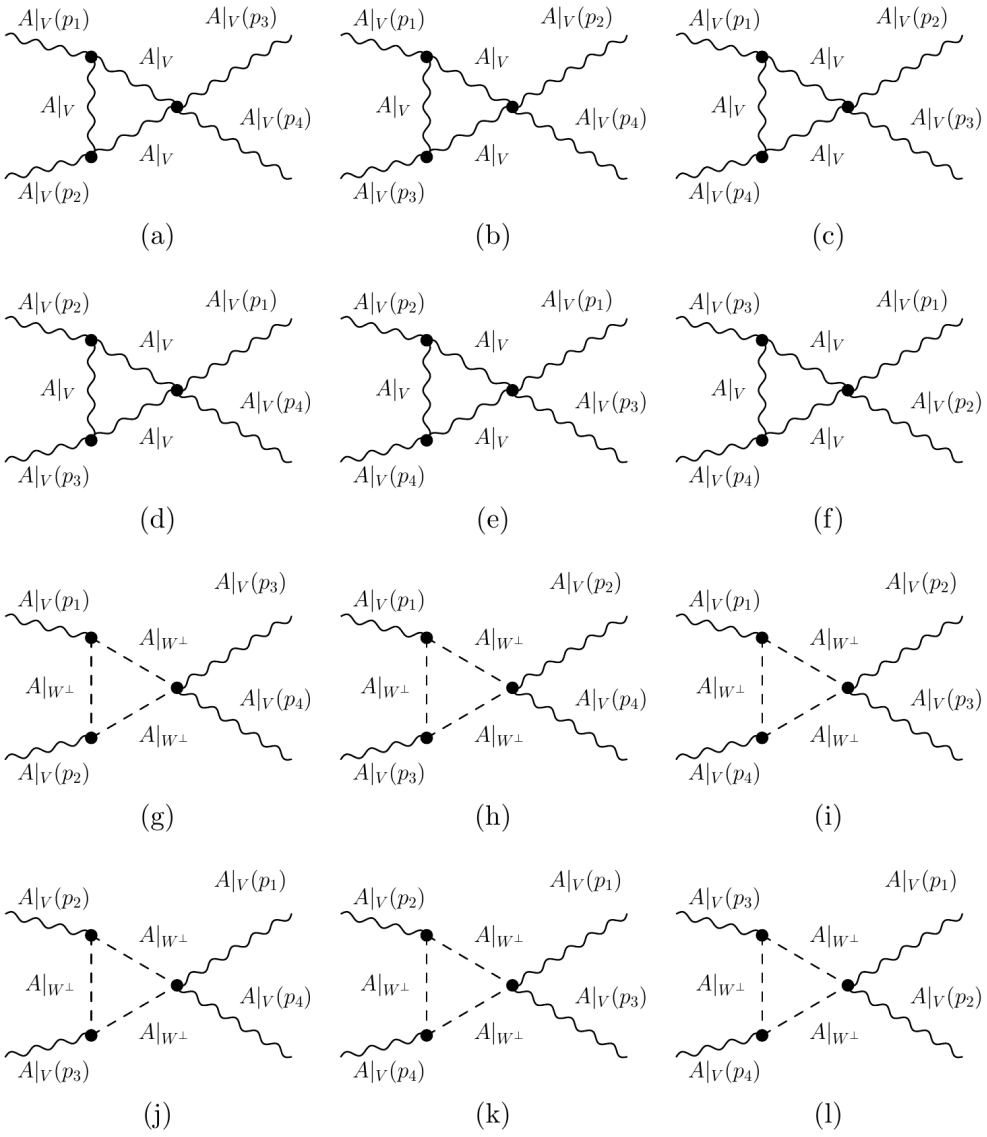
$$\begin{aligned}-\tfrac12\Big[&([X_1,X_2],[X_3,X_4])\big((v_1,v_2)(v_3,v_4)+(v_1,v_3)(v_2,v_4)-2(v_1,v_4)(v_2,v_3)\big)\\&+([X_1,X_4],[X_2,X_3])\big(2(v_1,v_2)(v_3,v_4)-(v_1,v_3)(v_2,v_4)-(v_1,v_4)(v_2,v_3)\big)\Big],\end{aligned}\tag{4.5}$$

and for $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$, where $(v_1,v_3)=(v_1,v_4)=0$, it is

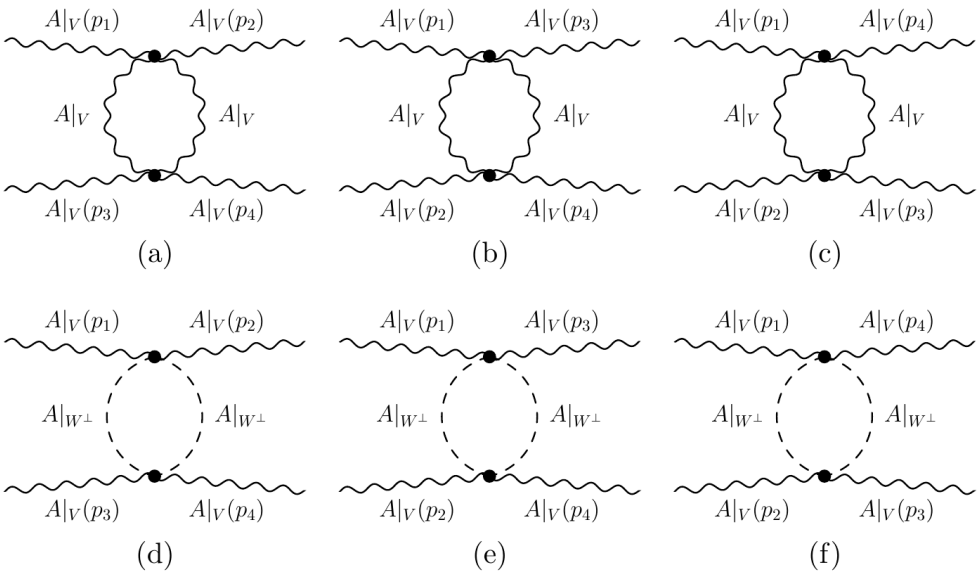
$$-\tfrac12\big[([X_1,X_2],[X_3,X_4])+2([X_1,X_4],[X_2,X_3])\big](v_1,v_2)(v_3,v_4).$$



Box graphs with external $(A|_V)^4$. Columns: the three cyclic orders of the legs. Rows: loops of $A|_V$ (a)–(c), $A|_{W_{\mathbb{R}}^\perp}$ (d)–(f), λ (g)–(i), and c–A^* in the two orientations (j)–(o).



Triangle graphs (one quartic vertex) with external $(A|_V)^4$: loops of $A|_V$ (a)–(f) and of $A|_{W^\perp_{\mathbb{R}}}$ (g)–(l).



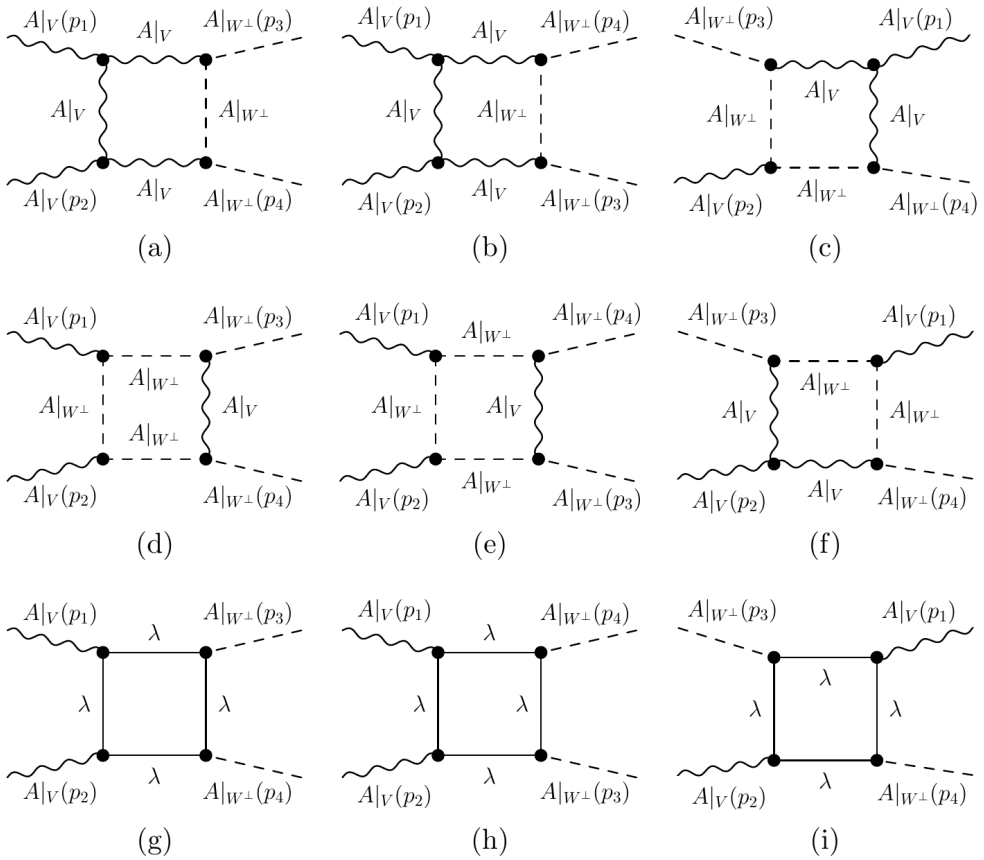
Bubble graphs (two quartic vertices) with external $(A|_V)^4$: $A|_V$ lines (a)–(c), $A|_{W_{\mathbb{R}}^\perp}$ lines (d)–(f).

graphs	vertices	lines	number	$1/\lvert\operatorname{Aut}\gamma\rvert$
box (a)–(c)	$(A\|_V)^3$, $(A\|_V)^3$, $(A\|_V)^3$, $(A\|_V)^3$	$A\|_V$, $A\|_V$, $A\|_V$, $A\|_V$	3	1
box (d)–(f)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	3	1
box (g)–(i)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$	λ, λ, λ, λ	3	1
box (j)–(o)	$A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$, $A\|_V\, c\, A^*\|_V$	c–A^*, c–A^*, c–A^*, c–A^*	6	1
triangle (a)–(f)	$(A\|_V)^3$, $(A\|_V)^3$, $(A\|_V)^4$	$A\|_V$, $A\|_V$, $A\|_V$	6	1
triangle (g)–(l)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	6	1
bubble (a)–(c)	$(A\|_V)^4$, $(A\|_V)^4$	$A\|_V$, $A\|_V$	3	$\frac{1}{2}$
bubble (d)–(f)	$(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	3	$\frac{1}{2}$

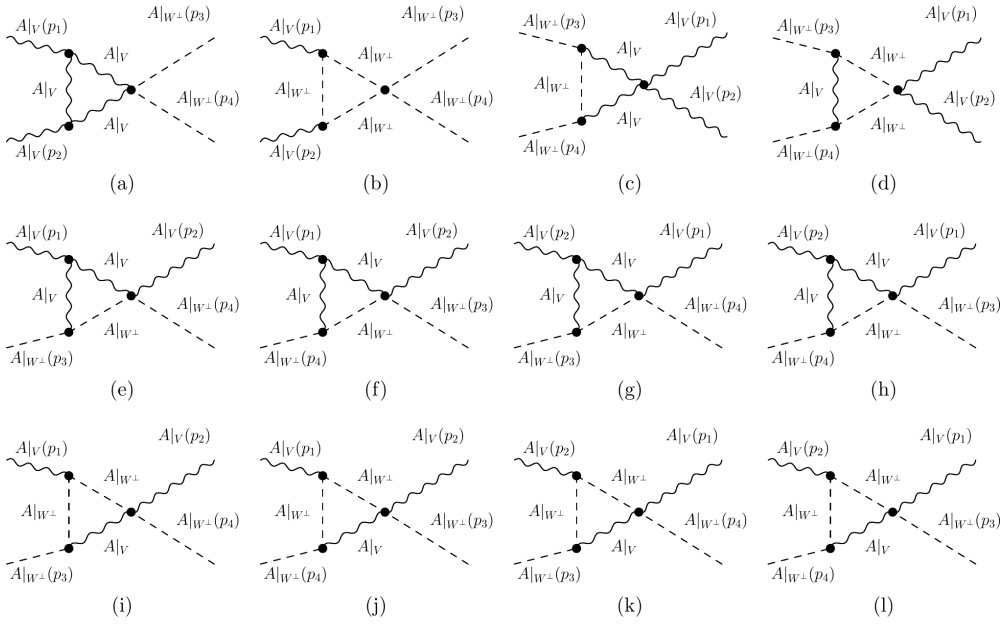
The classes of one-loop graphs with external $(A|_V)^4$; the panel letters refer to the box, triangle and bubble figures of this set.

colour	Lorentz	box (a)–(c)	box (d)–(f)	box (g)–(i)	box (j)–(o)	triangle (a)–(f)	triangle (g)–(l)	bubble (a)–(c)	bubble (d)–(f)	total	8×tree
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_2)(v_3,v_4)$	$\frac{5}{12}$	0	$-\frac{8}{3}$	0	$-\frac{13}{4}$	-2	$\frac{5}{2}$	1	-4	-4
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_3)(v_2,v_4)$	$\frac{5}{12}$	0	$-\frac{8}{3}$	0	$-\frac{13}{4}$	-2	$\frac{5}{2}$	1	-4	-4
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_4)(v_2,v_3)$	$-\frac{5}{6}$	0	$\frac{16}{3}$	0	$\frac{13}{2}$	4	-5	-2	8	8
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_2)(v_3,v_4)$	$\frac{5}{6}$	0	$-\frac{16}{3}$	0	$-\frac{13}{2}$	-4	5	2	-8	-8
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_3)(v_2,v_4)$	$-\frac{5}{12}$	0	$\frac{8}{3}$	0	$\frac{13}{4}$	2	$-\frac{5}{2}$	-1	4	4
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_4)(v_2,v_3)$	$-\frac{5}{12}$	0	$\frac{8}{3}$	0	$\frac{13}{4}$	2	$-\frac{5}{2}$	-1	4	4
averaged trace	$(v_1,v_2)(v_3,v_4)$	$-\frac{37}{4}$	-12	0	$\frac{1}{4}$	15	24	-6	-12	0	0
averaged trace	$(v_1,v_3)(v_2,v_4)$	$-\frac{37}{4}$	-12	0	$\frac{1}{4}$	15	24	-6	-12	0	0
averaged trace	$(v_1,v_4)(v_2,v_3)$	$-\frac{37}{4}$	-12	0	$\frac{1}{4}$	15	24	-6	-12	0	0

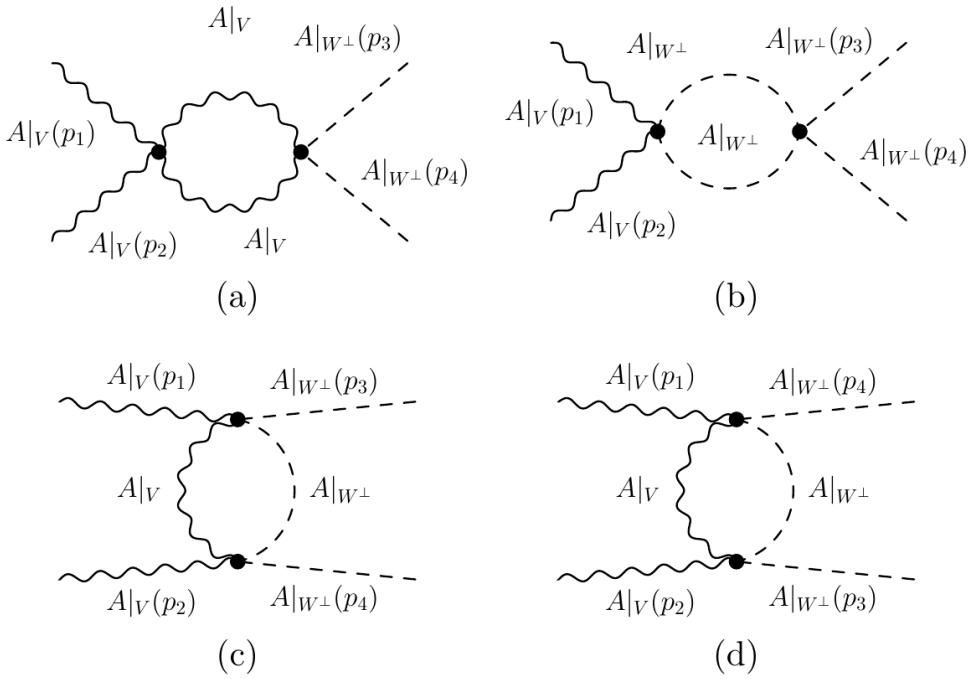
The $\log\varepsilon$ coefficients of the graphs with external $(A|_V)^4$ in the basis of colour and Lorentz factors, summed over each class, with the total and 8 times the tree vertex (4.5). Rows with a bracket pairing are in units of $\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(16\pi^2(X,Y))$, rows with the averaged trace in units of $\hbar/16\pi^2$.



Box graphs with external $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$; (g)–(i): gaugino loops in the three cyclic orders of the legs.



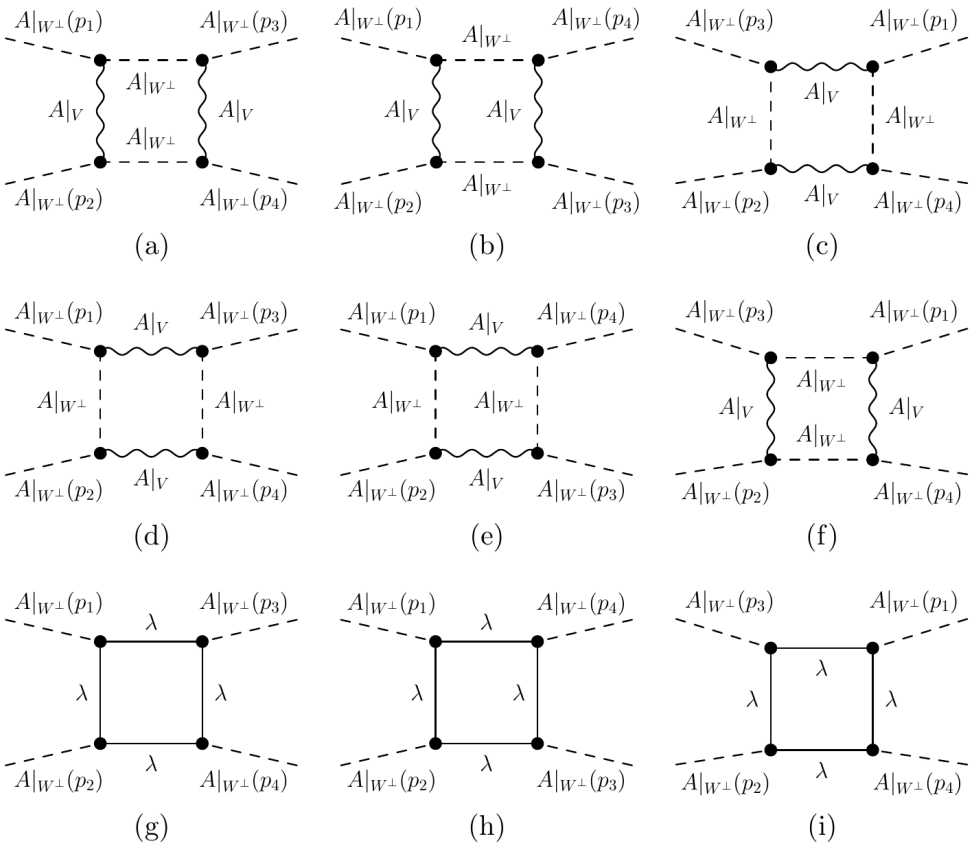
Triangle graphs (one quartic vertex) with external $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$.



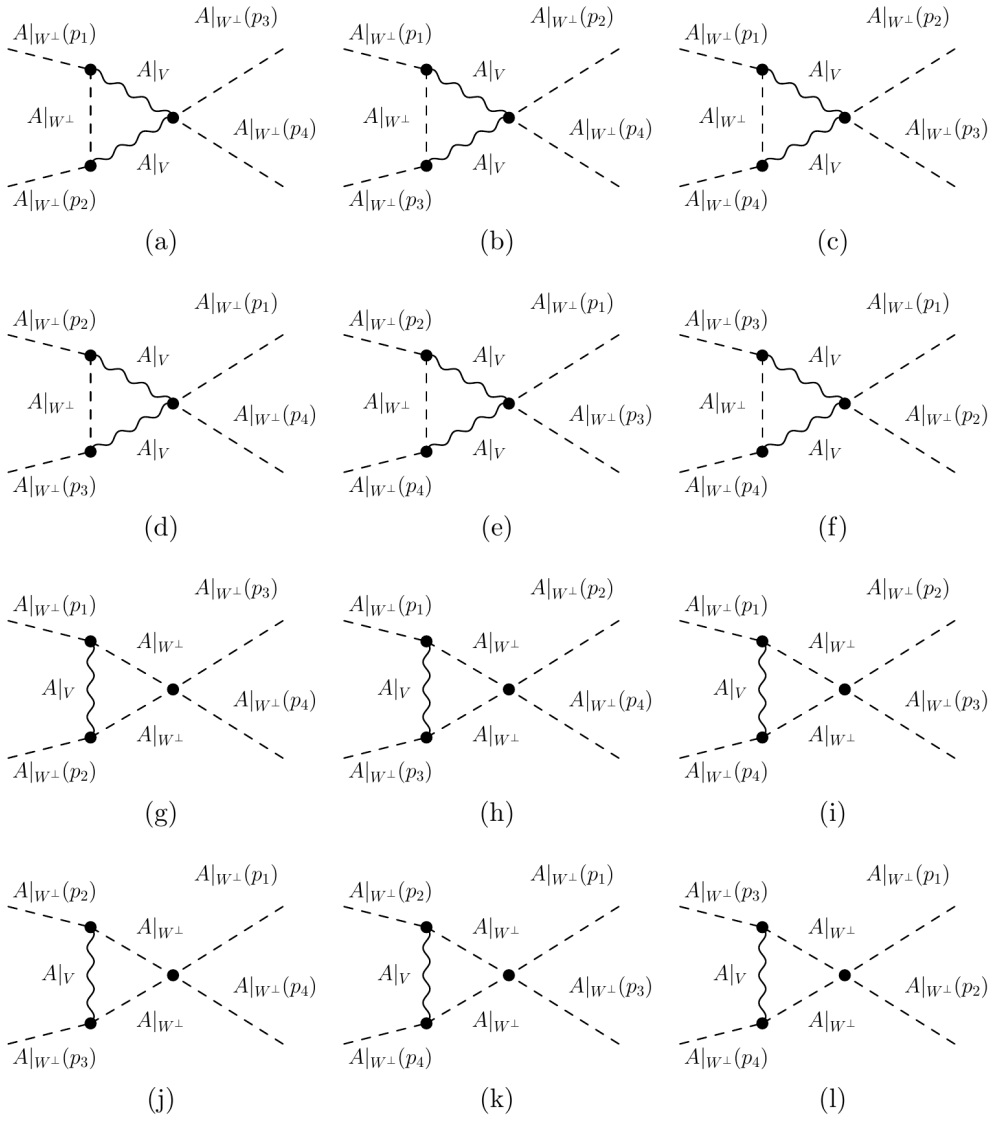
Bubble graphs (two quartic vertices) with external $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$.

graphs	vertices	lines	number	$1/\lvert\operatorname{Aut}\gamma\rvert$	$([X_1,X_2],[X_3,X_4])\times(v_1,v_2)(v_3,v_4)$	$([X_1,X_4],[X_2,X_3])\times(v_1,v_2)(v_3,v_4)$	averaged trace $\times(v_1,v_2)(v_3,v_4)$
box (a), (b)	$(A\|_V)^3$, $(A\|_V)^3$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	2	1	$\frac{1}{8}$	$\frac{1}{4}$	$-\frac{3}{2}$
box (c), (f)	$(A\|_V)^3$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	2	1	0	0	0
box (d), (e)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	2	1	$\frac{1}{24}$	$\frac{1}{8}$	-2
box (g)–(i)	$\lambda\lambda A\|_V$, $\lambda\lambda A\|_V$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	λ, λ, λ, λ	3	1	-4	-8	0
triangle (a)	$(A\|_V)^3$, $(A\|_V)^3$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_V$	1	1	$-\frac{3}{2}$	$-\frac{3}{2}$	9
triangle (b)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_{W_{\mathbb{R}}^\perp})^4$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	1	1	$-\frac{5}{6}$	$-\frac{5}{3}$	10
triangle (c)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^4$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	1	1	$-\frac{1}{8}$	$-\frac{1}{4}$	$\frac{3}{2}$
triangle (d), (i)–(l)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	5	1	$-\frac{5}{6}$	$-\frac{5}{3}$	4
triangle (e)–(h)	$(A\|_V)^3$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	4	1	$-\frac{5}{4}$	$-\frac{5}{2}$	-3
bubble (a)	$(A\|_V)^4$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$	1	$\frac{1}{2}$	$\frac{1}{2}$	1	-6
bubble (b)	$(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_{W_{\mathbb{R}}^\perp})^4$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	1	$\frac{1}{2}$	$\frac{5}{12}$	$\frac{5}{3}$	-10
bubble (c), (d)	$(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	2	1	$\frac{1}{24}$	$\frac{1}{4}$	-2
total					-4	-8	0
$8\times$tree					-4	-8	0

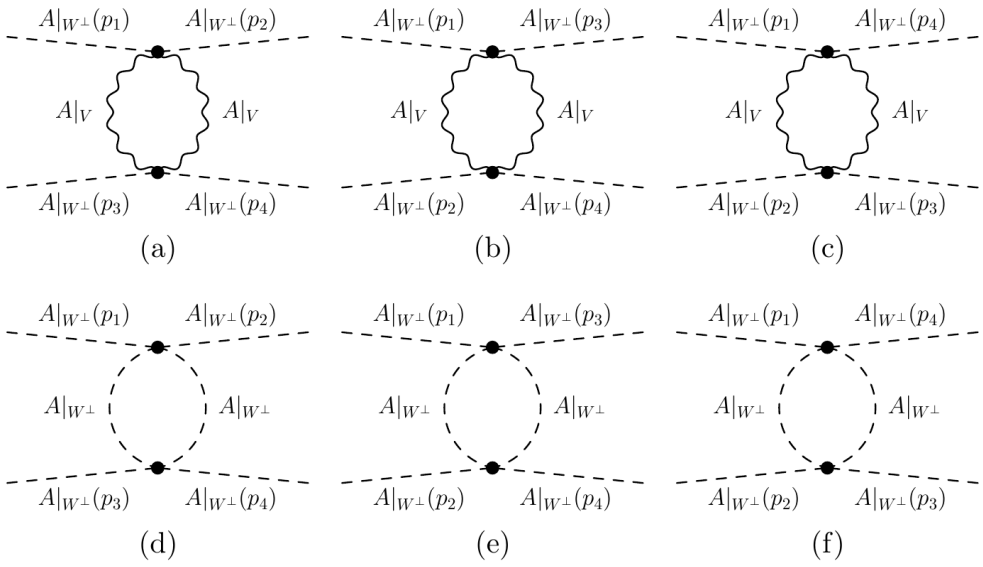
The $\log\varepsilon$ coefficients of the graphs with external $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$ (legs $1,2$ along V, legs $3,4$ along $W_{\mathbb{R}}^\perp$), summed over each class, with the total and 8 times the tree vertex; units as in the previous table.



Box graphs with external $(A|_{W_{\mathbb{R}}^\perp})^4$: alternating $A|_V$ and $A|_{W_{\mathbb{R}}^\perp}$ lines (a)–(f), gaugino loops (g)–(i).



Triangle graphs (one quartic vertex) with external $(A|_{W_{\mathbb{R}}^\perp})^4$: quartic vertex $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$ (a)–(f) and $(A|_{W_{\mathbb{R}}^\perp})^4$ (g)–(l).



Bubble graphs (two quartic vertices) with external $(A|_{W_{\mathbb{R}}^\perp})^4$: $A|_V$ lines (a)–(c), $A|_{W_{\mathbb{R}}^\perp}$ lines (d)–(f).

graphs	vertices	lines	number	$1/\lvert\operatorname{Aut}\gamma\rvert$
box (a)–(f)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	6	1
box (g)–(i)	$\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$, $\lambda\lambda A\|_{W_{\mathbb{R}}^\perp}$	λ, λ, λ, λ	3	1
triangle (a)–(f)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$	6	1
triangle (g)–(l)	$A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_{W_{\mathbb{R}}^\perp})^4$	$A\|_V$, $A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	6	1
bubble (a)–(c)	$(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$, $(A\|_V)^2(A\|_{W_{\mathbb{R}}^\perp})^2$	$A\|_V$, $A\|_V$	3	$\frac{1}{2}$
bubble (d)–(f)	$(A\|_{W_{\mathbb{R}}^\perp})^4$, $(A\|_{W_{\mathbb{R}}^\perp})^4$	$A\|_{W_{\mathbb{R}}^\perp}$, $A\|_{W_{\mathbb{R}}^\perp}$	3	$\frac{1}{2}$

The classes of one-loop graphs with external $(A|_{W_{\mathbb{R}}^\perp})^4$.

colour	Lorentz	box (a)–(f)	box (g)–(i)	triangle (a)–(f)	triangle (g)–(l)	bubble (a)–(c)	bubble (d)–(f)	total	8×tree
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_2)(v_3,v_4)$	$\frac{1}{6}$	$-\frac{16}{3}$	$-\frac{1}{3}$	-2	$\frac{2}{3}$	$\frac{17}{6}$	-4	-4
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_3)(v_2,v_4)$	$\frac{1}{6}$	$-\frac{16}{3}$	$-\frac{1}{3}$	-2	$\frac{2}{3}$	$\frac{17}{6}$	-4	-4
$([X_1,X_2],[X_3,X_4])$	$(v_1,v_4)(v_2,v_3)$	$-\frac{1}{3}$	$\frac{32}{3}$	$\frac{2}{3}$	4	$-\frac{4}{3}$	$-\frac{17}{3}$	8	8
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_2)(v_3,v_4)$	$\frac{1}{3}$	$-\frac{32}{3}$	$-\frac{2}{3}$	-4	$\frac{4}{3}$	$\frac{17}{3}$	-8	-8
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_3)(v_2,v_4)$	$-\frac{1}{6}$	$\frac{16}{3}$	$\frac{1}{3}$	2	$-\frac{2}{3}$	$-\frac{17}{6}$	4	4
$([X_1,X_4],[X_2,X_3])$	$(v_1,v_4)(v_2,v_3)$	$-\frac{1}{6}$	$\frac{16}{3}$	$\frac{1}{3}$	2	$-\frac{2}{3}$	$-\frac{17}{6}$	4	4
averaged trace	$(v_1,v_2)(v_3,v_4)$	-2	16	4	0	-8	-10	0	0
averaged trace	$(v_1,v_3)(v_2,v_4)$	-2	16	4	0	-8	-10	0	0
averaged trace	$(v_1,v_4)(v_2,v_3)$	-2	16	4	0	-8	-10	0	0

The $\log\varepsilon$ coefficients of the graphs with external $(A|_{W_{\mathbb{R}}^\perp})^4$ in the basis of colour and Lorentz factors, with the total and 8 times the tree vertex (4.5); units as in the $(A|_V)^4$ table.

In each quartic set the averaged traces cancel in the total, so that the total is a combination of the two bracket pairings, and the total equals 8 times the tree vertex.

4.3.3 The logarithmic part as a rescaling of the fields

Write $\int(A,A^*)$, $\int(\lambda^*,\lambda)$, $\int(c^*,c)$ for the local functionals $e\mapsto\langle e_A,e_{A^*}\rangle$, $\langle e_{\lambda^*},e_\lambda\rangle$, $\langle e_{c^*},e_c\rangle$, the pairing of Definition 1.5 applied to the components of the field e in the two slots of a pair, the even member first; the first uses all ten components of A. For a monomial in the fields, let $n_A,n_{A^*},n_\lambda,n_{\lambda^*},n_c,n_{c^*}$ denote the numbers of its factors of each kind.

Lemma 4.21 (The bracket as a field count)**.** *For every monomial F in the fields,*

$$\Big\{\int(A,A^*),F\Big\}=(n_A-n_{A^*})F,\qquad \Big\{\int(\lambda^*,\lambda),F\Big\}=(n_{\lambda^*}-n_\lambda)F,\qquad \Big\{\int(c^*,c),F\Big\}=(n_{c^*}-n_c)F.$$

Hence, on every monomial of S,

$$\Big\{2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c),\,F\Big\}=\big(2(n_A-n_{A^*})+4(n_\lambda-n_{\lambda^*})+(n_c-n_{c^*})\big)F.$$

Proof. By Definition 1.6, the kernel K_0 pairs each even member A,λ^*,c^* with its odd partner A^*,λ,c with the coefficient $-\delta(x-y)$ times the inverse fibre pairing, which is the structure $K_0=-\sum(x_i\otimes\xi_i+\xi_i\otimes x_i)$, $\Delta_0=\sum\partial_{x_i}\partial_{\xi_i}$ of [C07, §10.1] for the manifold a point, in the notation of loc. cit.: x_i an even coordinate and ξ_i its odd partner, applied pointwise at each position. For one pair, $x_i\xi_i$ is odd, and

$$\begin{aligned}\{x_i\xi_i,x_i\}&=\Delta_0(x_i^2\xi_i)-\Delta_0(x_i\xi_i)\,x_i+x_i\xi_i\,\Delta_0(x_i)=2x_i-x_i=x_i,\\ \{x_i\xi_i,\xi_i\}&=\Delta_0(x_i\xi_i\xi_i)-\Delta_0(x_i\xi_i)\,\xi_i+x_i\xi_i\,\Delta_0(\xi_i)=-\xi_i.\end{aligned}$$

The bracket with a fixed functional is a derivation, because Δ_0 is a second-order operator, so $\{x_i\xi_i,-\}=x_i\partial_{x_i}-\xi_i\partial_{\xi_i}$ counts x_i minus ξ_i. The three functionals of the lemma are the sums of $x_i\xi_i$ over the components, with x_i the even member, which gives the three identities; the combination follows by linearity. □

Theorem 4.22 (The logarithmic part of the one-loop counterterm)**.** *The* $\log\varepsilon$ *part of* $I^{CT}[\varepsilon]$ *is*

$$\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,\log\varepsilon\,\big(Q+\{I,-\}\big)\Big(2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)\Big).\tag{4.6}$$

Equivalently, it multiplies each monomial of S by $2(n_A-n_{A^*})+4(n_\lambda-n_{\lambda^*})+(n_c-n_{c^*})$: the bracket acts as a weighted field count with weights 2 for A, 4 for λ, 1 for c, and the opposite weights for the antifields.

Proof. The functional $2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)$ is odd. By Proposition 1.11, Q applied to it equals its bracket with $\frac12\langle e,Qe\rangle$, so

$$\begin{aligned}\big(Q+\{I,-\}\big)\Big(2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)\Big)&=\Big\{S,\,2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)\Big\}\\&=\Big\{2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c),\,S\Big\}\end{aligned}$$

by the graded symmetry of Definition 1.6, S being even. By Lemma 4.21, this bracket multiplies the terms of S by

$$\begin{aligned}&-\tfrac14\int\mathrm{dvol}\,(dA,dA):\ 4, \qquad \tfrac12\int\mathrm{dvol}\,(\lambda,\not{d}\lambda):\ 8, \qquad \int\mathrm{dvol}\,(dc,A^*):\ 1-2=-1,\\ &\int\mathrm{dvol}\,\big(-\tfrac12(dA,\tfrac12[A\wedge A])\big):\ 6, \qquad \int\mathrm{dvol}\,\tfrac12(\lambda,(\not{d}_A-\not{d})\lambda):\ 8+2=10,\\ &\int\mathrm{dvol}\,([A,c],A^*):\ 2+1-2=1, \qquad \int\mathrm{dvol}\,([c,\lambda],\lambda^*):\ 1+4-4=1,\\ &\int\mathrm{dvol}\,\tfrac12([c,c],c^*):\ 2-1=1, \qquad \int\mathrm{dvol}\,\big(-\tfrac14(\tfrac12[A\wedge A],\tfrac12[A\wedge A])\big):\ 8.\end{aligned}$$

These are the computed multiples: the quadratic ones by Theorem 4.16 and Remark 4.17 (with the coefficient 0 of $\int\mathrm{dvol}\,(d^*A|_V,d^*A|_V)$, absent from S), the cubic ones $6,6,10,10,1,1,1,1$ by the totals of the eight cubic tables, and the quartic ones $8,8,8$ by the totals of the three quartic tables. In the cubic and quartic sets the computed coefficient equals the multiple of the tree vertex

as a multilinear form, including the two chirality structures of $(\lambda, \lambda, A|_{W_{\mathbb{R}}^{\perp}})$ and of (c, λ, λ^*). By Proposition 4.20 these leg sets carry the whole $\log\varepsilon$ part of $I^{CT}[\varepsilon]$, so the two local functionals coincide. □

Corollary 4.23 (The coupling counterterm vanishes at one loop)**.** *Suppose the* $\log\varepsilon$ *part of* $I^{CT}[\varepsilon]$ *is written as* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon$ *times the sum of* $\{S,-\}$ *of an element of the span of* $\int(A|_V, A^*|_V)$, $\int(A|_{W_{\mathbb{R}}^{\perp}}, A^*|_{W_{\mathbb{R}}^{\perp}})$, $\int(c^*, c)$ *and the two chiral parts of* $\int(\lambda^*, \lambda)$, *and a multiple of* S. *Then the multiple of* S *is* 0 *and the element is* $2\int(A, A^*) - 4\int(\lambda^*, \lambda) - \int(c^*, c)$. *The same holds with* S *replaced by its classical part, the terms of* S *without antifields. Since the coupling constant sits in* $(-,-)$ *and enters* S *as an overall factor, the one-loop* $\log\varepsilon$ *counterterm is a rescaling of the fields with vanishing component along the coupling.*

Proof. By Lemma 4.21, $\{S,-\}$ of an element of the span multiplies each monomial of S by a sum of field counts weighted by the coefficients of the element, additive over the factors of the monomial, and a multiple of S adds the same number to every monomial. The terms of S built from $A|_V$ alone contain 2, 3, 4 factors $A|_V$, with computed multiples 4, 6, 8: an additive weight times $2, 3, 4$ plus a common constant fits $4, 6, 8$ with weight 2 and constant $3\cdot 4 - 2\cdot 6 = 0$. With the constant 0: the kinetic terms of $A|_{W_{\mathbb{R}}^{\perp}}$ and λ give the weight 2 of $A|_{W_{\mathbb{R}}^{\perp}}$ and the sum 8 of the two chiral weights of λ; the vertex $\lambda\lambda A|_{W_{\mathbb{R}}^{\perp}}$, whose two structures have both λ of the same chirality, gives each chiral weight $(10-2)/2 = 4$; the ghost kinetic term $\int \mathrm{dvol}\,(dc, A^*)$ gives the weight $-1+2 = 1$ of c; the remaining eight multiples are then reproduced. For the classical part: by Lemma 4.21 every term of S containing an antifield has $n_c - n_{c^*} = 1$ and every classical term has $n_c = n_{c^*} = 0$, so S minus its classical part equals $\{S, -\int(c^*, c)\}$, an element of the span; replacing S by its classical part shifts the element by a multiple of $\int(c^*, c)$ and leaves the multiple unchanged. □

Remark 4.24 (The mass term has nonzero BV variation)**.** The ε^{-1} part of $I^{CT}[\varepsilon]$, the mass term $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\frac{9}{4}\varepsilon^{-1}\int\mathrm{dvol}\,(A,A)$, has nonzero variation under $Q+\{I,-\}$. Indeed $\{I, \int\mathrm{dvol}\,(A,A)\}$ pairs the A^* of $\int\mathrm{dvol}\,([A,c],A^*)$ with A and is a multiple of $\int\mathrm{dvol}\,([A,c],A) = -\int\mathrm{dvol}\,(c,[A,A]) = 0$, the bracket $[A,A]$ contracting the symmetric metric with the antisymmetric bracket of $\mathfrak{g}$. By Proposition 1.11, Q inserts $Qc = -dc$ into the slot of $A|_V$, so

$$\big(Q+\{I,-\}\big)\Big(\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\frac{9}{4}\varepsilon^{-1}\int\mathrm{dvol}\,(A,A)\Big) = -\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\frac{9}{2}\varepsilon^{-1}\int\mathrm{dvol}\,(dc, A|_V).$$

This is the breaking of the gauge symmetry by the regulator. The next section shows how the master equation at the UV boundary absorbs it and what it leaves behind.

4.4 The quantum master equation at the UV boundary

Definition 4.25 (Defect vertex)**.** For an even functional J with values in $\mathbb{C}[[\hbar]]$, at least cubic modulo $\hbar$, the defect vertex at scale ε is the odd functional

$$QJ + \tfrac{1}{2}\{J,J\}_\varepsilon + \hbar\Delta_\varepsilon J,$$

with Δ_ε and $\{-,-\}_\varepsilon$ of Definition 3.15. By Corollary 3.18, its vanishing is the quantum master equation at scale ε.

Proposition 4.26 (The Q-variation at scale ∞ is the flow of the defect)**.** *On functionals without superconformal ghosts,*

$$Q\,W(P(\varepsilon,\infty),J) = e^{-W(P(\varepsilon,\infty),J)/\hbar}\,e^{\hbar\partial_{P(\varepsilon,\infty)}}\Big(\big(QJ + \tfrac{1}{2}\{J,J\}_\varepsilon + \hbar\Delta_\varepsilon J\big)\,e^{J/\hbar}\Big). \tag{4.7}$$

Thus $Q\,W(P(\varepsilon,\infty),J)$ is the sum of the connected graphs with exactly one vertex labelled by the defect vertex, all other vertices by J, and all edges by $P(\varepsilon,\infty)$, with the weights of Proposition 3.22 [C07, §15.4].

Proof. For an even J, the product rule of Definition 3.15 gives $(Q+\hbar\Delta_\varepsilon)e^{J/\hbar} = \hbar^{-1}\big(QJ+\frac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J\big)e^{J/\hbar}$, as in the proof of Corollary 3.18. Theorem 3.17 without superconformal ghosts gives

$$Q\,e^{\hbar\partial_{P(\varepsilon,\infty)}}e^{J/\hbar} = e^{\hbar\partial_{P(\varepsilon,\infty)}}\big(Q+\hbar\Delta_\varepsilon\big)e^{J/\hbar} = \hbar^{-1}e^{\hbar\partial_{P(\varepsilon,\infty)}}\Big(\big(QJ+\tfrac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J\big)\,e^{J/\hbar}\Big).$$

The left side is $Q\,e^{W(P(\varepsilon,\infty),J)/\hbar} = \hbar^{-1}\big(Q\,W(P(\varepsilon,\infty),J)\big)e^{W(P(\varepsilon,\infty),J)/\hbar}$, Q being a derivation; multiplying by $\hbar\,e^{-W(P(\varepsilon,\infty),J)/\hbar}$ gives (4.7). Expanding $e^{\hbar\partial_{P(\varepsilon,\infty)}}$ into graphs as in Proposition 3.22, the right side of (4.7) is the sum of the graphs with exactly one vertex labelled by the defect vertex; the factor $e^{-W(P(\varepsilon,\infty),J)/\hbar}$, the exponential of the connected graphs of J, removes the components without that vertex, which leaves the connected graphs. □

With $\Delta_\infty = 0$ and $\{-,-\}_\infty = 0$, the quantum master equation at scale ∞ for the renormalised flow is $Q\lim_{\varepsilon\to0}W(P(\varepsilon,\infty),J) = 0$ (Corollary 3.18). Its failure at order $\hbar$ is the order-$\hbar$ part of $\lim_{\varepsilon\to0}Q\,W(P(\varepsilon,\infty),I-I^{CT}[\varepsilon])$. By (4.7) it is a sum of graphs with one defect vertex, among them trees of I attached to that vertex; it is the flow to scale ∞ of an effective defect vertex, the one-loop obstruction. On a compact manifold with a finite upper scale the obstruction is the local functional of [C07, Prop 12.0.3], closed and independent of ε by [C07, Lemma 12.0.4]; on $\mathbb{R}^4$ with upper scale ∞ these properties are part of Step 2. In the sector with legs $(c,A|_V)$ no tree of I contributes at order $\hbar$, and the limit computed below is itself local.

Proposition 4.27 (The defect vertex at one loop)**.** *For $J = I - I^{CT}[\varepsilon]$, with I the classical interaction* (1.3)*,*

$$QJ+\tfrac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J = \tfrac12\big(\{I,I\}_\varepsilon-\{I,I\}\big)+\hbar\Delta_\varepsilon I-Q\,I^{CT}[\varepsilon]-\{I,I^{CT}[\varepsilon]\}_\varepsilon+(\textit{terms of order }\hbar^2). \quad (4.8)$$

The first term is the bracket built from $\Delta_\varepsilon-\Delta_0 = -\partial_{K_\varepsilon-K_0}$, i.e. two vertices of I joined by one special edge carrying

$$K_\varepsilon - K_0 = -\int_0^\varepsilon (H\otimes1)K_t\,dt,$$

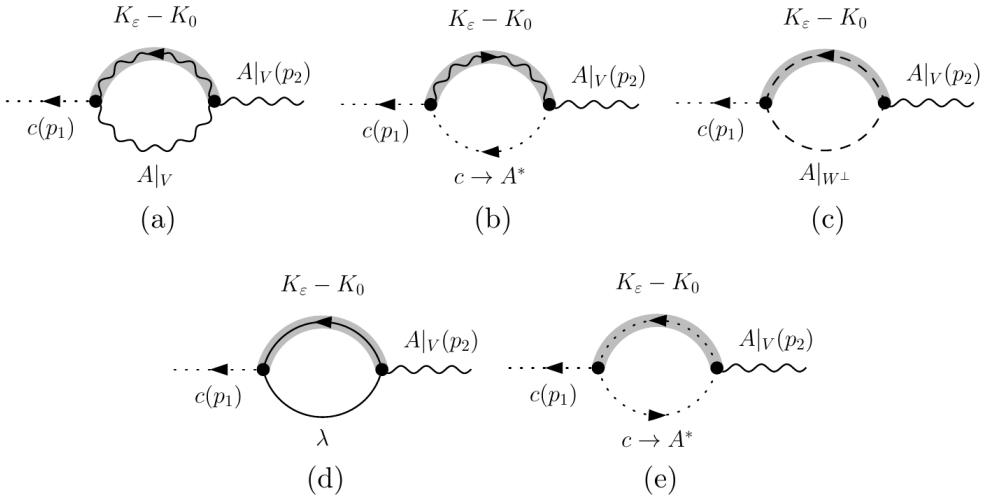
which in momentum space is $e^{-\varepsilon|p|^2}-1$ times the components of K_0; its Schwinger time lies in $[0,\varepsilon]$.

Proof. Expanding $QJ+\frac12\{J,J\}_\varepsilon+\hbar\Delta_\varepsilon J$ with $J = I - I^{CT}[\varepsilon]$ and $I^{CT}[\varepsilon]$ of order $\hbar$: at order $\hbar^0$, $QI+\frac12\{I,I\}_\varepsilon$; at order $\hbar$, $-QI^{CT}[\varepsilon]-\frac12\big(\{I,I^{CT}[\varepsilon]\}_\varepsilon+\{I^{CT}[\varepsilon],I\}_\varepsilon\big)+\hbar\Delta_\varepsilon I$, where the two brackets of even functionals coincide; the rest is $\frac12\{I^{CT}[\varepsilon],I^{CT}[\varepsilon]\}_\varepsilon-\hbar\Delta_\varepsilon I^{CT}[\varepsilon]$, of order $\hbar^2$. The classical master equation (Proposition 1.14) gives $QI = -\frac12\{I,I\}$, with $\{-,-\}$ built from Δ_0 (Definition 1.6). From $K_t = (e^{-tH}\otimes1)K_0$, $\frac{d}{dt}K_t = -(H\otimes1)K_t$, and integration from 0 to ε gives the special edge; in momentum space e^{-tH} is $e^{-t|p|^2}$. □

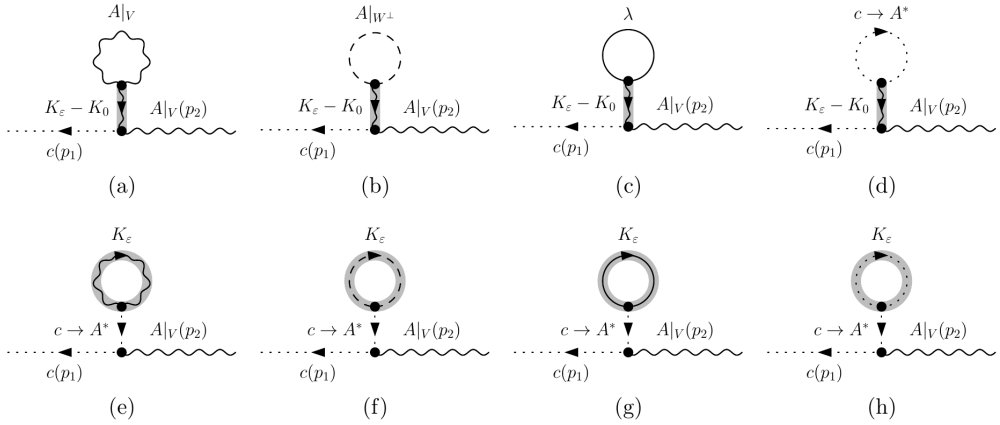
At order $\hbar$, the graphs of (4.7) with two legs are: the term $-QI^{CT}[\varepsilon]$ as a vertex; the bubbles formed by the two vertices of $\frac12(\{I,I\}_\varepsilon-\{I,I\})$, joined by the special edge and one edge $P(\varepsilon,\infty)$; and one-particle-reducible graphs in which a loop closes at a single cubic vertex. The term $\{I,I^{CT}[\varepsilon]\}_\varepsilon$ has at least three legs.

Lemma 4.28 (Two-leg sectors of degree one)**.** *An invariant local functional of degree* 1 *with two legs and weight at most* 4 *is a combination of $\int\mathrm{dvol}\,(dc,A|_V)$ (weight* 2*) and $\int\mathrm{dvol}\,(d^*dc,d^*A|_V)$ (weight* 4*). In particular the sectors with legs $(c,A|_{W_{\mathbb{R}}^\perp})$ and (c,λ) vanish.*

Proof. Degree 1 with two legs requires one c and one leg of degree 0, A or λ. The invariant pairings of a scalar with a spinor and with a vector of $W_{\mathbb{R}}^{\perp}$ vanish. With $A|_V$: one derivative in weight 2, $\int \mathrm{dvol}\,(dc, A|_V)$; three derivatives in weight 4, where $\int \mathrm{dvol}\,(dc, d^*dA|_V) = \int \mathrm{dvol}\,(d\,dc, dA|_V) = 0$ and $\int \mathrm{dvol}\,(dc, dd^*A|_V) = \int \mathrm{dvol}\,(d^*dc, d^*A|_V)$. □



The one-loop graphs of the defect vertex with external $c, A|_V$: the shaded edge is the special edge $K_\varepsilon - K_0$ of $\frac{1}{2}(\{I,I\}_\varepsilon - \{I,I\})$, the other edge carries $P(\varepsilon,\infty)$. Special edge between the slots (a) $A|_V$, $A^*|_V$ with an $A|_V$ line; (b) $A|_V$, $A^*|_V$ with a c–A^* line; (c) $A|_{W_{\mathbb{R}}^\perp}$, $A^*|_{W_{\mathbb{R}}^\perp}$ with an $A|_{W_{\mathbb{R}}^\perp}$ line; (d) λ^*, λ with a λ line; (e) c^*, c with a c–A^* line.



The one-particle-reducible graphs of the defect vertex with external $c, A|_V$. Row 1: from $\frac{1}{2}(\{I,I\}_\varepsilon - \{I,I\})$, with an edge $P(\varepsilon,\infty)$ closing at one cubic vertex. Row 2: from $\hbar\Delta_\varepsilon I$, a special edge K_ε closing at one cubic vertex. Each closed loop at a cubic vertex carries the colour factor $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X) = 0$.

Lemma 4.29 (Graphs with one special edge)**.** *The plane-wave value of each bubble of the first figure is ε^{-1} times a multiple of $\int \mathrm{dvol}\,(dc, A|_V)$ plus a multiple of $\int \mathrm{dvol}\,(d^*dc, d^*A|_V)$ plus a term bounded by a multiple of ε, with vanishing coefficient of $\log\varepsilon$.*

Proof. The special edge carries $-\int_0^\varepsilon (\cdots) e^{-t_1|p_1|^2} dt_1$ and the other edge $\int_\varepsilon^\infty (\cdots) e^{-t_2|p_1-p|^2} dt_2$. After the Gaussian integral over p_1 (Lemma 4.4, extended to the polynomial numerators of degree three of these graphs by the same moments), the integrand is $(t_1+t_2)^{-2} e^{-t_1 t_2 |p|^2/(t_1+t_2)}$ times a sum of terms

$(t_1+t_2)^{-m}$, $m \geq 0$, with polynomial coefficients in $\frac{t_1}{t_1+t_2}$ and p. On $[0,\varepsilon] \times [\varepsilon,\infty)$ the exponent satisfies $0 \leq \frac{t_1 t_2 |p|^2}{t_1+t_2} \leq t_1|p|^2 \leq \varepsilon|p|^2$, so its exponential is a power series in $\varepsilon|p|^2$ with uniform remainders. Substituting εt_1, εt_2 for t_1, t_2 maps the region to $[0,1] \times [1,\infty)$, on which $(t_1+t_2)^{-2-m}$ is integrable, and each term becomes an integer power of ε times a convergent integral. Hence the value is a series in integer powers of ε; by Lemma 4.28 the coefficients of ε^{-1} and ε^0 are the two listed functionals. □

graph	special edge $K_\varepsilon - K_0$	edge $P(\varepsilon,\infty)$	coefficient of $\varepsilon^{-1}\int \mathrm{dvol}\,(dc, A\|_V)$	coefficient of $\int \mathrm{dvol}\,(d^*dc, d^*A\|_V)$
(a)	$A\|_V$–$A^*\|_V$	$A\|_V$	$-\frac{9}{4}$	$-\frac{5}{8}$
(b)	$A\|_V$–$A^*\|_V$	c–A^*	$\frac{1}{8}$	$-\frac{1}{12}$
(c)	$A\|_{W_{\mathbb{R}}^\perp}$–$A^*\|_{W_{\mathbb{R}}^\perp}$	$A\|_{W_{\mathbb{R}}^\perp}$	$-\frac{9}{2}$	$-\frac{5}{4}$
(d)	λ^*–λ	λ	2	$-\frac{4}{3}$
(e)	c^*–c	c–A^*	$\frac{1}{8}$	$-\frac{1}{12}$
$-QI^{CT}[\varepsilon]$			$\frac{9}{2}$	0
total			0	$-\frac{27}{8}$

The one-loop part of $Q\,W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ with legs $(c, A|_V)$, graph by graph, in units of $\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(16\pi^2(X,Y))$. Each bubble has weight 1: the $\frac{1}{2}$ of $\frac{1}{2}\{I,I\}$ cancels against the two assignments of the two vertices. Every entry has vanishing $\log\varepsilon$ coefficient.

Theorem 4.30 (The one-loop obstruction in the sector $(c, A|_V)$)**.** *The part of $Q\,W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ of order $\hbar$ with legs $(c, A|_V)$ has the following ε-expansion: the special-edge bubbles contribute $-\frac{9}{2}\,\varepsilon^{-1}$ times $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(dc, A|_V)$, the vertex $-QI^{CT}[\varepsilon]$ contributes $+\frac{9}{2}\,\varepsilon^{-1}$ times the same functional, the coefficient of $\log\varepsilon$ is 0, and the limit is*

$$-\frac{27}{8}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*dc, d^*A|_V) = \frac{27}{16}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,Q\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V). \tag{4.9}$$

The sectors $(c, A|_{W_{\mathbb{R}}^\perp})$ and (c,λ) of the one-loop obstruction vanish.

Proof. By Proposition 4.26 and (4.8), the order-$\hbar$ part with two legs is the sum of the graphs listed before Lemma 4.28. The one-particle-reducible graphs vanish: a loop closing at one cubic vertex contracts two legs of a vertex of the form $([X,Y],Z)$ with $(-,-)^{-1}$ and gives $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X) = 0$ or the contraction of the antisymmetric bracket with the symmetric $(-,-)^{-1}$, both zero. The same argument gives $\Delta_\varepsilon I = 0$: K_ε pairs a field with its antifield, the vertices of I containing an antifield are $([A,c],A^*)$, $([c,\lambda],\lambda^*)$, $\frac{1}{2}([c,c],c^*)$, and each self-contraction is a trace of ad. The five bubbles are given in the table; their ε^{-1} coefficients sum to $-\frac{9}{4}+\frac{1}{8}-\frac{9}{2}+2+\frac{1}{8} = -\frac{9}{2}$, and by Lemma 4.29 they have vanishing $\log\varepsilon$ coefficient. By Theorem 4.16 and Remark 4.24, $-QI^{CT}[\varepsilon]$ restricted to $(c, A|_V)$ is $-\frac{9}{4}\cdot(-2) = \frac{9}{2}$ times $\varepsilon^{-1}\int \mathrm{dvol}\,(dc, A|_V)$ in the unit; its $\log\varepsilon$ part restricted to $(c, A|_V)$ vanishes, because $Q\int \mathrm{dvol}\,(dA, dA)$ contains $d\,dc = 0$, $Q\int \mathrm{dvol}\,(dc, A^*)$ restricted to $(c, A|_V)$ is $-\frac{1}{2}\int \mathrm{dvol}\,(dc, d^*dA|_V) = -\frac{1}{2}\int \mathrm{dvol}\,(d\,dc, dA|_V) = 0$, and the coefficient of $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$ is 0. The finite parts sum to $-\frac{5}{8}-\frac{1}{12}-\frac{5}{4}-\frac{4}{3}-\frac{1}{12} = -\frac{81}{24} = -\frac{27}{8}$. Finally $Q\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V) = 2\int \mathrm{dvol}\,(d^*A|_V, d^*(-dc)) = -2\int \mathrm{dvol}\,(d^*dc, d^*A|_V)$, so $-\frac{27}{8} = \frac{27}{16}\cdot(-2)$. The last statement is Lemma 4.28. □

Remark 4.31 (The obstruction as the longitudinal part of the two-point function)**.** Q preserves the polynomial degree of functionals, so the part of $Q\,W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$ with legs $(c, A|_V)$ at order $\hbar$ is Q applied to the order-$\hbar$ quadratic part of $W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon])$: the renormalised two-point function, whose limit as $\varepsilon \to 0$ at nonzero momentum exists by Theorem 4.16 and Proposition 4.8. Q annihilates its (c, A^*) part on $(c, A|_V)$, since ι_p of $|p|^2 A|_V - p\,\iota_p A|_V$ is 0, and its transverse $A|_V$ part, which depends on $A|_V$ through $dA|_V$. Hence the cancellation of ε^{-1} and the vanishing of $\log\varepsilon$ in Theorem 4.30 follow

from Theorem 4.16, and the obstruction is Q of the longitudinal part of the renormalised $A|_V$ two-point function at scale ∞. A longitudinal quadratic functional of $A|_V$ is $\int \mathrm{dvol}\,(d^*A|_V, G\,d^*A|_V)$ for a translation-invariant operator G on functions, and Q sends it to $-2\int \mathrm{dvol}\,(d^*dc, G\,d^*A|_V)$; comparing with (4.9) shows that G is the constant $\frac{27}{16}$ in the unit, so the longitudinal part is the local functional

$$\frac{27}{16}\,\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V).$$

Equivalently, Q applied to the longitudinal parts of the graphs of the $A|_V$ two-point function, with every edge cut off at ε, sums to $-\frac{9}{2}\varepsilon^{-1}$ and $-\frac{27}{8}$ times the functionals of (4.9); there the gauge bubble and the ghost loop carry separate $\log(\varepsilon|p|^2)$ and $\log 2$ terms in their longitudinal parts, which cancel in their sum. Theorem 4.30 is the evaluation of the same quantity through the UV-boundary identity, on graphs with one special edge confined to $[0,\varepsilon]$.

Theorem 4.32 (The finite counterterm in the sector $(c, A|_V)$)**.** *Replace the renormalised interaction $I - I^{CT}[\varepsilon]$ by*

$$I - \frac{27}{16}\,\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V) - I^{CT}[\varepsilon]. \tag{4.10}$$

Then $I^{CT}[\varepsilon]$ at one loop is unchanged, the order-$\hbar$ part of the flow has a limit at generic leg momenta, and the part with legs $(c, A|_V)$ of the one-loop obstruction vanishes. The added term is local, even, of degree 0 and weight 4, independent of ε, and invariant under translations, $\mathrm{Spin}(4,\mathbb{C})\times\mathrm{Spin}(6,\mathbb{C})$*, the adjoint action of $\mathfrak{g}$ and the dilatation. Among the quadratic functionals of $A|_V$ of weight 4 and degree 0, it is unique up to adding multiples of $\int \mathrm{dvol}\,(dA|_V, dA|_V)$, which is Q-closed.*

Proof. The added term is of order $\hbar$, so at order $\hbar$ it enters the flow as a tree vertex, with a limit as $\varepsilon\to 0$, and leaves the one-loop singular parts unchanged. In (4.8) it adds $-Q$ of itself at order $\hbar$, and $-\{I,\cdot\}_\varepsilon$ of itself, which has at least three legs. By Theorem 4.30 and $Q\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V) = -2\int \mathrm{dvol}\,(d^*dc, d^*A|_V)$, the $(c, A|_V)$ part becomes $-\frac{27}{8}+\frac{27}{16}\cdot 2 = 0$ in the unit. Invariance under the dilatation holds because the weight is 4. For uniqueness: by Lemma 4.19, the quadratic functionals of $A|_V$ of weight 4 are spanned by $\int \mathrm{dvol}\,(dA|_V, dA|_V)$ and $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$; the first is Q-closed, since $d\,dc = 0$, and the second has Q-variation $-2\int \mathrm{dvol}\,(d^*dc, d^*A|_V) \neq 0$, so the coefficient $\frac{27}{16}$ of the second is forced. □

Remark 4.33 (Renormalisation conditions belong to the definition of the theory)**.** $I^{CT}[\varepsilon]$ is purely singular, and unique among the translation-invariant and $\mathfrak{g}$-invariant local functionals of Proposition 4.10 (the one-loop analogue of [C07, Thm B], in the scope of Remark 4.9); the finite term of (4.10) is a finite change of the interaction, $I \mapsto I - \frac{27}{16}\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$, of the kind that lifts a theory across a Q-exact obstruction in [C07, Lemma 12.0.4]. The order-$\hbar$ part of the regulated flow with $I - I^{CT}[\varepsilon]$ has a finite limit at generic leg momenta (Proposition 4.8) and fails the master equation at scale ∞ by a finite local amount; the master equation forces the coefficient $\frac{27}{16}$ and leaves the Q-closed finite counterterms, such as multiples of $\int \mathrm{dvol}\,(dA, dA)$, free. This choice is a renormalisation condition, and it is part of the definition of the quantum theory: every observable $O[\infty]$ is built from the flow of the interaction, and changes with it. In this program it is the first instance of that statement; the same mechanism recurs for the equivariant counterterms of the superconformal ghosts.

Remark 4.34 (The cubic sector of the obstruction)**.** The one-loop obstruction also has components with three legs: $(c, A|_V, A|_V)$, $(c, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, (c,λ,λ) and $(c, c, A^*|_V)$. They are sums of triangles and quartic-vertex bubbles with one special edge, of the vertex $-\{I, I^{CT}[\varepsilon]\}_\varepsilon$, and of one-particle-reducible graphs in which $-QI^{CT}[\varepsilon]$ is joined to I by an edge $P(\varepsilon,\infty)$. The finite term of (4.10) changes them by $-\{I,\cdot\}$ of itself, a nonzero functional built from $\int \mathrm{dvol}\,([A,c], A^*)$. The full one-loop master equation, i.e. the vanishing of the obstruction in all sectors after a finite counterterm, is the subject of Step 2.

Proposition 4.35 (The finite counterterm under special conformal transformations)**.** *Let v be the special conformal field $x \mapsto (x,x)v_1 - 2(v_1,x)x$, $v_1 \in V$ (Definition 2.1), acting on $A|_V$ by the Lie derivative L_v (Definition 2.12). The derivative of $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$ along $A|_V \mapsto L_vA|_V$ is*

$$2\int \mathrm{dvol}\,(d^*A|_V, d^*L_vA|_V) = -8\int \mathrm{dvol}\,(d^*A|_V, \iota_{v_1}A|_V) \neq 0,$$

while for translations, rotations and the dilatation $2E$ the derivative vanishes.

Proof. For a conformal Killing field v of V, the symmetric part of the derivative of $(v,-)$ is $\frac{1}{4}\operatorname{div} v$ times the metric; taking one more divergence, d^*d applied to the components of v is $\frac{1}{2}\operatorname{grad}(\operatorname{div} v)$. The Lie derivative of the one-form $A|_V$ is $v(A|_V) + A|_V \circ$ (derivative of v), and the divergence of this expression, with the two identities above, is

$$d^*(L_vA|_V) = v(d^*A|_V) + \tfrac{1}{2}(\operatorname{div} v)\, d^*A|_V + \tfrac{1}{2}\,\iota_{\operatorname{grad}(\operatorname{div} v)}A|_V.$$

Hence

$$\begin{aligned}2\int \mathrm{dvol}\,(d^*A|_V, d^*L_vA|_V) = &\int \mathrm{dvol}\, v\big((d^*A|_V, d^*A|_V)\big) + \int \mathrm{dvol}\,(\operatorname{div} v)(d^*A|_V, d^*A|_V)\\ &+ \int \mathrm{dvol}\,(d^*A|_V, \iota_{\operatorname{grad}(\operatorname{div} v)}A|_V),\end{aligned}$$

and integration by parts cancels the first two terms, since $\int \mathrm{dvol}\, v(\cdot) = -\int \mathrm{dvol}\,(\operatorname{div} v)(\cdot)$. For the special conformal field, $\operatorname{div} v = -8(v_1,x)$ (Chapter 2) and $\operatorname{grad}(\operatorname{div} v) = -8v_1$, which gives the stated value. For translations and rotations $\operatorname{div} v = 0$, and for $2E$, $\operatorname{div} v = 8$ is constant. □

The finite counterterm of Theorem 4.32 is the Feynman-gauge longitudinal term, and Proposition 4.35 shows that it breaks the special conformal symmetry. In Step 2 the components of the equivariant counterterms along the special conformal generators have to absorb this variation, or exhibit it as part of an obstruction.

Remark 4.36 (Method of computation)**.** Each per-graph coefficient of this chapter is an exact rational number. It is obtained in an explicit realisation of V_{10}, Σ and $\mathfrak{g}$ over $\mathbb{Q}(i)$, with the loop momentum integrated as a Gaussian (Lemma 4.4) and the ε-expansion taken on the graphs with one pinned edge (Proposition 4.2), in exact rational arithmetic; for the two-point graphs the polarisations and momenta are symbolic. For the cubic and quartic sets each multilinear form is determined by exact linear algebra on a basis of the $\mathrm{Spin}(4,\mathbb{C}) \times \mathrm{Spin}(6,\mathbb{C})$-invariant multilinear forms of the given legs, including the parity-odd ones, from evaluations at integer polarisations and momenta drawn uniformly from a finite set of integers. A nonzero polynomial vanishes at such a point with probability at most its total degree divided by the size of the set (Schwartz–Zippel), so the sampled system has full rank with high probability; full rank is then established exactly, and the solution is exact. The colour factors are reduced exactly for a general simple $\mathfrak{g}$, and the resulting rational multiples of $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)/(X,Y)$ are the same in the explicit realisations $\mathfrak{g} = \mathfrak{su}(2)$ and $\mathfrak{g} = \mathfrak{su}(3)$. The sector $(c, A|_V)$ of the obstruction is also given by Q of the renormalised two-point function (Remark 4.31).

Chapter 5

Local operators: from functionals to local $O(\varepsilon)$

The insertion of a composite operator into a one-loop graph diverges as $\varepsilon \to 0$, and Costello's construction asks for a family $O(\varepsilon)$ whose flow to scale ∞ has a limit. At scale ∞, however, $K_\infty = 0$ and $P(0,\infty)$ is a contracting homotopy for Q on moment-free Schwartz sections (Chapter 3), so a general functional family $O(\varepsilon)$ produces the zero class on moment-free Schwartz sections. The content of an operator is therefore the requirement that $O(\varepsilon)$ be local, $O(\varepsilon) = O - \partial_\delta I^{CT}[\varepsilon]$ with coefficients in $\mathrm{span}\{\varepsilon^{-1}, \log\varepsilon\}$: locality and the existence of the limit fix the singular part, and the finite part is the scheme. This chapter sets up $O(\varepsilon)$ and $O[\infty]$, proves $O(\varepsilon) = O$ for single letters at one loop, and computes $O(\varepsilon)$ for the scalar bilinears graph by graph.

5.1 Observables as first-order deformations

Definition 5.1 (Background field and the family $I+\delta O$)**.** A background field is a compactly supported four-form $\omega \in \Omega^4_c(\mathbb{R}^4)$ [CG2, Lemma 10.1.1.3]. A local operator is a local functional

$$O = \int \omega\,(\text{density}),$$

linear in ω, whose density is a polynomial in the fields and their derivatives at a point; the Weyl weight w_O of O is the sum of the Weyl weights of the fields in the density (c: 0, A: 1, λ: $\frac{3}{2}$, A^*: 3, λ^*: $\frac{5}{2}$, c^*: 4) plus the number of derivatives. For O of cohomological degree i, δ is a square-zero parameter of degree $-i$ [CG2, Def 8.3.1.1], and $I + \delta O$ is a family of interactions over $\mathbb{C}[\delta]/\delta^2$; by [CG2, after Def 8.3.1.3] it satisfies the axioms of a family of theories over $\mathbb{C}[\delta]/\delta^2$ except locality of the flowed functionals. The scalar bilinears and the letters $\int \omega\,(X, \iota_w A)$, $\int \omega\,(X, \iota_v(A|_V))$, $\int \omega\,(\zeta, \lambda)$ have $i = 0$; the ghost letter $\int \omega\,(X, c)$ has $i = 1$, and then δ has degree -1.

The square-zero parameter δ is written without argument; the delta distribution is always written with its argument, as in $\delta(x - y)$.

Definition 5.2 (The local family and the observable at scale ∞)**.** Let $I^{CT}[\varepsilon]$ be the one-loop counterterm of I (Definition 4.1; the analogue of [C07, Thm B], Remark 4.9), and write the counterterm of the family $I + \delta O$ as $I^{CT}[\varepsilon] + \delta\,\partial_\delta I^{CT}[\varepsilon]$. Put

$$O(\varepsilon) := O - \partial_\delta I^{CT}[\varepsilon], \qquad O[\infty] := \lim_{\varepsilon\to 0} \partial_\delta W\big(P(\varepsilon,\infty),\, I - I^{CT}[\varepsilon] + \delta\, O(\varepsilon)\big), \tag{5.1}$$

with W the flow of Definition 3.16 [CG2, §7.2.7]. With the global ghosts of $\mathfrak{psl}(4|4)$, I is replaced by the equivariant interaction $I_{\mathfrak{psl}(4|4)} = I + S^{(1)} + S^{(2)}$ [ESW, Rem 1.29] (Definition 2.10) and $I^{CT}[\varepsilon]$ by

the counterterm of the family $I_{\mathfrak{psl}(4|4)} + \delta O$ over $C^\bullet(\mathfrak{psl}(4|4))$:

$$O(\varepsilon) := O - \partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon], \qquad O[\infty] := \lim_{\varepsilon\to 0} \partial_\delta W\big(P(\varepsilon,\infty),\, I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] + \delta\, O(\varepsilon)\big).$$

Both are cochains on $\mathfrak{psl}(4|4)$; their components of arity 0 are the functionals (5.1). This chapter computes the components of arity 0; the components with arguments in $\mathfrak{psl}(4|4)$ involve the counterterms of $I_{\mathfrak{psl}(4|4)}$ along the ghosts; Chapter 6 treats the dilatation through the scaling covariance, and the ghost components are part of Step 2 (Chapter 7).

At scale ∞ the differential on observables is $(d_{CE} + Q)O[\infty]$, since the scale-∞ structure is defined by the kernel $K_\infty := 0$, so that $\Delta_\infty = 0$ and $\{-,-\}_\infty = 0$ (Definition 3.15) [CG2, Def 8.3.1.3, Prop 13.4.0.1].

[C07, Thm B] is stated for interactions that are at least cubic modulo $\hbar$, and δO is quadratic for the operators below. The construction of [C07, Thm B] is applied graph by graph:

Proposition 5.3 (Counterterms of the family)**.** *Let O be a homogeneous local operator. At order $\hbar$ there is a unique local functional $\partial_\delta I^{CT}[\varepsilon]$ with coefficients in the span of the negative integer powers of ε and of $\log\varepsilon$, a subspace of $\mathscr{A}_{<0}$ [C07, Def 1.4.1; CG2, §13.7.2], such that the limit* (5.1) *exists at generic leg momenta. It is the singular part of the sum of the one-loop graphs in which the vertex O lies on the cycle, each evaluated with $P(\varepsilon,\infty)$; for $w_O \le 3$ its coefficients lie in* $\operatorname{span}\{\varepsilon^{-1}, \log\varepsilon\}$.

Proof. By Proposition 3.22, the order-$\hbar$ part of $\partial_\delta W(P(\varepsilon,\infty), I - I^{CT}[\varepsilon] + \delta O)$ is a sum over connected graphs with exactly one vertex O (since $\delta^2 = 0$) and either one cycle and vertices I, or no cycle and one vertex $-I^{CT}[\varepsilon]$. Every vertex of I is at least trivalent and O occurs once, so for a fixed number of legs the sum is finite. Three classes occur.

- O lies off the cycle. Removing the edges that join O to the rest leaves a one-loop graph without O; the corresponding tree graph with the vertex $-I^{CT}[\varepsilon]$ in its place cancels its singular part by Lemma 4.7, with O as the marked vertex, and Definition 4.1, and the trees joining O converge to trees with $P(0,\infty)$, because $P(\varepsilon,\infty) - P(0,\infty) = -\int_0^\varepsilon L_t\, dt$ is bounded by a multiple of ε at fixed nonzero momentum.
- O lies on the cycle. The vertex function of O is a polynomial in the momenta times the Fourier transform of ω at minus the total momentum of its legs (Lemma 5.8), and the numerator is homogeneous, O being homogeneous, so by Lemma 4.6 the singular part of such a graph is a local functional with coefficients in the span of the negative powers of ε and of $\log\varepsilon$, and the tree with the vertex $-\delta\,\partial_\delta I^{CT}[\varepsilon]$ in place of the cycle removes it (Lemma 4.7, with O as the marked vertex). Its superficial degree is w_O minus the sum of the Weyl weights of the legs (proof of Proposition 5.13), at most w_O, so for $w_O \le 3$ the powers ε^{-k}, $k \ge 2$, are absent.
- Trees with the vertex $-I^{CT}[\varepsilon]$ and with O: already paired in the first class.

Uniqueness: two such counterterms differ by a local functional with coefficients in $\mathscr{A}_{<0}$ whose tree-level flow has a limit. The part of that flow with as many legs as the component of the difference with the fewest legs is that component itself, which therefore has a limit and vanishes; by induction on the number of legs, as in Proposition 4.10, the difference vanishes. □

Proposition 5.4 (Local representatives)**.** *Let O be a homogeneous local operator.*

- *Two local families $O(\varepsilon)$, $O'(\varepsilon)$ for which the limit* (5.1) *exists have the same singular part.*
- *Adding to $O(\varepsilon)$ an ε-independent local functional of order $\hbar$ adds to $O[\infty]$, at order $\hbar$, the flow of that functional to scale ∞ through trees with vertices I.*

- *Let F be a continuous polynomial functional of moment-free Schwartz sections, without constant term, with $QF = 0$. Then F is Q-exact (Corollary 3.20). On compactly supported sections the Q-closed functionals carry in addition nonzero classes, among them the zero-mode classes of Theorem 3.19 and Remark 3.21, finitely many in each polynomial degree and infinitely many in all.*

Consequently a statement about $O[\infty]$ has content when it is written in the form

$$(d_{CE} + Q)\, O[\infty] = (a \text{ local functional})[\infty],$$

and, for an operator whose observable is a cocycle (Remark 5.6), an anomalous dimension is read off in the cohomology of local functionals.

Proof. The first item is the uniqueness part of Proposition 5.3. The second follows from linearity of (5.1) in the δ-component: at order $\hbar$ the added functional enters through its tree-level flow. The third item is Corollary 3.20, which rests on the homotopy identity $Q\,P(0,\infty)\star + P(0,\infty)\star\, Q = \mathrm{id}$ of Theorem 3.19 on moment-free Schwartz sections; the classes on compactly supported sections are those described in Theorem 3.19 and Remark 3.21. Every observable $O[\infty]$ that extends to a continuous functional on moment-free Schwartz sections, with $(d_{CE}+Q)O[\infty] = 0$ there, therefore lies in the image of Q on such functionals; what it records about O is the local functional $O(\varepsilon)$, through the first two items. □

Remark 5.5 (Where the content of an operator lies)**.** By Proposition 5.4, the singular part of $O(\varepsilon)$ is uniquely fixed and the finite part is the scheme freedom; this is the statement of Remark 3.21 for operators. In the rest of this chapter $\partial_\delta I^{CT}[\varepsilon]$ is computed; its $\log\varepsilon$ part is the input of the anomalous dimensions in Chapter 6.

Remark 5.6 (The cocycle condition)**.** Reading the dilatation component of Chapter 6 as the anomalous dimension of a cohomology class requires $O[\infty]$ to be a cocycle at arity zero,

$$Q\,O[\infty] = 0 \qquad \text{at order } \hbar,$$

the part linear in δ of the quantum master equation of the family $I + \delta O$ at scale ∞. For O with $(Q + \{I, -\})O = 0$, Proposition 4.26 applied to $J = I - I^{CT}[\varepsilon] + \delta O(\varepsilon)$ expresses $Q\,O[\infty]$ at order $\hbar$ as the limit of the graphs with one vertex labelled by the part of the defect vertex linear in δ: the special edge $K_\varepsilon - K_0$ between I and O, from $\{I, O\}_\varepsilon - \{I, O\}$; the vertices $-Q\,\partial_\delta I^{CT}[\varepsilon] - \{I, \partial_\delta I^{CT}[\varepsilon]\}_\varepsilon$ and $-\{I^{CT}[\varepsilon], O\}_\varepsilon$; the self-contraction $\hbar\Delta_\varepsilon O$, zero for operators without antifields; and the trees joining O to the action defect of Proposition 4.27, whose cubic sector is open (Remark 4.34). For the scalar bilinears the local output of Weyl weight 2, ghost degree 1 and colour singlet includes functionals of the type $\int \omega\,(dc, A|_V)$. A nonzero result of this kind is removed by a finite local counterterm of Weyl weight w_O added to $O(\varepsilon)$, for instance a multiple of $\int \omega\,(A|_V, A|_V)$, since

$$Q \int \omega\,(A|_V, A|_V) = -2 \int \omega\,(dc, A|_V).$$

Such a counterterm is independent of ε and of order $\hbar$; by Proposition 5.4 it adds to $O[\infty]$ the flow of a local functional of the same Weyl weight, and it leaves the $\log\varepsilon$ coefficients of this chapter and the dilatation coefficients of Chapter 6 unchanged. The evaluation of these graphs is part of Step 2 (Chapter 7).

5.2 Tools for one-loop insertions

The gauge algebra $\mathfrak{g}$ is simple. The inverse pairing $(-,-)^{-1} \in \mathfrak{g} \otimes \mathfrak{g}$ is symmetric, and a bilinear form on $\mathfrak{g}$ can be evaluated on it.

Lemma 5.7 (Colour contractions)**.** *For $X_1, X_2 \in \mathfrak{g}$:*

- *the bracket $[-,-]$ evaluated on $(-,-)^{-1}$ is 0;*
- *for every linear map of $\mathfrak{g}$, the bilinear form $(Z, Z') \mapsto (Z,$ the map applied to $Z')$ evaluated on $(-,-)^{-1}$ is the trace $\mathrm{tr}_{\mathfrak{g}}$ of the map; in particular $(-,-)$ evaluated on $(-,-)^{-1}$ is $\dim \mathfrak{g}$, and $(X_1, [-,-])$ and $(-, [X_1,-])$ evaluated on $(-,-)^{-1}$ are 0;*
- *the bilinear form $([X_1,-],[X_2,-])$ evaluated on $(-,-)^{-1}$ is $-\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})$;*
- *for all $X, Y \in \mathfrak{g}$ with $(X,Y) \neq 0$, the ratio $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)/(X,Y)$ is the same number, and $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}) = \frac{\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{(X,Y)}\,(X_1, X_2)$.*

Proof. The bracket is antisymmetric and $(-,-)^{-1}$ is symmetric. The inverse pairing is characterised by the property that contracting its second factor with $(Z,-)$ returns Z; hence the bilinear form of a linear map, evaluated on $(-,-)^{-1}$, is the trace of the map. By invariance, $(X_1, [Z, Z']) = ([X_1, Z], Z') = -(Z, [X_1, Z'])$, which is the case of the map $-\mathrm{ad}_{X_1}$, of trace 0 because ad_{X_1} is antisymmetric for $(-,-)$, and $(-, [X_1,-])$ is the case of ad_{X_1}; and $([X_1,Z],[X_2,Z']) = -(Z,[X_1,[X_2,Z']])$ is the case of $-\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}$. For simple $\mathfrak{g}$ every invariant symmetric bilinear form is a multiple of $(-,-)$, which gives the last item. □

In the following, the number

$$\frac{\hbar\ \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2\,(X,Y)}$$

is the unit of every one-loop coefficient; it is negative when $(-,-)$ is positive definite on the compact real form.

Lemma 5.8 (Vertex functions on decomposable legs)**.** *Give each leg of a vertex a fibre element $(v,-)\otimes Z$ with $v \in V_{10}$ and $Z \in \mathfrak{g}$; by linearity this determines every vertex function. The quartic vertex function (1.5) with legs $(v_j,-)\otimes Z_j$, $j = 1,\dots,4$, equals*

$$\begin{aligned} -\frac{1}{2}\Big[&\big((v_1,v_3)(v_2,v_4) - (v_1,v_4)(v_2,v_3)\big)\big([Z_1,Z_2],[Z_3,Z_4]\big) \\ &+ \big((v_1,v_2)(v_3,v_4) - (v_1,v_4)(v_3,v_2)\big)\big([Z_1,Z_3],[Z_2,Z_4]\big) \\ &+ \big((v_1,v_2)(v_4,v_3) - (v_1,v_3)(v_4,v_2)\big)\big([Z_1,Z_4],[Z_2,Z_3]\big)\Big], \end{aligned} \tag{5.2}$$

the three terms belonging to the pairings $(12|34)$, $(13|24)$, $(14|23)$ of the legs. The cubic vertex function (1.4) with two legs $(w,-)\otimes Z$, $(w',-)\otimes Z'$, $w, w' \in W^{\perp}_{\mathbb{R}}$, of incoming momenta p, p', and one leg $(v,-)\otimes Z''$, $v \in V$, equals

$$-\frac{i}{2}\,(v, p'-p)\,(w,w')\,\big(Z'', [Z,Z']\big); \tag{5.3}$$

The vertex function of $\int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$ on legs $(v,-)\otimes Z$, $(v',-)\otimes Z'$ is

$$\big[(w_1,v)(w_2,v') + (w_1,v')(w_2,v)\big](Z,Z'),$$

times the Fourier transform of ω at minus the total momentum of the legs.

Proof. With $A_j = (v_j, -) \otimes Z_j$, the two-form $[A_1 \wedge A_2]$ of Chapter 1 is $(v_1, -) \wedge (v_2, -) \otimes [Z_1, Z_2]$, because $[A_1 \wedge A_2](e, e') = (v_1, e)(v_2, e')[Z_1, Z_2] - (v_1, e')(v_2, e)[Z_1, Z_2]$, and likewise for every pair of legs. Polarising the normalisation of the Hodge pairing in Definition 1.8 gives

$$\big((v_1, -) \wedge (v_2, -), (v_3, -) \wedge (v_4, -)\big) = (v_1, v_3)(v_2, v_4) - (v_1, v_4)(v_2, v_3),$$

and likewise for the other two pairings, which gives (5.2). In (1.4), $\iota_p A(p_j) = (v_j, p) Z_j$ and $[A(p_i), A(p_j)] = (v_i, v_j)[Z_i, Z_j]$. With legs $1, 2$ along $W_{\mathbb{R}}^{\perp}$ and leg 3 along V, the momenta lie in $V \perp W_{\mathbb{R}}^{\perp}$, so $(v_1, p) = (v_2, p) = 0$ and $(v_2, v_3) = (v_3, v_1) = 0$; the second and third terms of (1.4) vanish and the first is $-\frac{i}{2}(v_3, p_2 - p_1)(v_1, v_2)(Z_3, [Z_1, Z_2])$, which is (5.3). The last item is the multilinear part of $(\iota_{w_1} A, \iota_{w_2} A)$ on two plane waves (Definition 1.13). □

Lemma 5.9 (Insertion on a cycle)**.** *Expand around a background consisting of two plane waves with fibre elements $A(p_1)$, $A(p_2)$, as in Remark 3.24. The part of the order-$\hbar$ term of $\partial_\delta W(P(\varepsilon, \infty), I + \delta O)$ that is bilinear in $A(p_1)$, $A(p_2)$ and comes from cycles through the vertex O whose edges all lie in the A slot is $\hbar$ times the sum of*

- $\frac{1}{2}$ *times each bubble: the cycle through O and one quartic vertex carrying both external legs;*
- 1 *times each triangle: the cycle through O, a cubic vertex carrying $A(p_1)$ and a cubic vertex carrying $A(p_2)$, traversed in one orientation.*

In each cycle the internal legs are contracted along the edges with the $A \otimes A$ component of $P(\varepsilon, \infty)$ (Proposition 3.11): an edge of momentum p replaces the fibre elements $(v, -) \otimes Z$, $(v', -) \otimes Z'$ at its two ends by 2 times the inverse metric in $v \otimes v'$ and $(-, -)^{-1}$ in $Z \otimes Z'$, times $\int_\varepsilon^\infty e^{-t|p|^2} dt = e^{-\varepsilon|p|^2} |p|^{-2}$, and the loop momentum p is integrated with $\int d^4p/(2\pi)^4$.

Proof. On the A slots the parity operator and the supertrace of Proposition 3.23 are trivial, and the one-loop operator is $P(\varepsilon, \infty)$ composed with the second derivative of $I + \delta O$ in the fluctuation. The δ-derivative of $\frac{1}{2n}\mathrm{STr}$ of its n-th power is $\frac{1}{2}\mathrm{STr}$ of the product of $n-1$ factors built from I and one factor built from O. The second derivative of O is its vertex function with both legs internal, since O is quadratic. For $n = 2$ the factor built from I is bilinear in the two external legs through the quartic vertex: by Definition 1.13 the part of the quartic term of I quadratic in the fluctuation and bilinear in $A(p_1)$, $A(p_2)$ is one half of (1.5) with two internal legs, whose second derivative is (1.5) with two legs open; this gives the weight $\frac{1}{2}$. For $n = 3$ the two factors built from I each carry one external leg through a cubic vertex; the two orders of these factors give equal traces, since all second derivatives and $P(\varepsilon, \infty)$ are symmetric and the trace is invariant under transposition and cyclic permutation, so $\frac{1}{2}$ times their sum is one triangle with weight 1. □

The singular parts are read off from the following lemma, which also locates them on the lower end $t = \varepsilon$ of the Schwinger integrals (Remark 3.13).

Lemma 5.10 (Logarithmic coefficient from the UV boundary)**.** *Let γ be a one-loop graph with edge set $E(\gamma)$ [C07, §15.5], $|E(\gamma)| \geq 2$, whose value is*

$$F_\gamma(\varepsilon) = \int_{[\varepsilon, \infty)^{E(\gamma)}} \prod_{e \in E(\gamma)} dt_e \int \frac{d^4p}{(2\pi)^4} \big(\textit{numerator of } \gamma\big)(p) \exp\Big(- \sum_{e \in E(\gamma)} t_e |p + p_e|^2 \Big) \qquad [\textit{C07, §15.4}],$$

with generic $p_e \in V$ and a numerator that is a polynomial in p and the p_e, homogeneous of degree $2|E(\gamma)| - 4$; let $\big(\textit{numerator of } \gamma\big)_{\mathrm{top}}$ be its part of degree $2|E(\gamma)| - 4$ in p, and write

$$\textit{(average)} = \frac{1}{2\pi^2} \int_{|p|=1} \big(\textit{numerator of } \gamma\big)_{\mathrm{top}}(p)\, d\mathrm{area}(p)$$

for its average over the unit sphere of V, of area $2\pi^2$. Then

$$\begin{aligned} \varepsilon\frac{dF_\gamma}{d\varepsilon} &= -\sum_{e\in E(\gamma)} \big(\varepsilon\cdot \textit{the value of } \gamma \textit{ with } t_e=\varepsilon \textit{ fixed}\big), \\ \lim_{\varepsilon\to 0}\varepsilon\frac{dF_\gamma}{d\varepsilon} &= -\frac{\textit{(average)}}{16\pi^2}, \end{aligned} \tag{5.4}$$

and each of the $|E(\gamma)|$ terms of the sum tends to $\frac{1}{|E(\gamma)|}\cdot\frac{\textit{(average)}}{16\pi^2}$. The coefficient of $\log\varepsilon$ in $F_\gamma(\varepsilon)$ is $-\frac{\textit{(average)}}{16\pi^2}$, and each edge carries the fraction $\frac{1}{|E(\gamma)|}$ of it.

Proof. The first identity is Proposition 4.2. Fix an edge e. As in the proof of Lemma 4.6, with $\tau=\varepsilon+\sum_{e'\neq e}t_{e'}$, the Gaussian integral over p writes the value of γ with $t_e=\varepsilon$ fixed as

$$\sum_{j=0}^{|E(\gamma)|-2}\int_{[\varepsilon,\infty)^{E(\gamma)\setminus e}}\frac{(\text{numerator})_j}{16\pi^2\,\tau^{2+j}}\exp\Big(-\tau\cdot\frac12\sum_{e',e''}\frac{t_{e'}}{\tau}\frac{t_{e''}}{\tau}|p_{e'}-p_{e''}|^2\Big)\prod_{e'\neq e}dt_{e'},$$

the exponent being ≥ 0. The term $j=|E(\gamma)|-2$ contracts all loop momenta of the top-degree part, so in polar coordinates

$$\begin{aligned}\int\frac{d^4p}{(2\pi)^4}\big(\text{numerator of }\gamma\big)_{\text{top}}(p)\,e^{-\tau|p|^2} &= (\text{average})\cdot\frac{2\pi^2}{(2\pi)^4}\int_0^\infty|p|^{2|E(\gamma)|-1}e^{-\tau|p|^2}\,d|p| \\ &= \frac{(|E(\gamma)|-1)!\,(\text{average})}{16\pi^2\,\tau^{|E(\gamma)|}},\end{aligned}$$

i.e. $(\text{numerator})_{|E(\gamma)|-2}=(|E(\gamma)|-1)!\,(\text{average})$, a constant. Substituting $t_{e'}\mapsto\varepsilon t_{e'}$ for $e'\neq e$, ε times this term becomes an integral over $[1,\infty)^{E(\gamma)\setminus e}$ of $\frac{(|E(\gamma)|-1)!\,(\text{average})}{16\pi^2}\big(1+\sum_{e'\neq e}t_{e'}\big)^{-|E(\gamma)|}$ times an exponential of ε times a nonnegative function, which is at most 1 and tends to 1. The function $\big(1+\sum_{e'\neq e}t_{e'}\big)^{-|E(\gamma)|}$ is integrable on $[1,\infty)^{|E(\gamma)|-1}$, so by dominated convergence the limit is

$$\begin{aligned}\frac{(|E(\gamma)|-1)!\,(\text{average})}{16\pi^2}\int_{[1,\infty)^{E(\gamma)\setminus e}}\Big(1+\sum_{e'\neq e}t_{e'}\Big)^{-|E(\gamma)|}\prod_{e'\neq e}dt_{e'} &= \frac{(|E(\gamma)|-1)!\,(\text{average})}{16\pi^2\,|E(\gamma)|!} \\ &= \frac{1}{|E(\gamma)|}\cdot\frac{(\text{average})}{16\pi^2}.\end{aligned}$$

The terms $j\leq|E(\gamma)|-3$, of lower degree in p, are bounded separately. Their integrands are bounded by a multiple of τ^{-2-j} times the exponential; on $\tau\geq1$ the integral is bounded as in the proof of Lemma 4.6, and on $\tau\leq1$ it is bounded by a multiple of $\int_0^1(\varepsilon+\rho)^{|E(\gamma)|-4-j}\,d\rho$, with $|E(\gamma)|-4-j\geq-1$, hence by a multiple of $1+|\log\varepsilon|$. Multiplied by ε these terms tend to 0. This proves the limit of each of the $|E(\gamma)|$ terms, and summing over the edges gives the limit in (5.4). Finally, by Lemma 4.6 with superficial degree 0, $F_\gamma(\varepsilon)$ is a multiple of $\log\varepsilon$ plus a function g with a limit and with $\varepsilon\frac{dg}{d\varepsilon}\to0$; hence the multiple is $\lim\varepsilon\,dF_\gamma/d\varepsilon$. □

For $|E(\gamma)|=2$ and numerator 1 the lemma gives the coefficient $-\frac{1}{16\pi^2}$ of $\log\varepsilon$ in

$$\int\frac{d^4p}{(2\pi)^4}\frac{e^{-\varepsilon|p|^2}}{|p|^2}\frac{e^{-\varepsilon|p+p_1|^2}}{|p+p_1|^2};$$

in position space this is the singular part $-\frac{\log\varepsilon}{16\pi^2}\,\delta(x-y)$ of the square of $\frac{1-e^{-|x-y|^2/4\varepsilon}}{4\pi^2|x-y|^2}$.

5.3 Single letters

Proposition 5.11 (Letters)**.** *Let $X \in \mathfrak{g}$, $w \in W_{\mathbb{R}}^{\perp}$, $v \in V$ and $\zeta \in \Sigma^* \otimes \mathfrak{g}$. For each of the letters*

$$\int \omega\,(X, \iota_w A), \qquad \int \omega\,(X, \iota_v(A|_V)), \qquad \int \omega\,(\zeta, \lambda), \qquad \int \omega\,(X, c),$$

one has $\partial_\delta I^{CT}[\varepsilon] = 0$ at one loop, hence

$$O(\varepsilon) = O. \tag{5.5}$$

Proof. The vertex O is univalent, and a cycle passes through every one of its vertices along two edges; hence O lies off the cycle in every one-loop graph, and by Proposition 5.3 $\partial_\delta I^{CT}[\varepsilon]$, the singular part of the graphs with O on the cycle, is 0. The graphs with O off the cycle consist of a one-loop graph with the vertices of I, joined to O by a tree. If the one-loop graph has one external leg, it is a cubic vertex of I with two of its legs joined by an edge; every cubic vertex of I has the form $(Z_1, [Z_2, Z_3])$ in its colour factors (Chapter 1), and joining two of its legs evaluates $(Z_1, [-,-])$ or $(-, [Z_2, -])$ on $(-,-)^{-1}$, which is 0 by Lemma 5.7. If the one-loop graph has more external legs, its singular part is cancelled by the tree with the vertex $-I^{CT}[\varepsilon]$ (first class in the proof of Proposition 5.3). □

5.4 Scalar bilinears

Definition 5.12 (Scalar bilinears and the symmetric square of $W_{\mathbb{R}}^{\perp}$)**.** For $w_1, w_2 \in W_{\mathbb{R}}^{\perp}$,

$$O = \int \omega\,(\iota_{w_1} A, \iota_{w_2} A),$$

a gauge-invariant local operator of Weyl weight $w_O = 2$, symmetric and bilinear in (w_1, w_2), hence linear in $w_1 w_2 \in \mathrm{Sym}^2 W_{\mathbb{R}}^{\perp}$. Under $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, with $\mathrm{Spin}(6;\mathbb{C}) \cong SL(4,\mathbb{C})$ [ESW, §10.3], the symmetric square of the six-dimensional space $W_{\mathbb{R}}^{\perp}$ is the sum of the traceless symmetric tensors, of dimension 20, and the line spanned by the inverse metric, of dimension 1. Correspondingly

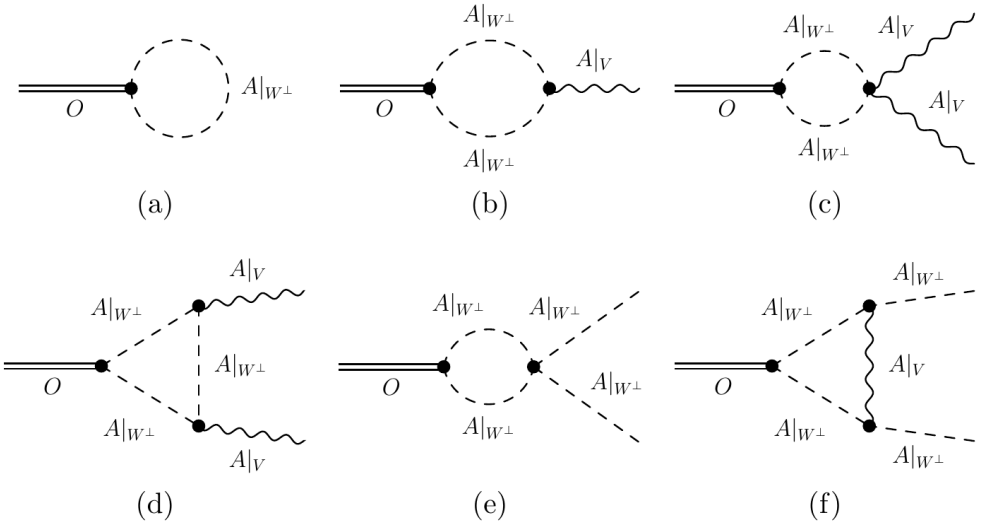
$$\begin{aligned} \int \omega\,(\iota_{w_1} A, \iota_{w_2} A) = \Big[& \int \omega\,(\iota_{w_1} A, \iota_{w_2} A) \\ & - \tfrac{1}{6}(w_1, w_2) \int \omega\,(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}}) \Big] \\ & + \tfrac{1}{6}(w_1, w_2) \int \omega\,(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}}), \end{aligned} \tag{5.6}$$

where $(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}})$ contracts the $W_{\mathbb{R}}^{\perp}$ factors with the metric and the $\mathfrak{g}$ factors with $(-,-)$. The bracket is the traceless part: replacing $w_1 w_2$ by the inverse metric of $W_{\mathbb{R}}^{\perp}$ turns (w_1, w_2) into 6 and the bracket into $(1 - \frac{1}{6} \cdot 6) \int \omega(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}}) = 0$. The last term is the trace part, a multiple of $\int \omega\,(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}})$.

Proposition 5.13 (The graphs)**.** *The one-loop graphs with the vertex O of Definition 5.12 on the cycle and superficial degree ≥ 0 are the six graphs of the figure:*

- *(a) the vacuum graph: one $A|_{W_{\mathbb{R}}^{\perp}}$ edge from O to itself, degree 2;*
- *(b) one external $A|_V$: a cycle of two $A|_{W_{\mathbb{R}}^{\perp}}$ edges through a cubic vertex, degree 1;*

- *(c), (d) two external $A|_V$: the bubble with the quartic vertex of type $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$, and the triangle with two cubic vertices of type $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$ and three $A|_{W_{\mathbb{R}}^\perp}$ edges, degree 0;*
- *(e), (f) two external $A|_{W_{\mathbb{R}}^\perp}$: the bubble with the quartic vertex of type $(A|_{W_{\mathbb{R}}^\perp})^4$, and the triangle with two cubic vertices of type $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$, edges $A|_{W_{\mathbb{R}}^\perp}$, $A|_V$, $A|_{W_{\mathbb{R}}^\perp}$, degree 0.*



The one-loop graphs (a)–(f) with one vertex $O = \int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$ (double line) on the cycle and superficial degree ≥ 0; dashed lines are $A|_{W_{\mathbb{R}}^\perp}$, wavy lines $A|_V$, every edge carries $P(\varepsilon, \infty)$.

Proof. All vertices of I have Weyl weight 4 (fields plus derivatives), so a one-loop graph with one vertex O has superficial degree w_O minus the sum of the Weyl weights of its external legs, i.e. 2 minus that sum. Its external legs have total cohomological degree 0; an external c (weight 0) needs an antifield (weight $\geq \frac{5}{2}$) to compensate its degree, and two λ have weight 3. A single external λ, of weight $\frac{3}{2}$, is excluded: every vertex of I and of O contains an even number of factors λ, λ^*, and every edge carrying the $\lambda \otimes \lambda$ component of $P(\varepsilon, \infty)$ joins two of them, so the number of external legs λ, λ^* is even; equivalently, the element -1 of $\mathrm{Spin}(4, \mathbb{C})$, which acts by -1 on $S_\pm$ and trivially on V, preserves the action and the regulator and changes the sign of every functional with an odd number of spinor legs. The external legs of degree ≥ 0 graphs are therefore none, one A, or two A. The element -1 of $SO(W_{\mathbb{R}}^\perp)$ lies in the image of $\mathrm{Spin}(6; \mathbb{C})$ and is a symmetry of the action and of the regulator (H and Q^{GF} commute with $\mathrm{Spin}(6; \mathbb{C})$); it fixes $w_1 w_2$ and changes the sign of every functional with an odd number of $A|_{W_{\mathbb{R}}^\perp}$ legs, so the singular parts with one $A|_{W_{\mathbb{R}}^\perp}$ leg, or with legs $A|_V$, $A|_{W_{\mathbb{R}}^\perp}$, vanish. Both edges at O are $A|_{W_{\mathbb{R}}^\perp}$ edges. A cycle that enters a Yukawa vertex along $A|_{W_{\mathbb{R}}^\perp}$ continues along λ and needs an external λ; a cycle that enters the vertex $([A, c], A^*)$ along $A|_{W_{\mathbb{R}}^\perp}$ leaves it along c or along $A^*|_{W_{\mathbb{R}}^\perp}$; $P(\varepsilon, \infty)$ has no component in the slot of $A^*|_{W_{\mathbb{R}}^\perp}$ (Proposition 3.11), and if the cycle leaves along c, through the $c \otimes A^*|_V$ component, the leg $A^*|_{W_{\mathbb{R}}^\perp}$ of that vertex is external, of weight 3, which is excluded. Hence the cycle passes through cubic vertices of type $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$ and quartic vertices, with at most two external legs: one cubic vertex gives (b), one quartic vertex gives (c) or (e), and two cubic vertices give (d) or (f), according to whether the edge between them is $A|_{W_{\mathbb{R}}^\perp}$ or $A|_V$. □

Proposition 5.14 (Singular parts graph by graph)**.** *The singular parts of the graphs (a)–(f) are:*

graph	$\varepsilon^{-1}\int\omega$	$\int\omega(\iota_{w_1}A,\iota_{w_2}A)$	$(w_1,w_2)\int\omega(A\|_{W_{\mathbb{R}}^\perp},A\|_{W_{\mathbb{R}}^\perp})$	$(w_1,w_2)\int\omega(A\|_V,A\|_V)$
(a)	$\frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2}$	0	0	0
(b)	0	0	0	0
(c)	0	0	0	-2
(d)	0	0	0	2
(e)	0	2	-2	0
(f)	0	2	0	0
total	$\frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2}$	4	-2	0

Singular parts of the graphs of the figure. The first column is the coefficient of $\varepsilon^{-1}\int\omega$; the other entries are coefficients of $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon$ times the functional at the head of the column.

Proof. External fibre elements are taken decomposable, which suffices by linearity; the external momenta are p_1,p_2 and ω absorbs $-p_1-p_2$. By Lemma 5.10, the $\log\varepsilon$ coefficient of a graph of degree 0 depends on the top-degree part of its numerator in the loop momentum p, so in numerators the external momenta are set to 0. The local functionals of the table have the following vertex functions (Lemma 5.8), the first two on legs $(w_3,-)\otimes X_1$, $(w_4,-)\otimes X_2$ and the third on legs $(v_3,-)\otimes X_1$, $(v_4,-)\otimes X_2$, with $w_3,w_4\in W_{\mathbb{R}}^\perp$, $v_3,v_4\in V$, $X_1,X_2\in\mathfrak{g}$:

$$\begin{aligned}\int\omega\,(\iota_{w_1}A,\iota_{w_2}A):&\quad\big[(w_1,w_3)(w_2,w_4)+(w_1,w_4)(w_2,w_3)\big](X_1,X_2),\\ (w_1,w_2)\int\omega\,(A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp}):&\quad 2(w_1,w_2)(w_3,w_4)(X_1,X_2),\\ (w_1,w_2)\int\omega\,(A|_V,A|_V):&\quad 2(w_1,w_2)(v_3,v_4)(X_1,X_2).\end{aligned}$$

The graphs are computed in the examples below; (a) and (b) are as follows.
Graph (a) is the term $n=1$ of Proposition 3.23, $\hbar\,\partial_{P(\varepsilon,\infty)}O$. The $A\otimes A$ component of $P(\varepsilon,\infty)$ at coincident points is $2(-,-)^{-1}\otimes$ (inverse metric of V_{10}) $\cdot\,\frac{1}{16\pi^2\varepsilon}$ (Proposition 3.11); contracting $\iota_{w_1}\otimes\iota_{w_2}$ with the inverse metric gives (w_1,w_2), and $(-,-)$ on $(-,-)^{-1}$ gives $\dim\mathfrak{g}$ (Lemma 5.7):

$$\hbar\,\partial_{P(\varepsilon,\infty)}O=\frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2\,\varepsilon}\int\omega.$$

Graph (b): by (5.3) the cubic vertex with external leg $(v_3,-)\otimes X_1$ and internal legs $(w,-)\otimes Z$, $(w',-)\otimes Z'$ is $-\frac{i}{2}(v_3,p'-p)(w,w')(X_1,[Z,Z'])$. Both edges end at O, whose vertex function is symmetric in its two legs, so the colour factor is $(X_1,[-,-])$ evaluated on $(-,-)^{-1}$, which is 0 by Lemma 5.7. □

Example 5.15 (Graph (e): the bubble with external $A|_{W_{\mathbb{R}}^\perp}$)**.** Take external legs $(w_3,-)\otimes X_1$, $(w_4,-)\otimes X_2$ and internal legs $(w_1,-)\otimes Z_1$, $(w_2,-)\otimes Z_2$ at the quartic vertex. The three pairings of (5.2), with legs $1,2$ external and $3,4$ internal, are

$$\begin{aligned}(12|34):&\quad\big[(w_3,w_1)(w_4,w_2)-(w_3,w_2)(w_4,w_1)\big]\big([X_1,X_2],[Z_1,Z_2]\big),\\ (13|24):&\quad\big[(w_3,w_4)(w_1,w_2)-(w_3,w_2)(w_1,w_4)\big]\big([X_1,Z_1],[X_2,Z_2]\big),\\ (14|23):&\quad\big[(w_3,w_4)(w_2,w_1)-(w_3,w_1)(w_2,w_4)\big]\big([X_1,Z_2],[X_2,Z_1]\big).\end{aligned}$$

The two edges join O to the internal legs. By Lemma 5.9 each edge contributes the factor 2, the inverse metric and $(-,-)^{-1}$; contracting the vertex function of O through the two edges, the internal legs of the quartic vertex receive $(w_1,-)\otimes(w_2,-)+(w_2,-)\otimes(w_1,-)$ tensored with $(-,-)^{-1}$ in

$Z_1 \otimes Z_2$, and the quartic vertex function is symmetric in legs 3, 4. By Lemma 5.7, $([X_1, X_2], [Z_1, Z_2])$ gives 0 and the other two colour factors give $-\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})$, so the quartic vertex evaluated on $(w_1, -) \otimes (w_2, -) \otimes (-, -)^{-1}$ is

$$-\frac{1}{2}\Big(-\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})\Big)\Big[2(w_1, w_2)(w_3, w_4) - (w_1, w_4)(w_2, w_3) - (w_1, w_3)(w_2, w_4)\Big].$$

With the bubble weight $\frac{1}{2}$ and the edge factor 2^2, and the factor 2 from the two terms of the vertex function of O,

$$\begin{aligned}\text{(e)} = \frac{\hbar}{2} \cdot 2^2 \cdot 2 \cdot \frac{1}{2} \operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})\Big[2(w_1, w_2)(w_3, w_4) - (w_1, w_3)(w_2, w_4) - (w_1, w_4)(w_2, w_3)\Big] \\ \cdot \int \frac{d^4p}{(2\pi)^4} \frac{e^{-\varepsilon|p|^2}}{|p|^2} \frac{e^{-\varepsilon|p+p_1+p_2|^2}}{|p+p_1+p_2|^2}.\end{aligned}$$

Lemma 5.10 with two edges and numerator 1 replaces the integral by $-\frac{1}{16\pi^2}\log\varepsilon$ in the singular part, and Lemma 5.7 replaces $\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})$ by $\frac{\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{(X,Y)}(X_1, X_2)$:

$$\begin{aligned}\text{singular part of (e)} &= \frac{\hbar \operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X, Y)} \log\varepsilon \\ &\quad \cdot \Big[2\big((w_1, w_3)(w_2, w_4) + (w_1, w_4)(w_2, w_3)\big) - 4(w_1, w_2)(w_3, w_4)\Big](X_1, X_2) \\ &= \frac{\hbar \operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X, Y)} \log\varepsilon \Big[2\int \omega\left(\iota_{w_1}A, \iota_{w_2}A\right) - 2(w_1, w_2)\int \omega\left(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp}\right)\Big],\end{aligned}$$

the second line read off from the vertex functions listed in the proof of Proposition 5.14.

Example 5.16 (Graph (f): the triangle with external $A|_{W_{\mathbb{R}}^\perp}$)**.** Route the loop momentum p from O to the vertex carrying $(w_3, -) \otimes X_1$, then along the $A|_V$ edge to the vertex carrying $(w_4, -) \otimes X_2$, then back to O. By (5.3), with the external momenta set to 0 in the numerator, the two cubic vertices are

$$\begin{aligned}\text{first vertex:} \quad &-\tfrac{i}{2}\,(v, 0-p)\,(w, w_3)\,\big(Z'', [Z, X_1]\big) = \tfrac{i}{2}\,(v, p)\,(w, w_3)\,\big(Z'', [Z, X_1]\big), \\ \text{second vertex:} \quad &-\tfrac{i}{2}\,(v', 0+p)\,(w', w_4)\,\big(Z''', [Z', X_2]\big),\end{aligned}$$

where $(w, -) \otimes Z$ and $(w', -) \otimes Z'$ are the internal $A|_{W_{\mathbb{R}}^\perp}$ legs (incoming momenta p and $-p$) and $(v, -) \otimes Z''$, $(v', -) \otimes Z'''$ the two ends of the $A|_V$ edge. The edges give:

- the $A|_V$ edge: 2, $(v, p)(v', p) \mapsto |p|^2$, and $(Z'', [Z, X_1])(Z''', [Z', X_2]) \mapsto ([Z, X_1], [Z', X_2])$;
- the two $A|_{W_{\mathbb{R}}^\perp}$ edges at O: the factor 2^2; the metric contractions of $(w, w_3)(w', w_4)$ with the vertex function of O, which give $(w_1, w_3)(w_2, w_4) + (w_1, w_4)(w_2, w_3)$; and $(-, -)^{-1}$ in $Z \otimes Z'$, so that $([Z, X_1], [Z', X_2])$ becomes $-\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})$ by Lemma 5.7.

With the triangle weight 1 and $\frac{i}{2} \cdot \big(-\frac{i}{2}\big) = \frac{1}{4}$, the numerator of the graph is

$$\begin{aligned}\big(\text{numerator of (f)}\big)(p) = 2^3 \cdot \frac{1}{4}\,|p|^2 \cdot \Big(-\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})\Big)\Big[(w_1, w_3)(w_2, w_4) + (w_1, w_4)(w_2, w_3)\Big] \\ + (\text{lower degree in } p),\end{aligned}$$

so its average in the sense of Lemma 5.10 is $-2\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})[(w_1,w_3)(w_2,w_4)+(w_1,w_4)(w_2,w_3)]$, and the lemma with three edges gives

$$\begin{aligned}\text{singular part of (f)} &= \hbar\cdot\frac{2\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})}{16\pi^2}\big[(w_1,w_3)(w_2,w_4)+(w_1,w_4)(w_2,w_3)\big]\log\varepsilon\\ &= 2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\int\omega\,(\iota_{w_1}A,\iota_{w_2}A).\end{aligned}$$

Example 5.17 (Graphs (c) and (d): the cancellation of $(A|_V,A|_V)$)**.** Take external legs $(v_3,-)\otimes X_1$, $(v_4,-)\otimes X_2$ with $v_3,v_4\in V$.
Graph (c). In (5.2) with legs $1,2$ external and $3,4$ internal along w_1,w_2, the pairing (12|34) carries $(v_3,w_1)(v_4,w_2)-(v_3,w_2)(v_4,w_1)=0$, since $V\perp W_{\mathbb{R}}^{\perp}$, and the pairings (13|24), (14|23) carry $(v_3,v_4)(w_1,w_2)$. By Lemma 5.7 the quartic vertex evaluated on $(w_1,-)\otimes(w_2,-)\otimes(-,-)^{-1}$ is $-\frac12\cdot2\cdot(-\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}))(v_3,v_4)(w_1,w_2)$, and

$$\text{(c)}=\frac{\hbar}{2}\cdot2^2\cdot2\cdot\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})(v_3,v_4)(w_1,w_2)\int\frac{d^4p}{(2\pi)^4}\,\frac{e^{-\varepsilon|p|^2}}{|p|^2}\,\frac{e^{-\varepsilon|p+p_1+p_2|^2}}{|p+p_1+p_2|^2},$$

$$\begin{aligned}\text{singular part of (c)} &= -\frac{4\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})}{16\pi^2}(v_3,v_4)(w_1,w_2)\log\varepsilon\\ &= -2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\,(w_1,w_2)\int\omega\,(A|_V,A|_V).\end{aligned}$$

Graph (d). Route p from O to the vertex carrying $(v_3,-)\otimes X_1$, to the vertex carrying $(v_4,-)\otimes X_2$, and back to O, all three edges along $W_{\mathbb{R}}^{\perp}$. In (5.3) the external leg is the V-leg and the internal legs have incoming momenta p and $-p$:

$$-\tfrac{i}{2}\,(v_3,-p-p)\,(w,w')\,\big(X_1,[Z,Z']\big)=i\,(v_3,p)\,(w,w')\,\big(X_1,[Z,Z']\big),$$

and likewise $i\,(v_4,p)(w'',w''')(X_2,[Z'',Z'''])$ at the second vertex. The metric contractions along the cycle give (w,w''') from the middle edge and then $(w_1,w_2)+(w_2,w_1)=2(w_1,w_2)$ at O. The colour factors, contracted along the middle edge, give $(X_2,[[X_1,Z],Z'''])=([X_2,[X_1,Z]],Z''')$, and the two edges at O contract $Z\otimes Z'''$ with $(-,-)^{-1}$, which gives $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_2}\mathrm{ad}_{X_1})$ by Lemma 5.7. With the edge factor 2^3, the average $\frac14(v_3,v_4)$ of $(v_3,p)(v_4,p)/|p|^2$ over the unit sphere of V and $i\cdot i=-1$, the average of the top-degree part of the numerator is

$$2^3\cdot(-1)\cdot\tfrac14(v_3,v_4)\cdot2(w_1,w_2)\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})=-4\,(v_3,v_4)(w_1,w_2)\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}),$$

$$\begin{aligned}\text{singular part of (d)} &= +\frac{4\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})}{16\pi^2}(v_3,v_4)(w_1,w_2)\log\varepsilon\\ &= 2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\,(w_1,w_2)\int\omega\,(A|_V,A|_V).\end{aligned}$$

The two singular parts cancel, and the coefficient of $(w_1,w_2)\int\omega\,(A|_V,A|_V)$ in $\partial_\delta I^{CT}[\varepsilon]$ is 0.

Theorem 5.18 ($O(\varepsilon)$ for the scalar bilinears)**.** *For $O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$, at one loop,*

$$\begin{aligned}\partial_\delta I^{CT}[\varepsilon] = \frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\Big[4\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)-2(w_1,w_2)\int\omega\,(A|_{W_{\mathbb{R}}^{\perp}},A|_{W_{\mathbb{R}}^{\perp}})\Big]\\ +\frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2\,\varepsilon}\int\omega.\end{aligned}\tag{5.7}$$

For the traceless part (the bracket in (5.6)*, and every traceless combination),*

$$O(\varepsilon) = \Big(1 - 4\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X,Y)}\,\log\varepsilon\Big)\,O, \tag{5.8}$$

and for the trace part $O = \int \omega\,(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}})$*,*

$$O(\varepsilon) = \Big(1 + 8\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X,Y)}\,\log\varepsilon\Big)\,O - \frac{12\,\hbar\,\dim\mathfrak{g}}{16\pi^2\,\varepsilon}\int\omega. \tag{5.9}$$

Proof. By Proposition 5.3, $\partial_\delta I^{CT}[\varepsilon]$ is the sum of the singular parts of the graphs (a)–(f), which is the last row of the table of Proposition 5.14; every other set of external legs gives 0 (proof of Proposition 5.13). This is (5.7). The map $w_1 w_2 \mapsto \partial_\delta I^{CT}[\varepsilon]$ is linear on $\mathrm{Sym}^2 W_{\mathbb{R}}^{\perp}$. On a traceless combination the terms with the factor (w_1, w_2) add up to zero, leaving 4 times the operator; for the bracket of (5.6) explicitly,

$$\begin{aligned}&\Big[4\int\omega\,(\iota_{w_1}A, \iota_{w_2}A) - 2(w_1,w_2)\,(\mathrm{trace})\Big] - \tfrac{1}{6}(w_1,w_2)\Big[4\,(\mathrm{trace}) - 12\,(\mathrm{trace})\Big]\\ &\qquad = 4\Big[\int\omega\,(\iota_{w_1}A, \iota_{w_2}A) - \tfrac{1}{6}(w_1,w_2)\,(\mathrm{trace})\Big],\end{aligned}$$

where (trace) stands for $\int\omega\,(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}})$ and the vacuum terms cancel in the same way: $2(w_1,w_2) - \frac{1}{6}(w_1,w_2)\cdot 12 = 0$. For the trace, replace $w_1 w_2$ by the inverse metric of $W_{\mathbb{R}}^{\perp}$: (w_1,w_2) becomes $\dim W_{\mathbb{R}}^{\perp} = 6$ and $\int\omega(\iota_{w_1}A, \iota_{w_2}A)$ becomes (trace), so the bracket of (5.7) becomes $4\,(\mathrm{trace}) - 12\,(\mathrm{trace}) = -8\,(\mathrm{trace})$ and the vacuum term becomes $\frac{12\hbar\dim\mathfrak{g}}{16\pi^2\varepsilon}\int\omega$. Then $O(\varepsilon) = O - \partial_\delta I^{CT}[\varepsilon]$ gives (5.8) and (5.9). □

Remark 5.19 (Method of computation)**.** The entries of the table of Proposition 5.14 are exact rational multiples of $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)/(X,Y)$. In explicit realisations of V_{10}, of the Clifford module and of $\mathfrak{g} = \mathfrak{su}(2)$ and $\mathfrak{g} = \mathfrak{su}(3)$, the Schwinger integrals of each graph are integrated in closed form in exact rational arithmetic and the singular part is read off from the exact ε-expansion; the multiples are the same for both gauge algebras, and the colour factors are reduced for a general simple $\mathfrak{g}$ by Lemma 5.7. The graphs (a)–(f) are all one-loop graphs with one vertex O on the cycle and superficial degree ≥ 0 (Proposition 5.13).

Chapter 6

The dilatation component and anomalous dimensions

The dilatation is the simplest superconformal component of the BV transformation of an operator. The classical theory is invariant under the rescaling of the fields by their Weyl weights, and the regulator breaks this invariance through ε alone: the rescaling maps $P(\varepsilon,\infty)$ to $P(s^2\varepsilon,\infty)$. Consequently the scale dependence of $O[\infty]$ is carried by the $\log\varepsilon$ parts of $I^{CT}[\varepsilon]$ and of $\partial_\delta I^{CT}[\varepsilon]$, and by Theorem 4.22 (the logarithmic part of the one-loop counterterm) the part coming from the action is a field redefinition. This chapter proves these statements, computes the resulting coefficients for the scalar bilinears and the single letters, shows how the boundary graphs with an L_ε edge produce them, and states the identification with the equivariant dilatation cochain as the task of Step 2, together with the graphs that Step 2 must evaluate.

6.1 Scaling covariance of the regulator

Definition 6.1 (Weyl scaling)**.** The Weyl weights of the slots are

slot	c	A	λ	A^*	λ^*	c^*
Weyl weight	0	1	$\frac{3}{2}$	3	$\frac{5}{2}$	4

Weyl weights; a field and its antifield have weights adding up to 4. They are one half of the weights $0,2,3$ of c, A, λ read off from $S^{(1)}(\Delta)$ in Example 2.13.

For a real number $s>0$ and a section e in a slot,

$$(\sigma_s e)(x) = s^{-(\text{Weyl weight of the slot})}\, e(s^{-1}x),$$

extended to $\mathscr{E}\otimes\mathscr{E}$ as $\sigma_s\otimes\sigma_s$ and to functionals by $(\sigma_s F)(e)=F(\sigma_s^{-1}e)$. For a local operator $O=\int\omega\,(\text{density})$ of Weyl weight w_O (Definition 5.1) we write

$$(E+w_O)\,O=\int\omega\,(E+w_O)(\text{density}),$$

where E is the Euler vector field of [SW, (3)] acting on the position dependence of the density.

Lemma 6.2 (Scaling of the data)**.**

- $\langle\sigma_s e_0,\sigma_s e_1\rangle=\langle e_0,e_1\rangle$ *and* $(\sigma_s\otimes\sigma_s)K_t=K_{s^2t}$.
- $\sigma_s Q=Q\sigma_s$, $\sigma_s Q^{GF}=s^2Q^{GF}\sigma_s$ *and* $\sigma_s H=s^2H\sigma_s$*; on functionals* σ_s *commutes with* Q *and with the bracket.*
- $\sigma_s S=S$ *and* $\sigma_s I=I$*. A local functional* $\int\mathrm{dvol}\,(\textit{density})$ *whose density has Weyl weight* k *(fields plus derivatives) satisfies* $\sigma_s\int\mathrm{dvol}\,(\textit{density})=s^{k-4}\int\mathrm{dvol}\,(\textit{density})$*; for example* $\sigma_s\int\mathrm{dvol}\,(A,A)=s^{-2}\int\mathrm{dvol}\,(A,A)$.

- *For a local operator O, $\sigma_s O$ is O with ω replaced by s^{w_O} times the pullback of ω along $x \mapsto s^{-1}x$, and $s\frac{d}{ds}\big|_{s=1}\sigma_s O = (E + w_O)\,O$.*

Proof. For the pairing substitute $x = sy$, $\mathrm{dvol}(x) = s^4\mathrm{dvol}(y)$; the two weights add up to 4. The scalar factor of K_t (Definition 3.8) becomes $s^{-4}(4\pi t)^{-2}e^{-|x-y|^2/4s^2t} = (4\pi s^2 t)^{-2}e^{-|x-y|^2/4s^2t}$. An operator with j derivatives from a slot of Weyl weight k to a slot of Weyl weight k' is multiplied by $s^{k+j-k'}$ under conjugation by σ_s. For the arrows of Q (Proposition 1.11) the exponent is $0+1-1$, $1+2-3$, $3+1-4$, $\frac{3}{2}+1-\frac{5}{2}$, all 0; for the arrows of Q^{GF} (Definition 3.1) it is $3+0-1$, $1+1-0$, $4+1-3$, $\frac{5}{2}+1-\frac{3}{2}$, all 2; hence also for $H = [Q, Q^{GF}]$. Every term of (1.1) has a density of Weyl weight 4: for example 1+1+2 for (dA, dA), $\frac{3}{2}+\frac{3}{2}+1$ for $(\lambda, \not{d}\lambda)$, 0+1+3 for (dc, A^*) and $([A,c], A^*)$, $0+\frac{3}{2}+\frac{5}{2}$, $0+0+4$, and 4 for the cubic and quartic gauge terms. For a density of Weyl weight k, $(\text{density})(\sigma_s^{-1}e)(x) = s^k(\text{density})(e)(sx)$, and $\int \mathrm{dvol}(x)\, s^k(\text{density})(sx) = s^{k-4}\int \mathrm{dvol}\,(\text{density})$. With ω in place of dvol the same substitution moves the factor s^{w_O} and the dilation onto ω, and $s\frac{d}{ds}\big|_{s=1}s^{w_O}(\text{density})(sx) = (w_O + E)(\text{density})(x)$. □

Proposition 6.3 (Scaling covariance)**.** *For every $s > 0$ and $\varepsilon > 0$,*

$$(\sigma_s \otimes \sigma_s)\,P(\varepsilon,\infty) = P(s^2\varepsilon,\infty), \qquad \sigma_s\,\partial_{P(\varepsilon,\infty)} = \partial_{P(s^2\varepsilon,\infty)}\,\sigma_s, \tag{6.1}$$

and for every even functional J as in Definition 3.16,

$$\sigma_s\,W\big(P(\varepsilon,\infty), J\big) = W\big(P(s^2\varepsilon,\infty), \sigma_s J\big). \tag{6.2}$$

Proof. By Lemma 6.2, $(\sigma_s \otimes \sigma_s)L_t = (\sigma_s Q^{GF} \otimes \sigma_s)K_t = s^2(Q^{GF}\otimes 1)K_{s^2t} = s^2 L_{s^2t}$, and

$$(\sigma_s \otimes \sigma_s)\,P(\varepsilon,\infty) = \int_\varepsilon^\infty s^2 L_{s^2t}\,dt = \int_{s^2\varepsilon}^\infty L_t\,dt = P(s^2\varepsilon,\infty).$$

For linear functionals ℓ_1, ℓ_2, the operator ∂_ϕ of [C07, §6.4] gives $\partial_\phi\big((\ell_1\circ\sigma_s^{-1})(\ell_2\circ\sigma_s^{-1})\big) = (\ell_1\otimes\ell_2)\big((\sigma_s^{-1}\otimes\sigma_s^{-1})\phi\big)$; since ∂_ϕ is determined by its values on products of two linear functionals and σ_s is an even algebra automorphism of functionals, $\partial_\phi\,\sigma_s = \sigma_s\,\partial_{(\sigma_s^{-1}\otimes\sigma_s^{-1})\phi}$. With $\phi = P(s^2\varepsilon,\infty)$ and the first identity for s^{-1} this is the second identity in (6.1). Then $\sigma_s e^{\hbar\partial_{P(\varepsilon,\infty)}}e^{J/\hbar} = e^{\hbar\partial_{P(s^2\varepsilon,\infty)}}e^{\sigma_s J/\hbar}$, and taking $\hbar\log$ gives (6.2). □

Lemma 6.4 (Covariance of the counterterms)**.** *For a local family $J = I + \delta O$ with O homogeneous, let $J^{CT}[\varepsilon] = I^{CT}[\varepsilon] + \delta\,\partial_\delta I^{CT}[\varepsilon]$ be its one-loop counterterm (Proposition 5.3). Then the counterterm of the family $\sigma_s J = I + \delta\,\sigma_s O$ at the cutoff $s^2\varepsilon$ is*

$$(\sigma_s J)^{CT}[s^2\varepsilon] = \sigma_s\big(J^{CT}[\varepsilon]\big) + 2\log s\;\sigma_s\big(\textit{coefficient of } \log\varepsilon \textit{ in } J^{CT}[\varepsilon]\big). \tag{6.3}$$

In particular σ_s maps the coefficient of ε^{-1} in $I^{CT}[\varepsilon]$ to s^{-2} times itself and fixes the coefficient of $\log\varepsilon$.

Proof. At one loop W is affine in the counterterm, which enters through its tree-level flow. By (6.2), $W(P(s^2\varepsilon,\infty), \sigma_s J - \sigma_s J^{CT}[\varepsilon]) = \sigma_s W(P(\varepsilon,\infty), J - J^{CT}[\varepsilon])$ has a limit as $\varepsilon \to 0$, and so has $W(P(s^2\varepsilon,\infty), \sigma_s J - (\sigma_s J)^{CT}[s^2\varepsilon])$. The difference of the two counterterms is a local functional whose tree-level flow has a limit, and its part without δ is invariant under translations and under the adjoint action of $\mathfrak{g}$, which σ_s preserves; so, as in the uniqueness parts of Propositions 4.10 and 5.3, their singular parts in ε coincide. Write $(\sigma_s J)^{CT}[\varepsilon']$ as ε'^{-1} times one local functional plus $\log\varepsilon'$ times another. At $\varepsilon' = s^2\varepsilon$, $\log\varepsilon' = \log\varepsilon + 2\log s$ and $\varepsilon'^{-1} = s^{-2}\varepsilon^{-1}$; matching the coefficients of ε^{-1} and $\log\varepsilon$ with those of $\sigma_s J^{CT}[\varepsilon]$ gives (6.3), the constant $2\log s$ times the $\log\varepsilon$ coefficient

being the remainder. For $J = I$, $\sigma_s I = I$ and the uniqueness of Proposition 4.10 give the last statement; it agrees with Lemma 6.2: the coefficient of ε^{-1} is a multiple of $\int \mathrm{dvol}\,(A|_V, A|_V) + \int \mathrm{dvol}\,(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ (Theorem 4.16), of density weight 2. □

6.2 The Callan–Symanzik form

Definition 6.5 (Tree insertion)**.** For local functionals F and X of either parity, the tree insertion of F into $X[\infty]$ is

$$\lim_{\varepsilon\to 0}\Big(\hbar\, e^{-W(P(\varepsilon,\infty),I)/\hbar}\, e^{\hbar\partial_{P(\varepsilon,\infty)}}\big(\hbar^{-1}F\,\hbar^{-1}X\, e^{I/\hbar}\big)\Big)_{\text{connected trees}},$$

the sum of the connected trees with one vertex F, one vertex X, any number of vertices I and edges $P(0,\infty)$, with the weights of Proposition 3.22, the vertex functions of F and X being multiplied in this order. For odd F and odd X this order fixes the sign; the opposite order multiplies the insertion by -1. For an even F of order $\hbar$ and $X = O$, at order $\hbar$ it is

$$\lim_{\varepsilon\to 0}\partial_\delta\Big[W\big(P(\varepsilon,\infty), I - I^{CT}[\varepsilon] + \delta O(\varepsilon) + F\big) - W\big(P(\varepsilon,\infty), I - I^{CT}[\varepsilon] + \delta O(\varepsilon)\big)\Big].$$

For a local functional F of order $\hbar$, $F[\infty]$ denotes at order $\hbar$ the construction (5.1) applied to F: the trees with one vertex F and vertices I.

Theorem 6.6 (Callan–Symanzik form)**.** *For a homogeneous local operator O, modulo $\hbar^2$,*

$$\begin{aligned} s\frac{d}{ds}\Big|_{s=1}\sigma_s\big(O[\infty]\big) = {}& \big((E+w_O)\,O\big)[\infty] + 2\,\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon]\big)[\infty] \\ & + 2\,\big(\textit{tree insertion of the coefficient of } \log\varepsilon \textit{ in } I^{CT}[\varepsilon] \textit{ into } O[\infty]\big). \end{aligned} \tag{6.4}$$

The coefficients of ε^{-1} in $I^{CT}[\varepsilon]$ and in $\partial_\delta I^{CT}[\varepsilon]$ are absorbed by the scaling of the cutoff, and the scale dependence of $O[\infty]$ is carried by the $\log\varepsilon$ parts.

Proof. Put $J(\varepsilon) = I - I^{CT}[\varepsilon] + \delta O(\varepsilon)$. Since σ_s is independent of ε and commutes with ∂_δ, (6.2) gives

$$\sigma_s\big(O[\infty]\big) = \lim_{\varepsilon\to 0}\partial_\delta W\big(P(s^2\varepsilon,\infty), \sigma_s J(\varepsilon)\big).$$

By $\sigma_s I = I$ and (6.3), with $\varepsilon' = s^2\varepsilon$,

$$\begin{aligned} \sigma_s J(\varepsilon) = {}& \Big(I - I^{CT}[\varepsilon'] + \delta\,(\sigma_s O)(\varepsilon')\Big) \\ & + 2\log s\,\sigma_s\big(\text{coefficient of } \log\varepsilon \text{ in } I^{CT}[\varepsilon] + \delta\,\partial_\delta I^{CT}[\varepsilon]\big), \end{aligned}$$

where $(\sigma_s O)(\varepsilon')$ is the local family of Definition 5.2 for the operator $\sigma_s O$ at the cutoff ε', by (6.3). The second term is of order $\hbar$, so modulo $\hbar^2$ it enters linearly and through trees. As $\varepsilon \to 0$, the first bracket gives $(\sigma_s O)[\infty]$, and the second gives $2\log s$ times the observable of the δ-part plus the tree insertion of the δ-independent part, both for $\sigma_s O$. Hence, modulo $\hbar^2$,

$$\begin{aligned} \sigma_s\big(O[\infty]\big) = (\sigma_s O)[\infty] + 2\log s\Big[{}& \sigma_s\big(\text{coefficient of } \log\varepsilon \text{ in } \partial_\delta I^{CT}[\varepsilon]\big)[\infty] \\ & + \big(\text{tree insertion of the coefficient of } \log\varepsilon \text{ in } I^{CT}[\varepsilon] \\ & \quad \text{into } (\sigma_s O)[\infty]\big)\Big]. \end{aligned}$$

The construction $O \mapsto O[\infty]$ is linear in ω, so $s\frac{d}{ds}\big|_{s=1}(\sigma_s O)[\infty] = ((E + w_O)O)[\infty]$ by Lemma 6.2; the derivative of $2\log s$ is 2 at $s = 1$. This is (6.4). The ε^{-1} coefficients are absorbed into $I^{CT}[\varepsilon']$ and $\partial_\delta I^{CT}[\varepsilon']$ by (6.3). □

6.3 The action counterterm as a field redefinition

By Theorem 4.22 (the logarithmic part of the one-loop counterterm), the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$ is $(Q + \{I, -\})$-exact:

$$\begin{aligned}&\text{coefficient of } \log\varepsilon \text{ in } I^{CT}[\varepsilon]\\ &\quad = \frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,(Q + \{I, -\})\Big(2\int(A, A^*) - 4\int(\lambda^*, \lambda) - \int(c^*, c)\Big).\end{aligned} \tag{6.5}$$

Here $\int(A, A^*)$, $\int(\lambda^*, \lambda)$, $\int(c^*, c)$ are the local functionals of Chapter 4, the pairing of Definition 1.5 applied to the two slots of a pair with the even member first; we call $2\int(A, A^*) - 4\int(\lambda^*, \lambda) - \int(c^*, c)$ the field-counting functional. The quadratic part of (6.5) is $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big(-\int \mathrm{dvol}\,(dA, dA) + 4\int \mathrm{dvol}\,(\lambda, \not{d}\lambda) - \int \mathrm{dvol}\,(dc, A^*)\big)$ (Theorem 4.16).

Lemma 6.7 (The field-counting functional)**.** *The derivation* $\big\{2\int(A, A^*) - 4\int(\lambda^*, \lambda) - \int(c^*, c), -\big\}$ *multiplies the coordinate functionals of* c, A, λ, A^*, λ^*, c^* *by* 1, 2, 4, −2, −4, −1, *hence a monomial by the corresponding weighted count of its fields. On the operators of Chapter 5 it acts as*

$$\int \omega\,(\iota_{w_1}A, \iota_{w_2}A) \mapsto 4\int \omega\,(\iota_{w_1}A, \iota_{w_2}A), \qquad \int \omega\,(X, \iota_w A) \mapsto 2\,(\cdot),$$
$$\int \omega\,(X, \iota_v(A|_V)) \mapsto 2\,(\cdot), \qquad \int \omega\,(\zeta, \lambda) \mapsto 4\,(\cdot), \qquad \int \omega\,(X, c) \mapsto (\cdot),$$

where $(\cdot)$ *is the operator on the left.*

Proof. This is Lemma 4.21 applied to the monomials of the operators: two factors A, one A, one A, one λ, one c, and no antifields. □

Lemma 6.8 (Insertion of a BV-exact functional)**.** *Let* F *be an odd local functional and* O *a homogeneous local operator. At order* $\hbar^0$*,*

$$\begin{aligned}&\textit{tree insertion of } (Q + \{I, -\})F \textit{ into } O[\infty] = Q\big(\textit{tree insertion of } F \textit{ into } O[\infty]\big) - \{F, O\}[\infty]\\ &\qquad\qquad + \textit{tree insertion of } F \textit{ into } \big((Q + \{I, -\})O\big)[\infty].\end{aligned} \tag{6.6}$$

Proof. Put $J = I + \delta O$. By Theorem 3.17, $Q\,e^{\hbar\partial_{P(\varepsilon,\infty)}}\big(F\,e^{J/\hbar}\big) = e^{\hbar\partial_{P(\varepsilon,\infty)}}(Q + \hbar\Delta_\varepsilon)\big(F\,e^{J/\hbar}\big)$. By the product rule of Definition 3.15 and $|F|$ odd,

$$\begin{aligned}(Q + \hbar\Delta_\varepsilon)\big(F\,e^{J/\hbar}\big) = &\big(QF + \{J, F\}_\varepsilon + \hbar\Delta_\varepsilon F\big)e^{J/\hbar}\\ &- F\,\hbar^{-1}\big(QJ + \tfrac{1}{2}\{J, J\}_\varepsilon + \hbar\Delta_\varepsilon J\big)e^{J/\hbar}.\end{aligned}$$

Divide both sides by $e^{W(P(\varepsilon,\infty),J)/\hbar}$. On the left, Q acts on the product $e^{W(P(\varepsilon,\infty),J)/\hbar}\cdot e^{-W(P(\varepsilon,\infty),J)/\hbar}e^{\hbar\partial_{P(\varepsilon,\infty)}}(Fe^{J/\hbar})$, and the term in which it acts on $e^{W(P,J)/\hbar}$ equals, by Corollary 3.18, the disconnected part of the last term on the right; the two cancel. In the remaining terms take the part linear in δ, connected, at order $\hbar^0$. The term $\hbar\Delta_\varepsilon F$ is of order $\hbar$. For even O the δ-part of $\{J, F\}_\varepsilon$ is $\{O, F\}_\varepsilon = \{F, O\}_\varepsilon$. For odd O the parameter δ is odd, and moving δ to the left through F, Q and the bracket gives $\{\delta O, F\}_\varepsilon = -\delta\{O, F\}_\varepsilon = \delta\{F, O\}_\varepsilon$, $Q(\delta O) = -\delta\,QO$,

$\{I,\delta O\}_\varepsilon = -\delta\{I,O\}_\varepsilon$ and $F\,\delta = -\delta\,F$; the signs agree term by term, and after removing δ the identities below are the same in both cases, with the vertex functions of F placed first as in Definition 6.5. The δ-part of $QJ+\frac{1}{2}\{J,J\}_\varepsilon$ is $QO+\{I,O\}_\varepsilon$, and its δ-independent part is $QI+\frac{1}{2}\{I,I\}_\varepsilon$. This gives, with edges $P(\varepsilon,\infty)$,

$$\begin{aligned} Q\big(\text{tree insertion of } F \text{ into } O\big) = {}& \text{tree insertion of } QF+\{I,F\}_\varepsilon \text{ into } O \;+\; \big(\{F,O\}_\varepsilon \text{ flowed}\big) \\ & - \text{tree insertion of } F \text{ into } QO+\{I,O\}_\varepsilon \\ & - \text{trees with } F,\ O \text{ and } QI+\tfrac{1}{2}\{I,I\}_\varepsilon. \end{aligned}$$

As $\varepsilon\to 0$, $\{-,-\}_\varepsilon$ tends to $\{-,-\}$ on local functionals, the edges tend to $P(0,\infty)$, and $QI+\frac{1}{2}\{I,I\}=0$ by Proposition 1.14. Rearranging gives (6.6). □

Corollary 6.9 (The field redefinition)**.** *For a homogeneous local operator O, modulo $\hbar^2$,*

$$\begin{aligned} s\frac{d}{ds}\Big|_{s=1}\sigma_s\big(O[\infty]\big) = {}& \big((E+w_O)O\big)[\infty] \\ & + 2\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon]\big)[\infty] \\ & - 2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big\{\textit{field-counting functional}, O\big\}[\infty] \\ & + Q\Big(2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\cdot \textit{tree insertion of the field-counting functional into } O[\infty]\Big) \\ & + 2\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)} \\ & \qquad \cdot \textit{tree insertion of the field-counting functional into } \big((Q+\{I,-\})O\big)[\infty]. \end{aligned} \tag{6.7}$$

Proof. Insert (6.5) into the last term of (6.4) and apply Lemma 6.8 with F the field-counting functional; the factor $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ makes the order-$\hbar^0$ identity (6.6) an identity at order $\hbar$. □

Remark 6.10 (Reading of the three terms)**.** By (6.5) the action counterterm is the change of S under the infinitesimal rescaling of the fields by the weights of Lemma 6.7; on $O[\infty]$ it acts through the bracket $\{-,O\}$ with the field-counting functional, the field redefinition proper, and through the image under Q of a tree. The tree insertion of the field-counting functional into $O[\infty]$ is a nonlocal functional whose image under Q is the scale-∞ observable of a local functional; by Proposition 5.4 such terms are the ones that a statement about classes of local functionals must specify, and the identification with the equivariant dilatation cochain, stated in the last section of this chapter, specifies how this term enters. The last term involves $(Q+\{I,-\})O$ and vanishes for gauge-invariant O.

6.4 The dilatation component

The dilatation is $\Delta = 2E$, the generator $\mathrm{diag}(1,-1\,|\,0)$ of [SW, (17)] (Remark 2.6). By Example 2.13 it acts classically on a field as $2E$ plus twice its Weyl weight, hence on a local operator as $2(E+w_O)$, and by Lemma 6.2 this is $2s\frac{d}{ds}\big|_{s=1}\sigma_s$.

Theorem 6.11 (Dilatation in Callan–Symanzik form)**.** *For a homogeneous local operator O, modulo*

$\hbar^2$,

$$
\begin{aligned}
2s\frac{d}{ds}\Big|_{s=1}\sigma_s\big(O[\infty]\big) - Q\Big(4\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\cdot \textit{tree insertion of the field-counting functional into } O[\infty]\Big)\\
= 2\big((E+w_O)O\big)[\infty]\\
+ 4\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon]\big)[\infty] - 4\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big\{\textit{field-counting functional}, O\big\}[\infty]\\
+ \big(\textit{terms built from } (Q+\{I,-\})O\big),
\end{aligned}
\tag{6.8}
$$

the last terms being $4\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ *times the tree insertion of the field-counting functional into* $\big((Q+\{I,-\})O\big)[\infty]$. *When the coefficient of* $\log\varepsilon$ *in* $\partial_\delta I^{CT}[\varepsilon]$ *minus the unit times the bracket with the field-counting functional is a multiple of* O, *the right side is*

$$
2\big(E + w_O + \textit{coefficient}\big)\,O[\infty] + \big(\textit{terms built from } (Q+\{I,-\})O\big),
$$

with the coefficient equal to twice that multiple; its values are given in the table.

operator	w_O	$\log\varepsilon$ *coefficient of* $\partial_\delta I^{CT}[\varepsilon]$, *per* O	*bracket with the field-counting functional, per* O	*coefficient*
traceless part of $\int\omega(\iota_{w_1}A,\iota_{w_2}A)$	2	4	4	0
$\int\omega\,(A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp})$	2	-8	4	-24
$\int\omega\,(X,\iota_wA)$	1	0	2	-4
$\int\omega\,(X,\iota_v(A|_V))$, *modulo terms built from* $(Q+\{I,-\})O$	1	0	2	-4
$\int\omega\,(\zeta,\lambda)$	$\frac{3}{2}$	0	4	-8
$\int\omega\,(X,c)$	0	0	1	-2

The coefficient in units of $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$; *the third column in the same units (Theorem 5.18, Proposition 5.11), the fourth from Lemma 6.7, and the coefficient is twice their difference. With the generator* $-E$ *of [SW, (3)], which is* $-\frac{1}{2}$ *times* Δ *(Remark 2.6), the same statement reads* $-\big(E+w_O+\textit{coefficient}\big)O[\infty]$. *The letters are gauge dependent (Feynman gauge); the ghost letter* $\int\omega\,(X,c)$ *has degree* 1.

Proof. Multiply (6.7) by 2 and move the Q-term to the left; this is (6.8). For the scalar bilinears, (5.8) and (5.9) give the $\log\varepsilon$ coefficients 4 and -8 per operator, and Lemma 6.7 gives 4; for the letters the $\log\varepsilon$ coefficient is 0 by (5.5) and the bracket gives $2, 2, 4, 1$. Twice the differences are $2(4-4)=0$, $2(-8-4)=-24$, $2(0-2)=-4$, -4, $2(0-4)=-8$, $2(0-1)=-2$. The operator $(E+w_O)$ acts on $O[\infty]$ through the linear dependence of $O[\infty]$ on ω. □

Remark 6.12 (The letter $A|_V$)**.** The letters are gauge variant, and $(Q+\{I,-\})O$ contributes. For $\int\omega\,(X,\iota_wA)$, $\int\omega\,(\zeta,\lambda)$ and $\int\omega\,(X,c)$ the linear part of Q on the coordinate functional vanishes (Q has no arrow into the slots of $A|_{W_{\mathbb{R}}^\perp}$, λ, c; Proposition 1.11), and $(Q+\{I,-\})O=\{I,O\}$ is quadratic in the fields: the terms built from $(Q+\{I,-\})O$ have at least two legs, and the coefficients -4, -8, -2 multiply O itself. For $\int\omega\,(X,\iota_v(A|_V))$ the arrow $c\mapsto -dc$ of Q gives the linear functional $(Q+\{I,-\})O\ni-\int\omega\,(X,\iota_v(dc))$; joined through the $c\otimes A^*|_V$ component of $P(0,\infty)$ (Proposition 3.11) to a vertex with an $A^*|_V$ leg, it produces linear functionals of $A|_V$ that depend on $A|_V$ through $d^*A|_V$, nonlocal projections onto the longitudinal part. The entry -4 of the table is therefore the coefficient of O modulo terms built from $(Q+\{I,-\})O$; the coefficient on the longitudinal part requires these terms, at order $\hbar^0$ (Proposition 6.18) and at one loop, and its evaluation is part of Step 2.

Remark 6.13 (The Konishi value)**.** For $\mathfrak{g}=\mathfrak{su}(N)$ with $(X,Y)=-\frac{4\hbar}{g^2}\,\mathrm{tr}(XY)$, tr the trace in the defining representation, $e^{S/\hbar}$ is the standard Euclidean Yang–Mills weight with coupling g, and $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)=2N\,\mathrm{tr}(XY)$. Then

$$
\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)} = -\frac{g^2N}{32\pi^2}, \qquad -24\cdot\Big(-\frac{g^2N}{32\pi^2}\Big)=\frac{3g^2N}{4\pi^2},
$$

so the trace bilinear has dimension $2+\frac{3g^2N}{4\pi^2}$ at one loop, the one-loop value for the Konishi operator, and the twenty traceless bilinears keep dimension 2.

Corollary 6.14 (Dilatation component)**.** *Assume the identification of Step 2 (last section of this chapter),*

$$\big[(d_{CE}+Q)\,O[\infty]\big](\Delta) = 2s\frac{d}{ds}\Big|_{s=1}\sigma_s\big(O[\infty]\big) \\ - Q\Big(4\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\cdot \textit{tree insertion of } 2\int(A,A^*) - 4\int(\lambda^*,\lambda) - \int(c^*,c) \\ \textit{into } O[\infty]\Big) + \big(\textit{terms built from } (Q+\{I,-\})O\big) \tag{6.9}$$

modulo $\hbar^2$, where the terms built from $(Q+\{I,-\})O$ are the trees with one vertex $(Q+\{I,-\})O(\varepsilon)$ and one vertex $S^{(1)}(\Delta)$, at order $\hbar^0$, with their one-loop corrections; they vanish for $(Q+\{I,-\})O=0$. Then, with $\Delta=2E$ of [SW, (17)],

$$\big[(d_{CE}+Q)\,O[\infty]\big](\Delta) = 2\big(E+w_O+\textit{coefficient}\big)\,O[\infty] + \big(\textit{terms built from } (Q+\{I,-\})O\big), \tag{6.10}$$

with the coefficient of the table of Theorem 6.11: 0 for the traceless scalar bilinears, $-24\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ for the trace, and -4, -4, -8, -2 times $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ for the letters $\iota_w A$, $A|_V$ (modulo terms built from $(Q+\{I,-\})O$, Remark 6.12), λ, c. With the generator $-E$ of [SW, (3)] the same statement reads $-\big(E+w_O+\textit{coefficient}\big)O[\infty]$.

Proof. Insert (6.9) into (6.8). □

6.5 The UV boundary at work: graphs with an L_ε edge

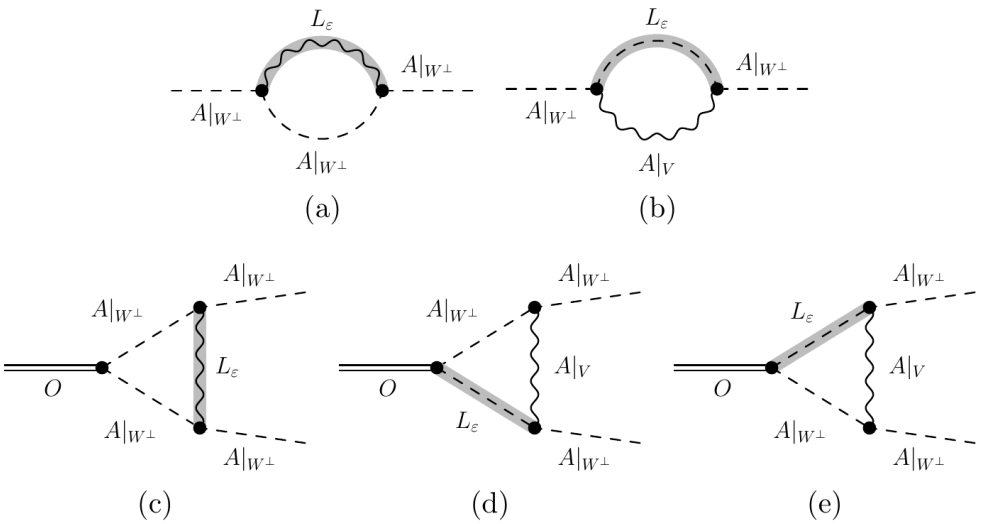
By (3.2), $s\frac{d}{ds}\big|_{s=1}P(s^2\varepsilon,\infty) = 2\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty) = -2\varepsilon L_\varepsilon$: the scaling acts on every edge as the boundary kernel $L_\varepsilon = (Q^{GF}\otimes 1)K_\varepsilon$ at $t=\varepsilon$ (Remark 3.13).

Proposition 6.15 (The coefficients on the boundary)**.** *Let γ be a one-loop graph whose value has the expansion $(\varepsilon^{-1}\text{-}\textit{coefficient})\,\varepsilon^{-1} + (\log\varepsilon\text{-}\textit{coefficient})\log\varepsilon + (\textit{a function with a limit})$. Then*

$$2\cdot(\log\varepsilon\text{-}\textit{coefficient}) = \lim_{\varepsilon\to 0}\Big(-2\varepsilon\sum_{e\in E(\gamma)}\big(\textit{value of } \gamma \textit{ with } t_e=\varepsilon,\ \textit{i.e. edge } e \textit{ carrying } L_\varepsilon\big) \\ + 2\varepsilon^{-1}(\varepsilon^{-1}\text{-}\textit{coefficient})\Big).$$

The term $2\varepsilon^{-1}(\varepsilon^{-1}\text{-}\textit{coefficient})$ is the scaling $s\frac{d}{ds}\big|_{s=1}$ of the counterterm $-(\varepsilon^{-1}\text{-}\textit{coefficient})\,\varepsilon^{-1}$ under σ_s (Lemma 6.4). Hence the terms $2(\cdot)$ of (6.4) are the limits of the sums of the graphs with one edge carrying $-2\varepsilon L_\varepsilon$, the ε^{-1} parts cancelling against the scaling of the ε^{-1} counterterms; for graphs of degree 0 each edge carries the fraction $1/|E(\gamma)|$ (Lemma 5.10).

Proof. By Proposition 4.2, $\varepsilon\frac{d}{d\varepsilon}(\text{value of }\gamma) = -\varepsilon\sum_e(\text{value with } t_e=\varepsilon)$, and $\varepsilon\frac{d}{d\varepsilon}$ of the expansion is $-(\varepsilon^{-1}\text{-coefficient})\varepsilon^{-1} + (\log\varepsilon\text{-coefficient}) + (\text{a function tending to 0})$. The last term tends to 0 by Lemma 4.6, which gives $\varepsilon\frac{d}{d\varepsilon}$ of the function with a limit tending to 0. Multiply by 2 and add $2\varepsilon^{-1}(\varepsilon^{-1}\text{-coefficient})$. □



Boundary graphs with one edge carrying $L_\varepsilon = (Q^{GF} \otimes 1)K_\varepsilon$ (shaded). (a), (b): the $A|_{W_{\mathbb{R}}^\perp}$ two-point graph with two cubic vertices of type $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$, each edge pinned in turn. (c)–(e): graph (f) of Chapter 5, the triangle with external $A|_{W_{\mathbb{R}}^\perp}$, each of its three edges pinned in turn.

Example 6.16 (The scalar bilinears on the boundary)**.** For $O = \int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$, the terms on the right of (6.4) other than $((E+w_O)O)[\infty]$ are, in their bilinear part and in units of $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$, the following limits of L_ε graphs (Proposition 6.15).

- Graph (e) of Chapter 5: each of its two edges carrying $-2\varepsilon L_\varepsilon$ tends to $2\int \omega(\iota_{w_1}A, \iota_{w_2}A) - 2(w_1,w_2)\int \omega(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, one half of twice the singular part of Example 5.15.
- Graph (f): each of its three edges (figure, (c)–(e)) tends to $\frac{4}{3}\int \omega(\iota_{w_1}A, \iota_{w_2}A)$, one third of twice the singular part of Example 5.16.
- Graphs (c), (d): each edge of (c) tends to $-2(w_1,w_2)\int \omega(A|_V, A|_V)$ and each edge of (d) to $\frac{4}{3}(w_1,w_2)\int \omega(A|_V, A|_V)$; the sum is 0 (Example 5.17).
- Graph (a): -2ε times its single edge carrying L_ε is $-2\varepsilon^{-1}$ times its ε^{-1}-coefficient, cancelled by the scaling of the vacuum counterterm; the limit is 0.
- The legs: the $A|_{W_{\mathbb{R}}^\perp}$ two-point graphs of Theorem 4.16 with an L_ε edge, joined to O by an edge $P(\varepsilon,\infty)$. Their ε^{-1} parts cancel against the scaling of $\frac{9}{4}\varepsilon^{-1}\int \mathrm{dvol}\,(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, and their limit is twice the tree insertion of $-\int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^\perp}, dA|_{W_{\mathbb{R}}^\perp})$ into O. The graphs in which the joining edge carries L_ε tend to 0, since $\varepsilon L_\varepsilon$ tends to 0 at fixed momentum and the renormalised two-point function has a limit.

The tree insertion is computed from the vertex function $2|p|^2(w,w')(Z,Z')$ of $\int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^\perp}, dA|_{W_{\mathbb{R}}^\perp})$ on legs $(w,-)\otimes Z$, $(w',-)\otimes Z'$ of momenta $\pm p$ and the edge $2(-,-)^{-1}\otimes(\text{inverse metric})|p|^{-2}$ of $P(0,\infty)$: each $A|_{W_{\mathbb{R}}^\perp}$ leg of O is multiplied by

$$-1\cdot 2|p|^2\cdot\frac{2}{|p|^2} = -4,$$

so the tree insertion of $-\int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^\perp}, dA|_{W_{\mathbb{R}}^\perp})$ into O has bilinear part $-8\,O$. Summing, with the trace written $\int \omega(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$:

contribution	general $w_1 w_2$	traceless part	trace
graph (e), two L_ε edges	$4O - 4(w_1, w_2)\,(\text{trace})$	4	-20
graph (f), three L_ε edges	$4O$	4	4
graphs (c), (d)	0	0	0
legs, $2 \times (-8)$	$-16O$	-16	-16
total	$-8O - 4(w_1, w_2)\,(\text{trace})$	-8	-32
removed Q-term	$8O$	8	8
coefficient	$-4(w_1, w_2)\,(\text{trace})$	0	-24

Contributions to $s\frac{d}{ds}\big|_{s=1}\sigma_s(O[\infty]) - ((E + w_O)O)[\infty]$ for the scalar bilinears, bilinear parts, in units of $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$; the traceless and trace columns give the multiple of the operator itself.

The row "removed Q-term" is the bilinear part of minus the Q-term of (6.7), the image under Q of twice the unit times the tree insertion of the field-counting functional into $O[\infty]$. Its value follows from the legs row and Lemmas 6.8, 6.7: the bilinear part involves the quadratic part of (6.5) alone, and twice its tree insertion into O equals the Q-term minus twice $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$ times the bracket of the field-counting functional with O. In units, $-16\,O =$ (bilinear part of the Q-term) $- 2 \cdot 4\,O$, so the Q-term has bilinear part $-8\,O$ and the removed term is $+8\,O$. The coefficient row agrees with Theorem 6.11: $2(4-4) = 0$ and $2(-8-4) = -24$.

Example 6.17 (The letter $\iota_w A$ on the boundary)**.** For $O = \int \omega\,(X, \iota_w A)$ there is no graph with O on the cycle (Proposition 5.11), and the boundary contributions are the legs. With the single leg multiplied by -4, the tree insertion of $-\int \mathrm{dvol}\,(dA|_{W_{\mathbb{R}}^\perp}, dA|_{W_{\mathbb{R}}^\perp})$ into O is $-4O$, and twice it is $-8O$ in units. The bracket with the field-counting functional is $2\,O$, so by the same argument $-8\,O =$ (linear part of the Q-term) $- 2 \cdot 2\,O$, the Q-term has linear part $-4\,O$, and the removed term is $+4\,O$. The result is $-8 + 4 = -4$ units, the coefficient $2(0-2) = -4$ of Theorem 6.11.

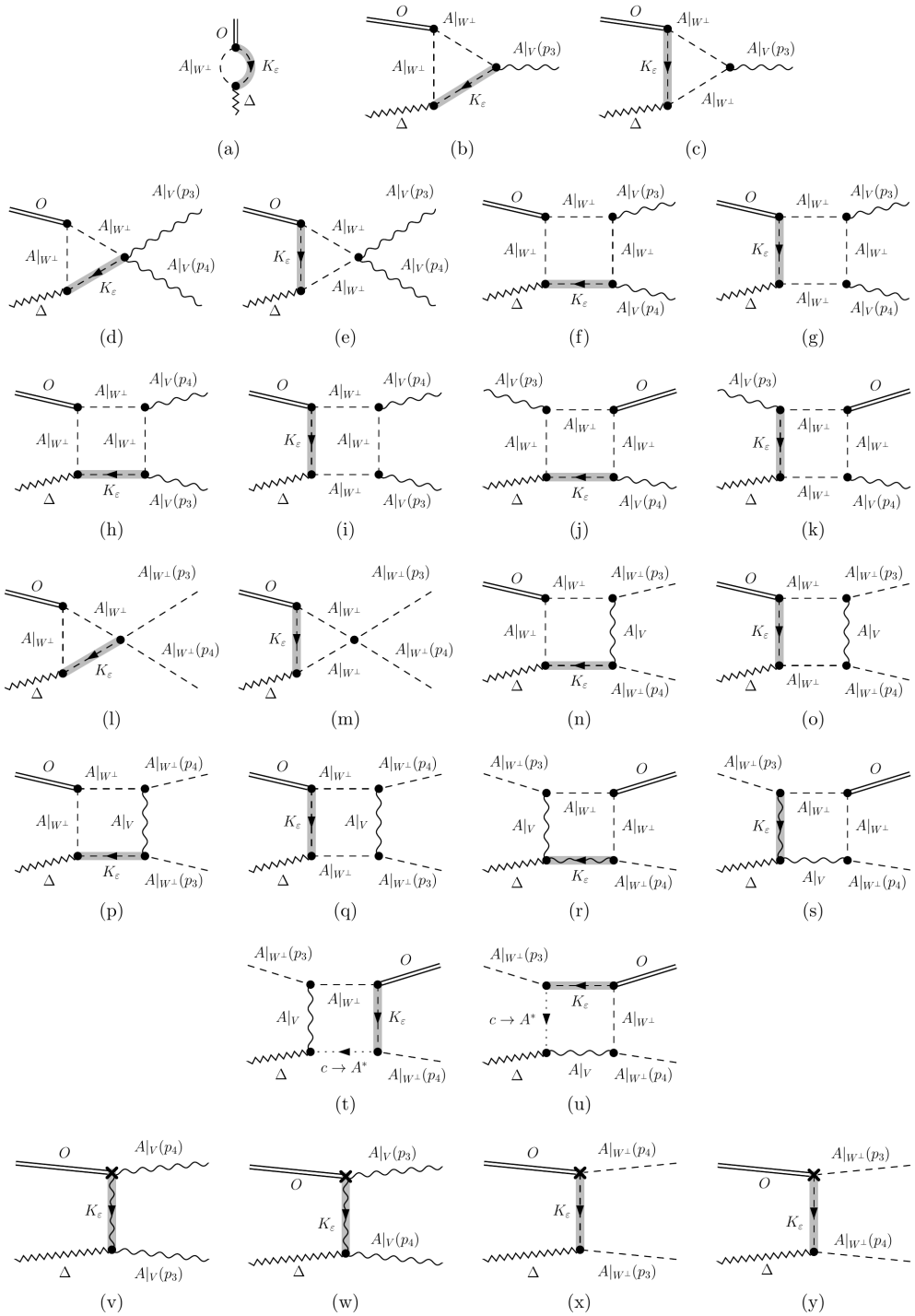
6.6 The equivariant dilatation cochain: the task of Step 2

The left side of (6.10) is defined by the equivariant construction of Definition 5.2. By Theorem 3.17 and the product rule used in Corollary 3.18, for every even J with values in $C^\bullet(\mathfrak{psl}(4|4))$,

$$(d_{CE} + Q)\,W\big(P(\varepsilon,\infty), J\big) = e^{-W(P(\varepsilon,\infty),J)/\hbar}\, e^{\hbar\partial_{P(\varepsilon,\infty)}}\Big(\big((d_{CE}+Q)J + \tfrac{1}{2}\{J,J\}_\varepsilon + \hbar\Delta_\varepsilon J\big)\, e^{J/\hbar}\Big). \quad (6.11)$$

For $J = I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] + \delta O(\varepsilon)$, applying ∂_δ and $\varepsilon \to 0$ gives $(d_{CE}+Q)O[\infty]$. Every bracket $\{-,-\}_\varepsilon$ and every $\Delta_\varepsilon = -\partial_{K_\varepsilon}$ is an edge carrying K_ε (Remark 3.13). Evaluated on Δ, the component of arity one contains the vertex $S^{(1)}(\Delta)$ of Example 2.13, or the component of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ along Δ, exactly once. Its terms are:

- the graphs with one vertex $O(\varepsilon)$, one vertex $S^{(1)}(\Delta)$ joined by an edge K_ε, and edges $P(\varepsilon,\infty)$ (figure): one-loop graphs (a)–(u), and trees (v)–(y) in which the vertex is $\partial_\delta I^{CT}[\varepsilon]$;
- the graphs with the component of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ along Δ, and with the finite counterterms that the equivariant master equation requires along Δ;
- the component along Δ of the equivariant master-equation defect $(d_{CE}+Q)J + \frac{1}{2}\{J,J\}_\varepsilon + \hbar\Delta_\varepsilon J$ at $\delta = 0$, inserted together with $O(\varepsilon)$;
- the graphs with the vertex $QO(\varepsilon) + \{I - I^{CT}[\varepsilon], O(\varepsilon)\}_\varepsilon + \hbar\Delta_\varepsilon O(\varepsilon)$ and one vertex $S^{(1)}(\Delta)$, which at order $\hbar^0$ are the trees of Proposition 6.18 and vanish for $(Q + \{I, -\})O = 0$.



The graphs with one vertex O, one vertex $S^{(1)}(\Delta)$ (zigzag leg Δ) and one edge carrying K_ε (shaded, arrow from the field to the antifield slot); all other edges carry $P(\varepsilon,\infty)$. (a)–(u): one loop; (v)–(y): trees with the vertex $\partial_\delta I^{CT}[\varepsilon]$ (cross).

Proposition 6.18 (Order $\hbar^0$)**.** *At order $\hbar^0$, the component $[(d_{CE}+Q)O[\infty]](\Delta)$ is the sum of $2\big((E+w_O)O\big)[\infty]$ and the trees with one vertex $(Q+\{I,-\})O$ and one vertex $S^{(1)}(\Delta)$. In particular, for O with $(Q+\{I,-\})O=0$, both sides of* (6.9) *equal $2\big((E+w_O)O\big)[\infty]$, and* (6.9) *holds at order $\hbar^0$ for every O, with these trees as its last term.*

Proof. At order $\hbar^0$ the counterterms and the field-redefinition term are absent, and the first term of the right side of (6.9) is $2((E + w_O)O)[\infty]$ by Theorem 6.6. On the left, by (6.11) at order $\hbar^0$, the arity-one component along Δ is the flow of graphs with one vertex given by the part of $(d_{CE} + Q)J + \frac{1}{2}\{J, J\}_\varepsilon$ that is linear in δ or contains $S^{(1)}(\Delta)$. The defect at $\delta = 0$ tends to 0 by the master equation of $S_{\mathfrak{psl}(4|4)}$ (Definition 2.10). The vertex $\{S^{(1)}(\Delta), O\}_\varepsilon$, with all other vertices I, tends to the tree flow of the classical bracket of $S^{(1)}(\Delta)$ with O; by Example 2.13 this bracket transforms each field by $2E$ plus twice its Weyl weight, hence O by $2(E + w_O)$. The vertex $QO + \{I, O\}_\varepsilon$, of arity zero, joined to a vertex $S^{(1)}(\Delta)$, gives the trees with one vertex $(Q + \{I, -\})O$ and one vertex $S^{(1)}(\Delta)$ in the limit. For $O = \int \omega\,(X, \iota_v(A|_V))$ these trees are nonzero: $QO = -\int \omega\,(X, \iota_v(dc))$ is linear, and its c leg joined to the $A^*|_V$ leg of $S^{(1)}(\Delta)$ through the $c \otimes A^*|_V$ component of $P(0, \infty)$ gives a linear functional of $A|_V$ through $d^*A|_V$. □

Remark 6.19 (Step 2)**.** Step 2 of the program consists of three computations at order $\hbar$: the component along Δ of the one-loop counterterm of $I_{\mathfrak{psl}(4|4)}$, with the finite counterterms fixed by the equivariant master equation along Δ; the graphs (a)–(y) of the figure together with the defect insertion; and the proof of (6.9) for every homogeneous local operator, with the trees built from $(Q + \{I, -\})O(\varepsilon)$ and $S^{(1)}(\Delta)$ included. Reading the result as the anomalous dimension of a class also needs the cocycle condition $Q\,O[\infty] = 0$ of Remark 5.6; a finite local counterterm of Weyl weight w_O that it may require is independent of ε, scales covariantly, and leaves the coefficients of Theorem 6.11 unchanged. The form of (6.9) is dictated by (6.5): the Weyl scaling and the dilatation cochain differ by the field redefinition generated by $2\int(A, A^*) - 4\int(\lambda^*, \lambda) - \int(c^*, c)$, whose image under Q in (6.7) is the term subtracted in (6.9). Until that computation, (6.10) is the dilatation component obtained from scaling covariance, with the coefficients of Theorem 6.11.

Remark 6.20 (Method of computation)**.** The limits of Proposition 6.15 are evaluated exactly in explicit realisations of $\mathfrak{g} = \mathfrak{su}(2)$ and $\mathfrak{g} = \mathfrak{su}(3)$: each boundary integral of the Schwinger parametrisation with one edge at $t = \varepsilon$ is computed in closed form in exact rational arithmetic, edge by edge, and each edge of a graph of degree 0 carries the fraction of Lemma 5.10; the resulting rational multiples of $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)/(X, Y)$ are the same for both gauge algebras.

Chapter 7

Other superconformal components and the road ahead

The dilatation component of Chapter 6 is one of the components $[(d_{CE}+Q)O[\infty]](x_1)$, one for each generator x_1 of $\mathfrak{psl}(4|4)$. How a generator acts on the regulator decides what a one-loop computation along it can produce. Translations, rotations and the R-symmetry commute with Q^{GF} and with $H=[Q,Q^{GF}]$, so every regulated kernel, every counterterm and the maps $O\mapsto O(\varepsilon)$, $O\mapsto O[\infty]$ are covariant, which is why these components can be chosen with no one-loop correction for gauge-invariant O, a choice realised in Step 2; the dilatation, the special conformal transformations, the supercharges and the special supercharges move Q^{GF}, and along them the one-loop output is confined by weight, Lorentz and R-representation, ghost degree and colour to a short list of local structures. This chapter proves the covariance statements and the selection rules, states exactly what relates the special conformal component to the dilatation component, and lists the objects and graphs of Steps 2–5.

7.1 Components along a generator

Definition 7.1 (Observables and their components)**.** Let O be a local functional of degree $|O|$ and δ a square-zero parameter of degree $-|O|$ [CG2, Def 8.3.1.1]. Let $I_{\mathfrak{psl}(4|4)}=I+S^{(1)}+S^{(2)}$ be the equivariant interaction of Chapter 2 [ESW, Rem 1.29] and $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ its one-loop counterterms, the singular parts, graph by graph, of the flow of $I_{\mathfrak{psl}(4|4)}$ with values in $C^\bullet(\mathfrak{psl}(4|4))$. Then

$$O(\varepsilon)=O-\partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon],\qquad O[\infty]=\lim_{\varepsilon\to 0}\partial_\delta W\big(P(\varepsilon,\infty),\,I_{\mathfrak{psl}(4|4)}-I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]+\delta\,O(\varepsilon)\big)$$

[CG2, Def 8.3.1.3, Prop 13.4.0.1]. The component of O along $x_1\in\mathfrak{psl}(4|4)$ is the arity-one part of the cochain $(d_{CE}+Q)O[\infty]$ evaluated on x_1 [ESW, Def 1.26–1.27], written $[(d_{CE}+Q)O[\infty]](x_1)$; as in Chapter 2, generators carry a subscript and x alone is the position. The component is represented by the local functional $\int\omega\,f_{x_1}$, with f_{x_1} a polynomial in the jets of the fields at a point, if

$$[(d_{CE}+Q)O[\infty]](x_1)=\Big(\int\omega\,f_{x_1}\Big)[\infty].$$

The counterterms computed in Step 1 are the arity-zero part of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$, i.e. $I^{CT}[\varepsilon]$ and $\partial_\delta I^{CT}[\varepsilon]$ of Chapters 4 and 5; the arity-one counterterms are part of Step 2 (last section of this chapter).

Lemma 7.2 (Components lie in the image of Q)**.** *For every $x_1\in\mathfrak{psl}(4|4)$, $[(d_{CE}+Q)O[\infty]](x_1)$ equals Q applied to the arity-one part of $O[\infty]$ evaluated on x_1, up to the Koszul sign of the evaluation. In particular it lies in the image of Q.*

Proof. The Chevalley–Eilenberg differential of Chapter 2 [ESW, Def 1.26] is a sum over pairs $i<j$ of arguments of terms $f([x_i,x_j],\dots)$; it maps cochains of arity k to arity $k+1$ and vanishes on cochains of arity zero, which have no pair of arguments. The arity-one part of $(d_{CE}+Q)O[\infty]$ is therefore d_{CE} of the arity-zero part, which is 0, plus Q of the arity-one part, Q acting on the values of the cochain. □

Remark 7.3 (Where the content of a component lies)**.** On moment-free Schwartz sections every Q-closed functional without constant term is Q-exact (Chapter 3); on compactly supported sections Q has in addition nonzero classes, among them the zero-mode classes, finitely many in each polynomial degree. Lemma 7.2 places every component in the image of Q. The content of a component is therefore its local representative $\int \omega\, f_{x_1}$, and the selection rules below are statements about f_{x_1}.

7.2 Translations, rotations and R-symmetry

Definition 7.4 (Action of an even generator on all slots)**.** Let x_1 be an even generator of $\mathfrak{psl}(4|4)$ with vector field v (Chapter 2). The linear operator $e\mapsto v.e$ on $\mathscr{E}$ is given on the slots of the fields by the transformations in $S^{(1)}(x_1)$ (Chapter 2, arity one):

$$v.A = L_vA + \tfrac14(\operatorname{div} v)A|_{W_{\mathbb{R}}^\perp},\qquad v.\lambda = v(\lambda)+\tfrac14\rho(d(v,-))\lambda+\tfrac38(\operatorname{div}v)\lambda,\qquad v.c=v(c),$$

and on the slots of the antifields by the condition

$$\langle v.e_0,e_1\rangle+\langle e_0,v.e_1\rangle=0\qquad\text{for all } e_0,e_1\in\mathscr{E}_c.$$

This extends the notation $v.\lambda$, $v.c$ of [ESW, (32)]. With it,

$$S^{(1)}(x_1)=\int \operatorname{dvol}\Big((v.A,A^*)-(v.\lambda,\lambda^*)-(v.c,c^*)\Big),$$

a functional linear in the fields and linear in the antifields, and the derivation $\{S^{(1)}(x_1),-\}$ of functionals is induced, as in [C07, §6.2], by $e\mapsto v.e$ times one sign fixed by the conventions of the bracket (Chapter 1). Every statement below about commutators of $v.$ is insensitive to this sign.

Lemma 7.5 (Weights on the antifields)**.** *For a conformal Killing vector field v of V and e_1 a section in the slot of $A^*|_{W_{\mathbb{R}}^\perp}$,*

$$v.e_1 = v(e_1)+\tfrac34(\operatorname{div}v)\,e_1.$$

For the dilatation Δ, $v=2E$, on every component,

$$\begin{aligned} v.A&=2E(A)+2A, & v.\lambda&=2E(\lambda)+3\lambda, & v.c&=2E(c),\\ v.A^*&=2E(A^*)+6A^*, & v.\lambda^*&=2E(\lambda^*)+5\lambda^*, & v.c^*&=2E(c^*)+8c^*, \end{aligned}$$

with $E(-)$ acting componentwise. Thus $v.$ is $2E(-)$ plus twice the Weyl weight, the weights of $A,\lambda,c,A^*,\lambda^*,c^*$ being $1,\frac32,0,3,\frac52,4$.

Proof. For compactly supported f_1,f_2, integration by parts gives

$$\int \operatorname{dvol} v(f_1)\,f_2=-\int\operatorname{dvol} f_1\,\big(v(f_2)+(\operatorname{div}v)\,f_2\big).$$

For the pair (A,A^*) the condition of Definition 7.4 reads $\int\operatorname{dvol}\big((v.A,A^*)+(A,v.A^*)\big)=0$, with the fibre pairing of Chapter 1, and likewise for the pairs (λ,λ^*) and (c,c^*). On $A|_{W_{\mathbb{R}}^\perp}$, $v.f_1=$

$v(f_1)+\frac14(\operatorname{div} v)f_1$, so $\int \operatorname{dvol}(v.f_1,f_2) = -\int \operatorname{dvol}\big(f_1,\, v(f_2)+(1-\frac14)(\operatorname{div} v)f_2\big)$, which gives the first formula. For $v=2E$, $\operatorname{div}(2E)=8$ and $v.A = 2E(A)+2A$, $v.\lambda = 2E(\lambda)+3\lambda$, $v.c = 2E(c)$ (Chapter 2, the dilatation example); the same computation gives $2E(A^*)+(8-2)A^*$, $2E(\lambda^*)+(8-3)\lambda^*$ and $2E(c^*)+8c^*$. □

Proposition 7.6 (Covariance of the regulator)**.** *Let x_1 be a translation, an element of $\mathfrak{so}(V)$ or an element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, with vector field v. Then:*

- $\operatorname{div} v = 0$*, and $v.$ is the infinitesimal action of the translation or rotation on every slot, through the spin representation $\frac14\rho\circ q$ on Σ and Σ^* [ESW, (29), (32)];*
- $[v.,Q]=0$*,* $[v.,Q^{GF}]=0$*,* $[v.,H]=0$*, and $v.$ preserves $\langle\,,\rangle$;*
- *for every $t>0$,*

$$(v.\otimes 1+1\otimes v.)K_t=0,\qquad (v.\otimes 1+1\otimes v.)L_t=0,\qquad (v.\otimes 1+1\otimes v.)P(\varepsilon,\infty)=0;$$

- *the derivation $\{S^{(1)}(x_1),-\}$ commutes with $\partial_{P(\varepsilon,\infty)}$ and with Δ_ε, and for every even functional J, at least cubic modulo $\hbar$,*

$$\{S^{(1)}(x_1),W(P(\varepsilon,\infty),J)\} = \partial_\delta W\big(P(\varepsilon,\infty),\, J+\delta\{S^{(1)}(x_1),J\}\big), \tag{7.1}$$

with δ a square-zero parameter of degree 0.

Proof. First item: a translation is constant, a rotation of V is the linear vector field of an antisymmetric endomorphism of V, and an element of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ is linear and antisymmetric on $W_{\mathbb{R}}^{\perp}$; each is divergence free, so the Weyl-weight terms of Definition 7.4 vanish, and the remaining terms are the Lie derivative of the components of A (for $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ the rotation of the index w of $\iota_w A$) and the spin-lifted Lie derivative $v(\lambda)+\frac14\rho(d(v,-))\lambda$. These actions preserve dvol and the fibre pairings, so on the antifields the condition of Definition 7.4 returns the same natural actions.
Second item: for a translation, $v.e$ is the constant-coefficient derivative $v_1(e)$ on every slot, which commutes with every constant-coefficient operator. For a rotation of V or of $W_{\mathbb{R}}^{\perp}$, the exterior derivative d, its adjoint d^*, the Clifford multiplication ρ (equivariant for the spin representation, [ESW, §2.1]) and the metric of V_{10} are invariant. The operator Q is built from d, d^* and $\not{d}=\rho\circ d$ (Chapter 1), and Q^{GF} from d, d^*, $\not{d}|_{\Sigma^*}$ and the identification $A^*\mapsto -2A^*$ through the metric of V_{10} (Chapter 3); both commute with $v.$, hence so does $H=QQ^{GF}+Q^{GF}Q$. The pairing is preserved by construction.
Third item: as for Q in Chapter 3, preservation of the pairing by the even operator $v.$ is $(v.\otimes 1+1\otimes v.)K_0=0$, and $K_t=(e^{-tH}\otimes 1)K_0$ with $[v.,H]=0$ gives the first identity. Then

$$(v.\otimes 1+1\otimes v.)(Q^{GF}\otimes 1)K_t = ([v.,Q^{GF}]\otimes 1)K_t + (Q^{GF}\otimes 1)(v.\otimes 1+1\otimes v.)K_t = 0,$$

and integration over $t\in(\varepsilon,\infty)$ gives the third.
Fourth item: by the computation of $[\partial_\phi,Q]=\partial_{Q\phi}$ in [C07, §6.4], with the even operator $v.$ in place of Q, the commutator of ∂_ϕ with the derivation induced by $v.$ is the operator ∂ of the kernel $(v.\otimes 1+1\otimes v.)\phi$; for $\phi=P(\varepsilon,\infty)$ and $\phi=K_\varepsilon$ this vanishes by the third item. Hence $\{S^{(1)}(x_1),-\}$ commutes with $e^{\hbar\partial_{P(\varepsilon,\infty)}}$, and, being a derivation of the product [C07, Lemma 4.3.2] of even degree ($S^{(1)}(x_1)$ is odd and the bracket is odd),

$$\{S^{(1)}(x_1),e^{\hbar\partial_{P(\varepsilon,\infty)}}e^{J/\hbar}\} = e^{\hbar\partial_{P(\varepsilon,\infty)}}\Big(\hbar^{-1}\{S^{(1)}(x_1),J\}\,e^{J/\hbar}\Big).$$

Dividing by $\hbar^{-1}e^{W(P(\varepsilon,\infty),J)/\hbar}$ gives (7.1), whose right side is $e^{-W(P(\varepsilon,\infty),J)/\hbar}e^{\hbar\partial_{P(\varepsilon,\infty)}}\big(\{S^{(1)}(x_1),J\}e^{J/\hbar}\big)$. □

Corollary 7.7 (Invariant counterterms, equivariant operators)**.** *For x_1 as in Proposition 7.6:*

- $\{S^{(1)}(x_1), I\} = 0$ *and* $\{S^{(1)}(x_1), I^{CT}[\varepsilon]\} = 0$*;*
- *for every local functional O, the counterterm $\partial_\delta I^{CT}[\varepsilon]$ of the family $I + \delta\{S^{(1)}(x_1), O\}$ equals $\{S^{(1)}(x_1), -\}$ applied to the counterterm $\partial_\delta I^{CT}[\varepsilon]$ of the family $I + \delta O$;*
- *consequently the arity-zero parts satisfy*

$$\big(\{S^{(1)}(x_1), O\}\big)(\varepsilon) = \{S^{(1)}(x_1), O(\varepsilon)\}, \qquad \big(\{S^{(1)}(x_1), O\}\big)[\infty] = \{S^{(1)}(x_1), O[\infty]\}.$$

Proof. The arity-one equation $\{S, S^{(1)}(x_1)\} = 0$ holds for these generators and every $\mathfrak{g}$ (Chapter 2). Its part of polynomial degree two is $\{\frac{1}{2}\langle e, Qe\rangle, S^{(1)}(x_1)\} = 0$, since $S^{(1)}(x_1)$ is quadratic and $\{I, S^{(1)}(x_1)\}$ has degree at least three; the remaining part is $\{I, S^{(1)}(x_1)\} = 0$, which by graded symmetry is the first identity. The counterterms are the singular parts, order by order and graph by graph, of $W(P(\varepsilon,\infty), J)$, for J equal to I minus the counterterms of lower order and for $J = I - I^{CT}[\varepsilon] + \delta O$ (Chapters 4 and 5). By (7.1), $\{S^{(1)}(x_1), -\}$ maps $W(P(\varepsilon,\infty), J)$ to the linearised flow of $\{S^{(1)}(x_1), J\}$, and it acts on the fields and leaves the coefficient functions of ε untouched, so it commutes with taking the singular part. Induction on the order gives $\{S^{(1)}(x_1), I^{CT}[\varepsilon]\} = 0$, and the same argument at first order in δ, with $\{S^{(1)}(x_1), \delta O\} = \delta\{S^{(1)}(x_1), O\}$ (the bracket with the odd functional $S^{(1)}(x_1)$ has even degree), gives the second item. The third follows from the definitions of $O(\varepsilon)$ and $O[\infty]$ and from (7.1) once more; the limit $\varepsilon \to 0$ commutes with the ε-independent derivation. □

Corollary 7.8 (No mixing between representations)**.** *Let $O = \int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$ with $w_1, w_2 \in W_{\mathbb{R}}^{\perp}$, and let w run over an orthonormal basis of $W_{\mathbb{R}}^{\perp}$. The one-loop counterterm $\partial_\delta I^{CT}[\varepsilon]$ of the family $I + \delta O$ is a linear combination of the four functionals*

$$\int \omega\,(\iota_{w_1}A, \iota_{w_2}A), \qquad (w_1, w_2)\sum_w \int \omega\,(\iota_w A, \iota_w A), \qquad (w_1, w_2)\int \omega\,(A|_V, A|_V), \qquad (w_1, w_2)\int \omega,$$

with coefficients independent of w_1, w_2, the first three multiples of $\log\varepsilon$ and the fourth a multiple of ε^{-1}. In particular the traceless part of O is renormalised into itself, and the trace part into the span of $\sum_w \int \omega\,(\iota_w A, \iota_w A)$, $\int \omega\,(A|_V, A|_V)$ and $\int \omega$. The component of the quadratic counterterm $I^{CT}[\varepsilon]$ pairing $A|_V$ with $A|_{W_{\mathbb{R}}^{\perp}}$ vanishes.

Proof. By Corollary 7.7 the map $O \mapsto \partial_\delta I^{CT}[\varepsilon]$ is equivariant for translations, $\mathfrak{so}(V)$ and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$; it is also equivariant for the adjoint action of $\mathfrak{g}$, which commutes with Q, Q^{GF}, H and preserves $\langle\,,\rangle$ and I, by the same argument. At one loop the coefficients lie in the span of ε^{-1} and $\log\varepsilon$, and under the scaling of Chapter 6, which multiplies ε in $P(\varepsilon,\infty)$ by the square of the scale factor, ε^{-1} multiplies a density of Weyl weight 0 and $\log\varepsilon$ a density of Weyl weight 2, the weight of $(\iota_{w_1}A, \iota_{w_2}A)$. The densities of weight 0 and 2, of ghost degree 0, invariant under $\mathfrak{g}$, $\mathfrak{so}(V)$ and translations, are the constant 1 (weight 0) and $(A|_V, A|_V)$, $(\iota_w A, \iota_{w'}A)$ with $w, w' \in W_{\mathbb{R}}^{\perp}$ (weight 2): a $\mathfrak{g}$-invariant needs two adjoint fields, a ghost c of degree $+1$ needs an antifield of weight at least $\frac{5}{2}$, and a derivative on a single field leaves it in the adjoint. As a representation of $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, $\mathrm{Sym}^2 W_{\mathbb{R}}^{\perp}$ is the sum of the irreducible traceless part and the trace line spanned by $\sum_w w \otimes w$; the span of the densities $(\iota_w A, \iota_{w'}A)$ is a copy of $\mathrm{Sym}^2 W_{\mathbb{R}}^{\perp}$, and $(A|_V, A|_V)$ and 1 are invariant. By Schur's lemma an equivariant linear map out of $\mathrm{Sym}^2 W_{\mathbb{R}}^{\perp}$ is a multiple of the identity on the traceless part and maps the trace line into the invariants, which gives the four functionals. For $I^{CT}[\varepsilon]$, a bilinear invariant of a vector of V and a vector of $W_{\mathbb{R}}^{\perp}$ is 0. □

The coefficients computed in Chapter 5 are

$$\partial_\delta I^{CT}[\varepsilon] = \frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\Big[4\int\omega\,(\iota_{w_1}A,\iota_{w_2}A) - 2(w_1,w_2)\sum_w\int\omega\,(\iota_wA,\iota_wA)\Big] + \frac{2\hbar\dim\mathfrak{g}\,(w_1,w_2)}{16\pi^2\varepsilon}\int\omega,$$

with coefficient 0 for $(w_1,w_2)\int\omega\,(A|_V,A|_V)$, and the quadratic $I^{CT}[\varepsilon]$ of Chapter 4 has vanishing $A|_V$–$A|_{W_{\mathbb{R}}^\perp}$ component, as Corollary 7.8 requires. Here $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(X,Y)$ is independent of $X,Y\in\mathfrak{g}$ for simple $\mathfrak{g}$.

Remark 7.9 (Internal insertions and the status of these components)**.** For x_1 as in Proposition 7.6, the arity-one part of $O[\infty]$ along x_1 consists of the classical tree with the vertex $S^{(1)}(x_1)$ joined to O, the graphs in which $S^{(1)}(x_1)$ sits on an internal line, and the arity-one counterterms along x_1. On an internal line, $(v.A,A^*)$ is the term of $S^{(1)}(x_1)$ with both legs joined by $P(\varepsilon,\infty)$ (every component of the propagator lies in the slots of A, A^*, c and λ), its A leg to an $A\otimes A$ component and its A^* leg to an $A^*\otimes c$ component. The resulting kernel is that of $P(\varepsilon,\infty)\star\circ v.\circ P(\varepsilon,\infty)\star = v.\circ\big(P(\varepsilon,\infty)\star\big)^2$, and since $[Q^{GF},H]=0$,

$$\big(P(\varepsilon,\infty)\star\big)^2 = (Q^{GF})^2\Big(\int_\varepsilon^\infty e^{-tH}dt\Big)^2,\qquad (Q^{GF})^2A^* = 2\,d^*(A^*|_V)\ \ \text{(in the slot of } c),$$

which is nonzero (Chapter 3). These graphs exist because $(Q^{GF})^2\neq 0$. By Proposition 7.6 every kernel, vertex and counterterm entering them is x_1-covariant; this is the reason these components can be chosen with no one-loop correction for O with $(Q+\{I,-\})O=0$,

$$[(d_{CE}+Q)O[\infty]](x_1) = \big(\{S^{(1)}(x_1),O\}\big)[\infty],$$

the classical transformation of O carried to scale ∞. For general O the right side acquires, already at order $\hbar^0$, the trees with one vertex $(Q+\{I,-\})O$ and one vertex $S^{(1)}(x_1)$, as in Proposition 6.18; for the letter $A|_V$ and a translation they are nonzero. The arity-one counterterms along x_1 and the internal-insertion graphs, which realise this choice, are evaluated in Step 2.

7.3 The generators that move the regulator

Proposition 7.10 (Dilatation and special conformal transformations)**.** *Let x_1 be the dilatation or a special conformal generator, with vector field v. Then $[v.,Q]=0$, and on the slot of $A^*|_{W_{\mathbb{R}}^\perp}$*

$$[v.,Q^{GF}]\,e_1 = (\mathrm{div}\,v)\,e_1\quad (\textit{in the slot of } A|_{W_{\mathbb{R}}^\perp}),\tag{7.2}$$

with $\mathrm{div}(2E)=8$ *and* $\mathrm{div}\,v=-8(v_1,x)$ *for* $v=\big(x\mapsto(x,x)v_1-2(v_1,x)x\big)$*. For the dilatation, $v=2E$, on every slot*

$$[v.,Q^{GF}] = -4\,Q^{GF},\qquad [v.,H]=-4\,H,\qquad (v.\otimes 1+1\otimes v.)P(\varepsilon,\infty) = 4\varepsilon L_\varepsilon = -4\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty).$$

Among the even generators, (7.2) *vanishes exactly for those with* $\mathrm{div}\,v=0$*: translations, rotations and* $\mathfrak{so}(W_{\mathbb{R}}^\perp)$*.*

Proof. $[v.,Q]=0$ is the part of polynomial degree two of the arity-one equation $\{S,S^{(1)}(x_1)\}=0$ (Chapter 2). On the slot of $A^*|_{W_{\mathbb{R}}^\perp}$, $Q^{GF}e_1=-2e_1$ in the slot of $A|_{W_{\mathbb{R}}^\perp}$ (Chapter 3); by Definition

7.4 and Lemma 7.5, $v.$ acts there by $v(-)+\frac{1}{4}\operatorname{div} v$ and $v(-)+\frac{3}{4}\operatorname{div} v$ respectively, plus the same rotation of the index for both. Hence

$$[v., Q^{GF}]e_1 = -2\big(v(e_1)+\tfrac{1}{4}(\operatorname{div} v)e_1\big)+2\big(v(e_1)+\tfrac{3}{4}(\operatorname{div} v)e_1\big) = (\operatorname{div} v)\,e_1.$$

The divergences are those of Chapter 2. For $v = 2E$ use Lemma 7.5 on each arrow of Q^{GF}, together with the homogeneity relations $[E,d]=-d$, $[E,d^*]=-d^*$ and $[E,\not{d}]=-\not{d}$ on componentwise E (each lowers the degree of homogeneity in x by one):

$$\begin{aligned}
A^*\mapsto -2A^*:&\quad (2E+2)(-2A^*)+2(2E+6)A^* = 8A^* = -4(-2A^*),\\
A\mapsto -d^*(A|_V):&\quad -2E\,d^*A + d^*(2E+2)A = 2d^*A + 2d^*A = -4\big(-d^*A\big),\\
c^*\mapsto dc^*:&\quad (2E+6)dc^* - d(2E+8)c^* = 2[E,d]c^* + 6dc^* - 8dc^* = -4\,dc^*,\\
\lambda^*\mapsto -\not{d}|_{\Sigma^*}\lambda^*:&\quad -(2E+3)\,\not{d}\lambda^* + \not{d}(2E+5)\lambda^* = 2\,\not{d}\lambda^* - 3\,\not{d}\lambda^* + 5\,\not{d}\lambda^* = -4\big(-\not{d}\lambda^*\big),
\end{aligned}$$

where in each line $A|_V$, $A^*|_V$ are handled componentwise (Lemma 7.5). Similarly $[2E, d^*d] = -4\,d^*d$ and the weight terms are constants, so $[v., H] = -4H$. Consequently $[v., e^{-tH}] = 4tHe^{-tH} = -4t\frac{d}{dt}e^{-tH}$; since $v.$ preserves the pairing, $(v.\otimes 1 + 1\otimes v.)K_t = ([v., e^{-tH}]\otimes 1)K_0 = -4t\frac{d}{dt}K_t$, and

$$(v.\otimes 1 + 1\otimes v.)L_t = -4L_t - 4t\frac{d}{dt}L_t.$$

Integrating over $t\in(\varepsilon,\infty)$, with $tL_t\to 0$ as $t\to\infty$ for $p\neq 0$,

$$(v.\otimes 1 + 1\otimes v.)P(\varepsilon,\infty) = -4\int_\varepsilon^\infty L_t\,dt - 4\big[tL_t\big]_\varepsilon^\infty + 4\int_\varepsilon^\infty L_t\,dt = 4\varepsilon L_\varepsilon,$$

which equals $-4\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty)$ by Chapter 3. The last statement holds because the even generators are the conformal Killing fields and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, and among the conformal Killing fields of Chapter 2 the divergence vanishes exactly on translations and rotations. □

The identity $(v.\otimes 1 + 1\otimes v.)P(\varepsilon,\infty) = -4\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty)$ is the infinitesimal form of the scaling covariance used in Chapter 6: the dilatation moves the regulator along its own scale.

Proposition 7.11 (Supercharges and special supercharges)**.** *Let $Q_1\in\Sigma$ be nonzero and write $Q_1.$, in the notation $v.$ of [ESW, (32)], for the odd linear operator on $\mathscr{E}$ that induces, as in [C07, §6.2], the derivation given by the bracket with the quadratic part*

$$\int \operatorname{dvol}\Big(-(\Gamma(Q_1,\lambda),A^*)+\tfrac{1}{2}(\rho(dA)Q_1,\lambda^*)\Big)$$

of $S^{(1)}(Q_1)$. Then:

- *$[Q_1., Q]=0$, $[Q_1., H]=0$, and $Q_1.$ preserves $\langle\,,\rangle$;*
- *the graded commutator $[Q_1., Q^{GF}]$ is nonzero; its component from the slot of A^* to the slot of λ is nonzero;*
- *consequently*
$$(Q_1.\otimes 1 + 1\otimes Q_1.)P(\varepsilon,\infty) = \int_\varepsilon^\infty \big([Q_1., Q^{GF}]\otimes 1\big)K_t\,dt \neq 0;$$
- *$S^{(1)}(Q_1)$ contains the cubic term $\frac{1}{2}\big(\rho(\frac{1}{2}[A\wedge A])Q_1,\lambda^*\big)$.*

For $Q_2\in\Sigma^*$, the corresponding operator contains multiplication by the function $x\mapsto\Gamma(\rho(x)Q_2,-)$, linear in x, and fails to commute with H.

Proof. First item: $[Q_1., Q] = 0$ is the part of polynomial degree two of $\{S, S^{(1)}(Q_1)\} = 0$ (Chapter 2); $Q_1.$ is a differential operator of order at most one with constant coefficients, and H is the scalar Laplacian d^*d on every component (Chapter 3), so they commute; the derivation of a quadratic functional preserves the constant pairing.
Second item. The term $-(\Gamma(Q_1, \lambda), A^*) = -(\rho(A^*)Q_1, \lambda)$ [ESW, (28)] pairs the slots of λ and A^*; it contributes to $Q_1.$ an arrow from λ to A and an arrow from A^* to λ^*, the latter a nonzero multiple of $A^* \mapsto \rho(A^*)Q_1$. The term $\frac{1}{2}(\rho(dA)Q_1, \lambda^*)$ contributes an arrow from A to λ, a nonzero multiple of $A \mapsto \rho(dA)Q_1$, and an arrow from λ^* to A^*. The operator Q^{GF} maps $A^* \mapsto -2A^*$ into the slot of A and $\lambda^* \mapsto -\not{d}|_{\Sigma^*}\lambda^*$ into the slot of λ (Chapter 3). Hence the component of $[Q_1., Q^{GF}]$ from A^* to λ is the sum of the two compositions $A^* \to A \to \lambda$ and $A^* \to \lambda^* \to \lambda$. On a plane wave $A^* = a\, e^{i(p,x)}$ with $p, a \in V$ real and nonzero, and with $\rho(q(p \wedge a)) = \rho(p)\rho(a) - (p, a)$ [ESW, (29)], these are nonzero multiples, fixed by the conventions, of

$$\big(\rho(p)\rho(a) - (p, a)\big)Q_1\, e^{i(p,x)} \qquad \text{and} \qquad \rho(p)\rho(a)Q_1\, e^{i(p,x)}.$$

Suppose their sum, with the sign of the graded commutator, vanished for all such p, a. When p and a are orthogonal, $(p, a) = 0$, the product $\rho(p)\rho(a)$ is invertible ($\rho(a)\rho(p)\rho(p)\rho(a) = |p|^2|a|^2$), and the two multiples would be opposite; at $a = p$ the sum would then reduce to the first multiple times $-|p|^2Q_1$, which is nonzero. Hence the component is nonzero.
Third item: as in Proposition 7.6, with $Q_1.$ odd, $(1 \otimes Q_1.)(Q^{GF} \otimes 1) = -(Q^{GF} \otimes 1)(1 \otimes Q_1.)$ and $(1 \otimes Q_1.)K_t = -(Q_1. \otimes 1)K_t$ by the first item, so $(Q_1. \otimes 1 + 1 \otimes Q_1.)L_t = \big((Q_1.Q^{GF} + Q^{GF}Q_1.) \otimes 1\big)K_t$. The integral is nonzero because $\int_\varepsilon^\infty e^{-tH}dt$ is $e^{-\varepsilon|p|^2}/|p|^2 \neq 0$ in momentum space and commutes with $[Q_1., Q^{GF}]$.
Fourth item: $F_A = dA + \frac{1}{2}[A \wedge A]$ in $\frac{1}{2}(\rho(F_A)Q_1, \lambda^*)$. For Q_2: the term $-(\Gamma(\rho(x)Q_2, \lambda), A^*)$ of $S^{(1)}(Q_2)$ (Chapter 2) gives the stated multiplication operator, and for a function f_1 linear in x, $d^*d(f_1 f_2) - f_1\, d^*df_2 = -2(df_1, df_2)$, which is nonzero for nonconstant f_1. □

Remark 7.12 (Large-distance behaviour of internal insertions)**.** The coefficients of $S^{(1)}(x_1)$ grow polynomially in the position x for the moving generators: linearly for Δ and for $Q_2 \in \Sigma^*$, quadratically for the special conformal fields. On an internal line, an insertion of $(v.A, A^*)$ at a point y is joined to the rest of the graph by an $A \otimes A$ component of $P(\varepsilon, \infty)$, of order $|y|^{-2}$ at large $|y|$, differentiated once by $v.$, and by an $A^* \otimes c$ component, of order $|y|^{-3}$ (Chapter 3). For the special conformal field the integrand is of order $|y|^2 \cdot |y|^{-3} \cdot |y|^{-3} = |y|^{-4}$, the borderline of integrability over $\mathbb{R}^4$; for Δ, and for Q_2 with its λ and A^* legs, it is of order $|y|^{-5}$. The convergence at large $|y|$ of the special conformal internal insertions, with no infrared scale available, is part of Step 2.

7.4 Allowed output structures

Throughout this section $O = \int \omega\, f$ with ω a compactly supported four-form [CG2, Lemma 10.1.1.3] and f a polynomial in the jets of the fields at a point. The Weyl weight of a monomial is the sum of the weights of its fields (A: 1, λ: $\frac{3}{2}$, c: 0, A^*: 3, λ^*: $\frac{5}{2}$, c^*: 4, Lemma 7.5) plus the number of derivatives; its ghost degree is its degree as a functional, the sum of the degrees of its factors (c: $+1$; A, λ: 0; A^*, λ^*: -1; c^*: -2, as in the proof of Lemma 4.19); it is a colour singlet if it is invariant under the adjoint action of $\mathfrak{g}$ on all fields. O has Weyl weight w_O if f is homogeneous of that weight.

Proposition 7.13 (Selection rules)**.** *Let $O = \int \omega\, f$ be a colour singlet of Weyl weight w_O and ghost degree $|O|$, and let x_1 be an odd generator, realised by the conformal Killing spinor $x \mapsto Q_1 + \rho(x)Q_2$ with $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$ (Chapter 2), or a special conformal generator, with vector field $v = \big(x \mapsto (x, x)v_1 - 2(v_1, x)x\big)$. Assume that the counterterms of Step 2 are chosen equivariantly for translations,*

$\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ *and* $\mathfrak{g}$, *which Proposition 7.6 permits, and that the component along* x_1 *is represented by a local functional* $\int \omega\, f_{x_1}$. *Then* f_{x_1} *can be chosen with the following properties.*

- *Ghost degree* $|O|$, *colour singlet.*
- *At a point* $x_0 \in V$, f_{x_1} *depends on* x_1 *through its conformal Killing data at* x_0*: for an odd generator,* f_{x_1} *is the sum of a term linear in the value* $Q_1 + \rho(x_0)Q_2$, *of Weyl weight* $w_O + \frac{1}{2}$, *and a term linear in the Dirac derivative* $\not{d}(Q_1 + \rho(x)Q_2) = 4Q_2$, *of weight* $w_O - \frac{1}{2}$*; for a special conformal generator,* f_{x_1} *is the sum of terms linear in* $v(x_0)$, *in the first derivative of* v *at* x_0 *and in its second derivatives, of weights* $w_O + 1$, w_O *and* $w_O - 1$.
- *The map* $(x_1, O) \mapsto f_{x_1}$ *is equivariant for translations,* $\mathfrak{so}(V)$ *and* $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, *with* x_1 *transforming by the bracket of* $\mathfrak{psl}(4|4)$.

Proof. Ghost degree: $(d_{CE} + Q)O[\infty]$ has degree $|O| + 1$, and evaluating an arity-one cochain on a generator, whose ghost coordinate has degree 1, lowers the ghost degree by one. Colour and equivariance: the adjoint action of $\mathfrak{g}$ and the action of the covariant subalgebra commute with Q, Q^{GF}, H and preserve the pairing (Proposition 7.6); the classical action $S + S^{(1)} + S^{(2)}$ is invariant under the simultaneous action on fields and on the arguments of the cochains, by the arity-two components of its master equation with one argument in the covariant subalgebra (Chapter 2); with equivariant counterterms every graph sum is equivariant, as in Corollary 7.7. Dependence on x_1: the vertices $S^{(1)}(x_1)$ and $S^{(2)}(x_1, -)$ depend on x_1 through the spinor $Q_1 + \rho(x)Q_2$ and its Dirac derivative $4Q_2$ (the last term of $S^{(1)}(Q_2)$ equals $-\frac{1}{4}(\rho(A|_{W_{\mathbb{R}}^{\perp}})\not{d}(\rho(x)Q_2), \lambda^*)$, Chapter 2), resp. through v, dv and $\operatorname{div} v$; the spinor has vanishing second derivatives and derivative $\rho(v_2)Q_2 = \frac{1}{4}\rho(v_2)\not{d}(Q_1 + \rho(x)Q_2)$ along $v_2 \in V$, and a conformal Killing field of V is a polynomial of degree two, so the data at a vertex point y are polynomials in $y - x_0$ with coefficients the data at x_0 listed in the statement. Weights: give the conformal Killing data the weights that make each vertex $S^{(1)}(x_1)$, $S^{(2)}(x_1, -)$ homogeneous of the weight 4 of the densities of the action: $-\frac{1}{2}$ to the value of the conformal Killing spinor and $+\frac{1}{2}$ to its Dirac derivative (in $(\Gamma(Q_1, \lambda), A^*)$ the fields have weights $\frac{3}{2} + 3$), and -1, 0, $+1$ to the value, the first and the second derivatives of a conformal Killing field (in $(L_v A, A^*)$ the derivative and the fields have weights $1 + 1 + 3$). Under the scaling of Chapter 6, which multiplies each field by the power of the scale factor given by its Weyl weight and ε in $P(\varepsilon, \infty)$ by the square of the scale factor, every vertex is then homogeneous, and the output has the weight of O; each datum therefore multiplies a density whose weight is w_O minus the weight of the datum, which gives the weights listed. A local term whose weight differs from the stated one by $2k$ carries a coefficient ε^k, possibly times powers of $\log \varepsilon$; for $k > 0$ it vanishes as $\varepsilon \to 0$, and singular coefficients are excluded by the existence of the limit defining $\big(\int \omega\, f_{x_1}\big)[\infty]$. □

Corollary 7.14 (Supercharges on the scalar bilinears)**.** *Let* $O = \int \omega\, (\iota_{w_1} A, \iota_{w_2} A)$ *(*$|O| = 0$, $w_O = 2$*) and* $Q_1 \in S_+ \otimes W$ *(the generators* Q_α^i *of [SW, (8)]) or* $Q_1 \in S_- \otimes W^*$ *(the generators* $\bar{Q}_{\dot{\alpha} i}$*). Under the hypotheses of Proposition 7.13,* f_{Q_1} *is a colour-singlet density of Weyl weight* $\frac{5}{2}$ *and ghost degree* 0, *odd in the fields, linear in* Q_1, *symmetric in* (w_1, w_2), *without derivatives, and* $\int \omega\, f_{Q_1}$ *lies in the span of the following four functionals, grouped in three types:*

- *type* $(A|_{W_{\mathbb{R}}^{\perp}}, \lambda)$ *(classical type):*

$$\int \omega \Big[(\iota_{w_1} A, (\rho(w_2)Q_1, \lambda)) + (\iota_{w_2} A, (\rho(w_1)Q_1, \lambda))\Big], \qquad (w_1, w_2) \int \omega\, (\rho(A|_{W_{\mathbb{R}}^{\perp}})Q_1, \lambda);$$

- *type* $(A|_V, \lambda)$ *(gauge-variant type):* $(w_1, w_2) \int \omega\, (\rho(A|_V)Q_1, \lambda)$*;*
- *type* (c, λ^*) *(BRST type):* $(w_1, w_2) \int \omega\, (c, (Q_1, \lambda^*))$.

Here $(\rho(w)Q_1,\lambda)$, (Q_1,λ^*) use the pairing $\Sigma^*\otimes\Sigma\to\mathbb{C}$ and are $\mathfrak{g}$-valued, paired with the remaining $\mathfrak{g}$-valued field through $(-,-)$, and $\rho(A|_V)$, $\rho(A|_{W_\mathbb{R}^\perp})$ are Clifford multiplication by the components of A with their $\mathfrak{g}$ factor kept. In particular, for $(w_1,w_2)=0$, where O lies in the traceless part, $\int\omega\, f_{Q_1}$ is a multiple of the first functional, which is the type of the classical component $\{S^{(1)}(Q_1),O\}$; the gauge-variant and BRST types occur in the trace part $\sum_w\int\omega\,(\iota_wA,\iota_wA)$.

Proof. By Proposition 7.13, $\not{d}Q_1=0$ and f_{Q_1} has weight $\frac{5}{2}$ and ghost degree 0. Field content: a half-integral weight requires an odd number of fields among λ,λ^*, the fields of half-integral weight. With one λ^* (weight $\frac{5}{2}$, degree -1) the rest has weight 0 and degree $+1$: one c. With one λ the rest has weight 1 and degree 0: one A, or one derivative; a single λ with a derivative is in the adjoint of $\mathfrak{g}$, and a colour singlet needs two adjoint fields joined by $(-,-)$, the invariant symmetric form of the simple $\mathfrak{g}$. Three spinor fields have weight at least $\frac{9}{2}$. Hence the types (c,λ^*), $(A|_V,\lambda)$, $(A|_{W_\mathbb{R}^\perp},\lambda)$, without derivatives.
Representations, for $Q_1\in S_+\otimes W$ (the case $S_-\otimes W^*$ is the same with $S_\pm$ and W, W^* exchanged). Under $\mathfrak{so}(W_\mathbb{R}^\perp)\cong\mathfrak{sl}(W)$ [ESW, §10.3], $\mathrm{End}(W)=W\otimes W^*\cong\mathbb{C}\oplus\wedge^2W_\mathbb{R}^\perp$ (dimensions $1+15$), and $\mathrm{Sym}^2W_\mathbb{R}^\perp$ is the sum of its irreducible traceless part, of dimension 20, and the trace line.

- Type (c,λ^*): c is a Lorentz and R-singlet, so the structure is $(c,(B\,Q_1,\lambda^*))$ with B an endomorphism of Σ depending bilinearly and equivariantly on (w_1,w_2). Lorentz invariance pairs Q_1 with the $S_+\otimes W^*$ component of λ^*, so B restricts to an element of $\mathrm{End}(W)$; by Schur's lemma the traceless part of $\mathrm{Sym}^2W_\mathbb{R}^\perp$, of dimension 20, has no image in $\mathbb{C}\oplus\wedge^2W_\mathbb{R}^\perp$, and the trace line maps to the identity: $B=(w_1,w_2)$ times a constant.

- Type $(A|_V,\lambda)$: $A|_V$ is an R-singlet and a vector of $V\cong S_+\otimes S_-$; Lorentz invariance pairs $\rho(A|_V)Q_1\in S_-\otimes W$ with the $S_-\otimes W^*$ component of λ, and the R-dependence is again an equivariant map from $\mathrm{Sym}^2W_\mathbb{R}^\perp$ to $\mathrm{End}(W)$: a multiple of (w_1,w_2).

- Type $(A|_{W_\mathbb{R}^\perp},\lambda)$: Lorentz invariance pairs Q_1 with the $S_+\otimes W$ component of λ through an odd element of the Clifford algebra of $W_\mathbb{R}^\perp$ built from the index w of ι_wA and from w_1,w_2. Equivariant multilinear expressions in w,w_1,w_2 with values in this Clifford algebra are combinations of products of $\rho(w),\rho(w_1),\rho(w_2)$, of the metric and of the volume element of $W_\mathbb{R}^\perp$, and the volume element acts on each of $S_+\otimes W$, $S_-\otimes W^*$ by a scalar (Chapter 1: Σ is a semi-spin representation and the four-dimensional chirality is fixed on each summand); the odd ones symmetric in (w_1,w_2) reduce, by $\rho(w_1)\rho(w_2)+\rho(w_2)\rho(w_1)=2(w_1,w_2)$ and $\rho(w)\rho(w_i)+\rho(w_i)\rho(w)=2(w,w_i)$, to the span of $(w,w_1)\rho(w_2)+(w,w_2)\rho(w_1)$ and $(w_1,w_2)\rho(w)$; contracting w with ι_wA gives the two functionals listed, using $\sum_w\iota_wA\,\rho(w)=\rho(A|_{W_\mathbb{R}^\perp})$. The term $\rho(q(w\wedge w_1\wedge w_2))$ is antisymmetric in (w_1,w_2).

For $(w_1,w_2)=0$ every functional with the factor (w_1,w_2) vanishes. The classical component replaces A by its transformation $\Gamma(Q_1,\lambda)$ in O (Chapter 2), and $\iota_w\Gamma(Q_1,\lambda)=(w,\Gamma(Q_1,\lambda))=(\rho(w)Q_1,\lambda)$ by [ESW, (28)], which gives the first functional. □

Corollary 7.15 (Special supercharges on the scalar bilinears)**.** *Let O be as in Corollary 7.14 and $Q_2\in S_+\otimes W^*$ (the generators S_i^α) or $Q_2\in S_-\otimes W$ (the generators $\bar S^{\dot\alpha i}$). Under the hypotheses of Proposition 7.13, $\int\omega\, f_{Q_2}$ lies in the span of the four functionals of Corollary 7.14 with Q_1 replaced by the value $\rho(x)Q_2$ of the conformal Killing spinor, which lies in $S_-\otimes W^*$, resp. $S_+\otimes W$. The term linear in $\not{d}(\rho(x)Q_2)=4Q_2$ is absent: it would require a colour-singlet density of Weyl weight $\frac{3}{2}$ and ghost degree 0, and the one field of weight $\frac{3}{2}$, λ, is in the adjoint of $\mathfrak{g}$ while every field of weight 0 has ghost degree $+1$.*

The coefficients of the four functionals along Q_2 are independent data; they are computed in Step

3 together with those along Q_1.

Corollary 7.16 (Special conformal components)**.** *Let $v(x) = (x,x)v_1 - 2(v_1,x)x$. Its first derivative at x_0 is $\frac{1}{4}\operatorname{div} v(x_0)\cdot\mathrm{id}$ plus an antisymmetric endomorphism of V, with $\frac{1}{4}\operatorname{div} v(x_0) = -2(v_1,x_0)$, and its second derivatives are constant and linear in v_1. Under the hypotheses of Proposition 7.13, for O a scalar bilinear or a single letter $\iota_w A$, $A|_V$, λ, c (with its colour index fixed by an element of $\mathfrak{g}$):*

- *the term linear in the second derivatives of v is absent: it would require a vector density of weight $w_O - 1$, ghost degree $|O|$ and the colour of O; for a scalar bilinear this is a colour-singlet vector of weight 1, and the field of weight 1 with a vector index, $A|_V$, is in the adjoint; for the letters the weight $w_O - 1 \in \{0, 0, \frac{1}{2}, -1\}$ with ghost degree $|O| \in \{0,0,0,1\}$ admits no monomial;*
- *the term linear in the first derivative is $\frac{1}{4}\operatorname{div} v(x_0)$ times a density of weight w_O with the R-representation of O, plus a density of weight w_O contracted with the antisymmetric part of $dv(x_0)$. For a scalar bilinear the latter lies in $\wedge^2 V$ and vanishes, since the one candidate $(A|_V \wedge A|_V)$ vanishes by the symmetry of $(-,-)$; for the letters $\iota_w A$ and c, which carry no index of V or of $S_\pm$, it vanishes. For the letter $\int\omega\,(X, \iota_{v_2}(A|_V))$, $v_2 \in V$, it is the rotation of the vector index, $\int\omega\,\big(X, \iota_{v_3}(A|_V)\big)$ with v_3 the antisymmetric part of $dv(x_0)$ applied to v_2; for the letter $\int\omega\,(\zeta,\lambda)$ it is the rotation of the spinor index, $\int\omega\,\big(\zeta, \frac{1}{4}\rho(d(v,-)(x_0))\lambda\big)$ [ESW, (29)]. Both are present classically, through $L_v A$ and $\frac{1}{4}\rho(d(v,-))\lambda$ in $S^{(1)}(v)$ (Definition 2.12), and are allowed by every selection rule;*
- *for a scalar bilinear with $(w_1, w_2) = 0$, the density of weight 2 in the first-derivative term is a multiple of $(\iota_{w_1}A, \iota_{w_2}A)$; for the trace it lies in the span of $\sum_w(\iota_w A, \iota_w A)$ and $(A|_V, A|_V)$;*
- *the term linear in $v(x_0)$ contracts $v(x_0)$, of degree two in x_0, with a vector density of weight $w_O + 1$.*

Proof. The derivative of v is $dv(x_0)v_2 = 2(x_0,v_2)v_1 - 2(v_1,v_2)x_0 - 2(v_1,x_0)v_2$; its trace is $\operatorname{div} v(x_0) = -8(v_1,x_0)$ and its symmetric traceless part vanishes, which is the conformal Killing equation. The remaining statements are the weight count of Proposition 7.13 with the field weights and degrees listed above, and the enumeration in the proof of Corollary 7.8. □

Remark 7.17 (The special conformal component and the dilatation)**.** The relation between the two components has three parts.

- Classical. At x_0 the one-jet of v is that of the translation by $v(x_0)$, plus a rotation, plus $\frac{1}{8}\operatorname{div} v(x_0)$ times the dilatation $2(x - x_0)$ centred at x_0; correspondingly the Weyl-weight terms of $S^{(1)}(v)$ at x_0, $\frac{1}{4}\operatorname{div} v(x_0)$ times the weight, are $\frac{1}{8}\operatorname{div} v(x_0)$ times those of $S^{(1)}(\Delta)$, twice the weight (Lemma 7.5).
- What the complex gives. The cochain $(d_{CE}+Q)O[\infty]$ has degree $|O|+1$, and $(d_{CE}+Q)\big((d_{CE}+Q)O[\infty]\big) = 0$. For even x_1, x_2 the differential of Chapter 2 gives $\big(d_{CE}(d_{CE}+Q)O[\infty]\big)(x_1,x_2) = (-1)^{|O|+1}[(d_{CE}+Q)O[\infty]]([x_1,x_2])$, and for the special conformal generator K_{v_1} of [SW, (3)] along $v_1 \in V$ and the translation P_{v_2} of [SW, (3)] along $v_2 \in V$, with the bracket of vector fields of Chapter 2,
$$[K_{v_1}, P_{v_2}] = (v_1,v_2)\,\Delta + \big(x \mapsto 2(v_1,x)v_2 - 2(v_2,x)v_1\big),$$
the second summand a rotation. The arity-two component of $(d_{CE}+Q)^2 O[\infty] = 0$ on (K_{v_1}, P_{v_2}) therefore reads
$$\begin{aligned}(-1)^{|O|+1}\Big((v_1,v_2)\,[(d_{CE}+Q)O[\infty]](\Delta) + [(d_{CE}+Q)O[\infty]]\big(x \mapsto 2(v_1,x)v_2 - 2(v_2,x)v_1\big)\Big)\\ = -\big(Q(d_{CE}+Q)O[\infty]\big)(K_{v_1}, P_{v_2}),\end{aligned}$$

which expresses the dilatation component, together with a rotation component, through Q of the arity-two part of $(d_{CE}+Q)O[\infty]$. The special conformal component $[(d_{CE}+Q)O[\infty]](K_{v_1})$ enters through no term.

- The conformal expectation. For O localised at x_0, the conformal algebra suggests that the anomalous part of the special conformal component is $\frac{1}{4}\operatorname{div} v(x_0) = -2(v_1, x_0)$ times the anomalous coefficient of the dilatation component (Chapter 6) times O, i.e. the first-derivative term of Corollary 7.16 with the weight w_O shifted by that coefficient. Its derivation requires the composition of components as a module structure on observables up to homotopy, the translation of O to x_0 by the components of Remark 7.9, and control of the second derivatives of v and of the large-distance behaviour of Remark 7.12 in the K_ε graphs. These are tasks of Steps 2 and 3.

generator	subspace	ad_Δ	weight of output	regulator	one-loop output for $\int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$
translations	V	-2	w_O+1	covariant	can be chosen classical (Step 2)
rotations	$\mathfrak{so}(V)$	0	w_O	covariant	can be chosen classical (Step 2)
R-symmetry	$\mathfrak{so}(W_{\mathbb{R}}^{\perp})$	0	w_O	covariant	can be chosen classical (Step 2)
dilatation Δ	$2E$	0	w_O	$-4\varepsilon\frac{d}{d\varepsilon}$	Chapter 6
supercharges	$S_+ \otimes W \subset \Sigma$	-1	$w_O+\frac{1}{2}$	moved	Corollary 7.14
supercharges	$S_- \otimes W^* \subset \Sigma$	-1	$w_O+\frac{1}{2}$	moved	Corollary 7.14
special supercharges	$S_+ \otimes W^* \subset \Sigma^*$	$+1$	$w_O+\frac{1}{2}$ at $\rho(x)Q_2$	moved	Corollary 7.15
special supercharges	$S_- \otimes W \subset \Sigma^*$	$+1$	$w_O+\frac{1}{2}$ at $\rho(x)Q_2$	moved	Corollary 7.15
special conformal	conformal Killing fields of degree 2	$+2$	$w_O+1,\ w_O,\ w_O-1$	moved	Corollary 7.16

Components along the ten named subspaces of $\mathfrak{psl}(4|4)$; in the names of [SW, (8)] the rows are P, M and $\bar{M}$, R, Δ, Q, $\bar{Q}$, S, $\bar{S}$, K, and the ad_Δ eigenvalues are those of [SW, (17)]. The column "regulator" records the action on $P(\varepsilon,\infty)$: covariant (Proposition 7.6), the infinitesimal change of scale (Proposition 7.10), or moved (Propositions 7.10 and 7.11). For the moving generators the output is a local functional of the listed weight and of ghost degree $|O|$, colour singlet, equivariant in (x_1, O).

7.5 The road ahead

Step 1 has produced the arity-zero part of the one-loop theory: $I^{CT}[\varepsilon]$ through quartic order, the quantum master equation in the sector $(c, A|_V)$ with the finite counterterm

$$\frac{27}{16}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$$

of Chapter 4, the local $O(\varepsilon)$ of the letters and scalar bilinears, and their dilatation through the scaling covariance. The next steps are organised by one identity. For an even functional J with values in $C^\bullet(\mathfrak{psl}(4|4))$, the UV-boundary identity of Chapter 3, applied to $e^{J/\hbar}$, shows that $(d_{CE}+Q)W(P(\varepsilon,\infty),J)$ is the sum of the connected graphs with edges $P(\varepsilon,\infty)$, one vertex labelled by

$$(d_{CE}+Q)J + \tfrac{1}{2}\{J,J\}_\varepsilon + \hbar\Delta_\varepsilon J, \tag{7.3}$$

and every other vertex labelled by J. For $J = I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$, the classical equation $(d_{CE}+Q)I_{\mathfrak{psl}(4|4)} + \frac{1}{2}\{I_{\mathfrak{psl}(4|4)}, I_{\mathfrak{psl}(4|4)}\} = 0$ [ESW, Rem 1.29] turns (7.3) at one loop into

$$\begin{aligned}\tfrac{1}{2}\Big(\{I_{\mathfrak{psl}(4|4)}, I_{\mathfrak{psl}(4|4)}\}_\varepsilon - \{I_{\mathfrak{psl}(4|4)}, I_{\mathfrak{psl}(4|4)}\}\Big) + \hbar\Delta_\varepsilon I_{\mathfrak{psl}(4|4)}\\ - (d_{CE}+Q)I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] - \{I_{\mathfrak{psl}(4|4)}, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]\}_\varepsilon,\end{aligned} \tag{7.4}$$

whose first term joins two vertices by a special edge carrying $K_\varepsilon - K_0$ and whose second is a K_ε self-contraction. The $\varepsilon \to 0$ limit of the graphs with one vertex (7.4) is the flow to scale ∞ of the one-loop obstruction to the quantum master equation, an effective defect vertex. On a compact manifold with a finite upper scale the obstruction is a local functional, closed for $d_{CE} + Q + \{I_{\mathfrak{psl}(4|4)}, -\}$, and a finite local counterterm removes it exactly when it is exact [C07, Prop 12.0.3, Lemma 12.0.4; CG2, §13.3.1]; on $\mathbb{R}^4$ with upper scale ∞ these properties are to be proved in Step 2. In the sector $(c, A|_V)$ of Chapter 4 the limit is itself local.

7.5.1 Step 2: equivariant counterterms and the full one-loop master equation

Objects.

- The cubic sector of the arity-zero obstruction, with external legs $(c, A|_V, A|_V)$, $(c, A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, (c, λ, λ) and $(c, c, A^*|_V)$, for the interaction $I - I^{CT}[\varepsilon]$ minus the finite counterterm above; then a local functional cubic in the fields whose image under Q equals it, or the class it defines; then the sectors with more legs.

- For each generator x_1 (translations, $\mathfrak{so}(V)$, Δ, special conformal, $\mathfrak{so}(W_{\mathbb{R}}^\perp)$, Σ, Σ^*): the arity-one counterterm along x_1, the singular part of the one-loop graphs with one vertex $S^{(1)}(x_1)$; and the arity-one obstruction along x_1, the limit of the graphs with one vertex given by the arity-one part of (7.4),
$$\begin{aligned}&\{I, S^{(1)}(x_1)\}_\varepsilon - \{I, S^{(1)}(x_1)\} + \hbar\Delta_\varepsilon S^{(1)}(x_1)\\ &\quad - Q\, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](x_1) - \{I, I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](x_1)\}_\varepsilon - \{S^{(1)}(x_1), I^{CT}[\varepsilon]\}_\varepsilon.\end{aligned}$$
  For translations, rotations and $\mathfrak{so}(W_{\mathbb{R}}^\perp)$ all these graphs are covariant (Proposition 7.6), and their evaluation realises the choice of Remark 7.9; for the special conformal fields the large-distance convergence of Remark 7.12 is part of the computation.

- The dilatation component $[(d_{CE} + Q)O[\infty]](\Delta)$ computed directly at the UV boundary, for the scalar bilinears, and its identification with the result of the scaling covariance of Chapter 6, including the trees built from $(Q + \{I, -\})O$.

- The cocycle condition $Q\,O[\infty] = 0$ at order $\hbar$ for the letters and the scalar bilinears (Remark 5.6): the graphs with a special edge $K_\varepsilon - K_0$ between I and O, the vertices $-Q\,\partial_\delta I^{CT}[\varepsilon] - \{I, \partial_\delta I^{CT}[\varepsilon]\}_\varepsilon$ and $-\{I^{CT}[\varepsilon], O\}_\varepsilon$, the trees of O with the action defect; and the finite counterterms of weight w_O that it requires.

- Finite counterterms in $C^\bullet(\mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E}))$ removing the whole one-loop obstruction, or the precise class that obstructs them.

Graphs. In the sector $(c, A|_V)$ the five one-loop graphs with one special edge $K_\varepsilon - K_0$ drawn in Chapter 4 (a bubble with an $A|_V$, an $A|_{W_{\mathbb{R}}^\perp}$, a λ and two c–A^* lines), together with the vertex $-QI^{CT}[\varepsilon]$, give the obstruction of Chapter 4. The cubic sector uses the same vertices with one more external leg: one-loop graphs with three external legs, one special edge $K_\varepsilon - K_0$ from the first term of (7.4) and the other edges $P(\varepsilon, \infty)$ (triangles with three cubic vertices, bubbles with one quartic vertex); the trees of $-\{I, I^{CT}[\varepsilon]\}_\varepsilon$ and of minus the bracket of I with the finite counterterm, with one K_ε edge joining a cubic vertex of I to the quadratic counterterm; and the vertex $-QI^{CT}[\varepsilon]$ restricted to the sector. The direct dilatation component of a scalar bilinear uses the twenty-five graphs (a)–(y) drawn in Chapter 6, with one O vertex, one $S^{(1)}(\Delta)$ vertex and one K_ε edge: twenty-one one-loop graphs (a)–(u), with the K_ε edge on the cycle and the other edges $P(\varepsilon, \infty)$ through cubic and quartic vertices of I (two of them, (t) and (u), with a c–A^* line ending at $S^{(1)}(\Delta)$), and four trees (v)–(y) with the vertex $\partial_\delta I^{CT}[\varepsilon]$; to them are added the graphs with the component of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ along Δ and the insertion of the defect (7.4) along Δ.

7.5.2 Step 3: supersymmetry and special supersymmetry of letters and scalar bilinears

Objects.

- For O a letter or a scalar bilinear and $x_1 = Q_1 \in \Sigma$ or $x_1 = Q_2 \in \Sigma^*$: the local functional representing $[(d_{CE} + Q)O[\infty]](x_1)$. By Corollaries 7.14 and 7.15 this is one number for the traceless bilinear (the multiple of the classical structure) and three numbers for the trace bilinear (on the trace the two classical-type functionals are multiples of $\int \omega\,(\rho(A|_{W^\perp_{\mathbb{R}}})Q_1, \lambda)$, and the gauge-variant and BRST-type functionals are added), for each of Q^i_α, $\bar{Q}_{\dot\alpha i}$, S^α_i, $\bar{S}^{\dot\alpha i}$.

- The arity-two components on pairs of odd generators, which contain $S^{(2)}$ [ESW, (34)] and express the closure of the quantum supersymmetry on O through $(d_{CE} + Q)^2 O[\infty] = 0$, as in Remark 7.17.

- The Konishi multiplet: the components of the trace bilinear along Σ and Σ^* and the coefficients of its gauge-variant and BRST-type structures.

Graphs. One-loop graphs with one $O(\varepsilon)$ vertex, one vertex of $S^{(1)}(x_1)$ and one K_ε edge, the other edges $P(\varepsilon, \infty)$. The vertices of $S^{(1)}(Q_1)$ are the bilinear vertices $-(\Gamma(Q_1, \lambda), A^*)$ and $\frac{1}{2}(\rho(dA)Q_1, \lambda^*)$ and the cubic vertex $\frac{1}{2}(\rho(\frac{1}{2}[A \wedge A])Q_1, \lambda^*)$; for Q_2 the spinor is $\rho(x)Q_2$ and the vertex $-(\rho(A|_{W^\perp_{\mathbb{R}}})Q_2, \lambda^*)$ is added. Trees with the vertex $\partial_\delta I^{CT}[\varepsilon]$ and with the arity-one counterterm along x_1 of Step 2 complete the list. For a scalar bilinear the external legs are those of Corollary 7.14: (A, λ) and (c, λ^*).

7.5.3 Step 4: all bilinears and covariant jets

Objects. For every gauge-invariant local functional bilinear in the letters $\iota_w A$, F_A, λ and their covariant derivatives d_A applied repeatedly: $\partial_\delta I^{CT}[\varepsilon]$ and $O(\varepsilon)$, a mixing matrix among operators of equal Weyl weight, ghost degree and representation of $\mathfrak{so}(V) \oplus \mathfrak{so}(W^\perp_{\mathbb{R}})$ (Corollary 7.7); the dilatation component by the scaling covariance of Chapter 6; the other components with the selection rules of this chapter.

Graphs. One-loop one-particle-irreducible graphs with one O vertex whose superficial degree

$$w_O - \sum_{\text{external legs}} (\text{Weyl weight of the leg})$$

is non-negative, the external legs ranging over A, λ, c and the antifields A^*, λ^*, c^* with the weights of Lemma 7.5.

7.5.4 Step 5: two loops

Objects. The counterterms at order $\hbar^2$, the singular parts of the order-$\hbar^2$ terms of $W\big(P(\varepsilon, \infty), J\big)$, with J equal to $I - I^{CT}[\varepsilon]$ minus the finite counterterms of Step 2, in the order of [C07, Thm B]; $\partial_\delta I^{CT}[\varepsilon]$ and $O(\varepsilon)$ at order $\hbar^2$; the two-loop obstruction, the order-$\hbar^2$ part of the graphs with one vertex (7.3), the UV-boundary identity behind (7.3) holding to all orders in $\hbar$; the dilatation at order $\hbar^2$, read from the $\log\varepsilon$ coefficients through the scaling covariance of Chapter 6. The singular ε-functions now include $(\log\varepsilon)^2 = \int_\varepsilon^1 \int_\varepsilon^1 (t_1 t_2)^{-1} dt_1 dt_2$, an element of the space of [C07, Def 7.1.1].

Graphs. Two-loop graphs with vertices of I and edges $P(\varepsilon, \infty)$; one-loop graphs with one vertex of $I^{CT}[\varepsilon]$ or of a finite counterterm, both of order $\hbar$, which subtract the subdivergences; trees with one vertex of order $\hbar^2$; and, for the obstruction, the same graphs with one vertex (7.3).